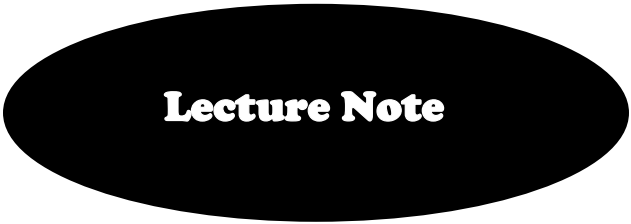


PAEDIATRICS AND CHILD HEALTH

FOR

Health Science students



Lecture Note

Jimma University

Paediatrics and Child Health

For

Health Science students

Lecture Note

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List of Acronyms

ABC	Airway, breathing and circulation
AC	Activated charcoal
Ach	Acetylcholine
ANC	Antenatal care
AFP	Acute flaccid paralysis
AGA	Appropriate for gestational age
AGN	Acute Glomerulonephritis
ARF	Acute renal failure
ARI	Acute respiratory infections
CHD	Congenital heart disease
CL	Cutaneous leishmaniasis
CMV	Cytomegalovirus
CNS	Central nervous system
CPS	Complex partial seizures
CRS	Congenital rubella syndrome
EPI	Expanded programme on immunization
GA	Gestational age
GFR	Glomerular filtration rate
HC	Head circumference
HSV	Herpes simplex virus
HMD	Hyaline membrane disease
Hx	History
IMCI	Integrated Management of Childhood Illnesses
ITP	Idiopathic thrombocytopenia purpura
IUGR	Intrauterine growth retardation (Infants weighing less than tenth percentile for gestational age)
LBW	Low birth weight
LGA	Large for gestational age
LP	Lumbar puncture
LRTI	Lower respiratory tract infection
MUAC	Mid-upper-arm-circumference

NCHS	National Centres for Health Statistics
NAC	N – acetylcysteine
NAPQI	N-acetyl p-benzoquinonemine
OPs	Organophosphates
PCP	Pneumocystis carinii pneumonia ()
PEM	Protein energy malnutrition
PMTCT	Prevention of mother to child transmission of HIV
PPV	Positive pressure ventilation
PTT	Partial Thromboplastin Time
PT	Prothrombin Time
RDA	Recommended daily allowance
RES	Reticulo endothelial system
RBP	Retinal binding protein
RPR	Rapid plasma reagin
RSV	Respiratory syncytial virus
RX	Treatment
SGA	Small for gestational age
SMR	Sexual maturity rating
SPS	Simple partial seizures
TT	Thrombin time
UTI	Urinary Tract Infection
VAD	Vitamin A deficiency
VDRL	Venereal Disease Research Laboratory
VLBW	Very Low birth weight
WBI	Whole bowel irrigation
WHO	World Health Organization
VL	Visceral leishmaniasis
VZV	Varicella zoster virus
VZIG	Varicella zoster immunoglobulin

PART ONE

General Paediatrics and Neonatology

CHAPTER ONE

GENERAL PAEDIATRICS

1.1. CHILD HEALTH IN ETHIOPIA

Tsinuel Girma

Objectives

At the end of this chapter, the student will be able to describe:

- The magnitude of childhood morbidity and mortality in Ethiopia
- Common causes of childhood morbidities and mortalities
- Child health interventions set in place at national level

The world population growth rate is very high compared with the socioeconomic development, particularly so in developing countries where 80% of the world's child deaths occur. This population is largely unhealthy. Each year 14 million children under five years old die and 2000 children die every hour. Child deaths in low income countries comprise 98% of the world's, of which 60% are preventable.

Main causes of morbidity are acute respiratory infections, diarrhoeal diseases, malaria and neonatal problems like tetanus. Emerging health problems such as substance abuse, street children and child labour make the situation worse.

Ethiopia is similar to many other low income countries in having a young population: those under the age of 15 years make up 44% of the population. Basic health indicators show unacceptably high figures. The infant mortality rate is 96.8 per 1000 live births. Low birth weight deliveries are 16%.

Neonatal deaths, especially in the first week of life, contribute a large fraction of infant deaths. Most of these deaths are easily preventable. But still 49 out of 1000 live births die in Ethiopia.

Data from health institutions and health surveys show the major causes of morbidity and mortality are PEM, anaemia and vitamin A deficiency, acute respiratory infections (ARI), malaria and vaccine preventable diseases.

Almost half of the children under five years of age are stunted and underweight, 52% and 47% respectively. The rate of wasted children (11%) is also very high. Surveys in different region show the rate of Bitot's spot and night blindness to be 5% and 7% respectively. The rate of breast-feeding is high especially in rural areas but in 60% of cases colostrum is avoided while 70% neonates get breast-feeding within 24 hours. ARI causes most deaths because in many of the cases treatment is not sought or delayed.

HIV/AIDS has had a major impact on maternal and child health survival. Each year 400,000 children are newly infected. The 2001 estimate for HIV/AIDS infection in Ethiopia showed 200,000 children living with the infection and 1.2 million orphans. A recent (2003) study in Jimma showed 9.8% sero-positive pregnant mothers during routine antenatal care (ANC). The national programme for the prevention of mother to child transmission of HIV (PMTCT) is at an early stage.

Various health interventions have been set in place to address the most important child health problems in Ethiopia. Some of these are EPI (expanded programme on immunization) since 1980, ARI case management (1990) and control of diarrhoeal disease (1983). EPI was launched with the intention of increasing the coverage by 10% annually and reach 100% of the people in 1990. The objective was not met for various reasons: the poor health infrastructure, the low levels of trained man power, the high turn over of staff and a decline in donor funding. In 2001 the national coverage for BCG, DPT3, and measles was 63%, 51% and 40% respectively.

The new strategy of integrating most aspects of child health care is the basic principle of IMCI (Integrated Management of Childhood Illnesses), which was launched since 1998 in three selected regions. The components of IMCI are improving health worker skills, improving health system and improving family and community practices.

Further Reading

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Integrated Management of Childhood Illnesses. (IMCI), Ministry of Health, Ethiopia. 1999.

Demographic and Health Survey. 2000.

1.2. IMMUNIZATION

Tsinuel Girma

Objectives

At the end of the topic, the student will be able to describe the:

- Basic mechanism of immunization
- Nature of vaccine antigens
- Route of vaccination and dose
- Major contraindications for vaccination
- Major side effects of vaccination
- EPI for Ethiopia
- Catch-up vaccination

Introduction

Infectious diseases lead to high mortality and morbidity in paediatrics patients. They range from mild self-limiting diseases to those that result in long-term disability like blindness, deafness and others. Maintaining good nutrition, environmental and personal hygiene, access to safe water supply and better living environments help to reduce the magnitude of communicable diseases. Immunization, pioneered 200 years ago by Jenner, represents another remarkably successful and very cost effective means of control. More than 50 immunobiologic products are available for clinical use. Sustainable financial support, well-organized infrastructure and epidemiology of disease will determine the type of vaccine used in given country.

The expanded programme on immunization (EPI), launched in Ethiopia in 1980 is targeted 6 diseases with the aim of universal coverage by 1990.. Logistics, access and management constraints contribute to the current coverage of less than 30%.

Definition

Vaccination is the administration of any vaccine or toxoid for prevention of disease.

Immunization is the process of inducing immunity artificially.

It can be active or passive

- Active immunization involves stimulating the immune system to produce antibodies and a cellular immune response against infectious agent through the use of a vaccine or toxoid.
- Passive immunization provides temporary protection through the administration of exogenous antibody or maternal antibody transfer to the foetus.

Determinants of the Immune response

The nature and magnitude of the response to a vaccine or toxoid depend on the following factors:

1. Age – presence of high concentration of maternal antibody and immature response to some vaccines in the first four months of life impair immunization. The measles vaccine is given at 9 months of age to reduce this effect.
2. Route of administration.
 - Vaccines given orally induce mucosal secretory IgA
e.g. OPV vaccine
 - Using an improper route to administer the vaccine may reduce the immune response e.g. where BCG is administered IM rather than intradermal
3. Nature of vaccine
 - Live attenuated vaccines induce immunity with a single dose which lasts longer than inactivated ones
4. Genetic
 - Individuals genetically vary in their ability to respond to the same vaccine.

5. Potency

Ensuring the potency of a vaccine, especially live attenuated, requires keeping the cold chain. This helps to determine the quality of EPI program (see Table 2.2).

Biologic nature of immunizing agents

Active immunization uses vaccines with different physical characteristics. They can be grouped based on their biologic nature (See Table 2.1.)

Table 1 The main antigenic preparations

Type of antigen		Vaccine example
Living organism	Natural	Small pox vaccine.
	Attenuated *	Polio (sabin) OPV, Measles, Mumps, Rubella, BCG
Intact but non-living organisms	Virus	Polio (Salk)
	Bacteria	Pertussis
Sub cellular	Capsular disaccharid	Meningococcal
Fragments	Surface antigen	Hepatitis B.
Toxoids		Tetanus, diphtheria

*Live attenuated vaccine (diminished virulence retaining desired antigens)

Live attenuated vaccines, particularly viral ones like measles, confer life long protection after a single immunizing dose, since they closely simulate natural infection and contain the greatest number of microbial antigen. Their drawbacks are:

- Reversion to wild type can lead to disease
- They can cause severe disease in immunocompromised children
- Some people exhibit hypersensitivity to viral antigens.

Vaccines for routine (EPI) use in children

Preparations

The vaccines targeted against the 6 major diseases are presented in different forms with a need to keep them in specified environmental conditions, particularly the temperature ranges.

Table 2: Recommended Storage Temperature of EPI

	Type of vaccine	Storage temperature
Most sensitive to heat	Oral polio	-15°C to – 25°C
	Measles (freeze dried)	-15°C to – 25°C
	DPT	2°C to 8°C
Least sensitive to heat	BCG (freeze dried)	2°C to 8°C
	Tetanus toxoid	2°C to 8°C

Vaccines

Administration

Proper technique of administering vaccines is needed to achieve acceptable immune response and avoid problems (see Table 2.3).

Injection safety procedures in immunization

Injections are safe when they are not a health threat to the recipient, health worker, the community or the environment.

Factors that contribute to unsafe injection are:

- Reuse of a single syringe and needle
- Inadequate supply of injection material
- Inadequate disposal of used syringes and needles

Guidelines on injection safety include:

- Use of a single sterile syringe and needle for each dose of vaccine

- Use of AD syringes preferably
- Provision of safety boxes for disposal of used syringes
- Incineration of full safety boxes
- Removal and proper burial of residue from the incinerator
- No disposal of used syringe, needle or full safety boxes in open garbage or dumped randomly.

Schedule

The strategies used to give vaccines can be adapted in three ways depending on the local situation:

1. Primary schedule for infants below one year in static health services:

Birth: OPV; BCG

6 weeks of age: OPV, DPT; BCG (if not given at birth)

10 weeks of age: OPV, DPT

14 weeks of age: OPV, DPT

9 months of age: Measles, Vit A. **

** Children less than 6 months –not recommended

- Children 6-11months -100, 000iu
- Children 12-59 months - 200,000iu
- Record on child's immunization card

Table 3 - Administrations of Vaccines

VACCINE	DOSE	ROUTE OF ADMINISTRATION	SITE OF ADMINISTRATION
BCG	Infants: 0.05 ml 0.1ml for children >1yr	Intradermal	Right deltoid region of the arm
DTP	0.5ml	Intramuscular	Upper, outer portion of the thigh
Polio	2 drops, or	Oral	Mouth
Measles	0.5 ml	Subcutaneous	Outer ,upper part of left arm
Tetanus Toxoid	0.5 ml	Intramuscular	Deltoid region of the upper arm

For children below 5yr but not vaccinated in the first one year (catch-up vaccination) the following should be administered:

<u>Visit</u>	<u>Vaccination</u>
1 st visit	DPT OPV Mantoux test, BCG if test negative in 3 days
2 nd visit (1 month after 1 st visit)	DPT OPV
3 rd visit (1month after 2 nd visit)	DPT OPV Measles, if not exposed
4 th visit (1 year later)	DPT OPV
5 th visit (4 years later)	(adult type)

2. Outreach (Mobile schedule for remote areas): This service gives coverage for people living within 20 km radius of the health facility.
3. Mass campaigns: National immunization days (NIDs) for polio (currently active) and measles.

Side effects of vaccines

BCG:

- Kochs phenomenon – self limiting acute inflammatory reaction four days after vaccination.
- Indolent ulcer – ulcer persisting 12 wk after vaccination or ulcer more than 10 mm, mainly resulting from deep injection or secondary infection.
- Deep abscess – abscess at site of injection or draining lymph nodes due to subcutaneous or deep injections.
- Disseminated disease with BCG – 1 per million vaccines results in BCG causing active disease especially in immunosuppressed children.

OPV: Paralytic polio from vaccine strain poliovirus

DPT: The major side effect is from the pertussis component

- Superficial injection can cause injection site abscess a week later.
- Encephalopathy (0.3 – 3/100,000 vaccines)
- Convulsion (0.3 – 90/100,000)
- Shock like state or collapse (hypotonic – hypo responsiveness episode) (0.5 – 30/100,000)
- Permanent brain damage (0.2 – 0.6/100,000)

Measles: May have mild reaction likes low grade fever irritability and allergic reactions.

Major side effects include (per 100,000 vaccinations):

- Encephalopathy (0.1)
- Convulsions (0.02 – 100)
- Sub acute sclerosing panencephalitis (0.01 – 0.05)

Contraindication and precautions

There are circumstances in which there should be a judgment of the risk and benefit of administering a vaccine.

Contraindications: there are circumstances in which a vaccine, either first dose or repeat, should not be given. The general contraindications for all vaccines include:

- Anaphylactic reaction
- Moderate to severe illness
- Live-attenuated vaccines for severely immunosuppressed patient (exception is measles) *Not contraindications*
- Moderate fever after prior vaccine dose
- Moderate local reaction after injectable vaccine
- Mild acute illness
- Prematurity (same dose as for full-term infants)
- Severe malnutrition (rather strong indication)
- Penicillin allergy (personal or family)

The contraindications for specific vaccines include:

BCG : Symptomatic AIDS

OPV:

- Severe immune suppression
- Diarrhea is not contraindication .Give OPV without registering it and repeat OPV 4 weeks later.

DPT:

- Encephalopathy within 7 days of Previous dose not attributable to another cause
- Seizures if poorly controlled or new onset, defer until control

The following precautions are necessary:

- Temperature $\geq 40.5^{\circ}\text{C}$ within 48 hr. of prior dose
- Collapses or shock like state
- Convulsion within 3 days after period close
- Persistent or inconsolable cry ($\geq 3\text{hr}$), within 48 hr of prior dose
- Gillian – Barrie syndrome within 6 wk after prior dose

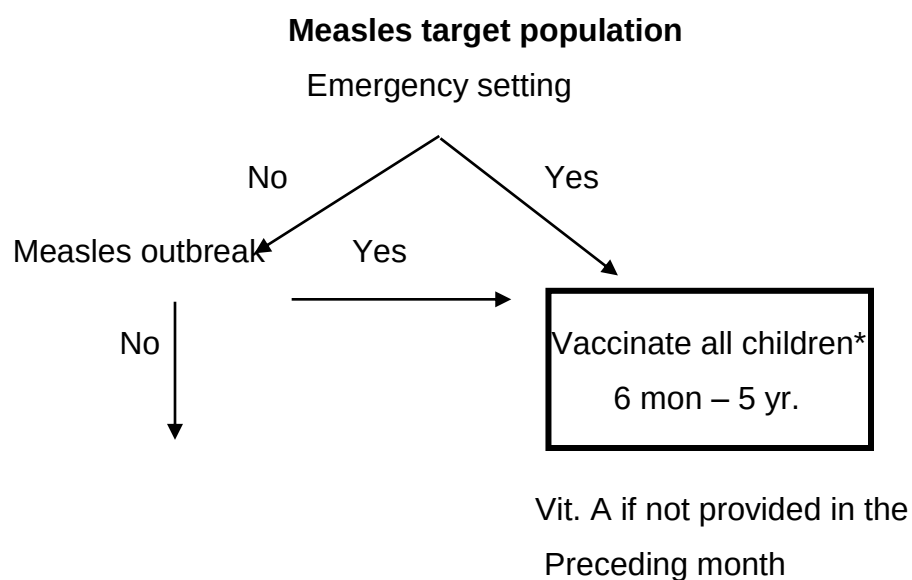
The following are not contraindications

- Family history of seizure

- Well controlled epilepsy
- History of febrile convulsion (personal or family). Give paracetamol for 24 hr after vaccination
- Measles: there are virtually no absolute contraindications

Vaccines in special circumstances

Disease outbreaks in normal situations or prevention of epidemics in emergency circumstances such as displacement or war, call for the use of different a schedule and vaccine antigens. the most common diseases are measles and meningococcal meningitis. Decisions about measles vaccination may be taken as follows:



* If limited vaccine supply, prioritize those who are undernourished or sick children

Meningococcal vaccine

The serotype A&C can cause major outbreaks in developing countries. The target group (population) for vaccination in epidemic situations includes:

- Classically persons aged from 2 to 30 yrs.
- If the attack rate is high among > 30 yrs, vaccinate entire population

Use the bivalent A & C vaccine within the meningitis belt

Further Reading

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Disease of Children in the Sub Tropics and Tropics, A/e, Edward Arnold, 1991

Manual for routine EPI, MOH, Ethiopia, 2003 (draft)

Study Exercises

1. Explain the concepts of active and passive immunization and compare the two
2. Select the live attenuated vaccine
 - a) BCG c) Diphtheria e) Polio (Salk)
 - b) Pertussis d) Tetanus
3. List conditions which are not contraindications for vaccination.

1.3. GROWTH AND DEVELOPMENT

Goitom G/Yesus

Objectives

By the end of this chapter, the student will be able to:

- Define growth and development
- Describe how to assess growth and development
- Describe the normal patterns of growth and development at different ages
- Describe factors affecting growth and development

Introduction

Growth and development in physical, intellectual, emotional and social terms are the essential biological characteristics of childhood. A child is not a small adult, but is a developing human being who changes from a new born baby through infancy, toddlerhood and early school years into an adolescent and finally, a mature adult. One needs to understand growth and development in three ways:

An understanding of the normative pattern of physical growth and the emergency of motor cognitive and emotional competence allows one to monitor children's progress.

Understanding of how biological and environmental forces interact to shape development allows one to target factors that increase or decrease risks.

Understanding of how parents conceptualize development facilitates guidance and intervention.

Definitions

- *Growth*: Is the process of growing – the increase in size and development of a living organism from a simple to a more complex form or from its earliest stage of being to maturity.
- *Development*: implies an increase in skill and complexity of function.

Factors Affecting growth

Genetic factors

The growth of a child is a result of complex interactions of genetic and environmental factors. As a result of multiple genetic differences, certain people are taller than others: for example, pigmies are one of the world's smallest races and appear to have genetically determined inability to respond to somatomedin; twin studies have shown that body size, shape and patterns of growth are strongly influenced by genetic factors.

Neural control of growth

It has been suggested that there may be a growth center in brain that keeps the child in his or her genetically determined growth curve: thus, if a child deviates from a growth pattern for a period of time because of malnutrition or illness, a period of accelerated or 'catch-up' growth brings him or her back to the 'predetermined' curve, implying some sort of central control mechanism.

Hormonal influence

Most of the endocrine glands influence growth significantly either by promoting protein synthesis, regulating substrate supply, or enhancing the effect of other hormones on specific organs. Hormones with a great influence on growth are: growth hormone, somatomedines, insulin, thyroid hormone, and sex hormones

Nutrition

Adequate food to provide substrate for energy and synthesis of proteins is essential for normal growth. Observations of growth in humans during famine and in economically disadvantaged countries have clearly shown a lack normal growth.

Environmental and social factors

The home, neighborhood, and school environments have profound effect on the child's psychosocial development. Socio-economic status also significantly affects the growth of a child: thus, children in top socioeconomic groups are taller than children of lower socioeconomic class.

Assessment of Growth and Development

Disease in childhood affects growth and development. Measurement of growth and development is therefore an essential skill to monitor the health of the individual child and the community at large.

Assessment of Growth

Growth assessment is an essential component of paediatric health surveillance because almost any problem with in the physiologic, inter-personal and social domain can adversely affect growth.

Physical growth varies in rate and pattern in different tissues and different regions of the body. In practice, several measures such as body weight, height or length, head circumference, skin fold thickness and mid-upper arm circumference may be used.

The most powerful tool in growth assessment is the *growth chart* (see Annex 1). The growth chart provides most of the information needed to assess growth. The standard charts are based on data collected from 1963 to 1975 by the National Centres for Health Statistics (NCHS) on weight for age, height for age, head circumference for age, and weight for height.

Separate charts are provided for boys and girls. Each chart is composed of seven percentile curves, which indicate the percentages of children at a given age on the X-axis whose measured values fall below the corresponding value on the Y-axis. The

normal values for age are represented between the fifth and ninety-fifth percentiles, and the fiftieth percentile represents the median (standard) value.

In children under 5 years old an additional chart (Harvard standard) is used to measure weight for age as percentages of the median or standard value in the NCHS curve.

Weight for age

Weight is usually measured using beam balance or spring balance for small children and a standing scale for older children. When weighing, children should be undressed or wear light clothing. Children falling below the 5th centile of NCHS or below the 80% of Harvard Standard are labelled as malnourished.

Height or length for age

Height is measured in the upright position for children more than three years of age, while for those younger than three years, length should be taken. Length in children less than three years should be taken by two examiners. One should position the child (with the child supine on a measuring board). Then the child's head should be held with the ear/eye plane vertical, the ankles gently pulled to stretch the child, and the feet turned up vertically.

The older children should stand straight against a wall or the erect measuring scale; and their head, shoulder, buttocks and heels should touch the wall or scale. Then, they should take a deep breath to relax the shoulders and with a flat object such as a book, the upper level of their head should be marked against the scale.

A height for age measure shows linear skeletal growth. A height for age that is below normal shows stunting and indicates chronic states of malnutrition.

Weight for height

Getting an accurate age may be a problem in countries like Ethiopia, where the majority of people are from rural areas and illiterate. Weight for height measurements can overcome this problem. However, it only identifies children with acute malnutrition, whose height has not been as affected as their weight. Those with weight for height falling below the 5th percentile of NCHS are labelled as stunted.

Head circumference for age

Head circumference is measured by crossed tape technique. Placing a measuring tape just above the brow anteriorly and the occiput, measure the occipito-frontal circumference. Values below the -3 standard deviation (SD) are labelled as microcephally and above +3 SD as macrocephally.

Mid-upper arm circumference

The circumference of left upper arm is taken at mid point while the arm is hanging by the side. This is used as an age independent criterion for detecting malnutrition in children of age one to five years. A mid upper arm circumference of less than 12.5cm indicates malnutrition and when less than 11cm, severe malnutrition.

Developmental Assessment

Assessment of development helps to identify children with developmental delay. For this purpose the following developmental assessments may be used.

Gross motor development

Denotes gross muscular activity and neuro-development including posture, independent nobilities, progresses from head control when the child is sitting alone to standing, walking and running.

Manipulation (fine motor development)

Fine motor development denotes the ability of the child to reach for, grasp and manipulate objects. It follows the loss of the primitive grasp reflex, and progressively becomes more sophisticated through infancy and childhood.

Cognition and social skill

These include social smile, watching a mirror, waving goodbye, general alertness, and curiosity about the surrounding.

Language

Development of speech is closely linked with general understanding.

General principles

- The sequence of each pathway of development does not vary, although the rate of development is variable
- The rate of development along different pathways is variable
- Motor development proceeds in a *cephalo- caudal* direction (e.g. Hand control develops before trunk control)
- Some primitive reflexes must be lost before voluntary movement occurs (e.g. primitive grasp must disappear before voluntary grasp can develop)

Growth and developmental milestones at different ages

Infancy (0-12 months)

This is a period of rapid physical growth. Normally this is manifested as follows:

- *Physical growth:* babies weigh an average of 10 kg at 12 months, length increases by 25 cm, and head circumference increases by 12 cm.
- *Gross motor:* at 3 months babies need their head supported when pulled to sit, at 6 months sit without support, at 8 months crawl, and at around 12 months they can stand without support.
- *Fine motor:* at 3-4 months babies can grasp objects, at 5-6 months they can transfer objects from hand to hand, at 9 months they can achieve a thumb-finger (pincer) grasp, and at around 12 months they start to scribble.
- *Cognition and social skills:* at 6-8 weeks babies start to smile in response to faces or voices, at 4 months they laugh, at 6-8 months they develop stranger anxiety, and at 9-12 months they discover that objects are permanent.
- *Language:* during their first 6 months babies pronounce vowels; at 7 months they start to voice consonants, at 8-9 months they pronounce “papa” and “mama”, and at 12 months they can say a few words besides “papa” and “mama”.

Toddlerhood (12 - 24 months)

In this period the growth rate slows down compared to infancy. Normal development is as follows:

- *Physical growth:* toddlers gain an average weight of 2 kg, height of 12cm, and HC of 3cm.
- *Gross motor:* toddlers start to walk independently between the age of 12 to 15 months; at 18 months, they can run stiffly and walk upstairs with one hand held, at 24 months they can run well and walk up and down stairs one step at a time.
- *Fine motor:* at 15 month toddlers can make a line with crayon and insert pellet in a bottle; at 18 month they can imitate a vertical stroke, and at 24 months they can imitate a horizontal stroke.
- *Cognition and social skills:* at 15 month toddlers can indicate some desires by pointing; at 18 month they can feed themselves and they may complain when wet or soiled; at 24 months they can help to undress and listen to stories with pictures.

- *Language*: at 15 month toddlers start to speak in jargon; at 18 months they can say 10 to 15 words; at the end of 24 months they can say two word sentences (e.g. mama shoe)

Pre school years (2-5 years of age)

The normal pattern of development in the pre school years is as follows:

- *Physical growth*: at this time the average weight gain is 2 kg per year and height increases by 7 cm per year. By the end of three years all primary teeth have erupted.
- *Gross and fine motor*: at two and half years children can go upstairs alternating feet, at 3 years they can ride tricycles, at 4 years they can hop on one foot and throw a ball overhead, and at 5 years they can skip.
- *Cognition and Language*: language development occurs most rapidly between the ages of 2 - 5 years. Vocabulary increases from 50 - 100 to about 2000 words. Sentence structure advances and usually the number of words in a sentence equals the child's age in years. At two and half years most children know their full name and engage in pretend play, at 3 years they know their age and sex, count three objects correctly, wash their hands and put on shoes. At 4 years children can tell a story and play with several children; at five years they can name 4 colors, count 10 objects correctly, and dress and undress.

Table 4: Developmental milestones in the 1st yr of life

Milestone	Average age of Attainment (in months)	Developmental implication
Gross motor		
Head steady in sitting	2	Allows more visual intention
Pull to sit, no head lag	3	Muscle tone
Hands together in mid line	3	Self discovery
Sits without support	6	Increasing exploration
Rolls back to stomach	6.5	Truncal flexion, risk of falls
Walks alone	12	Exploration, control of proximity to parents
Runs	16	Supervision more difficult
Fine motor		
Grasps rattle	3.5	Object use
Reaches for object	4	Visual motor coordination
Transfers objects	5	Comparison of objects
Thumb -finger grasp	8	Able to explore small object
Scribbles	13	Visual-motor coordination

Communication and Language		
Smile in response to face, voice	1.7	More active social participation
Monosyllabic babble	6	Experiment sound
Follows one step command gesture	7	non verbal communication
Follows one step out gesture	10	Verbal receptive language
Speaks 1 st real word	12	Beginning of labeling
Speaks 4-6 word	15	Acquisition of object & personal name
Speaks 10-15 word	18	Acquisition of objects & personal name
Speaks two word sentences	19	Beginning of grammatization
Cognitive		
Stares momentarily at spot where object disappeared	2	Lack of objective permanence
Uncovers toy (after seeing i.e. hidden)	8	Object permanence
Egocentric pretend play	12	Beginning symbolic thoughts

School age (6-12 years)

Normal development in the school years is as follows:

- *Physical growth*: the average weight and height gain per year is 3-3.5kg and 6cm respectively. Head circumference increases by only 2-3 cm. Loss of primary teeth starts at 6 years of age.
- *Motor*: coordination and stamina increase progressively, and children are able to perform complex movements such as dancing.
- *Cognition and language*: children start thinking in concrete logical operations.

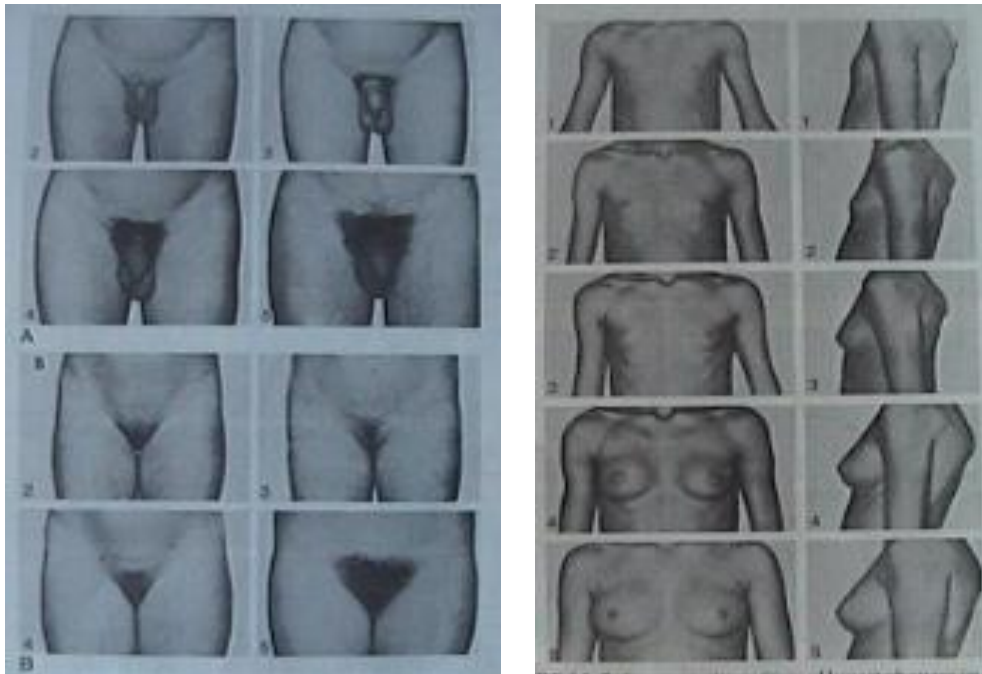
Adolescence

Children undergo rapid changes in body size, shape, physiology, and psychology and social functioning. Adolescence proceeds across three distinct periods: early, middle and late, each marked by a characteristic set of salient biologic, psychological and social issues. The common changes that occur during the early, middle, and late adolescence period are described in the following table.

Table 5: Changes in early, middle, and late adolescence

Variable	Early adolescence	Middle adolescence	Late adolescence
Age	10-13 yrs	14-16	17-20 & beyond
Somatic	2 ^o sex char, beginning of rapid growth	Height peak, acne & odor, menarce spermarche	Slower growth
Sexual	Sexual interest	Sex drive urges experimentation	Consolidation of sexual identity
Cognitive and moral	Concrete operation conventional morality	Abstract thought, self centered	Idealism; absolution
Family	Bids for independence	Continued struggle for acceptance of greater autonomy	Practical independences family remains secure base
Peers	Same-sex groups, conformity, cliques	Dating, peer groups less important	Intimacy, possibly commitment

The sequences of somatic and physiologic changes that occur can be used to assess developmental maturity according to Tanners 'Staging: Sexual Maturity Rating (SMR) as described in the following tables and figures.

Figure 1: Sexual changes in boys and girls during adolescence**Table 6: Classification of sex maturity stages in girls**

SMR Stage	Pubic Hair	Breasts
1.	Preadolescent	Preadolescent
2.	Sparse, lightly pigmented, straight, medial border of labia	Breast and papilla elevated as small mound; areola diameter increased
3.	Darker, beginning to curl, increased amount	Breast and areola enlarged, no contour separation
4.	Coarse, curly, abundant, but amount less than in adult	Areola and papilla form secondary mound
5.	Adult feminine triangle, spread to medial surface of thighs	Mature, nipple projects, areola part of general breast contour

Table 7: Classification of sex maturity stages in boys

SMR Stage	Pubic Hair	Penis	Testes
1.	None	Preadolescent	Preadolescent
2.	Scanty, long and, slightly pigmented	Slight enlargement	Enlarged scrotum, pink, texture altered
3.	Darker, beginning to curl, small amount	Longer	Larger
4.	Resembles adult type but less in quantity; coarse, curly,	Larger; glans and breadth increase in size	Larger, scrotum dark
5.	Adult distribution, spread to medial surface of thighs	Adult size	Adult size

Exercise

1. What does a centile curve describe in the standard growth charts?
2. How old is a normal child if he can crawl, say mama and has developed stranger anxiety?
3. Describe the physical characteristics of a boy and girl with SMR stages of 3 and 4 respectively?

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1.4. NUTRITION

Habtamu Fekadu

Objectives

By the end of this topic, the reader should be able to:

- Describe malnutrition, its magnitude, and consequences.
- Describe infant and young child feeding patterns in Ethiopia.
- Enumerate the benefits of breast feeding.
- Recite optimal breast feeding and complementary feeding behaviors.
- Know feeding options for an infant born to HIV positive mother.
- Describe methods to assess nutritional status.

Introduction

Nutrition is the provision of adequate energy and nutrients (in terms of amount, mix and timeliness) to the cells for them to perform their physiological function (of growth, reproduction, defense, repair, etc). Impaired nutritional status is called malnutrition.

There are two types of malnutrition:

1. *Undernutrition*

Undernutrition is due to inadequate nutrient in terms of quality and quantity. It takes two forms:

Protein energy malnutrition (PEM) arises from inadequate energy and protein supply to cells. Currently micronutrient deficiencies are incriminated as one important cause. It includes mild and moderate malnutrition, and the severe forms kwashiorkor, marasmus and marasmic-kwashiorkor.

Micronutrient deficiencies arise from inadequate vitamin and mineral intake. They include deficiency of vitamin A, iron, iodine, vitamin D, zinc etc. Deficiencies frequently occur in populations with a high infectious disease burden and monotonous, poor quality diets characterized by limited consumption of animal products and seasonal and periodic food insecurity.

Most micronutrient deficiencies and the majority of PEM (mild and moderate forms) are 'hard to see': i.e they don't show obvious signs. This is called the hidden hunger. That is why the enormous consequences of malnutrition are not often appreciated.

If an infant is not properly fed, he or she will grow poorly, be at increased risk of illness and death, have impaired intellectual development and be less productive in later life.

2. *Overnutrition (obesity)*

Obesity is due to excess intake of nutrients

Epidemiology

Undernutrition is prevalent in developing countries: one-third to one-half of all children are stunted and 20-40% are underweight in most countries of eastern, central, and southern Africa. One out of ten children suffer from acute malnutrition. Fifty four percent of the deaths in children under-five in developing countries are directly or indirectly attributable to malnutrition.

In Ethiopia, 51%, 47%, and 11% of under-five children are stunted, underweight and wasted respectively. Malnutrition is the major cause of child mortality and is responsible for 58 % of under-five mortality. The peak age for malnutrition is 12 to 24 months.

In Ethiopia, 27% of under 5 children suffer from sub clinical vitamin A deficiency and 17% of childhood deaths are attributable to vitamin A deficiency. Anemia and iodine deficiency are also very common cause of morbidity and mortality.

The cause of malnutrition is multifactorial. The commonest causes of malnutrition are suboptimal infant and young child feeding practices such as suboptimal breastfeeding practices, poor quality complementary foods (too few calories, protein and/or micronutrients), detrimental feeding practices (inappropriate timing of the introduction of complementary foods as well as infrequent, unsupervised feedings), and contamination of food and feeding utensils.. In Ethiopia infant feeding practices are suboptimal. According to 2000 DHS survey, only 54% of babies are exclusively breast fed from 0-6

months and 57% of infants 6-9 months are sub-optimally fed. 24% of infants deaths are due to poor breast feeding.

Infant and Young Child Feeding

Breast-feeding is an unequalled way of providing ideal food for the healthy growth and development of all infants.

Optimal feeding of infants and young children means exclusive breastfeeding from birth to about six months, followed by introduction of complementary foods drawn from the local diet at about six months. Breastfeeding should be sustained well into or beyond the second year of life, with increasing amounts of complementary foods.

Children are at highest risk of nutritional deficiency and growth retardation between the ages of 6 and 24 months. The prevalence of malnutrition usually peaks at 12 to 24 months of age. Micronutrient deficiency, such as anemia, vitamin A deficiency, also peaks at this age.

Complementary feeding means the provision of other foods or liquids along with breast milk. At six months of age, breastmilk alone does not provide sufficient calories and nutrients to sustain optimal growth and at this age infants are developmentally ready to take soft and semi-solid foods in addition to breastmilk.

Appropriate complementary feeding involves a combination of practices to continue on-demand breastfeeding and at the same time, improvements in the quantity and quality of foods, hygienic food preparation and active feeding of the child. The best complementary foods for children are foods that are:

- High in energy (calories) and protein
- Good in variety
- Easy to digest
- Pure and clean
- Easy and inexpensive to prepare

The advantages of breast-feeding

Breast milk is the best nutrition for infants and also young children. For the details of breast milk composition see Table 8.

The benefits of breastfeeding for the child include:

- It is the best natural food for babies. It contains sufficient amount and the right mixture of fats, sugars, proteins, minerals, and most vitamins for a growing baby. It changes in composition to meet baby's changing needs.
- It is easy to digest and nutrients are well absorbed.
- It contains enough fluids for the first 6 months of life (breast milk is 90 % water)
- It protects the baby from disease. Breast milk, esp colostrum, contains protective substances against the germs that cause diarrhoea respiratory and other infections. Breast-fed children have fewer diarrheal, respiratory, and ear infections.
- It is always clean and available 24 hours a day and requires no special preparation. It does not cost anything.
- It creates bonding between the mother and baby that leads to better psychomotor and social development.
- It protects against allergies because it contains no substance that could provoke allergic reactions.
- Because of its above benefits, breast milk promotes normal growth and development and saves lives of children.

The benefits of breast-feeding for the mother include:

- Breast-feeding provides an efficient contraceptive method during the first 6 months if breast-feeding is exclusive and frequent. This has health benefits for the mother and child.
- It reduces the risk of post-partum hemorrhage.
- It prevents breast engorgement and its consequences.
- It reduces the mother's workload since breast milk is available at any time and anywhere and it does not need any processing.

Table 8 Composition of Mature Human Milk and Cow's Milk

Component per Liter	Human Milk	Cow's Milk
Energy, kcal	680	680
Protein, g Whey/casein, %	10 72/28	33 18/82
Fat, g Medium chain and long chain triglycerides, %	39 2/98	38 8/92
Carbohydrate, g Lactose, %	72 100	47 100
Calcium, mg	280	1200
Phosphorus, mg	140	920
Magnesium, mg	35	120
Sodium, mg	180	480
Potassium, mg	525	1570
Chloride, mg	420	1020
Zinc, µg	1200	3500
Copper, µg	250	100
Iron, µg	400	460
Vitamin A, IU	2230	1000
Vitamin D, IU	22	24
Vitamin E, IU	2.3	0.9
Vitamin K, mg	2.1	4.9
Thiamine (vitamin B ₁), µg	210	300
Riboflavin (vitamin B ₂), µg	350	1750
Pyridoxine (vitamin B ₆), mg	93	470
Niacin, mg	1.5	0.8

Common problems of breast feeding

Retractile nipples: This condition should be detected during prenatal care. The solution is to stretch the tip of the breast often during pregnancy and later make a hole in the bra at nipple level.

Sore nipples: The causes of sore nipples are wrong attachment of the baby's mouth to the nipple and candidiasis. The solutions include:

- good attachment and frequent feeding.
- applying breastmilk on the infected zone
- not washing the breast too often.
- treating the candidiasis.

Mastitis and abscess: These conditions are caused by cracks, untreated ductal obstruction, or engorgement. Solutions include:

- Breast hygiene
- Treat the abscess i.e. antibiotic and drainage
- Warm compress before breast-feeding to let the milk flow
- Cold compress after breast feeding to relieve pain and reduce edema
- Treat plugged duct or engorgement if they are the causes.

Cleft lip or palate in the baby

It should still be possible to breastfeed if the health practitioner:

- Tells the mother breast milk is important.
- Tries to fill the cleft (with mother's finger or special tooth plaque (obturator), sometimes the mother's breast can fill the cleft) in order for the child to keep the nipple and areola in his/her mouth during feeds.
- Expresses milk and gives it to the child using a cup or teaspoon

Poor production of milk

It should still be possible to breastfeed if the mother and health practitioner:

- Increases the number and duration of feeds; the more the child sucks, the more the mother produces milk.
- Breast feeds day and night
- Properly positions and attaches the baby on the breast
- Builds the mother confidence that she can breast feed

Recommended Breastfeeding Practices for Infants and Young Children:

Optimal breastfeeding practices from 0 to 6 months

Optimal breastfeeding practices from 0 to 6 months include:

1. Initiating breastfeeding within one hour of birth. The first milk (colostrum) is of particular nutritional and health value to the infant because of its high content of proteins and fat-soluble vitamins and its anti-infective properties.
2. Do not give any prelacteal feeds such as water, other liquids, or ritual foods.
3. Establish good breastfeeding skills, i.e. proper positioning, attachment and effective feeding. Correct positioning means mother's back is supported whether sitting or lying down and the baby is held close to the mother, facing the breast with the baby's ear, shoulder, and hip in a straight line. Proper attachment signs are:
 - infant's mouth is open wide
 - the lower lip is turned outwards
 - the chin touches mother's breast
 - the entire nipple and a good portion of the areola (dark skin around the nipple) are in infant's mouth.
 - More areola shows above than below.

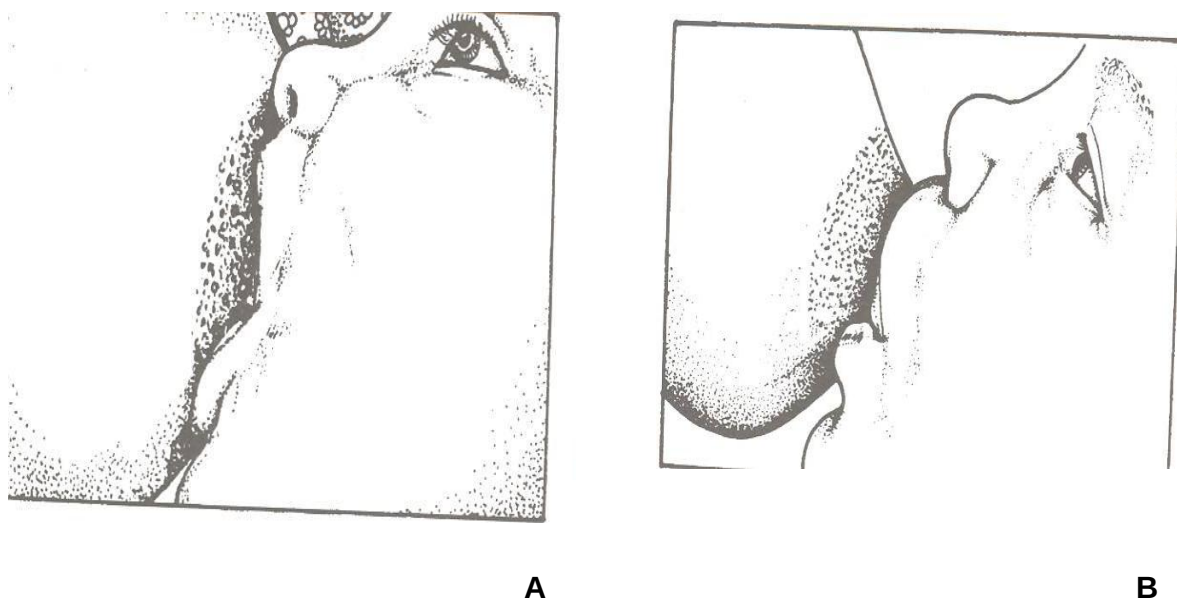


Figure 2 – well attached baby, notice – the **Figure 3** this baby is not well attached. Mouth is wide open, the lower lip turned outward **and** the chin touches mother's breast, a good portion of the areola are in the infant's mouth.

4. Exclusive breast feeding for the first 6 months. Do not give any fluids such as water, other liquids and foods other than breast milk before 6 months of age.
5. Frequent and on-demand breastfeeding, including night feeds (as often as infant wants). Feeding every 2 to 3 hours (8 to 12 times per 24 hours) or more. frequent suckling is important for milk production.
6. Offer second breast after infant empties the first. The infant receives both 'fore' milk (which has a high water content to quench the thirst) and 'hind' milk (which is rich in fat and nutrients).
7. Mother continues breastfeeding more often when the infant is ill.
8. A mother who will be away from her infant for an extended period expresses her breast milk and the caregiver feeds the milk from a cup.

Recommended complementary feeding practices

Recommended complementary feeding practices include:

1. Introducing complementary feeding at six months of age.
2. Continuing frequent and on-demand breastfeeding until 24 months. Breastmilk is still major source of energy and key source of fat, vitamin A, calcium, riboflavin.

3. Increasing food quantity and frequency as the child grows. A healthy breastfed child needs:
 - 2-3 meals per day at 6-8 months
 - 3-4 meals per day at 9-12 months and 1-2 snacks
 - 3-4 meals per day at 12-24 months and 2 snacks
4. Gradually increasing food consistency and variety as the child gets older:
 - Feed mashed and semi-solid (thick gruel) foods by 6 months of age.
 - Feed energy-dense soft foods to 6-11 months.
 - Introduce snacks that can be eaten by children alone by 8 months and family foods by 12 months
5. Diversifying the diet to improve quality and micronutrient intake:
 - Feed vitamin A rich fruits and vegetables daily.
 - Feed meat, poultry, or fish daily or as often as possible
 - Use fortified foods or other staples, when available
 - Give vitamin-mineral supplements when animal products and/or fortified foods are not available (see Fig 4).
6. Practising active feeding:
 - Feed infants directly and assist older children when they feed themselves
 - If they refuse, experiment with different food combinations, tastes, textures and methods.
 - Feed slowly and patiently. Do not force.
7. Frequent and active feeding during and after illness:
 - During illness, increase fluid intake by more frequent breastfeeding and patiently encourage children to eat favorite foods.
 - After illness, breastfeed and give foods more often than usual, and encourage the child to eat more food
8. Practicing good hygiene and proper handling of foods:
 - Wash caregivers and children's hands before and after food preparation and feeding.
 - Serve food immediately after preparation.
 - Use clean utensils to prepare and serve food.

- Serve using clean cups and bowls, and never use feeding bottles

Additional measures and advice

1. Give iron supplements daily (12.5 mg/day) to infants 6 months to 1 year of age, if daily vitamin-mineral supplements or iron-fortified foods are not being given. If the prevalence of anemia is known to be very high (40 percent or more), continue supplementation until 24 months of age. For low birth weight infants, start supplementation at 3 months.
2. Give every 6 months, high-dose vitamin A supplements after 6 months (100,000 IU for infants 6-12 months and 200,000 IU for children 12 months and older)

Infant feeding and HIV

The overwhelming source of HIV infection in young children is mother-to-child transmission. Infants can acquire HIV infection from their mothers during pregnancy, labor and delivery or after birth through breastfeeding. Between 25-45% (1 in 4 or 3) of HIV + women living in resource-poor settings pass on the virus to their babies. Conversely, 55 - 77% (two-thirds) of babies born to HIV+ mothers are not infected, even if breastfed. Fourteen percent (i.e. 1 out of 7 infants) are infected through breastfeeding. Timing of mother-to-child HIV transmission with breastfeeding and no ARV.

The risks of not breastfeeding in poor countries include the increased risk of malnutrition, infectious diseases, and mortality. Failure to breastfeed is associated with a 3-5 fold increase in infant mortality. In addition, the other benefits of breastfeeding (psychological, birth spacing etc.) are lost.

This delicate balance between breastfeeding's lifesaving benefits and the risk of HIV transmission complicates optimal infant feeding in communities affected by HIV, especially developing countries. For this reason, counseling on infant feeding is one of the most important interventions to reduce mother to child transmission (MTCT) of HIV and to increase HIV-free survival.

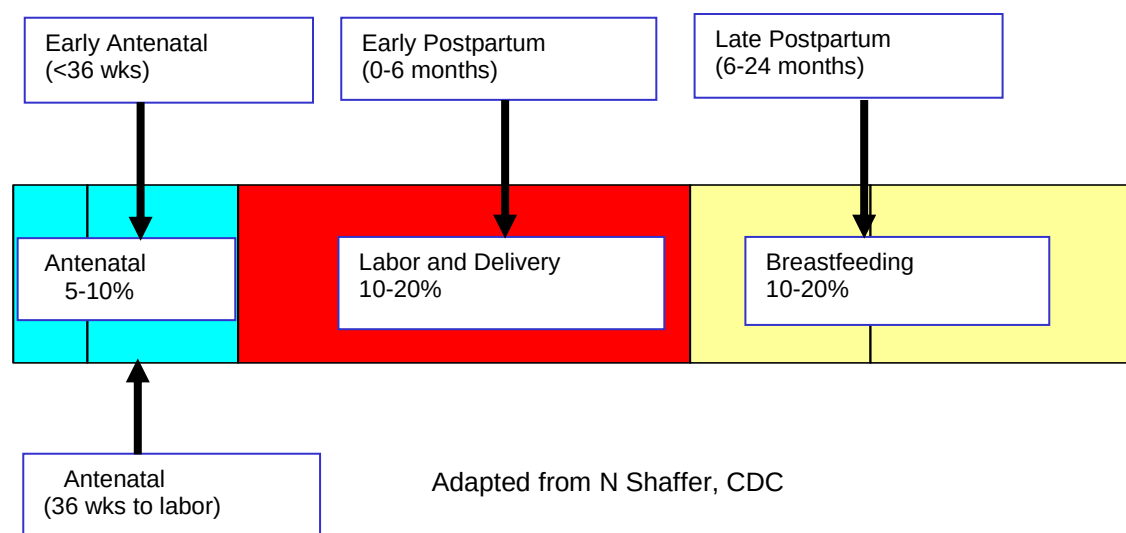
Figure 4: Feeding recommendations adapted for Ethiopia (IMCI)

Up	6 Months up to 12 Months	12 Months up to 2 Years	2 Years and Older
 • Breastfeed as often as the child wants, day and night, at least 8 times in 24 hours. • Do not give other foods or fluids. • Expose child to sunshine for 20 to 30 minutes daily	 • Breastfeed as often as the child wants. • Add complementary: potatoes softened in gruel, porridge	 • Breastfeed as often as the child wants. • Give adequate servings of: porridge made of cereal and legume mixes shiro, kik, merrek fitfit, mashed potatoes and carrot, gommen, undiluted milk and egg and fruits. • Add some extra butter or oil to child's food. • Give these foods 5 times per day.	 • Give family foods at least 3 meals each day. • Also, twice daily, give nutritious food between meals, such as: Egg, milk, fruits, kitta, dabo

The infant feeding options in communities with HIV/AIDS are based on the United Nations policy of informed choice (WHO, UNICEF, UNAIDS 2000).

1. Mothers whose HIV status is unknown or who have been tested and HIV negative can feed their infant as described above in optimal feeding practices:
 - Exclusive breastfeeding 0-6 months.
 - Introduction of complementary feeding at 6 months.
 - Continue breastfeeding to 24 months.
2. Mothers who have been tested and HIV positive should be provided with information on the risks and benefits of different infant feeding options so that she chooses and is advised to implement the safest feeding option for her situation.

Figure 5: Timing of Mother-to-Child HIV Transmission with Breastfeeding and No ARV



When replacement feeding is *acceptable, feasible, affordable, sustainable, and safe*, avoidance of breastfeeding is recommended.

ACCEPTABLE means taking account of:

- Social and cultural norms about infant feeding.
- Concerns about stigma associated women who do not breastfeed, suspicion of HIV

FEASIBLE means means taking account of:

- Economic, behavioral, psycho-social aspects for the care-giver and infant
- Resources and skills that are required

AFFORDABLE means:

- Able to purchase adequate amount of the formula over the duration it is needed
- Able to purchase necessary fuel and equipment

SUSTAINABLE means:

- Formula feeding can be practiced every day and night
- Resources are available throughout
- Formula feeding will be exclusive over first 6 months

SAFE means:

- Formula free from contamination
- Nutritious
- Free from stigma
- Does not spillover to general population

There are four options if a HIV positive mother chooses breastfeeding:

Option 1- safer breastfeeding

Safer breastfeeding involves:

1. Exclusive breastfeeding from 0-6 months. Avoid completely mixed feeding because it may increase risk of HIV transmission.
2. Maintain good breast health by correct positioning, attachment etc. because mastitis and cracked nipples increase risk of HIV transmission.
3. The mother should practice safe sex.
4. Mothers with breast problems (mastitis, sore/cracked nipples, abscess) should stop breastfeeding from the infected breast and seek prompt treatment.
5. Stop breastfeeding as soon as complementary feeding is acceptable, feasible, affordable, sustainable and safe or when the baby is started on complementary feeding at the age of 6 months.
6. Stop breastfeeding if the mother develops full-blown AIDS.
7. Introduce complementary feeding at 6 months.

Option 2 – wet nursing

Wet nursing is breastfeeding by a woman who is not the infant's mother. If this option is taken it is important to:

1. Ensure the wet nurse is HIV negative.
2. Maintain exclusive breastfeeding for 0-6 months.
3. Introduce complementary feeding at 6 months.

Option 3- Heat treated breast milk

In this option, the mother expresses her breastmilk safely and heats the breastmilk to boiling point before feeding the breastmilk to her infant with a cup. The following points should be observed:

1. The heat treated milk should be used within one hour.
2. Introduce complementary feeding at 6 months or if complementary feeding is acceptable, feasible, affordable, sustainable, and safe.

Options if the mother chooses not to breastfeed:

If the mother chooses not to breastfeed, the following points are important

1. Replacement feeding from birth with breastmilk substitutes
2. It should be exclusive replacement feeding (never mix-feed)
3. The most important thing is the hygienic preparation.
4. There are 2 groups of breastmilk substitutes options:
 - Home prepared formula
 - Commercial infant formula

For more detail about the formula read the references at the end of this section.

An artificially fed baby needs about 150 milliliters of milk for each kilogram of his or her own weight each day.

If cow's, or goat's, or camel's milk is used for artificial feeding:

- During the first 15 days of life give 1 part of boiled milk diluted with 1 equal part of boiled and cooled water.

- From 2 weeks to 6 months of age give 2 parts of boiled milk diluted with 1 part of boiled and cooled water.
- After 6 months of age give undiluted boiled and cooled milk.

Nutritional Assessment

Assessment of nutritional status is the primary step in the evaluation of childrens' growth and should be an integral part of the evaluation and management of children with acute and chronic disease. This chapter discusses nutritional assessment methods and their practical application. There is no easy and satisfactory way to measure the nutritional status of a child or community at a single examination. Four methods that are widely used are described below. However, none of them is ideal.

Clinical method

The clinical method assesses nutritional status by examining the changes in the skin, hair, buccal mucosa or thyroid glands which are believed to be due to inadequate nutrition. It is a cheap, noninvasive and quick method.

The main drawbacks are it is very subjective, less specific (other diseases may have similar picture e.g. edema can be due to cardiac disease) and usually the signs cannot be attributed to be deficiency of a particular nutrient (e.g. glossitis may be observed in riboflavin, vitamin B12, or iron deficiency.)

Table 9: Some clinical signs of malnutrition that are used for nutritional assessment.

Nutritional deficiency	Clinical signs
	Edema, easily pluckable and gray hair, flacky
PEM	paint dermatosis, muscle wasting etc.
Anemia	Pale conjunctivae and palms
Vitamin A deficiency	Corneal xerosis, Bitot spot keratomalacia, hypertkeratosis of skin
Iodine deficiency	Thyroid enlargement
Vitamin D deficiency (Rickets)	Wrist widening, rachitic rosary, Harrison's sulcus

Anthropometric assessment

Anthropometry measures the variations in physical growth at different ages and the degree of nutrition. It evaluates the long term nutritional history. It is a widely used, simple, accurate, rapid, quantitative, inexpensive and non-invasive measure of the general nutritional status of an individual or a population group. It is applicable to larger sample sizes and it can be used both in clinical set up and fieldwork. If children are measured more than once, growth velocity data are obtained that can be more valuable because they reflect change.

The major pitfalls are it cannot detect impaired nutritional status of short duration nor can it identify specific nutrient deficiencies.

Particular care should be taken to use appropriate equipment and techniques for measuring stature/length and weight. This can significantly improve the assessments that are performed.

Length or Stature

Length or stature is the most useful indicator of growth status. In infants and children younger than 2 years, recumbent length is measured. Two people are required to accomplish this measurement. The measuring table or board should consist of a fixed headboard, a movable footboard, and a rule attached at one side. One of the measurers should hold the crown of the infant's head against the headboard so the external auditory meatus and the lower margin of the eye orbit are aligned perpendicular to the table. The second measurer grips both ankles of the infant with one hand and positions the heels firmly against the footboard, which is manipulated with the other hand. The knees, which are slightly flexed, are then pressed down on the table with the lateral edge of the hand. The recumbent length should be recorded to the nearest 0.1 cm.

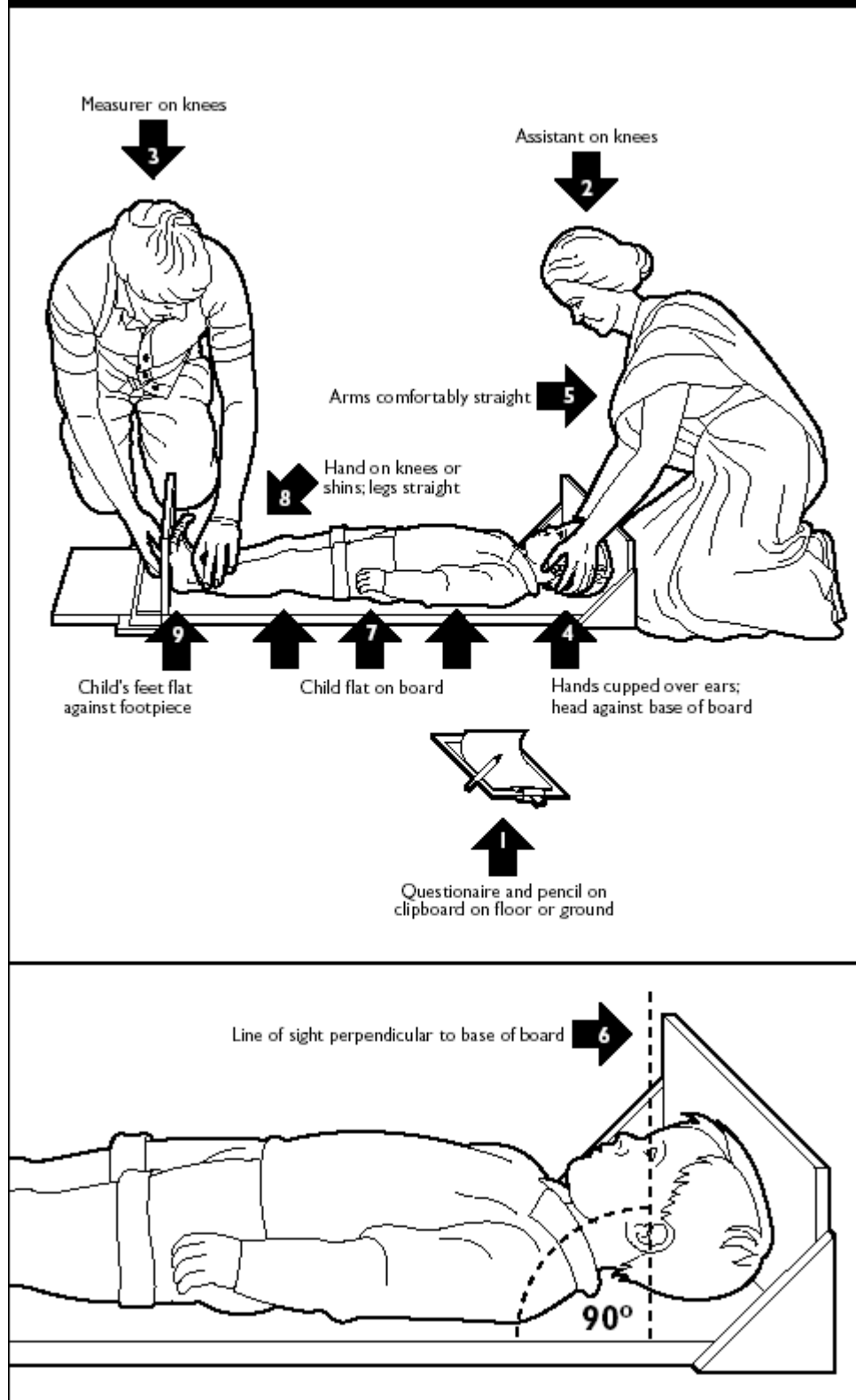
Height is measured for children older than 2 years. Measurements of height are made with the child's feet bare. The child should stand erect with the heels, buttocks, shoulders, and head all touching the measuring board. The feet should be positioned at a 90-degree angle. The child's axis of vision should be horizontal, with the child looking ahead and the external auditory meatus and lower margin of the orbit aligned

horizontally. The stature is recorded to the nearest 0.1 cm (see the figures on the following pages).

Weight

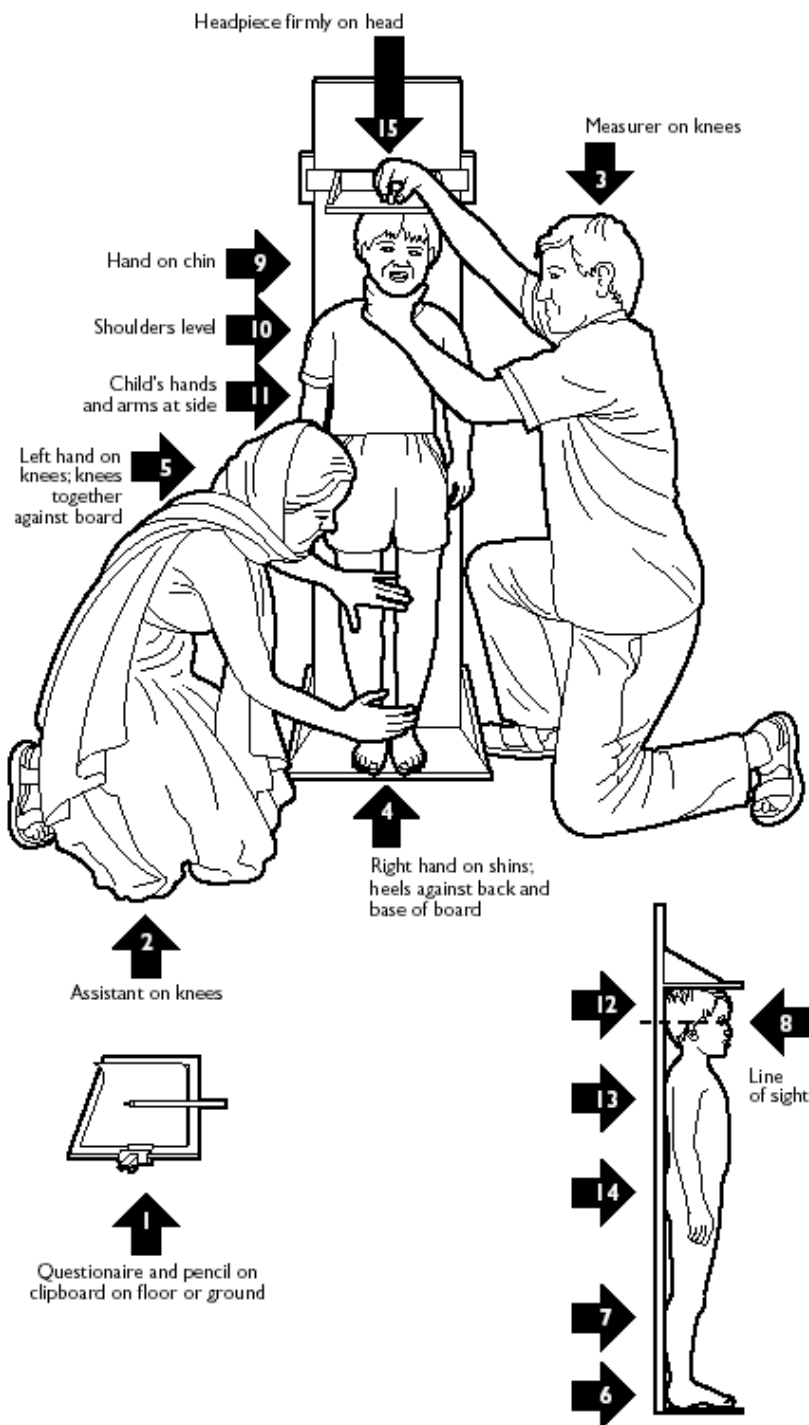
Various types of apparatus, such as infant scales, beam-balance scales and readout scales are available to measure body weight. The type used needs to be regularly calibrated to maintain accuracy. Weight should always be recorded as nude weight. If the subject is not weighed nude, then the estimated weight of the clothing should be subtracted from the total weight.

Figure 5.2. Child length measurement - length for infants and children 0-23 months



Source: How to Weigh and Measure Children: Assessing the Nutritional Status of Young Children, UN 1986.

Figure 5.1. Child Height Measurement - Height for Children 24 Months and Older



Source: How to Weigh and Measure Children: Assessing the Nutritional Status of Young Children, UN 1986.

Head Circumference

Head circumference is a useful measurement until about 3 years of age, when head growth slows. It must be measured with a narrow and nonstretchable measuring tape. To obtain an accurate measurement, the tape must cross the forehead just above the supraorbital ridges, passing around the head at the same level on both sides to the occiput. It is then moved up or down slightly to obtain the maximum circumference. The tape should have sufficient tension to press the hair against the skull.

The commonly used four building blocks or *measures* used to undertake anthropometric assessment are age, sex, weight and height. When two of these variables are used together they are called an *index/indices*. There are many anthropometric measures including mid-upper-arm-circumference (MUAC), head circumference, and skin fold thickness but three indices are commonly used in assessing the nutritional status of children

- Weight-for-age
- Length-for-age or height-for-age;
- Weight-for-length or weight-for-height.

The values of these measurements will be interpreted using standard charts (Harvard curve, NCHS) of weight-for-age, height-for-age, weight-for-height, and head circumference-for-age. The three indices are used to identify three nutritional conditions: underweight, stunting and wasting.

Weight-for-age

For monitoring growth of children over a limited period, weight-for-age is the simplest method. A low weight-for-age index identifies the condition of being *underweight*, for a specific age. The advantage of this index is that it may reflect both past (chronic) and/or present (acute) under nutrition (although it is unable to distinguish between the two). Underweight, based on weight-for-age, is a composite measure of stunting and wasting.

Height-for-age

This index is an indicator of past undernutrition or chronic malnutrition. It cannot measure short term changes in malnutrition. For children below 2 years of age, the term is length-for-age; above 2 years of age, the index is referred to as height-for-age. Deficits in length-for-age or height-for-age is called *stunting*. Stunting is an indicator of past growth failure.

Weight-for-height

The ratio of actual weight to the ideal weight for height (usually referred to as weight for height in children under 2 years of age or weight-for-height in children over 2 years of age) helps to identify children suffering from current or acute under nutrition or *wasting* and is useful when exact ages are difficult to determine. It can be used to differentiate stunted growth from wasting and is independent of age. Wasting results from acute or subacute nutritional deprivation and by acute medical conditions such as diarrhea. Body weight is depleted out of proportion to length, making the weight-for-height ratio low.

Mid upper arm circumference (MUAC)

The mid upper arm circumference is an indicator of muscle growth. It is age independent, relatively easy to measure and a good predictor of immediate risk of death. It is used for rapid screening of acute malnutrition in children from 1-5 years of age. MUAC can be used for acute undernutrition screening in emergency situations and for estimating prevalence of undernutrition at the population level.

The left arm is usually measured by convention. A point is marked midway between the acromion (shoulder) and the olecranon (elbow) on the vertical axis of the upper arm with the arm bent at a right angle and between the lateral and medial surface of the arm. The child should then stand or sit with the arm hanging loose at the side. The tape is passed around the arm at the level marked and is tightened so it touches but does not compress the skin or alter the contour of the arm. Because the arm on cross section is not an exact circle, some difficulty is usually met in ensuring that the tape touches the arm on the medial surface. To accomplish this, the middle finger of the examiner's left hand can be used to gently press the tape to the skin (see the figure on the next page).

Classification

The various methods of classification are described below.

Gomez:
$$\text{Wt. for age} = \frac{\text{Wt. of subject}}{\text{Wt. of normal child of same age}} \times 100$$

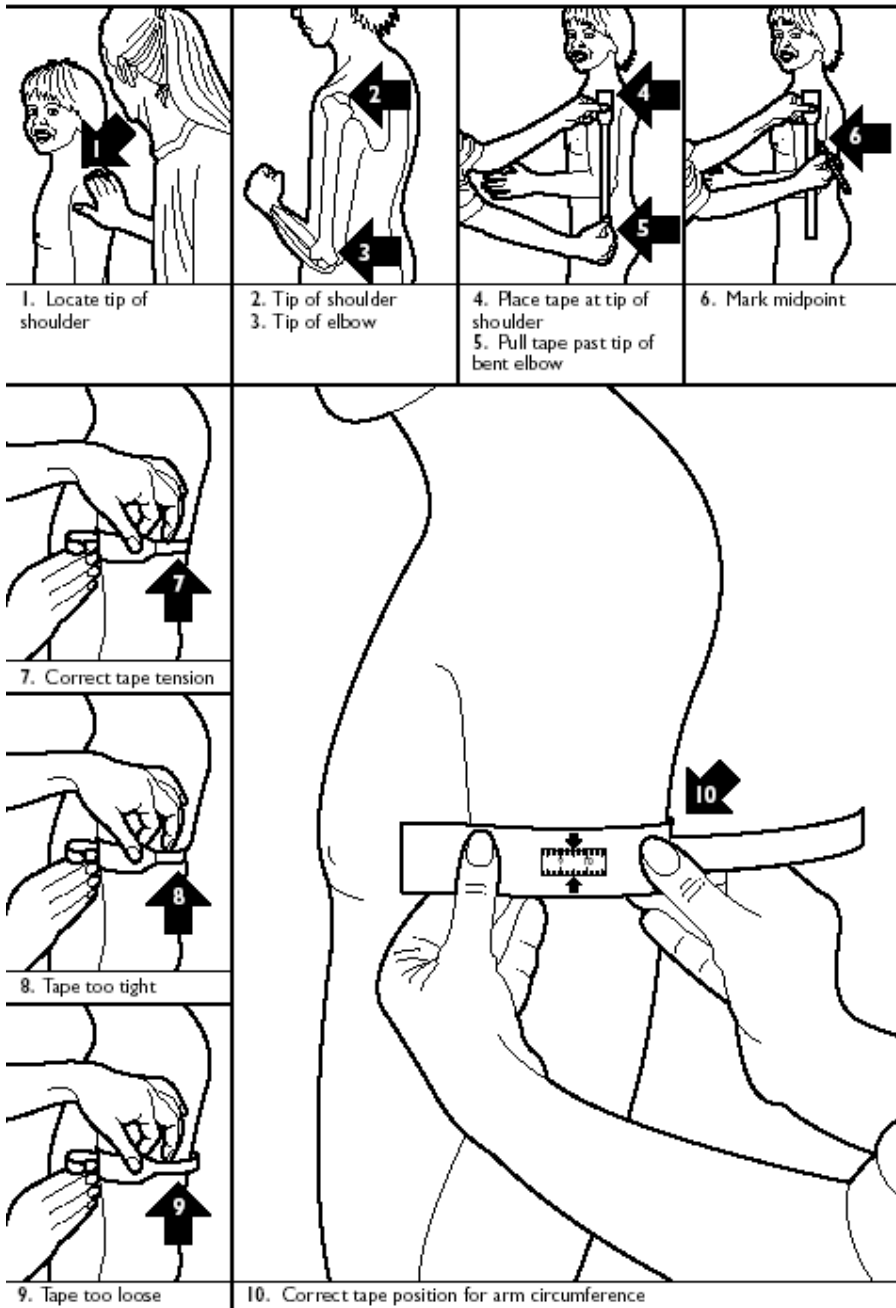
<u>Wt. for age</u>	<u>Degree of malnutrition</u>
90-109%	normal
75-89%	mild (grade I)
60-74%	moderate (grade II)
< 60	severe (grade III)

The Gomez system is used in public health screening and to evaluate the impact of public health intervention. Its shortcomings are:

- It relies on age which is usually unreliable, especially in developing countries.
- It fails to differentiate acute vs chronic malnutrition.
- It fails to differentiate between types of severe PEM

Figure 5.5. Child Mid-Upper Arm Circumference Measurement

Arm circumference "insertion" tape



How to weigh and measure children: assessing the nutritional status of young children in UN (1986)

Welcome uses the weight for age, measured by the Harvard Curve

Table 10: The Welcome system for measuring weight for age

Edema	wt. for age	
	60-80%	60%
Absent	Underweight	Marasmus
Present	Kwashiorkor	Marasmic- Kwashiorkor

This system is preferable for clinical setups to distinguish different forms of PEM its shortcomings includes that it:

- Relies on age
- Does not consider height
- Does not differentiate acute VS chronic malnutrition

Water low

In this system terms suggest the following:

- Wasting suggests acute malnutrition
- Stunting suggests chronic malnutrition

The system has the following advantages:

- It is the best method for screening malnourished children in the community
- It can detect mild forms of PEM
- It can distinguish acute from chronic malnutrition
- It does not rely on age, which is sometimes difficult to ascertain in rural areas

Its short coming is that it does not differentiate severe forms of PEM.

Table 11: The Water low system for measuring nutritional status

Wt. for Ht.	Nutritional status	Height for age	Nutritional status
90-100%	Normal	> 95%	Normal
85-90%	Mild wasting	90-95%	Mild stunting
75-85%	Moderate	85-90%	Moderate
< 75 %	Severe wasting	< 85%	Severe stunting

Biochemical methods

Biochemical methods measure the total amount or concentration of nutrient in body fluids (blood, serum, urine) and organs (storage sites) of the body. The tests reflect recent nutritional status and identify specific nutrient abnormalities. They also detect sub clinical deficiencies. The most commonly used tests are hemoglobin, serum albumin and others like vitamin A, thiamine in urine, serum ferritin etc.

Dietary survey

Dietary survey is an important component of nutritional assessment. It assesses through recording the food intake and translating it into nutrients consumed. Dietary data can be collected at national, household, and individual level. It can be estimated in individual level by:

- Dietary history: to record the usual food intake pattern over a relatively long period of time and is obtained by interview.
- 24 hours recall: the subject is asked to recall all food consumed during the previous day
- Food record: the subject keeps a record of the food consumed during a specific period of 3-7 days, quantities being estimated in household measures. This is the most accurate method of dietary assessment because it accounts for most of the daily variation in diet and eliminates the subjectiveness of diet recall records
- Weighed intake: the subject or trained person weighs all food consumed during a given period of time.

Protein Energy Malnutrition (PEM)

Objectives

By the end of this topic, the reader should know:

- the magnitude of PEM.
- the basic, underlying and immediate causes and the pathogenesis of PEM.
- the clinical pictures of marasmus and kwashiorkor.
- the pathophysiological changes and their implication on the manifestations and treatment.
- the common complications and the principles of management.

Definition

PEM is a clinical syndrome mainly due to deficiency of macronutrients (protein and energy) and micronutrients.

Epidemiology

PEM is frequently observed in children between 6 months and 5 years of age. Marasmus peaks in the first year and Kwashiorkor between 1 and 3 years of age. Stunting is most common between 2.5 and 3 years of age. PEM is more common during the weaning period and the rainy season.

Causes

The nutrition status of an individual, including any of the four major forms of malnutrition (PEM, IDD, vitamin A deficiency and iron deficiency anemia) is an outcome of complex biological and social processes, as summarized in the conceptual framework of malnutrition (see Figure 9).

The causes are classified as the follows:

Immediate causes: inadequate dietary intake and disease.

Underlying causes: insufficient food available to families (household food insecurity), inadequate care of women and children, insufficient health care, and an unhealthy environment.

Basic causes: inadequacies in educational, political, and economic systems and problems with the availability and control of resources.

Pathogenesis

Pathogenesis is a multi-deficiency state but there are different theories for the exact pathogenesis. Currently the most accepted pathogenesis, especially for edematous malnutrition, is the free radical theory.

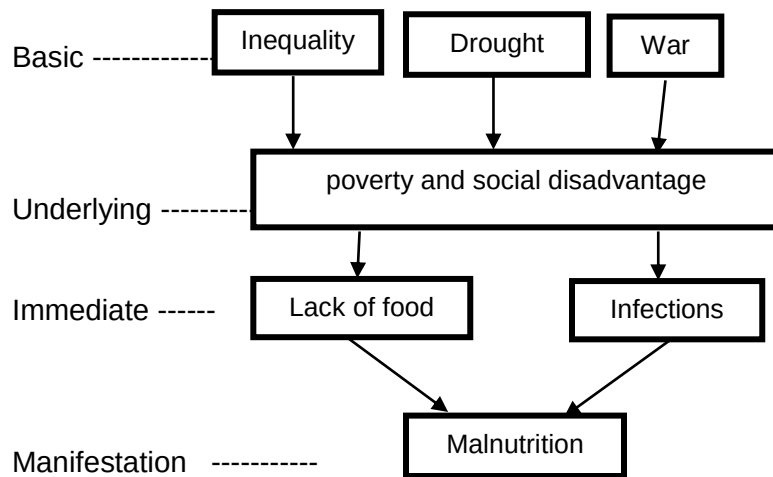


Figure 9 - Conceptual framework for causes of malnutrition.

Protein and calorie deficiency

Marasmus is due to both calorie and protein deficiency, predominantly calorie. Kwashiorkor is due to more marked protein deficiency than calorie which leads to hypoalbuminemia and edema. Currently less accepted cause.

Adaptation

Marasmus is well adapted to the deficiency due to the high cortisol and growth hormone levels and depression of insulin and TH. Kwashiorkor is a poorly adapted form of PEM.

Free radical theory

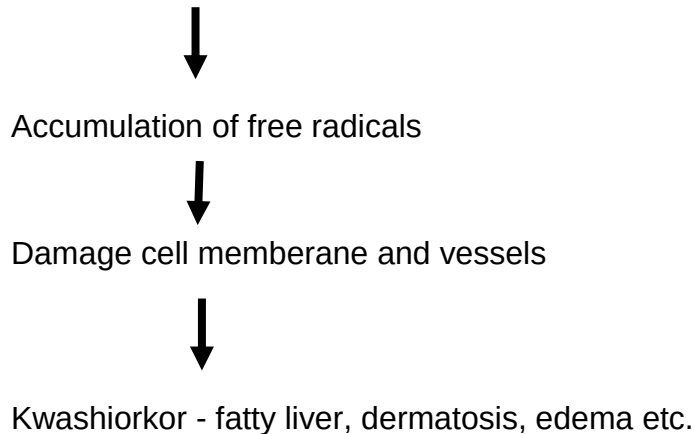
Free radical theory can be described using the figure on the next page:

Figure 10: The free radical pathogenesis

Increased production of free radicals (oxidants) - peroxides, epoxides, due to NOXA - infection, bowel bacteria, contaminated foods, aflatoxin

AND

Decreased scavenger mech. that removes free radicals because of deficiency of micronutrients (vitamins, zinc, selenium, etc) and glutathione



Classification – There are different classification but only the Welcome has clinical importance and is mentioned here. For the other classifications, see the nutritional assessment section (page 52).

Under this system, even if the weight-for-age is more than 80% of what it should be, a child will be labeled as kwashiorkor if he /she has clinical signs of Kwashiorkor.

It is important in clinical setups to distinguish between different forms of PEM. These are described below.

Severe malnutrition

A child is labeled as having severe malnutrition if any one of the following is present:

- Weight for height less than 70 % of what it should be;
- MUAC less than 11cm for a child with a length less than or equal to 75cm;
- Bilateral edema; or

- Based on the Welcome classification, if the child has Kwashiorkor, Marasmus, or Marasmic-Kwashi

Clinical features

Marasmus

The clinical features of marasmus include marked loss of weight, almost no subcutaneous tissue and atrophic muscles. The child will have the face of an old man, distended abdomen. The child will usually have a good appetite

Pathophysiological Changes

Body Composition

- TBW and ECF increased
- Increased ICF Na⁺
- Decreased body K⁺ and Magnesium (Mg⁺)
- Marked loss of fat and muscle

Liver

- Fatty liver
- Reduction in synthesis of proteins
- Ability to take up, metabolize, and excrete toxins is limited
- Impaired gluconeogenesis

Diagnostic and therapeutic implication

- Assessment of dehydration is difficult.
- Need large doses of K⁺ and magnesium
- Hepatomegaly
- Decrease protein load during initial phase
- Reduce drug dose if hepatotoxic or needs hepatic disposal.
- Predispose to hypoglycemia. This effect is not severe enough to interfere recovery during diet

GIT

- Villi atrophy
- Reduced disaccharidase
- Bacterial overgrowth
- Decreased biliary secretion
- Pancreatic acinar cells atrophy
- Decreased digestive pancreatic enzymes

Temperature

- Heat generation is impaired
- Sweating is also impaired

Hematology

- Anemia is common

Metabolic

- Hypoglycemia
- Decreased BMR

CVS

- Atrophied myocardium
- Reduced cardiac output and stroke volume.
- Blood pressure is low
- Start low calorie and protein
- Limited digestive and absorptive capacity.

malabsorption eg lactose intolerance and diarrhea.

- So feed small and frequent meals
- Prone both to hypothermia and hyperthermia.
- Keep emperature to normal
- Vulnerable to both increase and decrease blood volume
- Easily develop heart failure.
- Restrict IV fluid, rehydrate slowly

Immunity

- All aspects of immunity are impaired except humera I- CMI profoundly affected:
- Reduced secretory IGA

- Impaired phagocytic function Low complement

Inflammatory response

- Impaired acute phase response
- WBC do not migrate to area of infection
- Susceptible for infection
- Mild and severe infection spread
- Give antibiotics
- Signs of infection are subtle
- Raised WBC and fever are uncommon

Kwashiorkor

Kwashiorkor is an acute illness. Bilateral edema is a sine qua non for diagnosis. Bilateral edema is pitting and starts in the lower legs and later involves the upper limbs and face. Large ascites is uncommon so consider other causes.

The hair is red or brown, straight, sparse, lusterless, easily and painlessly pluckable. The face appears moon-shaped and puffy (jowls). The skin is hyperpigmented at first and become dry and peels off leaving hypopigmented and sometimes ulcerated areas (flaky paint dermatosis): the skin appears like old paint flaking off the surface of the wood. The lesions frequently occur on pressure areas like the buttocks, flexure areas, behind the ears and legs.

Hepatomegaly occurs in one third of patients. They are miserable, disinterested to their environment when undisturbed and irritable when disturbed.



Figure 11: A 5 year old child with kwashiorkor

Notice the edema, hair loss and dermatosis over the thigh, shoulder, wrist and scalp, the wasting of the shoulders and arms and the prominent ribs.



Figure 12: The same child with hyperpigmented and hypopigmented dermatosis over the knee



Figure 13: The same child with hyperpigmented and hypopigmented dermatosis over the foot. Note the pitting edema.

Investigations

Investigations for kwashiorkor include:

- Hct, blood glucose, urine analysis, blood and urine culture,
- blood film if febrile,
- Serum albumin or protein if edematous malnutrition is suspected.

Complications

The complications of kwashiorkor include:

Hypoglycemia

- All are at risk and most are asymptomatic.
- An important cause of death in the 1st 2 days of treatment.
- impaired gluconeogenesis may lead to infections
- if the patient is not fed for more than 4-6 hours they may suffer from hypothermia.
- Patients may present with hypothermia, limpness, lethargy or convulsions
- possible do random blood glucose

Hypothermia

Hypothermia is diagnosed by a temperature of less than 35.5 C.

Patients with kwashiorkor are highly susceptible to hypothermia. Mostly this is due to a low environmental temperature and lack of covers. Kwashiorkor patients are

susceptible because they have impaired heat production and loss of subcutaneous tissue. This may lead to infection and hypoglycemia.

Dehydration

Most of the signs of dehydration in normal children occur in a severely malnourished child who is not dehydrated. Therefore, recent sunkenness of the eyes, eagerness to drink, pulse volume, blood pressure, urine output and lethargy or coma are used to assess for dehydration.

Giving fluid for a malnourished child who is not dehydrated leads to congestive heart failure and death because they are very sensitive for excesses sodium and fluid. Inappropriate fluid therapy is the commonest cause of death especially after 24 hours of admission.

If there is recent fluid loss (diarrhea or vomiting) and recent sunkenness of the eyes, the child is dehydrated.

In a child with fluid loss, the diagnosis of severe dehydration or shock (if there are signs of shock) is made:

- fast and weak pulse,
- cold extremities and
- disturbed consciousness (drowsiness, unconscious)

Septic shock is considered if the above signs of shock are present in the absence of major fluid loss or if the child with severe dehydration or shock does not respond to the first 15 ml/kg/hr intravenous fluid.

Infections

Nearly all patients have bacterial infection at admission and they are difficult to detect. They present with apathy, drowsiness, hypothermia, and hypoglycemia.

Common infections include pneumonia, diarrheal diseases, UTI and sepsis

Etiologies follow those of common pathogens in other normal children and *E.coli*, *klebsiella*, *Staph. aureus* or *epidermidis*

Congestive heart failure

- Usually occurs after 24 – 48 hrs.
- Causes - commonly as a complication of over hydration
 - Severe anemia
- High Na⁺⁺ content, calorie and protein diet

The earliest sign is tachypnea. Other signs are weight gain, increased liver size, engorged neck veins, respiratory distress, tachycardia, gallop etc.

Treatment of PEM

All patients with severe PEM (Wt/Ht < 70%, or bilateral edema or MUAC < 11cm) should be admitted to hospital.

The management includes three phases of treatment

Phase I

- Prevent and RX. Complications
- Start to feed
- Give routine medications
- Monitor the patient

Transition phase

- Continue to feed
- Continue the medication started in the phase I

Phase 2

- Feed
- Start to give iron supplements
- provide psychosocial support
- Monitor the response
- Teach the mother about feeding

Phase1

Feeding: Feed the patient with F-75 which has 75 kcal/100 ml. It is low in sodium, protein, energy, and sodium. Give 100 kcal (130 ml)/Kg/day or use the look up table. Feed 8 times daily using cup and saucer.

Nasogastric tube is indicated in the following cases:

- Pneumonia with rapid breathing
- Painful oral lesions
- Disturbances of consciousness
- The Patient is taking less than 75% of the daily milk

Routine medication

Routine medication includes:

- *Vitamin A* given on day 1, 2, and 14. For children 6-11 months of age give 100,000 iu and for children of 12 months or older (more than 8 Kg in weight) give 200,000 iu
- *Folic acid* given as a 5 mg single dose
- *Antibiotics:* During the phase I + 4 days the first line antibiotics are amoxicillin or cotrimoxazole and the second line antibiotics are gentamicin or chloramphenicol
 - Treat orally as much as possible.
 - If there are no apparent signs of infection and complication, give oral cotrimoxazole or amoxicillin
 - If there are signs of infection or complication, give ampicillin and gentamicin or penicillin with gentamicin.
- Measles vaccination if the child is older than 6 months and not vaccinated
- Treat for malaria according to the national guideline

Preventing and treating complications

Hypoglycemia can be prevented by frequent feeding and keeping temperature normal. If the patient is conscious, give 50 ml 10% sugar water (5 g or one teaspoon of sugar in 100 ml of water) or F-75 by mouth. If they are unconscious or convulsing, 5 ml/kg of 10% glucose IV or give sugar water by nasogastric tube. **Start second line antibiotic.**

Hypothermia): The environmental temperature must be monitored and kept between 8-

32. Do not wash a severely malnourished child. Treatment includes:

- Keeping the patient warm with kangaroo method for young infants.
- Putting a hat on the child and wrapping the mother and child together with double blanket.
- Feeding frequently
- Treating for hypoglycemia and starting second-line antibiotic
- Taking the patient's temperature every 30 minutes.

Dehydration

The treatment for dehydration is different from normal children. It involves:

- Rehydrating as much as possible orally.
- Infusions are almost never used unless clearly indicated i.e. severe dehydration or septic shock.
- ReSoMal, rather than standard ORS, is preferred because it has low sodium, and osmolarity and high potassium.
- Rehydrate slowly over 12 hrs.
- Before treatment take weight, pulse rate, respiratory rate, liver size.
- Feed during rehydration.
- Monitor closely (every 1 hour) for both under and over hydration.
- If the child has recent sunken eyes or is eager to drink and conscious, give ReSoMal 5 ml/kg every 30 min. for the first 2 hrs PO or NGT and then 5 – 10ml/kg/hr for 10 hrs.
- If the child is unconscious, start I.V. infusion to give:
 - Ringer Lactate or half normal saline with 5% D/W, add 20 mmol KCl/l
 - Rate- 15ml/kg over 1 hr and reassess. If improving give 15 ml/kg over the next 1 hr.
- If the child regains consciousness or pulse rate drops, continue the rest of the dehydration with 10ml/kg/hr of ReSoMal.
- Monitor weight, Pulse rate, Respiratory rate, liver size, and heart sound (gallop rhythm)

Infections

If there are no apparent signs of infection and complication, give PO cotrimoxazole or Amoxicillin.

If there are signs of infection and complication, give gentamicin or penicillin + gentamicin I.M. or IV.

Congestive heart failure

Treatment

Treatment involves

- Stopping all PO intakes and intravenous fluids.
 - giving small amount of sugar-water
 - Furosemide – 1-2 mg/Kg

Anemia

If hemoglobin is less than 4 gm% or hematocrit is less than 12% and it is within 48 hours of admission, transfuse with 10 ml/kg of packed cell volume or whole blood over 3 hours and give furosemide.

If hemoglobin is equal to or less than 4 gm% or hematocrit is equal to or less than 12% or any level of hematocrit 2-14 days after admission, give iron during phase II.

Monitoring

Take the patient's temperature twice daily. Measure/observe their weight, edema, diarrhea, vomiting, signs of dehydration, cough, respiratory rate, pulse rate and liver size daily.

Transition phase

Progress to transition phase if the edema starts to decrease, the appetite return and there are no NGT, infusion, and severe medical problems.

For marasmic children, if the patient tolerates the diet for 2 days, the appetite returns and there are no NGT, infusion, and severe medical problems, treat them in the same way as phase I (routine medication, feed and monitoring) except F-100 is given instead of F-75. The same volume is given as F-75 so that the energy intake increases by 30% and the child starts to gain tissue without causing fluid overload or congestive heart failure.

Indication for a return to phase I are as follows:

- Increasing edema
- Rate of weight gain $>10\text{g/kg/day}$ which is a sign of fluid retention.
- Any sign of fluid overload, heart failure or respiratory distress.
- Tense abdominal distention.
- Development of complications that require intravenous drugs or rehydration therapy or poor appetite.
- Refeeding diarrhea that results in weight loss.

Phase 3

The criteria to enter this phase should all be fulfilled:

- Good appetite
- At least 2 days for wasted (marasmic)
- When the edema disappears
- No other medical problems

Feeding

This phase is the period of catch up growth so the patient needs a high calorie and protein diet.

Feed the patient F-100 five times per day based on the weight, refer the chart for the amount. Additionally give porridge if the child's weight is more than 8 kg.

Routine medication

Routine medication includes the following:

- Start ferrous sulfate.
- Treat with mebendazole or albendazole.
- Give the third dose of vitamin A at day 14.

Monitoring

Monitoring involves measuring/observing the following:

- Weight 3 times/week.
- Temperature daily.
- Diarrhea, vomiting, Dehydration, Cough, pulse, and respiratory rate daily.

A good response is indicated by weight gain of more than 10 gm/kg/day.

Return to phase I there is any sign of morbidity.

Discharge criteria

The child may be discharged if he/she fulfills the following criteria.

- Weight /height $\geq 85\%$.
- No edema for the last 10 days.
- Complications are adequately treated.
- Education completed.
- Immunization up-to-date.

Vitamin A Deficiency (VAD).

Objectives

By the end of this topic the learner is expected to know:

- The function and main sources of vitamin A.
- The magnitude of VAD in children in Ethiopia.
- The causes and consequences of VAD.
- The eye manifestations and staging of VAD.
- The principle of treatment and the strategies to eliminate VAD.

Vitamin A is a fat soluble vitamin which is required by the body in small amounts. It is essential for the functioning of the immune system and needed for vision and eye health. It is now also recognized as a critical factor in child health, growth, and survival. Any intervention that proves effective in improving vitamin A status in deficient populations will improve markedly childhood mortality and morbidity.

Epidemiology

Vitamin A deficiency is a public health problem in more than 75 countries including Ethiopia and affects as many as 228 million children subclinically worldwide, at a severe or moderate level. It is responsible for as many as one out of four (25%) child deaths globally. Vitamin A deficiency is the most preventable cause of corneal ulceration and blindness in developing countries.

The prevalence of subclinical vitamin A deficiency in children under five years of age in Ethiopia is 27%. The prevalence of Bitot's spots, corneal xerosis and ulceration and corneal scar in the hyper endemic areas of Bale and Arsi are 5.0%, 0.8%, and 0.5% respectively.

Vitamin A deficiency and risk of childhood mortality and morbidity

Infants and young children who are vitamin A deficient are at increased risk of more frequent and severe episodes of diarrhea, ARI and measles. Improving vitamin A status:

- Improves resistance to common childhood infections and decreases severity of infections.
- Reduces child mortality from measles by 50% and mortality from diarrheal disease by 40%.

The overall mortality in infants and children between the ages of 6 months and 5 years can be reduced by 23%.

Children who are at high risk for vitamin A include:

- Children 6 to 59 months
- Children with PEM, measles, diarrheal disease, ARI.

- Infants less than 6 months old who are not breastfed.

Biochemistry and Physiology of Vitamin A

Sources

The two most important sources of vitamin A are animal foods and breast milk. The best sources of preformed retinol (active Vitamin A) are liver, fish, eggs and cheese. Breast milk is an excellent source for infants but the quality dependent on the mother's vitamin A status.

Some plants contain beta carotene or provitamin A which can be converted to retinal in intestinal wall. Red palm is the best plant source. Others are green leafy vegetables such as spinach and colored vegetables such as carrot, mango and papaya. Cereals and roots have little or no vitamin A.

Absorption/ storage

The ingested vitamin A is emulsified and absorbed. In the mucosal cells it will be esterified with palmitic acid and transported by intestinal lymphatics to the liver and stored as retinyl palmitate. From the liver it is transported in blood bound to Retinal Binding Protein (RBP) to the eyes/other tissues.

RDA for infants is 375 µg and for children it is 400 -500µg.

Functions

The functions of vitamin A are as follows:

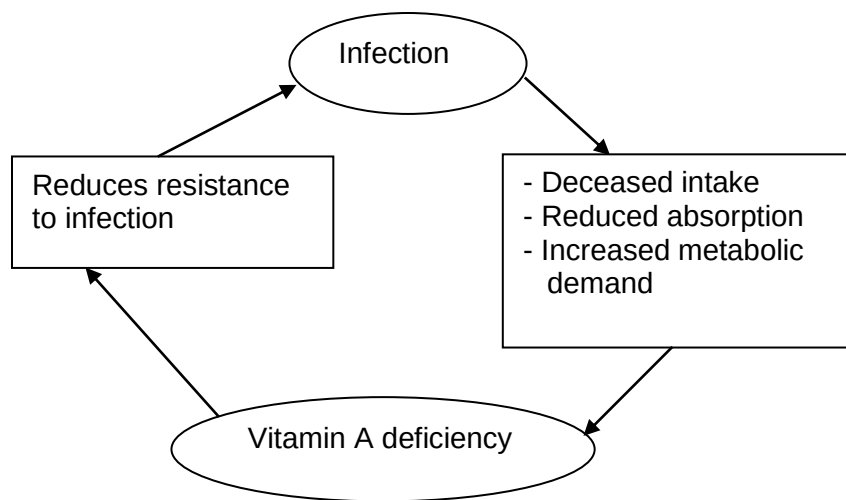
- Maintain a healthy epithelial tissue
- Has a role in keratinization
- Important for the functioning of the immune system.
- Forms visual pigment so important for night vision

Causes

The causes of vitamin A deficiency are mainly nutritional and infectious and include:

- Decreased intake due to too little vitamin A in the foods (dietary deficiency) or anorexia.
- Decreased absorption because of problems such as diarrhea, chronic GIT disorder.
- Infections such as measles, diarrhea, ARI etc.

Figure 14: infection and Vitamin A deficiency interaction



When body stores of vitamin A are lost because of the above causes, blood levels fall, damaging the immune system and later the eyes are affected.

Clinical Features

Ocular signs develop insidiously/ bilaterally. The posterior segment is initially affected.

WHO Classification of Xerophthalmia

Primary Signs

X 1A - Conjunctival Xerosis

X 1B - Bitot's spot

X 2 - Corneal Xerosis

X3A - Corneal ulceration

Secondary signs

XN- Night blindness

XF- Xerophthalmia fundus

Xs- Xerophthalmia scars.

X 3B - Keratomalacia.

Night blindness (XN)

Night blindness is characterized by poor night vision. It is the most early and sensitive sign of xerophthalmia.

Conjunctival xerosis (X 1A)

Conjunctival xerosis occurs in bulbar conjunctivae and is characterized by dry conjunctivae which:

- loses its shiny luster and becomes unwettable
- loses its transparency
- wrinkles when the eyes move is fully reversible with treatment.

Bitot's spot (x1B)

Bitot's spot is characterized by a foamy /cheese like plaque which is oval /triangular in shape on bulbar conjunctivae. It usually occurs on the temporal side and is reversible with treatment

Corneal xerosis (X₂)

Is characterized by a hazy, dull and rough corneal surface, which is a highly specific sign and indicates greatest risk of corneal damage. Early treatment reverse it.

Corneal ulceration (X 3A)

The usual site of corneal ulceration is the central or lower part of the cornea. It is not fully reversible, but vigorous treatment decreases the extensive ulceration.

Keratomalacie (X₃B)

Keratomalacie is characterized by very rapid onset. It is the most severe form of xerophthalmia. Part or whole cornea is destroyed leading to phthisis bulbi or staphyloma. It is not reversible and results in blindness.

Note: The above stages reveal the earlier the diagnosis the better the reversibility. So it is preferable to diagnose at night blindness, conjunctival xerosis, or Bitot's spot stages.

Dry and scaly skin, retarded mental and physical growth are other manifestation of vitamin a deficiency.

Diagnosis

Diagnosis is mainly clinical

Treatment

Treatment takes two forms:

1. Massive P.O dose of vitamin A
 - 100,000 for children from 6mths to 1year old and 200,000 for children of more than 1year of age on the 1st, 2nd and 14th day
2. Local treatment for all with corneal ulcers.
 - Topical antibiotics
 - Eye pad.

Prevention

VAD is considered as a public health problem if the following biological factors are present in a community survey:

<i>Biological indicators</i>	<i>WHO cutoff points</i>
Night blindness	$\geq 1.0\%$
Bitot's spot	$\geq 0.5\%$
Corneal xerosis/ulceration/keratomalacia	$\geq 0.01\%$
Corneal scars	$\geq 0.05\%$
Serum retinol	$\geq 10\%$ ($< 0.35\mu\text{mol/l}$)
Breast milk retinol	$\geq 10\%$ ($8\mu\text{g milk}$)

There are 4 strategies for the virtual elimination of vitamin A deficiency and to decrease child morbidity and mortality.

1. Breastfeeding – exclusive breastfeeding for 0-6 months.
 - Continued breastfeeding for 2 years.
 - Vitamin A supplementation (200,000 or 400,000 iu) to mothers within the first 8 weeks postpartum
2. Supplementation - Massive P.O dose of vitamin A.
 - For children 6-59 months: 2 times a year.
 - For children 6-59 months with different diseases:
 - Pneumonia, persistent diarrhea: one dose.
 - Severe PEM, measles: day 1, day 2, and day 14.

The dose in both cases is:

 - 6-12 months – 100,000 iu.
 - > 12 months – 200,000 iu.
3. Food diversification – access and consume Vitamin A rich foods.
4. Food fortification

Vitamin D Deficiency (Rickets)

Objective

By the end of this topic, the learner will be able to describe:

- The function, main sources and metabolism of vitamin D.
- The causes and consequences of vitamin D Deficiency.
- The clinical manifestations and complication of vitamin D deficiency.
- The principle of treatment and prevention.

Definition

Rickets is the failure of calcification of bone matrix of growing bones.

Osteomalacia is failure of calcification of bone matrix of mature bonea.

Sources of vitamin D

Vitamin D is a fat soluble vitamin and its two main sources are:

1. Dietary: vitamin D₂ (ergocalciferol) or vitamin D₃ (cholecalciferol):
 - Fish liver oil -very rich, others are eggs, milk etc.
 - The recommended daily allowance (RDA) is 10µg or 400 I.U. daily.
2. Vitamin D₃ is present in the skin as 7 dehydrocholesterol that will be converted to 1,25 dehydrocholecalciferol by the action of sunlight and by undergoing hydroxylation the liver and kidney.

Function of 1,25 (OH)₂ D₃ (1,25 dihydrocholecalciferol)

Dihydrocholecalciferol functions as a hormone. It:

- Facilitates calcification of bone by increasing Intestinal ca⁺ & phosphrus (P⁺) absorption and reabsorption of p⁺ in kidney
- Controls serum ca⁺ & p⁺ with parathahormone (PTH) and calcitonin.

Causes of Vitamin D Deficiency

Causes of Vitamin D Deficiency include:

- Inadequate exposure to sunlight – the commonest cause in Ethiopia.

- Inadequate dietary intake
- Others: - Chronic GIT disorders which leads to malabsorption.
 - Liver and renal disorders
 - Anticonvulsants (phenytoin, phenobarbitone)
 - Hereditary (e.g. familial hypophosphatemia)

Children at high risk for rickets include:

- Low birth weight, preterms, twins.
- Breast fed and inadequately exposed to sunlight.
- Infants born from mothers with osteomalacia.

The Effects of Vitamin D Deficiency (Pathology)

Effects on the bone

Deposition of excessive uncalcified osteoid tissue at the metaphysis (rachitic metaphysis) results in irregular and frayed metaphysis. When the uncalcified osteoid tissue is compressed, it results in rachitic rosary and enlargement of ends of long bones (wrist or ankle widening).

Subperiosteal bone demineralization as resorption leads to a thin and less rigid cortex which results in long bone deformity (bow legs etc) and pathological fracture.

Chemical Changes

The chemical changes in rickets are normal or low serum Ca^{+} , Low serum Phosphorus, and increased Alkaline phosphatase.

Clinical features

The peak age is 6 months to 2 years.

Early signs.

The early signs of vitamin D deficiency include:

- Craniotabes - ping pong feeling on pressing the skull.
- Rachitic rosary- palpable enlargement of costochondral junction.
- Enlarged ends of long bones of arm and legs (wide wrist, double aleoli)

Late/Advanced signs

The late/advanced signs of vitamin D deficiency include:

- Frontal bossing, pigeon chest, bow legs/knocked knees, hypotonia, caput quadratum (box like head), Harrison's groove, fractures, etc.
- Wide fontanel /delayed closure and delayed teeth eruption.

Diagnosis

Diagnosis is mainly clinical and involves:

- Wrist X-ray – shows decreased bone density and irregular and cuped ends of bones.
- Normal/low serum Ca^{++} , low phosphorus, elevated alkaline phosphatase

Complications

Complications include the following:

- Pneumonia
- Atelectasis
- Hypocalcemic tetany
- Cor pulmonale
- Pathologic fracture

Treatment

There are two forms of treatment:

- Vitamin D₃ (cholecalciferol) is the best treatment
Dose - 600,000, i.m stat.
Healing occurs after 2-4 wks which can be proved with a wrist X-ray.
- Sunlight exposure is useful if no vitamin D₃ is available.

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CHAPTER TWO

NEONATOLOGY

Solomon Amsalu

2.1 PHYSICAL EXAMINATION OF THE NEWBORN

Objectives:

By the end of this section, readers will be able to

- Discuss steps in examining the newborn
- Identify life-threatening conditions in the newborn
- Identify common malformations that occur in the new born
- Estimate postnatal gestational age of a new born

The initial examination of a newborn infant should be performed as soon as possible after delivery to detect abnormalities.

Table 12: Neonatal growth and normal birth measurements

Neonatal growth parameters	Normal birth measurments
weight	2500gm – 4000gm
Height	
Head circumference	

Pulse (normal 120-160 beats/min), respiratory rate (normal 30-60 breaths/min), temperature (normal 36.5 37.5°C), weight, length and head circumference should be recorded.

General appearance

- Observing the infant is the most important aspect of physical examination.
- The state of alertness, the muscle tone, the activity, obvious anomalies or injuries, respiration, and the skin are assessed during inspection.
- The muscle tone should be sufficient for the hips, knees, and elbows to be flexed while the infant is lying supine.

- Some spontaneous movement should be evident, and the well infant should appear alert at some point during the examination.

Skin

Vernix usually covers the skin, especially in the skin folds of the axillae, neck and groin at birth. Post term infants characteristically have little vernix, and the skin is dry, crackled and wrinkled.

Generalized cyanosis may indicate significant heart or lung disease. Pallor may represent severe asphyxia (asphyxia pallida) or anemia. Erythema toxicum, which is a benign small, white occasionally vesiculopustular papules on an erythematous base appear 1 to 3 days after birth. It may persist for as long as one week.

Head

- The commonest abnormalities are caput succedaneum and cephalhematoma.
- Caput succedaneum is edema of the scalp skin and crosses suture lines.
- Subgaleal hemorrhage is bleeding below the aponeurosis, It is a boggy swelling that crosses suture lines.
- Cephalhematoma is a subperiosteal hemorrhage and does not cross suture line.
- Macrocephally, microcephaly, hydrocephalus.
- Craniosynostosis is the premature fusion of sutures. There is a hard non-movable ridge over the suture and an abnormally shaped skull.
- Craniotables (softening of the skull) are occasionally found in the parietal bones at the vertex near the sagittal suture. It is commonly a normal variant.

Eyes

- Conjunctival and retinal hemorrhages are usually benign.
- The cornea should be clear.
- The presence of bilateral red reflexes suggests the absence of cataract or of intraocular pathology/.
- Leukokoria (white papillary reflex) suggest cataract, tumor or chorioretinitis.

Nose

- The naris should be symmetric and patent. Newborn infants breathe preferentially through the nose so that bilateral choanal atresia may cause severe respiratory distress.
- Passing a catheter through each naris to the stomach excludes the diagnosis of choanal atresia.

Mouth

- The soft and hard palate should be inspected for cleft lip or cleft palate.
- Natal teeth occur rarely and are usually shed in a few days.
- On the hard palate on either side of the raphe may be temporary accumulations of epithelial cells called Epstein pearls.

Neck

- The neck of newborn always seems short. Very short and webbed neck may be associated with Turner syndrome.
- A thyroglossal duct cyst is usually palpable in the midline and retracts with tongue protrusion.
- The infant may have congenital torticollis with or without a fibrous mass in the sternocleidomastoid muscle, associated with head tilt.
- A cystic hygroma is a spongy mass that may increase with increased intrathoracic pressure.
- Hemangiomas are similar to hygromas upon palpation but are frequently associated with skin discoloration.
- Thyroid masses are usually visible and easily palpable with the head extended.

Chest and Lungs

- The appearance of flaring, grunting, tachypnea or retractions (lower chest indrawing intercostal retractions) is called respiratory distress
- Look for chest deformity. Breast hypertrophy, under maternal hormonal influence, is common. Milk may be present for a few days or weeks.
- A respiratory rate consistently over 60/min during periods of regular breathing usually indicates pulmonary, cardiac or metabolic disease.
- Look for apnea which is cessation of breathing for more than 15 seconds.

Heart

- The pulse rate may vary from 90/min in relaxed sleep to 180/min during activity.
- Radial, brachial and femoral pulses should be palpated and compared. Poor or diminished quality of pulses is an indication of inadequate cardiac output.
- Heart murmurs may be transient or may indicate significant heart disease.

Abdomen

- The liver is usually palpable and less commonly the tip of the spleen may be felt.
- The abdomen should be palpated for masses. Renal pathology is a usual cause of neonatal abdominal mass.
- Abdominal distention may be due to intestinal obstruction.
- A scaphoid abdomen in a newborn suggests diaphragmatic hernia
- The umbilical vessels should be inspected to exclude the presence of single umbilical artery. Approximately 10-15% of infants with a single umbilical artery have other major congenital malformations.

Genitalia

- Males should be checked for hypospadias, to determine the location of the testes and to exclude inguinal hernias or hydrocele.
- Non purulent vaginal discharge may be present in females secondary to transplacentally passed maternal hormones.
- A large penis or clitoris should raise the question of adrenogenital syndrome.

Anus

- Some passage of meconium usually occurs within the first twelve hour after birth. Imperforate anus should be excluded by gentle insertion of the little finger.

Musculoskeletal system

- The hips should be examined to rule out congenital dislocation
- Barlow Test should be undertaken: as the knees are brought together (adducted) the examiner pushes laterally on the upper inner thigh. A click or clunk indicates that the femoral head slips over the lateral ridge of the acetabulum.
- Ortolani test should be undertaken: the thigh is flexed and abducted, and the femoral head is lifted in to the acetabulum. If reduction is possible, the relocation will be felt as a clunk.

- The feet should be checked for the presence of clubfoot.

Neurologic system

Examination should include:

- Level of consciousness
- The cry (high pitched or inconsolable)
- Posture of the newborn
- Muscle tone
- Spontaneous motor activity
- Neonatal (primitive) reflexes

Neonatal reflexes

Testing the neonatal reflexes allows an assessment of CNS integrity in the newborn. They are primitive reflexes that should disappear as the baby's CNS develops and matures. Asymmetry, absence or persistence beyond a given time frame signifies dysfunction of the CNS or CNS immaturity (prematurity).

Sucking reflex

When an object is placed in the infant's mouth or touches the lips, a sucking reflex is elicited. This reflex disappears by 12 months of age.

Rooting reflex

Stimulating one corner of the mouth or the lower lip causes the infant to turn towards the touch and open his/her mouth. This reflex disappears by 4 to 7 months of age.

Moro reflex

The infant is placed in a semi-upright position. The head is momentarily allowed to fall back wards, with immediate resupport by the examiners hand. A complete response consists of abduction of the upper limbs at the shoulder, extension of the forearms at the elbow and extension of the fingers followed by adduction and flexion. This reflex disappears by 4 to 6 months of age.

Absent or weak moro response indicate serious CNS disturbance.

Asymmetries occur with Erb palsy and clavicular fractures.

Plamar grasp reflex

The plamar grasp reflex is elicited by placing a finger in the open palm of each hand; normal infants grasp and hold the object. This reflex diminishes by 3 to 4 months of age.

Estimation of post natal gestational age

The Ballard scoring system for the estimation of post natal gestational age uses 6 neurologic and 6 physical criteria:

- Neurologic criteria include posture, square window (wrist), arm recoil, popliteal angle, scarf sign and heel to ear test.
- Physical criteria includes assessment of the skin, lanugo hair, plantar creases, breast, ear and genitalia
- Assessment of gestational age using physical and neurologic criteria is accurate only ± 2 weeks.
- If the discrepancy of gestational age by LMP and post natal assessment is less than or equal to 2 weeks, take the gestational age obtained by LMP. If it is greater than 2 weeks, take the gestational age determined by post natal assessment.

Table 13: The Ballard chart

Physical Maturity sign	Score						Record score here
	0	1	2	3	4	5	
Skin	Gelatinous, Red, translucent	Smooth, pink, visible veins	Superficial peeling and/or rash, few veins	Cracking, pale areas, rare veins	Parchment, deep cracking, no vessels	Leathery, cracked, wrinkled	
Lanugo	None	Abundant	Thinning	Bald areas	Mostly bald		
Plantar Creases	No Crease	Faint red markds	Anterior transverse crease only	Creases on ant. 2/3	Creases over entire sole		
Breast	Barely perceptible	Flat areolano bud	Stripped areola, 1-2 mm bud	Raised areola, 3-4 mm bud	Full areola, 5-10 mm bud		
Eye/ear	Pinna flat, stays folded	Slightly curved pinna; soft; slow recoil	Well-curved pinna, soft but ready recoil	Formed & firm. Instant recoil	Thick cartilage, ear stiff		
Genitals male	Scrotum empty, no rugae		Testes descending, few rugae	Testes down, good rugae	Testes pendulous, deep rugae		
Genitals female	Prominent clitoris and labia minora		Majora & minora equally prominent	Majora large, minora small	Cilloris and minora completely covered		
Total Physical Maturity score							

Physical criteria for maturity rating, Ballard scoring.

Neuromuscular Maturity (Ballard). (From Ballard J et.al: J pediatr 95:769, 1979.)

NEUROMUSCULAR MATURITY SIGN	SCORE						RECORD SCORE HERE
	0	1	2	3	4	5	
POSTURE							
SQUARE WINDOW (WRIST)							
ARM FLEXION							
POPITEAL ANGLE							
SCARF SIGN							
HEEL TO EAR							
TOTAL NEUROMUSCULAR MATURITY SCORE							

Maturity rating

Score	Weeks
-10	20
-5	22
0	24
5	26
10	28
15	30
20	32
25	34
30	36
35	38
40	40
45	42
50	44

Maturity rating as calculated by adding the physical and neurological score, thus calculating the gestational age (Ballard scoring)

Classification of the newborn

Classifying neonates based on their weight and gestational age is important because newborns, in the different classifications, differ in the cause, expected complications and prognosis.

Newborns are classified based on gestational age and birth weight.

Based on gestational age they are classified as:

Term – if the baby is born after 37 and before 42 completed weeks.

Post term – if the baby is born after 42 completed weeks.

Preterm - if the baby is born before 37 completed weeks.

Based on birth weight, babies are classified as:

Normal weight – if the birth weight is between 2500gm and 4000gm.

Low birth weight (LBW) – if the weight is between 2500gm and 1500gm.

Very Low birth weight (VLBW) – An infant with a birth weight of less than 1500gm.

LBW and VLBW includes preterm and small for gestational age.

Finally, newborns are classified by comparing the birth weight and gestational age using the weight for gestational age graph.

A newborn is classified as:

Appropriate for gestational age (AGA) if the birth weight lies between 10%- 90% for the specified gestational age.

Large for gestational age (LGA) if the weight lies above 90% for the specified gestational age, the newborn has a high weight for gestational age.

Small for gestational age (SGA) if the weight lies below 90% for the specified gestational age, the infant has a low weight for gestational age. Intrauterine growth is retarded

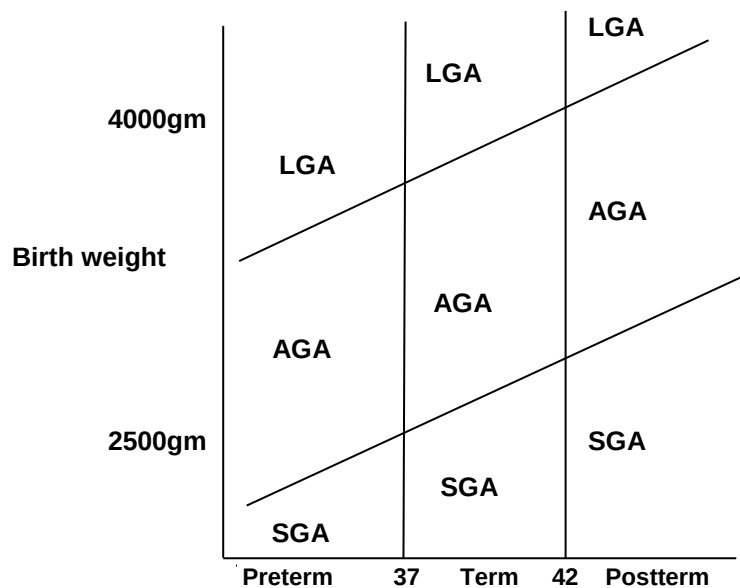


Figure 15: Chart showing weight for gestational age

Exercise

1. What is the aim of examining newborn?
2. How do you assess gestational age of a newborn baby?

2.2 RESUSCITATION OF THE NEWBORN

Objectives

By the end of this section the reader should be able to:

- Identify newborn babies needing resuscitation
- Know the steps of neonatal resuscitation

Goals of neonatal resuscitation include:

- Prevention of morbidity and mortality associated with hypoxic - ischemic tissue (brain, heart, kidney) injury
- Establishment of adequate spontaneous respiration and cardiac output.
- Prevention of heat loss (hypothermia).

The Apgar score is a practical method of systematically assessing the newborn infant immediately after birth to help identify infants requiring resuscitation for hypoxic acidosis.

Assessment is carried out in the first minute and resuscitation measures are made (if necessary) to improve the score at 5 minutes.

Table 14: The Apgar Score

Sign	0	1	2
Heart rate	absent	<100/min.	>100/min.
Resp. rate	absent	slow, irregular	good, crying
Muscle tone	limp	some flexion of extremities	active motion
Response to nasal catheter	no response	grimace	cough or sneeze
Color	blue, pale	body pink extremities blue	completely pink

A score of 0-3 → endotracheal intubation (if no skilled health worker bag and mask assistance is usually adequate)

A score of 4-6 → bag and mask assistance

A score of 7-10 → minimal interference

Infants who are born limp, cyanotic, apneic or pulseless require immediate resuscitation prior to the assessment of 1-minute Apgar score. Waiting until a one minute Apgar score is assigned before initiating resuscitation only increases the chance of permanent damage in a severely asphyxiated infant.

The elements of resuscitation include the following:

1. Initial steps: thermal management, clearing the airway, tactile stimulation
2. Establishment of ventilation
3. Chest compression
4. Medication

Initial steps

Immediately after birth the neonate should:

- be placed under a radiant heater, dried, positioned down and slightly extended,
- the airway cleared by suctioning and
- gentle tactile stimulation provided

If after suctioning and tactile stimulation there is no respiration or the heart rate is less than 100/min, positive pressure ventilation (PPV) with 100% oxygen must be initiated.

Positive pressure ventilation

- Usually can be provided quickly by bag and mask.
- Endotracheal intubation should be performed in any infant who does not respond to initial bag and mask ventilation or who was born apneic, pulseless, cyanotic, and limp with signs of fetal distress.
- In the vast majority of cases ventilation alone will be sufficient to resuscitate the infant.
- If after 30 seconds of positive pressure ventilation (PPV) with 100% oxygen the heart rate remains low, chest compression may be needed to assist a hypoxic and a failing myocardium.

Chest compression

Chest compression is recommended if:

- after ventilation with 100% oxygen for 15 - 30 seconds the heart rate is below 60 or
- between 60 and 80 and not rising.

The chest should be compressed between 100-120 times per minute with ventilation occurring 40 to 60 times per minute.

If chest compressions and ventilation do not raise the heart rate above 80 within 30 seconds, support for the cardiovascular system with medications is needed.

Medications

Epinephrine, 1:10,000 (0.1- 0.3 ml/kg) intratracheal or intravenous, should be the first drug to be given. The dose may be repeated every 5 minutes. Epinephrine stimulates the heart and increases non-cerebral peripheral vascular resistance.

10 to 20 ml/kg of volume expanders (normal saline, ringers lactate or blood) should be given for hypovolemia, history of blood loss, hypotension or poor response to resuscitation.

Sodium bicarbonate (1-2 meq/kg) should be given slowly (1 meq/kg/min) if there is a documented metabolic acidosis or if there is no adequate response to proper ventilation.

Dopamine and dobutamine (5-20 µg/kg/min) and volume expanders should be started after the initial resuscitation effort to improve cardiac output in an infant with poor peripheral perfusion, weak pulses, hypotension, tachycardia and poor urine output.

If resuscitative efforts do not result in spontaneous respiration within 30 minutes the prognosis is uniformly bad with respect to gross neurologic impairment. It is therefore entirely appropriate to discontinue active resuscitative efforts after 30 minutes if no spontaneous respiration has been observed, provided that the baby has been given intermittent opportunities to demonstrate spontaneous breathing.

EXERCISE

1. First minute evaluation of a newborn showed that he is bradycardic, has slow and irregular breathing, limp, no response to nasal catheter and is cyanosed. What is the Apgar score? How would you resuscitate this newborn?

2.3. Preterm, Low Birth Weight and Small for Gestational Age Infants

Objectives

At the end of this topic, the student will be able to:

- Describe problems of a preterm newborn
- Discuss the care of a preterm newborn
- Discuss discharge criteria for a preterm newborn from a hospital

In developing countries 20% and more of newborns are low birth weight.

Causes of preterm birth

Causes of preterm birth include the following:

Fetal - Fetal distress

- Multiple gestation

- Erythroblastosis

Placental - Placenta previa

- Abruptio placentae

Maternal - Preeclampsia

- Chronic medical illness

- Infections

Other - Premature rupture of membranes

- Polyhydramnios

- Iatrogenic

Main problems of preterm newborns

The problems of preterm newborn babies include:

- | | |
|--------------------------------------|----------------|
| - Hyaline membrane disease | - Hypothermia |
| - Hemorrhagic disease of the newborn | - Apnea |
| - Hyperbilirubinemia | - Hypocalcemia |
| - Bacterial sepsis | - Hypoglycemia |
| - Necrotizing enterocolitis | - PDA |
| - Intraventricular hemorrhage | - Anemia |
| - Asphyxia | |

In addition preterm infants frequently have

- Weak and uncoordinated ability to feed
- Prolonged failure to gain weight and
- Late metabolic acidosis

Main causes of death in low birth weight infants

The main causes of death in low birth weight infants are:

- Hyaline membrane disease
- Intraventricular hemorrhage
- Septicemia
- Asphyxia

Problems of SGA or IUGR infants

The problems of SGA or IUGR infants include:

- Perinatal asphyxia
- Hypoglycemia
- Hypothermia
- Meconium aspiration
- Necrotizing enterocolitis
- Polycythemia

Nursery care

At birth the measures needed for clearing the airway, initiating breathing, care of the cord and the eyes and administering vitamin K are the same in immature infants as in those of normal weight and maturity.

Additional considerations

There is a need to

- monitor temperature, and respiration
- for increased oxygen
- pay special attention to the details of feeding

Thermal management

The preterm newborn encounters the problem of severe heat loss for several reasons:

- the baby's large body surface area relative to body mass
- the tiny baby's small size presents a much smaller heat sink to store thermal reserve

- minimal shivering during exposure to cold and
- poor reserve of brown fat

Heat loss can be minimized by keeping infants in a thermoneutral environment. This could be achieved by using radiant warmers, blankets, heating lamps and by controlling the temperature and humidity of the room.

Kangaroo mother care is one method of keeping the baby warm. It involves placing the LBW/VLBW infants between the mother's breasts, covered and held in a prone or vertical position by the woman's clothes.its advantages are:

- It provides a physiological thermoneutral environment and protect the newborn from hypothermia,
- It encourages early contact between mother and newborn and early breast feeding and
- It prevents apnea because the mother breathing movement stimulates the baby

Supplementation with oxygen

Administering oxygen is indicated in newborns with respiratory distress or recurrent apnea.

Initiation of Feeding

The main principle in feeding premature infants is to proceed cautiously and gradually.

Weight less than 1500 g

Infants weighing less than 1500 g and GA less than 34 weeks require tube feeding because they are unable to coordinate breathing, sucking and swallowing.

Intestinal tract readiness for feeding may be determined by:-

- active bowel sounds
- passage of meconium
- absence of abdominal distention and
- no bilious aspirate or emesis

For these infants feed breast milk every 2-3 hrs. If breast milk is not possible, give preterm formula.

Weight more than 1500 g

Start breast milk within one hour of birth and feed at least 8 times per day.

Weight gain may not be achieved for 10-12 days because of the physiologic loss of weight for the first 10 -12 days.

Fluid requirements

Fluid therapy is indicated for preterm newborns as follows:

- Day 1: 70 -100 ml/kg/d
- After day 2: 150 ml/kg/d
- Type of fluid: day 1 - 10 % dextrose, thereafter, normal saline in 10 % dextrose.

This higher fluid requirements in preterms is due to high insensible water loss and they are less able to concentrate urine; thus, their fluid intake required to excrete solutes increases.

Discharge from Hospital

Prerequisites for discharge from a hospital include that the baby:

- is able to breast feed
- has sustained steady increments in weight of approximately 10-30 g in 24 hours
- stabilization of body temperature
- resolution of acute life-threatening illnesses

The required weight for discharge is 1500 grams

Exercise

1. What are the main problems of a preterm newborn?
2. What nursery care does a preterm newborn require?
3. Describe fluid management in a preterm newborn

2.4. Respiratory distress in the newborn

Objectives

By the end of this section the reader will be able to:

- Discuss main causes of respiratory distress in the newborn
- Know their clinical manifestation and management

Hyaline Membrane Disease

Hyaline membrane disease (HMD) is one of the most important causes of mortality and morbidity in preterm neonates.

An increased frequency is associated with infants of diabetic mothers, delivery by cesarean section, multiple pregnancies, precipitous delivery, asphyxia, cold stress and a prior history of affected infants.

Etiology

Surfactant deficiency is the primary causes of HMD. The failure to develop a functional residual capacity and the tendency of affected lungs to become atelectatic correlate with high surface tensions and absence of pulmonary surfactant.

Clinical features

- Signs of HMD usually appear within minutes of birth, although they may not be recognized for several hours in larger premature infants.
- The late onset of tachypnea should suggest other conditions.
- Characteristically, tachypnea, nasal flaring, subcostal and intercostal retractions, cyanosis and expiratory grunting are noted.
- Decreased bilateral air entry may be noted on auscultation.
- The natural course is characterized by progressive worsening of cyanosis and dyspnea. In most cases, the symptoms and signs may reach a peak within 3 days, after which improvement is gradual.

Investigation

CXR - The typical roentgenographic features consist of a diffuse reticulogranular pattern in both lung fields with superimposed air bronchograms.

Treatment

Thermoregulation

Infants with respiratory difficulty require an optimum thermal environment to minimize oxygen consumption and oxygen requirements. Body temperature should be maintained between 36.5 and 37°C.

Fluid

Maintenance of fluid requirements usually begin at 60 to 80 ml/kg/days as a 10 % dextrose solution and increases gradually to 150 ml/kg/day by the fifth day of life.

Antibiotics

The potentially fulminant course of congenital pneumonia and the difficulty in distinguishing it from HMD have led to the recommendation that all infants with significant respiratory distress receive antibiotic after appropriate cultures.

A penicillin combined with an aminoglycoside is the antibiotic regimen of choice.

Ventilatory support

Warm humidified oxygen should be provided to keep arterial levels between 55 and 70 mmHg.

Infants with severe HMD or those who develop complications resulting in persistent apnea require assisted mechanical ventilation.

Surfactant therapy

Multi-dose endotracheal instillation of exogenous surfactant for the treatment of HMD has dramatically improved survival and reduced the incidence of pulmonary air leaks.

Prevention

- Most important is the prevention of prematurity
- Administration of dexamethasone to women 48 hours before delivery of fetuses less than 34 weeks of gestation significantly reduces the incidence and the mortality and morbidity of HMD.

Meconium Aspiration Syndrome

Definition

Meconium staining of the amniotic fluid is usually considered to be indicator of fetal distress.

The passage of meconium in utero accompanies 8 to 20% of all deliveries and is seen predominantly in small for gestational age and post mature infants. Five percent of such infants develop meconium aspiration pneumonia.

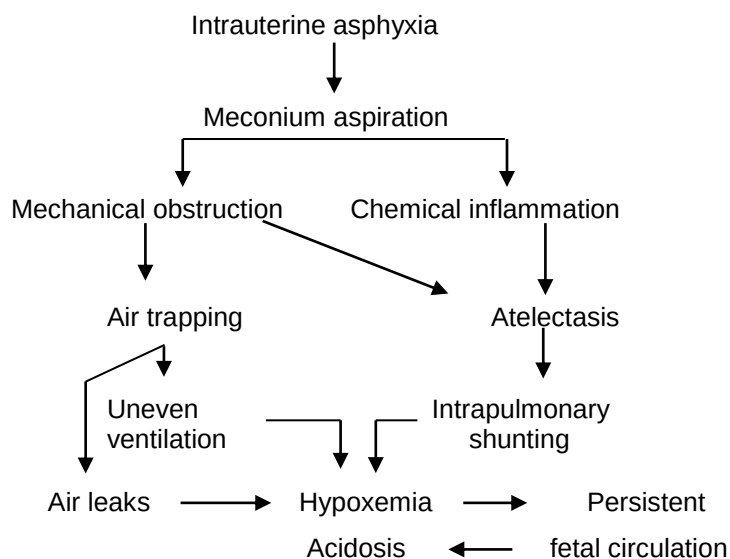
Clinical features

The infant frequently exhibits the classic signs of post maturity with evidence of weight loss together with nails, skin and umbilical cord that are heavily stained with a yellowish pigment.

The infants are often depressed at birth secondary to the hypoxic insult precipitating the passage of meconium.

Respiratory distress with cyanosis, grunting, flaring, retractions and marked tachypnea soon ensues.

Characteristically the chest acquires an over inflated appearance, and rales may be audible on auscu

Figure 16: Pathophysiology of meconium aspiration syndrome

Investigation

Chest X-ray is indicated by coarse, irregular pulmonary densities, increased anterior posterior diameter and flattening of the diaphragm.

Treatment

Depressed infant should undergo endotracheal intubation and suction should be applied directly to the endotracheal tube to remove meconium from the airway.

Treatment includes supportive care and standard management for respiratory distress.

Supportive management may include the use of antibiotics, because bacterial sepsis may have precipitated the passage and subsequent aspiration of meconium and bacterial pneumonia may be indistinguishable roentgenographically from meconium aspiration syndrome. Further, meconium enhances bacterial growth in vitro.

A vigorous infant with lightly stained amniotic fluid may not require treatment.

Prevention

The risk of meconium aspiration syndrome can be decreased by:

- Paying careful attention to fetal distress and initiating prompt delivery in the presence of fetal distress.
- Careful, thorough suctioning of the oropharynx after the head is delivered.

- Suctioning via endotracheal tube is indicated for depressed infants before initiation of positive pressure ventilation.

Pneumonia

Pneumonia could be acquired in the following ways:

Transplacentally

- Cytomegalovirus, rubella, toxoplasma gondii, listeria or treponema pallidum

Natally

- By aspiration of infected amniotic fluid or birth canal secretions.
- Onset occurs during the first few days of life.
- Commonly associated with group B streptococci, gram negative enteric bacilli, chlamydia and herpes simplex virus.

Postnatally

- It may be caused by a nosocomial or community acquired infection.
- Etiologies: *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Klebsiella*, *Serratia*, *B. pertussis* and respiratory viruses.

Congenital pneumonia

- Is pneumonia acquired transplacentally or perinatally.
- Frequently associated with premature rupture of membranes (PROM), chorioamnionitis, labor, premature labor or fetal distress.

Clinical features

The clinical features include respiratory distress, poor feeding, lethargy, irritability and a rise or a drop in temperature.

Chest X-Ray may be typical for bronchopneumonia or show diffuse atelectasis.

Treatment

Treatment is with parenteral ampicillin or penicillin and gentamycin for 7 to 10 days.

Supportive treatment includes giving intravenous fluids and oxygen and keeping a thermo neutral environment.

EXERCISE

1. What is the cause of hyaline membrane disease?
2. What are the manifestations of meconium aspiration syndrome?
3. How do you manage a newborn with congenital pneumonia?

2.5. Neonatal Jaundice and Hyperbilirubinemia

Objectives:

By the end of this section the reader will be able to:

- Discuss the etiology of jaundice in the newborn
- Identify physiologic and pathologic jaundice
- Discuss manifestations of kernicterus and management of hyperbilirubinemia

Introduction

Hyperbilirubinemia is a common and in most cases a benign problem in neonates.

The most common type is unconjugated (indirect reacting) hyperbilirubinemia. It is the type of pigment found in physiologic jaundice and in pathologic states in which there is increased production, decreased hepatic conjugation, or decreased hepatic uptake of bilirubin.

Conjugated hyperbilirubinemia (direct reacting) is far less common in neonates and most often denotes potentially serious disorder. However, it is not neurotoxic.

The unconjugated form is neurotoxic.

Definition

Neonatal jaundice - it is visible jaundice due to elevated serum level of bilirubin.

Hyperbilirubinemia - It is a markedly elevated serum bilirubin usually >15 mg/dl which predispose neonates to kernicterus.

Causes

Causes of unconjugated hyperbilirubinemia include:

- Increased load of bilirubin eg. hemolytic anemia following Rh or ABO incompatibility disease, polycythemia, subgaleal hemorrhage.
- Damage or reduction of the activity of transferase enzyme (genetic deficiency, hypoxia, prematurity, and infection)
- Blockage of the transferase enzyme (drugs)
- Absence or decreased amount of the enzyme (genetic defect, prematurity)

Causes of conjugated hyperbilirubinemia include:

- Sepsis
- Intrauterine infection
- Biliary atresia
- Hepatitis

Clinical Manifestations

- Jaundice may be present at birth or may appear anytime in the neonatal period.
- Jaundice usually begins on the face, and as the serum level increases, progresses to the abdomen and then the feet.
- Affected infants may appear lethargic or may feed poorly.

Physiologic Jaundice

- Physiologic jaundice is the result of increased bilirubin production following breakdown of fetal red blood cells combined with transient limitation in conjugation of bilirubin by the liver.
- Jaundice becomes visible on the second or third day, usually peaking between the second and fourth days at 5 to 6 mg/dl, and decreasing between the fifth and seventh day of life.

Pathological Jaundice should be considered if:-

- It appears in the 1st 24 hours of life
- Serum bilirubin is rising at rate greater than 5 mg/dl/24 hr.
- Serum bilirubin is greater than 12 mg/dl in full term or 14 mg/dl preterm infants
- Jaundice persists after 10-14 days of life or
- Direct reacting bilirubin is greater than 2 mg/dl

Investigation

Investigation involves testing for blood group, Rh,Hct, reticulocyte count, peripheral morphology, bilirubin (total and direct) and coomb's test.

Kernicterus

Defintion

Kernicterus is a neurologic syndrome resulting from the deposition of unconjugated bilirubin in the brain cells.

The risk of kernicterus increases with a rise in serum unconjugated bilirubin. Preterms and newborns with infections (sepsis, meningitis) have a greater susceptibility to kernicterus.

Clinical manifestation

Signs and symptoms usually appear 2 to 5 days after birth in term infants and as late as seventh day in preterm infants.

Common initial manifestations include lethargy, poor feeding and loss of moro reflex. Subsequently, the infant appears gravely ill with diminished deep tendon reflexes and respiratory distress.

It is manifested by opisthotonus, bulging fontanel, a high pitched cry and convulsions occur in advanced cases. Many infants who progress to these severe neurologic signs die.

Treatment

The goal of therapy is to avoid kernicterus. The principles of treatment are:

- Phototherapy
- Exchange transfusion
- Treatment of underlying disease.

Exercise

- 1 What is the difference between physiologic and pathologic jaundice?
- 2 What are the causes of pathologic jaundice?

2.6 Neonatal Sepsis and Meningitis

Objectives

By the end of this section the reader will be able to:

- Know the etiologies of sepsis and meningitis in newborns
- Discuss the clinical manifestations
- Describe the management of sepsis and meningitis

Neonatal Sepsis

Definitions

Bacteremia is bacteria in the blood.

Septicemia is caused by bacteremia and is a constellation of signs and symptoms caused by microorganisms or their toxic products in the circulation.

Neonatal sepsis (sepsis neonatorum) is a clinical syndrome resulting from the pathophysiologic effects of local or systemic infection in the first month of life.

Epidemiology

Predisposing factors for neonatal sepsis:

- Presence of maternal (obstetric) risk factors such as:
 - prolonged rupture of membranes (more than 18 hrs)
 - maternal chorioamnionitis (maternal intrapartum fever, maternal leukocytosis, uterine tenderness, fetal tachycardia)
 - Prolonged labor that lasts for more than 24 hours
 - Repeated vaginal examination
- Fetal risk factors include:
 - prematurity, LBW
 - asphyxia
 - developmental or congenital immune defects

- congenital anomalies

Etiology

Group B streptococci and *escherichia coli* are the leading causes of early and late onset neonatal sepsis. Other causes include *listeria monocytogenes*, *klebsiella* and enterococci.

Nosocomial infections are often related to the use of procedures, intravenous lines and other risk factors. They are most often due to coagulase negative staphylococci, *staphylococcus aureus* and gram-negative enteric organisms.

Early onset sepsis presents at birth, or within the first 7 days of life, usually within the first 3 days and is associated with obstetric risk factors.

Late onset sepsis: the onset is from 8 - 28 days of life and may extend to 60 days.

- Obstetric risk factors are unusual.
- Frequently associated with meningitis.

Clinical features

The initial signs of infection are often subtle or minimal.

There may be temperature instability (hypo or hyperthermia), poor sucking, depressed reflexes, irritability, lethargy and/or convulsion.

Late manifestations include apnea, cyanosis, hypotension, sclerema, and DIC.

The clinical signs of infection in the newborn are often nonspecific, vary greatly in intensity and severity and are seen in many non-infectious disorders as well.

Investigations

Evaluation includes blood culture, lumbar puncture (not indicated for those less than 12 hours), urinalysis and urine culture and total WBC count and differential

Immature neutrophil to total neutrophil ratio (more than 0.16) suggests bacterial infection)

Chest x-ray is indicated for those with respiratory problems

Treatment

- Initial empiric therapy of early and late onset community acquired infections should consist of parenteral ampicillin or penicillin and an aminoglycoside (usually gentamycin).
- In nosocomial infection, an anti-staphylococcal drug like cloxacillin should be substituted for ampicillin.
- Once the pathogen has been identified and antibiotic sensitivities determined, the most appropriate drug(s) should be selected.
- Third generation cephalosporins are valuable additions for treating neonatal sepsis and meningitis but have modest activity against *staphylococcus aureus* and *listeria monocytogens*, and enterococci are uniformly resistant
- Therapy should be continued for a total of 10 to 14 days.
- Supportive therapy includes should be carefully monitoring and correcting fluids, electrolytes, and glucose. The newborn should be kept in thermoneutral environment.

Neonatal Meningitis

Neonatal meningitis is often associated with sepsis and is a cause of significant morbidity and mortality. Agents that produce sepsis also commonly cause neonatal meningitis.

Initial signs and symptoms may be indistinguishable from those of neonatal sepsis.

Neurologic manifestations include lethargy, bulging fontanel, seizure, nuchal rigidity and signs of increased intracranial pressure.

The diagnosis is confirmed by examination of the CSF and the identification of bacteria, virus or fungus by culture or antigen detection.

Blood culture is positive in 70-85% of neonates with meningitis.

Contraindications for doing a lumbar puncture (LP) include:

- Increased intracranial pressure
- Infection on the lumbar space

- Severely ill infant if the lumbar puncture will further compromise the respiratory status.

CSF findings suggestive of meningitis include:

- WBC $> 30/\text{mm}^3$, particularly with an increased in the number of PMNL.
- CSF protein $> 150 \text{ mg/dl}$.
- CSF glucose $< 50\%$ of blood glucose.

CSF should be gram stained and cultured. Appropriate latex agglutination test can be done.

Therapy

Therapy will include intravenous ampicillin and gentamycin or ampicillin and cephalosporin for 21 days for gram's negatives or unknown status and for 10-14 days for group B strptococcus.

Exercise

1. Describe risk factors for neonatal sepsis.
2. What is early and late onset neonatal sepsis?
3. What is the difference in terms of management between neonatal sepsis and meningitis?

2.7. Congénital Infections (TORCH syndrome)

Objectives

By the end of this section the reader will be able to

- Know congenital infections that are commonly transmitted from the mother to the infant
- Discuss the manifestation of common congenital infections

TORCH stands for:

T - toxoplasmosis

O - other (includes other viruses and congenital syphilis)

R - rubella

C - cytomegalovirus

H - herpes simplex virus

Toxoplasmosis

Toxoplasmosis is caused by the protozoa *toxoplasma gondii*. Infection may be transmitted from the mother to the fetus transplacentally or during vaginal delivery.

Overall risk of transmission from the mother to the fetus is about 50%. The incidence of transmission is least in early gestation and greater later in gestation. The most severe involvement results from maternal infection that occurs during the first and second trimesters.

GOD IS GOOD

Clinical manifestations

Toxoplasmosis may present as mild or severe disease. Manifestations include IUGR (intrauterine growth retardation), prematurity, hepatosplenomegaly, jaundice and thrombocytopenia.

The classical triad consists of hydrocephalus, chorioretinitis and cerebral calcification.

Diagnosis

Diagnosis is usually based on:

- Toxoplasma dye test
- Immunofluorescent IgG toxoplasma antibody test or
- IgM antibody test.

Treatment

A combination of pyrimethamine and sulfonamide (sulfadiazine or triple sulfonamides) should be given for one year.

Prevention

The best way to prevent congenital toxoplasmosis is by preventing acute maternal infection in pregnancy. These include:

- Avoid contact with cats
- Avoid eating undercooked meat in pregnancy
- Wash vegetables thoroughly before eating
- Wear gloves when handling meat and vegetables

Rubella

The risk of a congenitally infected infant varies with the gestational age of the fetus when maternal infection occurs. Between 50 and 80% are infected if maternal infection occurs prior to the 8th week of gestation, but fetal infection is uncommon if maternal infection is in the third trimester.

Clinical manifestations

Rubella may produce

- Transient neonatal manifestations,
- Permanent organ malformations and tissue injury or
- Delayed late onset disease.

Manifestations include IUGR, microcephaly, microphthalmia, cataract, thrombocytopenia, congenital heart disease (patent ductus arteriosus and peripheral pulmonary artery stenosis), linear bone lesions, retinitis, convulsions and sensorineural deafness.

The majority of manifestations however present much later in life. These include:

- Hearing loss (87%)
- Congenital heart disease (46%)
- Mental retardation (39%) and
- Cataract and glaucoma (34%)

Diagnosis

- Viral isolation from nasopharyngeal secretions, urine or CSF
- Detection of rubella specific IgM

Treatment

- There is no specific treatment for rubella infection.
- Seronegative women of childbearing age should be offered vaccination either after pregnancy test or immediately postpartum.

Prevention

Prevention includes:

- Rubella vaccine for all nonimmune individuals 12 months of age and older and child bearing age mothers. Conception also should be avoided for 3 months following immunization and it is contraindicated during pregnancy.
- Avoid contact with anyone known to have acute or recent rubella infection.
- Termination of pregnancy if active maternal infection is detected

Cytomegalovirus (CMV)

CMV is a member of the herpes group of viruses.

Mode of transmission include:

- transplacentally
- through genital secretions
- breast milk and
- blood transfusion

CMV produces characteristic intranuclear and intracytoplasmic inclusion bodies and an enlarged cell size (cytomegalic).

Fetal damage may follow primary infection or recurrent infection at any stage of pregnancy

Clinical manifestations

In about 40% of primary infections of the mother, the fetus will be infected but the majority of neonatal infections are asymptomatic.

Clinical manifestations include IUGR, jaundice, hepatosplenomegaly, petechial rash, chorioretinitis, pneumonia, periventricular calcifications and microcephaly.

Infants who have symptoms in the newborn period nearly always have subsequent handicap, the most common problem being sensory neural hearing loss.

Diagnosis

Diagnosis involves:

- Isolation of the virus from the urine or saliva
- Demonstration of CMV specific IgM antibodies

Treatment

There is no specific therapy. The use of ganciclovir is being studied.

Herpes simplex virus (HSV)

Three quarters of HSV neonatal infection is due to HSV type 2 and the rest is due to type 1.

Modes of infection include

- Mostly from maternal genital tract infection
- Intrauterine infection (5%)
- Postpartum (10%)
- Transmission from a primary lesion is about 50%
- Transmission from recurrent lesions is only 1-3%.

Clinical manifestations

There are three major categories

1. Disseminated form commences at about 1 week of age with constitutional signs and symptoms of sepsis and the characteristic erythematous based vesicles

2. Skin, eye or mouth disease
3. Encephalitic form
 - Usually of later onset (3 wks)
 - Has a lower incidence of viremia and vesicles

Diagnosis

Diagnosis is based on:

- cytologic examination,
- direct viral culture of skin, mouth or eye lesions or
- analysis of IgM specific antibody response

Treatment and prevention

Treatment and prevention involves intravenous acyclovir for 14 -21 days.

Prevention

Delivery by cesarean section may reduce the risk of neonatal infection.

Congenital syphilis

Syphilis is a chronic, generalized infectious disease caused by a slender spirochete, *treponema pallidum*.

Congenital syphilis results from transplacental transmission of spirochetes.

Pregnant women with 1^o or 2^o syphilis and spirochetemia are more likely to transmit infection to the fetus than are women with latent infection.

Transmission may occur throughout pregnancy. Syphilis during pregnancy has a transmission rate approaching 100%. Fetal and perinatal deaths occur in 40% of affected infants.

Clinical manifestations

Early congenital syphilis

Early congenital syphilis appears during the first 2 years of life.

Approximately two-thirds of the infants are asymptomatic at birth, if they are untreated, symptoms develop within weeks or months.

Manifestations include hepatosplenomegaly, lymphadenopathy, hemolytic anemia, thrombocytopenia and mucocutaneous rash. Rhinitis (snuffles) and condylomatous lesions are highly characteristic features of mucous membrane involvement. CNS abnormalities, failure to thrive, chorioretinitis, nephritis and nephrotic syndrome may also be seen. Roentgenographic abnormalities include:

- Multiple sites osteochondritis at the wrist, elbow, ankle and knees and
- Periostitis of the long bones.

Late congenital syphilis

The manifestations result primarily from chronic inflammation of bone, teeth, and the CNS. Skeletal changes include frontal bossing, thickening of the sternoclavicular portion of the clavicle and an anterior bowing of the mid portion of the tibia (Saber shins).

Hutchinson's triad includes:

- Hutchinson's teeth (peg shaped upper central incisors)
- 8th nerve deafness and
- interstitial keratitis

The clutton joint represents synovitis involving the lower extremities (usually the knee). Saddle nose is a result of rhinitis.

Diagnosis

Serology: Non treponemal tests include:

- VDRL (Venereal Disease Research Laboratory)
- RPR (Rapid Plasma Reagin)

These tests are rapid and non-expensive but they are not specific. Maternal antibody crosses the placenta and accordingly a false positive VDRL can occur in an uninfected infant delivered to a VDRL positive mother. This passively transferred antibody is not demonstrable by 3 months of age.

Treponemal tests include:

- TPI (Treponema Pallidum Immobilization test)
- FTA-ABs (Fluorescent Treponemal Antibody Absorption test)

Dark field microscopy or direct fluorescent antibody test of scrapings from primary lesions can reveal *T. pallidum* and often permits a definitive diagnosis.

Because doing these investigation is not always possible, it is recommended:

1. A presumptive diagnosis of congenital syphilis should be made and be treated in an infant whose mother had untreated or inadequately treated syphilis or whose treatment status is unknown. Inadequate treatment means she was treated ≤ 30 days before delivery or treated with non penicillin regimen.
2. All infants with a presumptive diagnosis of congenital syphilis should be treated with regimen effective for neurosyphilis because this cannot be reliably excluded.

Treatment

Treatment involves:

- Crystalline penicillin (100,000-150,000 IU/kg/d **iv**) or
- Procaine penicillin (50,000 IU/kg/d **im**) for 10-14 days.

Treatment is recommended for any infant if maternal treatment is unknown, inadequate or given during the last 4 wks of pregnancy or if the treatment was by any drug other than penicillin.

Exercise

1. What are the manifestations of congenital rubella?
2. What is the difference between early and late congenital syphilis?
3. Describe the management of congenital syphilis?

CHAPTER THREE

SHOCK

Goitom G/Yesus

Definition

Shock is an acute syndrome characterized by inadequate circulatory perfusion of tissue to meet the metabolic demands of vital organs.

Etiology

Most common triggering conditions are hypovolemia, sepsis, cardiac failure, trauma and toxin.

Epidemiology

Shock is a clinical state which may result in high mortality rate of 20 to 50%. Most patients die in the acute hypotensive or hypoxemic phase of shock, as a result of one or more complications associated with shock states. Multiple organ system failure increases the probability of death. In developed countries, the mortality rate of shock in children has declined as result of educational efforts and pediatric advanced life support, which emphasizes early recognition and intervention.

Pathophysiology

Shock may result in tissue injury through a number of pathways and factors such as:

- loss of fluid from the body,
- decreased myocardial contraction or obstruction to out flow of the heart,
- shift of fluid from intravascular to extra-vascular compartment, or
- accumulation or stasis of body fluid in the periphery as a result of vasodilation

This may result from the different etiologies of shock leading to inadequate tissue perfusion. In the early phase of shock, a number of compensatory physiologic mechanisms act to maintain blood pressure and preserve tissue perfusion. These responses include increase of heart rate, stroke volume and vascular smooth muscle tone to help preserve blood flow to vital organs such as the brain, heart and kidneys.

When the compensatory mechanisms cannot maintain adequate tissue perfusion, the resulting hypoxia and ischemia causes cell injury and ultimately cell death results.

Clinical manifestations

Shock is classified clinically as shown below:

- Hypovolemic: characterized by loss of body fluid; e.g. diarrhea and hemorrhage
- Cardogenic: cardiac dysfunction; e.g. congenital heart disease, arrhythmias, myocarditis, cardiomyopathy and mechanical defects.
- Obstructive: obstruction to inflow or outflow of blood to or from the heart; e.g. cardiac tamponade, pulmonary embolism and pneumothorax.
- Distributive: abnormal vasodilation or increased permeability due to loss of sympathetic vasomotor tone and/or presence of vasodilating substance in the blood; e.g. anaphylaxis, sepsis and trauma.

The clinical manifestations of shock depend, in part, on the underlying causes.

General manifestations are tachycardia, weak or impalpable pulses, low blood pressure, cold extremities, mottled extremities, sluggish or absent capillary refill, tachypnea, reduced urine output, irregular respiration or apnea, peripheral or central cyanosis and lethargy ranging up to coma.

Specific manifestations are:

- In hypovolemia- poor skin turgor. Sunken eye balls, dry mouth, signs of multiple trauma
- In sepsis- fever, flushed appearance, hepatomegally, chest crepitations

Investigation

The investigations that are needed are usually specific to the suspected underlying cause and may include:

- Complete blood count, cross-match, and culture of blood, urine, and cerebro spinal fluid, coagulation profile and serum electrolytes may be needed
- Renal function test
- Chest X-ray and EKG

Management

Early recognition of shock is a key step in the success of treatment.

Initial step include:

- Early attention to patency of airways and breathing
- Give oxygen.
- Maintain intra-venous access and give 20ml/Kg of Normal Saline or Ringer Lactate as rapidly as possible. Reassess the patient and repeat another 20ml/Kg if needed. Give subsequent deficit replacement therapy.
- In a patient not responding to the 40ml/Kg dose of fluid or if there is loss of blood, give 10ml/Kg of blood, 5% albumin, or plasma.
- Catheterize the patient to monitor urine output.
- If sepsis is suspected, give empiric antibiotic coverage.
- Assess response to therapy by checking vital signs, and signs of end organ perfusion such as skin color and warmth, mental status, and urine out put.
- In patients cardiogenic or septic shock if there is no response to fluid therapy vasoactive drugs such as Dopamine, Epinephrine, or Dobutamine may be needed
- Treat the underlying cause of shock.

Exercise

1. Define shock.
2. What is the clinical classification of shock?

Further Reading

Cleary, T.G.. *Shock*. Behrman, Kliegman, Jenson, Nelson Text Book of Pediatrics, 16th edition, 2000.

PART TWO

Specific Paediatrics

CHAPTER FOUR

INFECTIOUS DISEASES

Objectives

At the end of this chapter, the student will be able to:

- describe epidemiology of infectious diseases, emphasising on Ethiopian context
- identify etiologic agents and their basic nature
- explain pathogenesis
- describe the diagnosis and important lab investigations
- describe the treatment, preventive measures as well as prognosis

4.1. COMMON CHILDHOOD INFECTIONS

MEASLES

Habtamu Fekadu

Definition

Measles is a highly contagious viral disease characterized by fever, coryza, conjunctivitis and maculopapular rash.

Etiology

Measles is caused by an *RNA virus* that belongs to the *polymyxoviridae* family

Epidemiology

Measles is a leading cause of morbidity and mortality among vaccine preventable diseases and is endemic in most countries. It accounts for 42 million cases and 1 million deaths each year. Most occur in developing countries. The highest incidence is between 9 months and 2 years of age and 80% occurs in children under 5. Both sexes are equally affected. Measles is severe and carries a high fatality rate in malnourished and immunocompromised children.

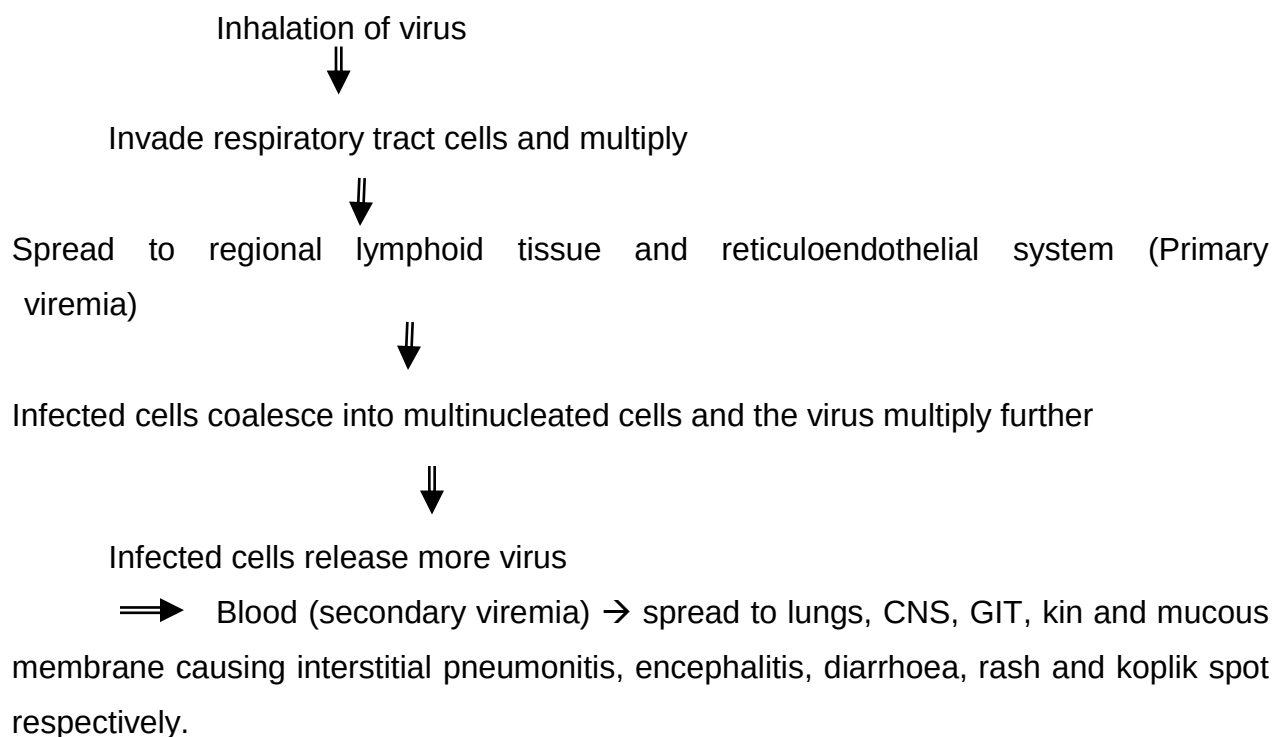
Measles is spread by the respiratory route via small droplets. It is highly contagious: 90% of susceptible contacts contract measles if they are exposed to infected patient. The infected child is infectious 1 week before and 5 days after the rash appears.

Neutralizing antibodies and activated lymphocytes control infection and provide life long immunity against subsequent infection. Transplacental immunity is protective for 6 to 9 mths.

Important features for the eradication of the measles include:

- Humans are the only reservoir and there are no carriers
- Vaccination is effective
- Measles' characteristic clinical feature is it is only one serotype

Figure 17: Pathogenesis and pathology



Clinical features

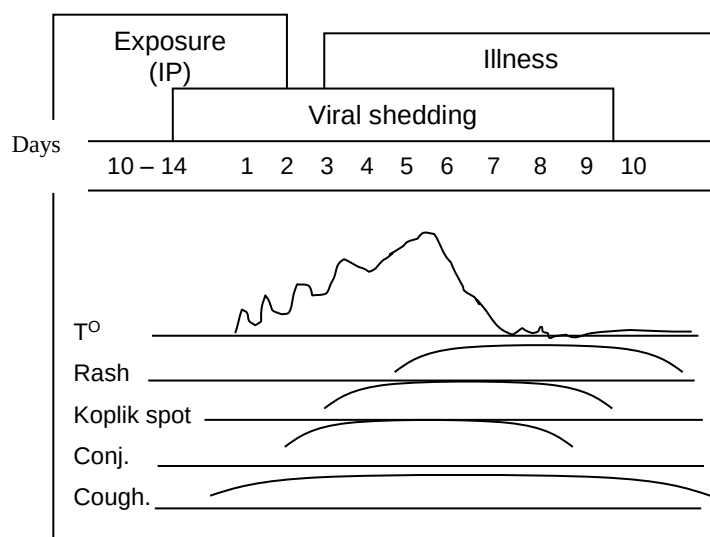
The incubation period lasts 10 – 12 days. Typically, measles has 2 phases:

1. Prodromal phase lasts 3 to 5 days. During this phase the child presents with fever, cough, coryza and conjunctivitis. Koplik spots appear. These are greyish or whitish spots surrounded by red base that appear on the buccal mucosa opposite to lower

molar on the second day and disappear 2 days after onset of the rash. It is pathognomonic of measles.

2. Eruptive phase during which the child is very sick and manifested with fever, which rises markedly as the rash appears and peaks by the second or third day and then starts to decrease. The rash is initially faint erythematous macules that progresses to a maculopapular rash, first appearing behind ears, along hairline and cheek and spreading centrifugally to involve the neck, trunk and lastly the feet by the second or third day. This fades in 4 to 7 days in the same sequence in which it appeared, leaving desquamation. Diarrhoea and cough are common especially in infants and young children

Figure 17: The natural course of measles



Diagnosis

Typically, the diagnosis of measles presents no difficulties so it is mainly clinical: a child who has generalized rash and one of coryza symptoms (cough, runny nose or red eyes) may be assumed to have measles. It is rarely that a laboratory test is needed.

Complications

The complications from measles include:

- Respiratory complications are the most common complication and cause of death. Pneumonia is the most common and carries a high mortality. Pneumonia is caused

by measles interstitial pneumonitis (giant cell pneumonia) caused by the measles virus or bacterial pneumonia (*pPneumococcus*, *H. influenzae*, *staph. aureus*) which has a high mortality. Otitis media is also common.

- Gastroenteritis is the second commonest complication. The child present with diarrhoea, vomiting, or persistent diarrhoea.
- Malnutrition is a common sequel of measles due to anorexia, oral lesions, persistent diarrhoea or increased catabolism. Measles often precipitates acute kwashiorkor.
- Corneal ulceration can be caused by the exacerbation of xerophthalmia or direct viral keratoconjunctivitis and leads to blindness.
- Encephalitis is a rare complication which presents with fever, seizure, neck
- Other complications include measles croup and the reactivation of tuberculosis stiffness, confusion, and coma.

Treatment

Treatment is symptomatic and includes:

- Bed rest
- Antipyretic
- Increase fluid intake
- Vitamin A on first, second and fourteenth day decreases corneal ulceration and morbidity and mortality from measles
- Treat complications with antibiotics for pneumonia and otitis media and treat croup as any croup

Prognosis

Most children recover completely but young and malnourished children and those with encephalitis have a poor prognosis. The commonest cause of death is pneumonia.

Prevention

Prevention involves

- Isolation for the infectious period
- Vaccination at 9 months has more than 95% efficacy
- Post exposure prophylaxis giving measles vaccine within 72 hours for susceptible children older than 6 months.

Pertussis

Tsinuel Girma

Definition Pertussis is an acute respiratory infection caused by *bordetella pertussis*.

Epidemiology

Pertussis occurs as sporadic and epidemic cases. Most patients are under six years old, but any age including neonatal age groups are susceptible, since there is no transplacental protection. Overcrowding favors the spread of the disease, because transmission is via droplet infection. The highest infectivity is in the early stages of the disease and the attack rate is close to 100%.

Etiology

B. pertussis and *B. parapertussis* are gram-positive rods which grow well on border gengenon agar (glycerin-potato-blood) media.

Pathogenesis

Transmission is through aerosol droplets at close range. Bacterias attach to ciliated epithelium of the respiratory tract, resulting in congestion, excessive mucous production and infiltration with lymphocytes. The lumen of the air ways especially the bronchioles narrow, producing atelectasis and emphysema.

B. pertussis produces many toxins. Pertussis toxin (DT) plays a central role in the pathogenesis.

Clinical Manifestations

The clinical manifestations are divided into three stages each lasting about two weeks: catarrhal, paroxysmal and convalescent.

Catarrhal stage

After the incubation period of 3 to 12 days non distinctive symptoms of low grade fever, nasal discharge, sneezing and lacrimation are seen. This stage usually simulates simple upper respiratory infections

Paroxysmal stage

Progressively increasing repetitive series of forceful coughs in a single expiration is the hallmark of this stage. Children above the age of 2 years have massive inspiratory effort (whoop) as air passes through narrowed bronchial tree. The cough is followed by vomiting (post tussive vomiting) and profuse sweating. The intense cough and its associated increased pressure may result in hernias, rectal prolapsed, orbital edema, ulceration of frenulum and **epistaxis** which are helpful in the diagnosis. Immunized children and adults have no distinct stages.

Convalescent stage

Severity and frequency of paroxysms decrease but cough and whoop may persist for months.

Diagnosis

Diagnosis is confirmed by positive culture of *B. pertussis* on bordet – gagnon medium.

Leukocytosis due to absolute lymphocytosis greatly supports the diagnosis. An absolute increase in neutrophils suggests a different diagnosis or secondary bacterial infection. Chest X-rays show perihilar infiltration and variable atelectasis.

Treatment

Therapy includes:

- Limit number of paroxysms
- Clearance of airways and respiratory support if paroxysms are life threatening
- Maximizing nutrition
- Adequate rest

Admission criteria

- Infants below 3 months of age
- Witnessed severe paroxysms
- Children with significant underlying cardiopulmonary or neuromuscular disorders

Supportive care

- Small volume frequent feeds between cough episodes when patient is at rest
- Suction airways when necessary
- Oxygen if in distress

Antibiotics

- Always indicated for suspected or confirmed cases for potential clinical benefit and to limit spread of infection
- Standard treatment is erythromycin 40 – 50mg. kg/24h p.o. in four divided doses for 14 days.

Prognosis

Young age (less than 2 months) and severe malnutrition are poor prognostic factors.

Complications

Complications include apnea, pneumonia (*S. aureus* and *S.pneumoniae*), otitis media, seizure and physical squeale of forceful coughing and the reactivation of tuberculosis.

Prevention

Primary prevention involves 3 doses of pertussis (DPT) vaccine (EPI) Secondary prevention involves isolating the patient and treating them, vaccination and giving erythromycin for contacts

Rubella

Habtamu Fekadu

Definition

Rubella is a common and mild exanthematous disease associated with fever, rash and lymphadenopathy but which can result in serious damage to infants born from infected mothers.

Etiology

Rubella is an *RNA virus* of the genus *rubivirus* in the family *Togaviridae*

Epidemiology

Rubella is a widespread and highly contagious disease. It mainly affects children, though it affects all age groups. Transmission occurs mainly by respiratory droplets and occasionally by direct close contact. Transplacental spread results in congenital rubella syndrome (CRS). The highest risk of transmission, which is greater than 50%, and serious damage occurs in the first trimester. The patient is infectious 1 week before and 1 week after the rash.

Natural infection confers lifelong immunity. The transplacental immunity protects the infant for 6 to 9 months.

Clinical features

The incubation period is 14 to 21 days (usually 17 days). Most infections are subclinical. The characteristic presentations are lymphadenopathy, skin rash, and low-grade fever.

A tender posterior cervical, occipital, and retro-auricular lymphadenopathy appears 2 days before the onset of rash.

A discrete macular / maculopapular rash appears that starts from the face and spreads centrifugally to involve the trunk and extremity. It fades on the third day in the same order it appeared.

Foetal infection leads to abortion, stillbirth or congenital rubella.

Congenital Rubella Syndrome (CRS)

Congenital Rubella Syndrome (CRS) affects all organ systems. The infant presents with IUGR, jaundice, cataract, microcephaly, congenital heart disease (persistent ductus arteriosus), purpura, hepatosplenomegaly, anemia and meningoencephalitis which manifests with convulsion and lethargy.

Late sequelae include sensorineural deafness and mental retardation .

Diagnosis

Diagnosis is mainly clinical although viral culture (nasopharynx, blood) may be used.

Treatment

Treatment is supportive and involves no specific treatment.

Prognosis

Childhood rubella has an excellent prognosis, but CRS has a poor prognosis

Prevention

Prevention is mainly through the rubella vaccine which is given in combination with measles and mumps as MMR. It has a 90% efficacy.

Mumps

Habtamu Fekadu

Definition

Mumps is a viral illness characterized by painful swelling of salivary gland.

Etiology

Mumps is caused by a RNA virus that belongs to *paramyxovirus* group.

Epidemiology

Mumps is endemic worldwide, especially in unvaccinated and heavily populated areas and It occurs in epidemics. Transmission occurs through airborne droplets and direct contact of saliva. The patient is infectious two days before parotid swelling and up to three days after the swelling subsided.

Mumps affects any age but the peak age is 5 to 9 years and 85% occur in children under 15 yrs of age. Infants are protected by transplacentally passed antibodies.

Pathogenesis

The inhaled virus multiply in the respiratory tract and enter the blood (viremia). They spread to many tissues (mainly salivary glands, gonads, pancreas and brain).

Clinical features

The incubation period is 14 to 24 days. 10 to 50% of cases are subclinical. Mumps presents with:

- Painful swelling of parotid gland that rapidly grows and attains the maximum size by 3rd day. Unilateral at onset and in 24 to 48hr becomes bilateral in most cases.
- Submandibular glands are affected in 10 – 15% of cases.

Complications

Meningoencephalitis is the most common complication. It occurs in 30% of cases. Symptoms and signs are as any meningoencaphalitis. It carries a mortality of 2%.

Other Complications are orchitis/epididymitis, pancreatitis, oophoritis and myocarditis,

Treatment

Treatment is supportive. There is no specific treatment.

Prevention

Prevention is mainly through the mumps vaccine, which is given at the age of 15 months in combination with rubella and measles as MMR, has a 97% efficacy.

Chickenpox

Habtamu Fekadu

Definition

Chickenpox is a highly contagious disease which is caused by varicella zoster virus and characterized by fever, mild constitutional symptoms and centripetal skin rash appearing on the first day of the illness.

Etiology

Chickenpox is caused by a herpes virus called *varicella zoster virus* (VZV) which is a DNA, and neurotropic virus. It is the acute invasive phase of VZV whereas herpes zoster is the reactivation form.

Epidemiology

Chickenpox is common and worldwide. The virus is transmitted mainly through airborne droplets. Other modes of transmission are direct contact and the transplacental route that leads to neonatal chickenpox. It is highly contagious with home contact attack rate of 80 to 90%.

Chickenpox occurs in epidemics. The period of infectivity is 24 to 48 hours before and 3 to 7 days after the onset of rash. Infection confers lifelong immunity.

Chickenpox mostly occurs in children under 10 years of age and in most cases it is benign, but it tends to be severe and life threatening in neonates and immunocompromised.

Pathogenesis

The inhaled virus replicates in respiratory tract which leads to subclinical viremia which in turn leads to secondary viremia and causes wide spread skin lesion and can involve different organs (lung, brain etc.)

Host immune responses limit viral replication and facilitates recovery. This VZV primary infection establishes lifelong latent infection of sensory ganglion neurons, which when reactivated results in herpes zoster.

Clinical features

The prodromal phase is not present in young children but it is mild in older children and adults. there is a characteristic rash that:

- Starts as macule, becomes papule and turns into vesicle and pustules, leading to a crust.
- Occurs in crops and all stages are present at the same time which is characteristic of chickenpox. It is pruritic.
- Has centripetal distribution and mainly involves the trunk.

Complications

Complications include super infection of the skin lesions, encephalitis and pneumonia

Treatment

Treatment involves supportive care and acyclovir for neonates and children that are immunocompromised.

Prognosis

The prognosis is good for healthy children but poor for neonates and those that are immunocompromised.

Prevention

Prevention involves isolation of the patient during the infectious period, active immunization with varicella vaccine and post exposure prophylaxis with varicella zoster immunoglobulin (VZIG)

Poliomyelitis

Habtam Fekadu

Definition

Poliomyelitis is an acute viral infection caused by poliovirus.

Etiology

The poliovirus is one of the *enteroviruses* and there are three serotypes (1, 2, and 3). Paralysis and epidemic are often caused by type 1. Humans are the only reservoir: there are no carriers.

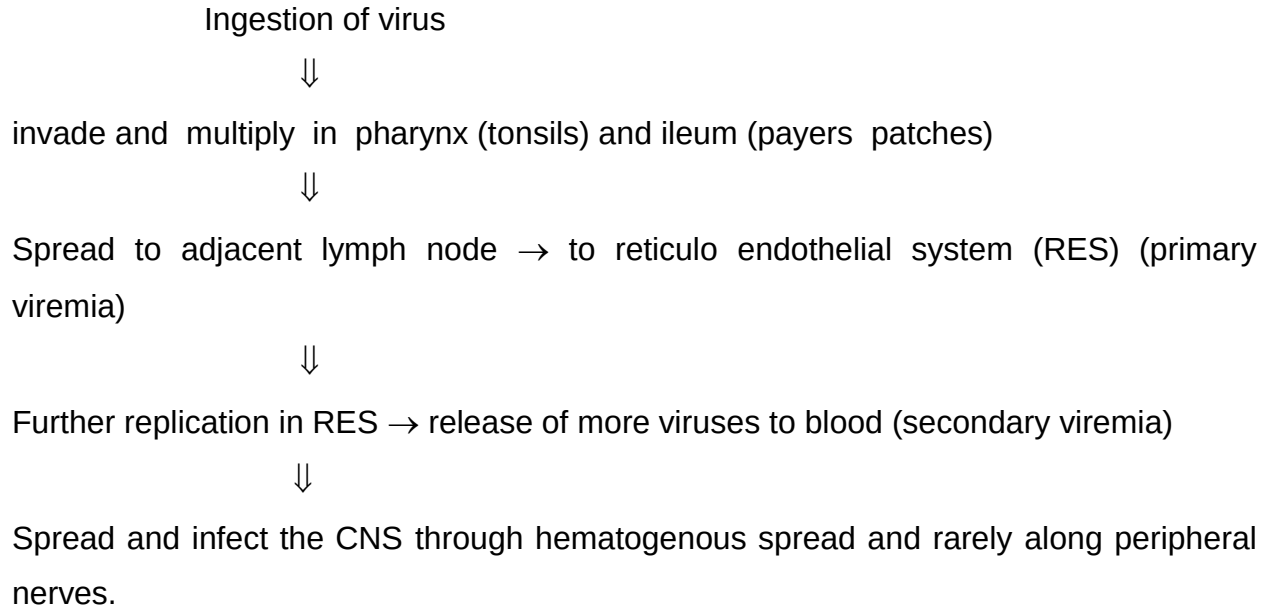
Epidemiology

Poliomyelitis is eradicated from developed countries but it still occurs in developing countries. However, the incidence is decreasing. Transmission occurs by fecal-oral route. It is very infectious: nearly 100% of susceptible exposed children acquire the virus. Poliovirus is excreted in stools for 6 to 8 weeks following infection.

the infected child transmits the disease for a few days before and 7 to 10 days after symptoms appear. All unimmunized children are susceptible. It largely affects infants and young children. Ninety percent occurs in children less than 5 years of age.

Active immunity after infection or vaccine is life long.

Figure 18: Pathogenesis of Poliomyelitis



Primary targets are the motor neurons in the anterior horn of the spinal cord, cranial nerve motor nuclei and vital centers in brain stem. This selective attack is characteristic of poliomyelitis.

Clinical features

The clinical features are as follows:

- Incubation period – 7 - 20 days
- Most are asymptomatic – 90 -95%
- It has four courses:
 - Inapparent (asymptomatic): in 90 to 95% of cases.
 - Abortive poliomyelitis (minor illness): in 5% of cases the infection presents with non-specific symptoms like fever, headache, sore throat and no neurologic symptoms. Most recover in 48 to 72 hours. Sometimes this form merges with non-paralytic poliomyelitis.
 - Non paralytic poliomyelitis: in 2% of cases. This form is due to aseptic meningitis. Besides the above symptoms, the patient manifest with severe headache, muscle pain, tenderness, signs of meningitis and spinal rigidity which

is the basis for diagnosis. The symptoms and signs last for 1 to 2 days and most cases the patients recover. However, it progresses to paralysis in a few patients.

- Paralytic poliomyelitis occurs in less than 1% of cases, following non-paralytic poliomyelitis or it may appear 4 days after symptoms of abortive poliomyelitis subsided.

Paralysis is the hallmark of this last type, characteristically it

- is asymmetric and flaccid paralysis
- begins rapidly, progress markedly in 48 hrs. and no progress after 3 days.
- Sensory involvement is rare.

This type of infection has 4 forms: spinal, bulbar, bulbospinal and encephalitic poliomyelitis. The first two are discussed below, since they are more common than the later two.

Spinal poliomyelitis is the most common form and has the following characteristics:

- The cervical and thoracic cords are affected
- It results in paralysis of neck, trunk, limbs or respiratory muscles, so patients present with flaccid monoplegia, paraplegia or quadriplegia, with or without breathing difficulty.
- Cranial nerves, sensation and autonomic nervous system are spared.

Bulbar poliomyelitis.

Bulbar poliomyelitis is rare but it is life threatening.

- It is characterized by cranial nerve motor nuclei lesion with or without vital center involvement.
- Usually cranial nerves IX, X, and XI are affected and patients present with difficulty of swallowing, speech, breathing etc.
- There is vital center (respiratory and circulatory) involvement.
- It manifests with shallow, irregular breathing, labile blood pressure usually hypertension and rapid change in temperature.

Diagnosis

History and characteristic findings are important in diagnosis. A flaccid limb paralysis or cranial nerve palsies without sensory loss occurring during the course of an acute febrile illness is highly suggestive of poliomyelitis. Other methods of diagnosis include: Serology fourfold rise in antibody titer between acute illness and convalescence confirm the diagnosis.

- Viral isolation from stool.
- CSF -increased WBC of 20-300/mm³ (largely lymphocytes), increased protein, and normal glucose.

Differential Diagnosis

- Non paralytic poliomyelitis – Aseptic or bacterial meningitis
- Paralytic poliomyelitis Gullian Barrie Syndrome, transverse myelitis

Treatment

Treatment is supportive:

- Make the child comfortable.
- Maintain adequate fluid and dietary intake.
- Minimize skeletal deformity.
- Anticipate and treat complications.
- Maintain airway, ensure adequate ventilation

Prognosis

The prognosis for paralytic poliomyelitis depends on its type and extent. Generally, mortality is between 2 and 20%. Most paralysis recover. However, if the paralysis persists for 3 months or if it is extensive paralysis in the first 10 days, the patient will remain paralyzed. The major cause of death is respiratory failure.

Prevention

Prevention includes routine EPI (OPV). The strategies for polio eradication include:

1. Strengthen routine EPI

2. National immunization days (NID) Two doses of OPV should be given to all children under 5 years of age annually irrespective of their immune status. This will boost herd immunity and stop wild poliovirus spread
3. Mopping-up immunization - in high risk areas, house-to-house vaccination should be undertaken of all children less than 5 years of age, irrespective of their immune status.
4. Acute flaccid paralysis (AFP) surveillance- to detect and investigate all cases of AFP in children under 15 years of age.

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4.2 TYPHOID FEVER

Goitom G/Yesus

Definition

Typhoid fever is a febrile systemic syndrome caused by *salmonella typhi*.

Epidemiology

It is endemic in most tropical countries. Humans are the only reservoir for salmonella typhi; hence, direct or indirect contact with an infected person (sick or apparently healthy chronic carrier) is necessary for infection.

The most common mode of transmission is ingestion of water or food that has been contaminated with human feces.

Pathogenesis

After ingestion, the organism invades intestinal epithelium and multiplies in the intestinal lymph nodes. From the lymph nodes it reaches the blood stream causing transient bacteremia. Circulating organisms reach the reticuloendothelial cells in the liver, spleen and bone marrow where proliferation occurs and leads to bacteremia. The gall bladder is particularly susceptible to infection and local multiplication of the organism. The large numbers of organisms in the bladder are passed to the intestine through the bile.

Clinical Manifestation

The incubation period for typhoid fever is usually 7 to 14 days. The clinical manifestations depend on the age of the patient.

In school age children and adolescents

Initial (first week) symptoms are fever, malaise, anorexia, myalgia, headache, and abdominal pain. Diarrhea may be present in the early course of the disease, but later on constipation becomes a more prominent symptom. Fever rises in step-ladder fashion. The fever is high grade and unremitting, and may be followed by cough and **epistaxis**.

In the second week there will be sustained high grade fever, fatigue, cough, and the abdominal symptoms increase in severity. Patients appear acutely ill, disoriented and lethargic. Delirium and stupor may also be observed. On physical examination, relative brady-cardia, hepatomegally, splenomegally and distended abdomen with diffuse tenderness are very common.

In about 50% of patients a reddish rash (rose spot) occurs in the lower part of the chest and abdomen. If uncomplicated, the symptoms usually resolve in two to four weeks. Nausea and vomiting occurring in the second week suggest a complication.

Infants and young children usually present with mild fever and malaise, which may be misinterpreted as a viral syndrome. Diarrhea is more common in this age group than in adults, suggesting the diagnosis of acute gastroenteritis.

Complications

Complications include intestinal hemorrhage and perforation, sepsis, and pneumonia.

Diagnosis

Laboratory diagnosis involves:

Isolation of the organisms (gold standard)

- blood culture will be positive within the first few days,
- In the second and later weeks, urine and stool cultures.

Serologic tests

Widal test: diagnosis depends on demonstration of antibodies against 'O' and 'H' antigens of the organism. In areas where the disease is endemic, an 'O' antibody titer of 1/250 or above is suggestive, but more important is a rising titer in two consecutive tests done within 7 to 10 days.

Another laboratory finding commonly associated is a low leukocyte count

Treatment

➤ Supportive

- Maintenance of appropriate fluid and electrolytes
- Surgical intervention for intestinal perforation

➤ Antibiotics

- Chloramphenicol 50-75 mg/Kg/24hr.PO/IV in four divided doses or,
- Ampicillin 200 mg/Kg/24hr. IV in four divided doses or,
- Amoxicillin 100 mg/Kg/24hr.PO in four divided doses
- Duration of antibiotic is 14 days

Prognosis

In developing countries mortality is higher than 10% because of the delay in diagnosis, hospitalization and treatment.

Prevention

Prevention involves:

- Improved sanitation
- Provision of clean water and safe waste disposal
- Hygienic food preparation practices
- Eradication of salmonella typhi from carriers

Exercise

1. Characterize the classic fever of typhoid fever.
2. What percentage of patients with typhoid fever manifest with rose spots?
3. What are the common complications of typhoid fever?
4. When do blood, urine and stool culture become positive in the clinical course of typhoid fever?

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4.3. TUBERCULOSIS

Goitom G/Yesus

Definition

Tuberculosis (TB) is a chronic infectious disease with diverse clinical manifestations caused in most cases by *mycobacterium tuberculosis*; occasionally it can also be caused by *mycobacterium bovis* or *mycobacterium africanum*.

Epidemiology

The main sources of infection are untreated pulmonary TP patients discharging the bacilli. Transmission of infection occurs through the air-borne spread of droplets containing bacilli expelled by coughing and sneezing, and inhaled by healthy persons. People who are in frequent contact with an infectious patient are at greatest risk of being exposed to the bacilli. In addition, consumption of raw milk containing *M. bovis* is a possible way of transmission in Ethiopia.

TB affects individuals of all ages and both sexes within every socio-economic group. However, factors making people more vulnerable to develop the disease are poverty, malnutrition, crowded living conditions, measles, pertussis and HIV infection.

The natural history of TB is such that, in the majority of exposed people, their immunological defense either kills all the inhaled or ingested bacilli or more often keeps them suppressed (silent focus) causing latent *M. tuberculosis* infection. Under normal circumstances, only about 10% of all individuals who are infected by the bacilli will develop the disease throughout their lifetime (in HIV infected persons the risk is much higher).

Pathogenesis

Primary infection occurs on the first exposure to tubercle bacilli. Infection begins with the multiplication of the bacilli in the lungs. This is the *ghon focus*. Lymphatics drain the bacilli to the hilar lymph nodes. The ghon focus and related hilar lymphadenopathy form the *primary complex*. The immune response (delayed hypersensitivity reaction) develops about 4 to 6 weeks after the primary infection. What happens next is

determined by the size of the infecting dose of bacilli and the strength of the immune response. Mostly, the immune response stops the multiplication of bacilli. However, a few dormant bacilli may persist. In a few cases the immune response is not strong enough to prevent multiplication of bacilli and disease occurs within a few months.

In children, TB is usually primary TB. The age at which the child is infected determines the pattern of primary disease. Up to puberty, blood-borne spread is common, resulting in disseminated (military and extra pulmonary) disease, whereas after puberty pulmonary spread is more common.

Post-primary TB occurs after a latent period of months or years after the primary infection. It may occur either by reactivation or by reinfection. Reactivation occurs mainly in response to a trigger, such as weakening of the immune system by HIV infection, malnutrition, measles and pertussis. Post-primary TB usually affects the lungs but can involve any part of the body. The characteristic features of post-primary TB are: extensive lung destruction with cavitations, positive sputum smear, and upper lobe involvement. The different types of post-primary TB are:

Pulmonary TB – involves the cavities, upper lobe infiltrates, fibrosis, progressive pneumonia, endo-bronchial TB etc.

Extra-pulmonary TB – involves pleural effusion, lymphadenopathy (usually cervical), central nervous system (meningitis, cerebral tuberculoma), pericarditis (effusion, constriction) gastro-intestinal, spina, bones and joints and other less common ones.

Clinical Manifestations

Primary pulmonary disease

This includes the lung parenchymal focus and regional lymph nodes. About 70% of lung foci are sub-pleural and localized pleurisy is common. All lobar segments are at equal risk of initial infection. The hallmark of 1^o PTB is the relatively large size of the regional lymphadenitis compared with the relatively small size of the initial lung focus. In some children progressive enlargement of the hilar lymph nodes may compress the bronchus leading to obstruction and atelectasis. Rarely inflamed nodes attach to the endobronchial wall causing endobronchial TB.

The symptoms and physical findings of primary pulmonary disease in children are surprisingly meager. Infants are more likely to manifest with signs and symptoms such as: cough, mild dyspnea, failure to gain weight or a true failure to thrive; localized wheezing or decreased breath sounds in those with bronchial obstruction.

Reactivation of TB

Pulmonary TB

In adults this usually represents endogenous reactivation of TB infection established previously in the body. Pulmonary TB is uncommon in childhood but may occur in adolescence. More patients may experience fever, anorexia, malaise, weight loss, night sweats, productive cough and hemoptysis and chest pain than those with 1^o PTB. Most common radiographic presentation is extensive infiltrates or thick walled cavities in the upper lobes.

The other forms of reactive tuberculosis are the extra pulmonary tuberculosis. The signs and symptoms depend mainly on the organ(s) involved. The most common forms are as follows:

Lympho- hematogenous (disseminated) disease

Tubercle bacilli are disseminated to distant sites, including liver, spleen, skin, and lung apices in all TB infection. The clinical manifestation of disseminated TB depends on the number of organisms released from the primary focus and the strength of the host immune response.

Although the clinical picture may be acute, usually it is indolent and prolonged. Multiple organ involvement is common. The most significant form is *miliary disease* which occurs when massive bacilli are released into blood stream causing disease in 2 or more organs. It usually complicates the primary disease within 2 to 6 month of initial infection, and is common in infants and young children. The clinical manifestations are protean depending on load of bacilli that disseminate and where they lodge. Patients are usually very ill. The common findings are as follows:

- Lesions on CXR are smaller than 2 to 3 mm in diameter initially but later coalesce to form large lesions.
- Generalized lymphadenopathy and hepatosplenomegaly occur in 50%
- Meningitis and peritonitis occur in 20 – 40%
- Diagnosis may be difficult and high index of suspicion is required; most important clue is recent exposure to infectious adult.

CNS disease

CNS disease is the most serious and fatal complication in children. It may arise from metastatic focus in cerebral cortex or meninges. The brain stem is often the site of greatest involvement (frequent dysfunction of Cranial nerves III, VI and VII. Combinations of vasculitis, infarction, cerebral edema and hydrocephalus result in severe damage and complicates about 0.3% of untreated TB in children. It is most common between 6 month and 4 years of age. Clinical progression may be rapid or gradual.

More commonly, the signs and symptoms progress slowly over several weeks and can be divided into 3 stages:

- 1st Stage lasts 1 to 2 weeks and is characterized by non specific symptoms. Focal neurologic signs are absent but infants may show stagnation or loss of development mile stones.
- 2nd Stage: common features are lethargy, meningeal signs, seizure, hypertonicity, vomiting, cranial nerve palsies and other focal neurologic signs. Some patients may have signs of encephalitis: disorientation, movement disorders or speech impairment.
- 3rd stage is marked by coma, hemiplegia or paraplegia hypertension, decerebrate posturing, deterioration of vital signs and even finally death

Prognosis correlates with stage of disease at the time of initiation of treatment. Stage 3 survivors have permanent disabilities

Diagnosis involves:

CSF analysis shows cell count is in hundred ($< 500/\text{mm}^3$) with moderate lymphocyte predominant, a high protein concentration (usually greater than 200mg/dl) and a low glucose concentration.

Examination and culture of CSF is most important for diagnosis.

Complete investigation for TB is essential.

Tuberculous lymphadenitis

It is a slowly developing and painless enlargement of lymph nodes, followed by matting and eventually drainage of pus. It is the most common form of tuberculosis in children.

Investigation involves fine needle aspiration or biopsy of lymphnode and stain and culture for AFB and histological examination

Tuberculosis of bones and/or joints

The vertebrae are involved most of the time, and manifest classically by gibbus deformity and kyphosis. It is usually a late manifestation.

Diagnosis of TB

Diagnosis of TB involves:

- History and physical examination
- WBC with differential ESR
- Microscopic examination of sputum or gastric aspirate
- Radiological examination (chest x-ray, x-ray of bones etc)
- Culture of organism
- Histo-pathological examination of biopsied tissues
- Tuberculin test

Children should be strongly suspected of having TB when they have contact with a known adult case of pulmonary TB and have clinical signs and symptoms.

Criteria for the diagnosis of tuberculosis in children

Suspected tuberculosis

An ill child with a history of contact with a confirmed case of pulmonary tuberculosis OR

Any child:

- Not regaining normal health after measles or whooping cough
- With loss of weight, cough and wheeze not responding to antibiotic therapy for respiratory disease
- With painless swelling of the superficial lymph nodes

Probable tuberculosis

A suspected case and any of the following:

- Positive (>10 mm in diameter) induration on tuberculin testing
- Suggestive appearance on chest X-ray (e.g. unilateral hilar/ mediastinal lymph node enlargement with or without lobar or segmental opacity, military pattern, pleural effusion, infiltrates and cavitations)
- Suggestive histological appearance of biopsy material

Confirmed tuberculosis

- Detection by microscopy or culture of tubercle bacilli from secretions or tissues
- Identification of tubercle bacilli as *mycobacterium tuberculosis* by culture

Standardized Tb Case Definitions

On making the diagnosis of TB, you must also decide on the TB case definition. There are four determinants for this: site of TB, result of sputum smear, previous TB treatment and severity of TB.

Case definition by site and result of sputum smear for PTB

1. *Smear positive case:* At least 2 sputum smears positive for AFBs or 1 sputum smear positive and CXR abnormalities consistent with TB.
2. *Smear negative case:* At least 2 (preferably 3) sputum smears negative for AFBs and Chest X-ray consistent with TB.

Case definition by previous treatment

- *New case (N)*: a patient who has never taken treatment for TB or has been on anti-TB treatment for less than one month.
- *Relapse(R)*: a patient who has been declared cured or treatment completed of any form of TB in the past, but who reports back to the health service and is found to be AFB smear-positive or culture positive.
- *Treatment failure(F)*: a patient who, while still on treatment remains smear-positive or comes again sputum smear-positive 5 months or more after starting treatment
- *Return after default (D)*: a patient who has previously been recorded as defaulted from treatment (completed at least one month of treatment and interrupted for at least 2 months) and returns to the health service with smear-positive sputum.
- *Transfer in (T)*: a patient registered for treatment in one district (woreda) and is transferred to another.
- *Chronic case (C)*: a patient who remains smear-positive after completing a supervised re-treatment regimen. This is usually seen in adults and is rare in childhood but may occur in adolescents.
- *Other (O)*: a patient who does not easily fit into one of the above case definition (e.g. a smear-negative PTB who returns after default)

Management

Principles of management include:

1. Chemotherapy
2. Nutritional rehabilitation.
3. Family screening
4. Follow up

Chemotherapy

1. Drugs used for chemotherapy of tuberculosis:

- Streptomycin (S) - 1gm (vial)
- Ethambutol (E) - 400 mg
- Isoniazid (H) -100 mg, 300 mg
- Rifampicin (R) - 150 mg, 300 mg
- Pyrazinamide (Z) - 500 mg

2. Phases of chemotherapy

The treatment for tuberculosis has two phases:

Intensive (initial) phase: this phase consists of three or more drugs for the first 8 weeks. This has the advantage of making the patient non-infectious by rapidly reducing the load of bacilli in the sputum and minimizing the danger of development of drug resistance.

Continuation phase: this phase immediately follows the intensive phase and is important to ensure that the patient is permanently cured and does not relapse after completion of treatment. This phase at least two drugs to be taken for 6 months.

3. Drug regimens and treatment categories

In order to treat the patient with the correct drug regimen, it is important to determine whether or not the patient has previously taken treatment for TB.

There are three different categories of treatment regimens.

Category I – Short course chemotherapy for smear- positive PTB and seriously ill smear-negative PTB and EPTB cases

The treatment regimen for this category is:

- S (RHZ) for 2 months and then (EH) for 6 months OR
- (ERHZ) for 2 months and then (EH) for 6 months

Children who are 6 years or below get RH 150/75 mg in the continuation phase for a total period of 4 months. Also, children older than 6 years but are seriously ill receive 4 months RH in the continuation phase.

This regimen is prescribed to:

- New smear-positive PTB patients
- New smear-negative PTB patients, who are seriously ill
- New EPTB patients, who are seriously ill
- Returns after default from DOTs, who have smear-negative PTB (case definition = “other”)

Category II – Re-treatment regimen

The regimen for this category is:

- SE(RHZ) for 2 months, then E(RHZ) for 1 month, and then E(RH) three times a week for 5 months

This treatment regimen is prescribed for:

- Smear-positive relapses
- Smear-positive treatment failures
- Smear-positive return after defaults
- Smear-negative PTB patients who become smear-positive after 2 months of treatment (case definition: other)
- Return after default from re-treatment should be given the re-treatment regimen once again, but only once
- Relapses after re-treatment should also be given the re-treatment regimen only once

Category III – short course chemotherapy for smear-negative PTB, EPTB and TB in children

The treatment regimen for this category is:

- (RHZ) for 2 months and then (EH) for 6 months

This regimen is prescribed to:

- New adult patients with smear-negative PTB
- New adult patients with EPTB
- Children between 7 and 14 years old with any type of TB, who are *NOT* seriously ill

Prevention

Prevention comprises:

- Adult case finding and treatment
- Chemoprophylaxis
- BCG vaccination
- Adequate nutrition

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4.4. Tetanus (Lockjaw)

Tsinuel Girma

Definition

Tetanus is acute, spastic paralytic illness caused by tetanus toxin (tetanospasmin), a neurotoxin produced by *clostridium tetani*.

Etiology

Clostridium tetani is anearobic, gram positive, spore forming motile bacteria found worldwide in soil, dust and excreta of animals.

It lives in two forms. The spore form resists damage like boiling. The vegetative form secretes toxin in hypoxic conditions and is killed by standard disinfections.

Epidemiology

Tetanus, particularly neonatal tetanus (tetanus neonatorum), is a major public health problem in developing countries where it kills half a million infants each year. A household survey in Ethiopia showed incidence rate of 4.3 per 1000 live birth and with mortality rate of 86%. In Jimma Hospital neonatal tetanus accounted to 2.9% of admissions.

Spores get into the body through wounds such as penetrating injuries, animal bites, jigger and tooth extraction by local healers, otitis media and injections (quinine i.m.) Neonates, born to unimmunized mother, will acquire it via umbilical stump due to unhygienic cord care, and dressing with herbal medication or and applying cow dung on the cord

Pathogenesis

The vegetative form produces tetanospasmin under anaerobic environment of injured /infected site.

Clinical manifestations

Neonatal tetanus manifests few days (3 to 12 days) after birth mainly with progressive difficulty in sucking and swallowing, hunger and cry, diminished movements, stiffness to touch and spasms. Tetanus may be localized or generalized in postnatal period.

Generalized tetanus, the most common form, manifests with spasm in different muscles. Masseter muscle spasm (trismus) is the presenting symptom in 50% of cases. Progressively neck rigidity, opisthotonus, laryngeal and pharyngeal spasms appear. Continuous spasm of facial muscles result in sardonic smile (risus sardonicus), pathognomonic for tetanus. Urinary retention results from sphincter spasm.

Tetanic seizures, characterized by sudden tonic first clenching arm flexion and hyperextension of the legs in a conscious patient, may occur spontaneously or induced by touch, sound or light.

Autonomic nervous symptom over activation leads to tachycardia, fever, labile (fluctuating) hypertension and sweating.

Localized tetanus results in painful spasms of muscles adjacent to wound site and may precede generalized tetanus.

Cephalic tetanus is a rare form of localized tetanus arising from wounds above the neck. Retracted eye lids, trismus, risus sardonicus and spastic paralysis of the tongue and pharyngeal muscle characterize it.



Diagnosis

Diagnosis greatly depends on clinical manifestations. No lab test is needed. Trismus may result from dental abscess, retropharyngeal abscess, tonsillitis or parotitis.

Rabies may give trismus and seizures, but hydrophobia, clonic seizures and CSF pleocytosis are present in rabies. Neck stiffness from meningitis may confuse. However impaired consciousness and CSF pleocytosis are present.

Treatment

Main objectives of treatment are:

- Supportive care and proper nursing care
- Control of spasms and maintain airways
- Neutralization of toxin,
- Eradication of bacteria
- Prevention of recurrence

Eradication of bacteria

This involves surgical wound excision and debridement after administration of TAT and antibiotics.

Excision of the umbilical sump is not recommended. Penicillin 100,000 V/kg/24 divided into four doses should be given for 10 day to eradicate vegetative forms from wound site. For penicillin allergic patients, erythromycin can be used.

Neutralization of toxin

Tetanus antitoxic (TAT) is used in a dose of 10,000U, half given intravascular and half intramuscular after a skin test. High dose (50,000- 100,000 U) can be used.

Supportive care

- Patients should be cared for in a quiet dark and separate room
- Avoid unnecessary sounds and manipulations
- Feeding should be given via nasogastric tube

- Frequent cardiorespirator monitoring is mandatory.
- Meticulous nursing care with attention to mouth, skin, bladder and bowel prevents bed ulcers, infection and constipation.
- Physiotherapy

Control of spasms and airway maintenance

Diazepam (0.25 – 0.5 mg/kg, I.V) and chlorpromazine (2 -4 mg/kg, I.V) given every three hour alternatively are still in use to control spasms. The intravenous route may be changed and nasogastric tube used after adequate control of spasms. Two to six weeks may be needed before tapered withdrawal of there drugs.

In selected patients tracheotomy can be done to decrease the deaths from laryngeal spasm and ventilatory support may be needed.

Active immunization with DPT (DT) at discharge is recommended for all patients because tetanus doesn't result in production of toxin neutralizing antibodies that prevents recurrence

Complications

Complications include:

- Pneumonia
- Laceration of the mouth and tongue
- Long bone fractures
- Bed sores
- Malnutrition

Prognosis

The most important factor determining outcome is quality of supportive care. The following factors indicate unfavorable prognosis:

- Short incubation period (≤ 1 wk)
- Short period of onset (≤ 3 days) - the time between appearance of trismus and onset of generalized tetanic spasms.
- Extremes of age (e.g, neonatal tetanus)

- Cephalic tetanus
- Fever (if part of autonomic effect)
- Pneumonia
- Frequent spasms

Prevention

- Tetanus is completely preventable disease
- There are three major ways:
- Vaccinate all child bearing age group females with TT
- Primary vaccination of infants and children with DPT
- Proper wound management. See table 3.1.

Table 15 Tetanus prophylaxis in wound management

Prior tetanus doses	Minor Wounds	clean	Other wounds
	DPT +	TAT	DPT TAT
Uncertain or < 3 doses	Yes	No	Yes Yes
≥ 3 doses	No*	No	No” No

+ TT can be given for children >7years (Td not available in Ethiopia)

*Yes, if >10years since last dose.

“Yes, if >5 years since last dose.

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4.5 PAEDIATRIC HIV/AIDS

Goitom G/Yesus

Definition

HIV/AIDS is an acquired immune deficiency syndrome with a fatal clinical course caused by viral etiologic agents.

Etiology

- ◆ HIV-1 and HIV-2 come from the *retroviridae* family, *lentivirus* genus, single stranded RNA
- ◆ HIV-2 is rare in children and is prevalent in western and southern Africa

Epidemiology

Worldwide HIV-AIDS has created an enormous challenge on the survival of humankind. Since its recognition, the virus has infected close to 65 million individuals and over 25 million have already died due to AIDS. Currently, over 42 million people are living with the virus, and of these more than 70% are in Sub Saharan Africa. HIV-AIDS has caused major impact on maternal and child health and survival. Most infected children are in developing countries. Each year around 400,000 children are newly infected. Currently between 5 and 8 million children are infected globally. As a result of maternal death due to AIDS, between 10 and 15 million children are orphaned.

The year 2001 estimate for HIV /AIDS infection in Ethiopia was:

- Total people living with HIV infection is around 2.2 million
- adult prevalence rate is 6.6%
- The number of children with HIV-AIDS is 200,000
- Estimated number, of death due to AIDS in 2001 is 160,000
- Estimated number. of AIDS orphans currently living is 1.2 million

Transmission

Transmission occurs via:

- vertical transmission from mother to child
- parenteral exposure to blood

➤ sexual contact

The primary route of infection in children is by vertical transmission from the mother to the child, accounting for virtually all new cases. This may occur in the intrauterine, intrapartum or postnatal period via breast feeding. The rate of transmission from mother to child in sub-Saharan Africa ranges from 25 to 52%.

Risk factors that may influence rate of vertical transmission are:

- high maternal viral load
- Low maternal antenatal CD4⁺ count (< 29%)
- Preterm delivery (< 34 wks of gestation)
- More than 4hr interval between membrane rupture and delivery
- Low birth weight
- Second twin

Pathogenesis

In adults and adolescents, after HIV has entered the circulation intense viremia ensues, causing flu like symptom in 50 to 70% of cases. This primary viremia results in widespread seeding of the virus to various organs including the brain and lymphoid tissues. HIV selectively binds to cells expressing CD4⁺ molecules on their surface, primarily helper T- lymphocytes (CD4⁺ cells) and cells of the macrophage lineage. The CD4⁺ cells migrate to the lymph nodes, where they become activated and proliferate. This may contribute to the dramatic decrease in the number of circulating CD4⁺ cells and the generalized lymphadenopathy (LAP) characteristic of *Acute Retroviral Syndrome* in adolescents and adults.

Following this, there will be establishment of cellular and humeral immune response with in 2 to 4 months, with the level of cultivable virus from blood declining markedly and patients enter a phase characterized by lack of symptoms and a return of CD4⁺ cells to only moderately decreased levels. This phase is called *latency period* and lasts 8 to 12 years. During this seemingly quiescent period, there is a very high turnover of virus and CD4⁺ cells, which gradually causes deterioration of the immune system. As the disease progresses, the lymph nodes disrupt and free the virus to the blood, producing high level of viremia during the later stage of the disease.

In young children, unlike in adolescents and adults, the latency period is short. There are three patterns of clinical course in these younger children:

a) Rapid course of progression

- 15 to 25% of new born babies present with the onset of symptoms and AIDS in the first few months of life
- if untreated median survival is between 6 and 9 months
- such babies are mostly positive for HIV culture / PCR in the first 48 hours of life

b) Slower progression of diseases

- The majority of perinatally infected newborn (60 – 80%) show a slower progression
- the median survival time is six years
- Many patients have negative culture PCR in the first week of life

c) Long term survivors

- Less than 5% of perinatally infected children have minimal or no progression of disease with relatively normal CD4⁺ counts and very low viral load for more than 8 years.

Clinical manifestations

Clinical manifestations vary widely among infants, children and adolescents. Initial symptoms may be subtle or non-specific such as lymphadenopathy, hepatosplenomegally, failure to thrive, chronic or recurrent diarrhoea, interstitial pneumonia or oral thrush. These symptoms may be distinguishable only by their *persistence*. Chronic diarrhea, wasting and severe malnutrition are predominating symptoms in African children compared to developed countries.

Symptoms found more commonly in children than adults are recurrent bacterial infection, chronic parotid swelling, lymphoid interstitial pneumonitis (LIP), and early onset of progressive neurological deterioration.

Opportunistic infections are generally seen with severe depression of the CD4⁺ count. The most common opportunistic infections in paediatric age group are as follows:

Bacterial infection

More than 20% of AIDS defining illnesses in children are recurrent bacterial infection caused primarily by encapsulated organisms such as *Streptococcus pneumoniae* and salmonella species. Other commonly seen pathogens are staphylococci, enterococci, pseudomonas, H. influenza and other gram negative and gram positive bacteria. The most common serious infections are sepsis and pneumonia, accounting for more than 50% of infections in children with HIV. The common moderate infections are otitis media, sinusitis and skin and soft tissue infections.

Fungal infection

Most common fungal infection is oral candidiasis, which involves about 20% the oesophagus presenting with anorexia, dysphagia, vomiting and fever.

Pneumocystis carinii pneumonia (PCP)

The peak incidence is at 3 to 6 months of age. Its clinical manifestations include acute onset of fever, tachypnea, dyspnea and marked progressive hypoxemia. Chest X-ray findings most commonly consist of interstitial infiltrates or diffuse alveolar infiltrates.

Parasitic infection

Cryptosporidiosis, microsporidiosis, isosporiasis or giardiasis are common opportunistic infections causing significant morbidity.

Viral infections

Herpes virus causes significant morbidity in the form of recurrent gingivo-stomatitis. Varicella zoster may cause chicken pox and recurrent herpes zoster. Measles may occur despite immunization and without the typical rash. Other viral infections are disseminated CMV disease, RSV and adenovirus infections.

No organ system is spared by HIV infection. The common organs that are affected by the disease are:

CNS

The CNS is involved in 40 to 90% of children infected perinatally. The common manifestations are progressive encephalopathy with loss of developmental milestones, cognitive deterioration, acquired microcephaly and motor dysfunctions.

Respiratory tract

Recurrent respiratory tract infections (otitis media and sinusitis) are very common.

Lymphoid interstitial pneumonitis is the most common lower respiratory tract disease, affecting 30 to 50% of HIV infected children. It manifests with insidious onset of tachypnea, cough, and mild to moderate hypoxemia with normal or minimal auscultatory findings. Characteristic reticulonodular pattern is seen on chest X-ray.

Bacterial pneumonia and pulmonary and extra-pulmonary tuberculosis are also common in HIV infected children.

Cardiovascular system

About 20% of children with HIV have some degree of cardiac involvement.

Common manifestations are gallop rhythm, tachypnea and hepatosplenomegaly, which are clinical indicators of congestive heart failure in HIV-infected children. It is due to cardiomyopathy, myocarditis etc.

GIT

Oral manifestations include candidiasis, ulcerative gingivitis and salivary gland disease.

Lower intestinal include persistent or recurrent diarrhea, malabsorption, abdominal pain, dysphagia and chronic liver inflammation.

Skin

Most problems involve inflammatory or infectious disorders that are not unique to this disease

Seborrheic dermatitis or eczema that is severe form and unresponsive to treatment is commonly associated with HIV-AIDS And are recurrent or chronic episodes of herpes simplex, herpes-zoster moluscum contagiosum, anogenital warts and candidial infections.

Hematologic manifestations

Hematologic manifestations include anaemia, leucopenia, and thrombocytopenia.

Malignancy

Non-Hodgkin's lymphoma, 1^o CNS lymphoma and leiomyosarcomas are commonly reported

Diagnosis

Diagnosis may involve the following:

Serologic test

In a child older than 18 month, demonstration of antibodies to HIV by ELISA and western immunoblot test establish diagnosis. However, in infants less than 18 months of age, ELISA can be positive due to transplacentally passed antibodies.

Viral detection

Viral detection is very useful in young patients less than 18 month of age and may involve:

- HIV culture
- HIV DNA or RNA by Polymerase chain reaction

WHO clinical case definition of aids in children

This detects only children with symptomatic HIV infection

Major signs include:

- Weight loss or abnormally slow growth
- Chronic diarrhea for more than 1 month
- Prolonged fever for more than 1 month
- Recurrent bacterial lower respiratory infections

Minor signs include:

- Generalized lymph adenopathy and hepato splenomegally
- Oropharyngeal candidacies
- Repeated common infections
- Generalized dermatitis
- Confirmed maternal HIV

The presence of 2 major + 2 minor suggest the diagnosis of AIDS

The Ethiopian AIDS Case Definition (revised in February 2000)

Major signs

1. Failure to thrive (when the weight and height for age are less than the fifth percentile or if either one crosses two major growth percentiles.
2. Repeated (2 or more within a period of 1 year or persistent (persisting for 1 month or more) lower respiratory tract infections (LRTI)
3. Chronic/recurrent diarrhea for more than 1 month
4. Unexplained prolonged fever for more than 1 month

Minor signs

1. Generalized lymphadenopathy
2. Repeated or persistent common infections (Otitis media, pharyngitis, oropharyngeal candidiasis, purulent rhinitis, sinusitis, impetigo, etc
3. Unexplained neurological disorders or developmental delay and/or microcephaly
4. Hepatosplenomegaly or splenomegaly
5. Extensive varicella infections or molluscum contagiosum
6. Confirmed maternal HIV infection

Fever should not be counted as a major sign in the presence of LRTI.

The purpose of this case definition is for surveillance and not for diagnosis unless there is laboratory evidence of HIV infection.

AIDS in a child of up to 12 years of age is defined with evidence of positive HIV test in the presence of 3 major signs alone in the absence of other known causes of immunosuppression *OR*

AIDS in a child of up to 12 years of age is defined without laboratory evidence of HIV infection in the presence of:

- 2 major and 2 minor signs, or
- 3 major and 1 minor signs

In the absence of other known causes of immunosuppression.

Treatment

Treatment involves:

- Supportive care:
 - Counselling of care-giver on the options for infant feeding (see Infant feeding section)
 - Proper feeding
 - Immunization (standard)
 - Growth surveillance and social support
 - Early access to medical care
 - Treatment of opportunistic infections
- Antiretroviral drugs (according to the national guide)

Prevention

Prevention includes:

- Prevention of HIV among parents to be
- Prevention of unwanted pregnancy in HIV infected women
- Provision of antiretroviral drugs (nevirapine) to the mother and infant during pregnancy and after delivery
- Safe delivery practice - C/S delivery may have a protective effect compared to vaginal delivery
- Combining elective c/s delivery with perinatal administration of nevirapine reduces the transmission significantly.
- Counselling on infant feeding options.

Exercise

1. What are the modes of transmission of HIV infection?
2. Describe the 3 patterns of clinical course of HIV-AIDS in children.
3. What are the common manifestations of PCP?
4. Describe the clinical case definition of AIDS in children.
5. List 4 preventive measures for HIV.

References

1. WHO/TB/96.2000, TB/HIV clinical manual, 1996.
2. Behrman, Kliegman, Jenson, Nelson Text Book of Pediatrics, 16th edition, 2000.

4.6 Viral Hepatitis

Dula Benti & Yezna Temesgen

Definition

Viral hepatitis is an inflammation of the hepatocytes due to viral causes.

Epidemiology and Etiology

Viral hepatitis continues to be a major health problem in both developing and developed countries. Viral hepatitis is caused by hepatotropic viruses designated as hepatitis A, B, C, D, E and other viruses like herpes simplex virus, cytomegalovirus, varicella virus, HIV, rubella and others which cause hepatitis as part of their clinical spectra. All the hepatitis viruses own RNA except the B variant, which is a DNA virus. Hepatitis A and E do not cause chronic infection but the rest are capable of causing chronic liver disease. The D virus is a defective virus and thus needs co-infection with hepatitis B virus to cause infection.

Pathogenesis and pathology - see specific viral hepatitis.

Clinical features

Most are asymptomatic. Those that are symptomatic present in three forms:

Anicteric form

The symptoms are non specific. Patients present with vomiting, fever and malaise. The diagnosis is suspected if a child has history of contact with a patient suffering from hepatitis or if the liver enzymes (SGOT, SGPT) are elevated.

Icteric form

The patients present with fever, nausea, vomiting, abdominal discomfort, jaundice and tender and enlarged liver.

Fulminant form

This is due to massive hepatic necrosis. Patients present with deep jaundice, impaired mental status that progresses to coma, and death. The prognosis is poor.

Diagnosis

Diagnosis includes:

- Elevated liver enzymes SGOT, SGPT
- Increased serum bilirubin
- Prolonged prothrombin time, and low serum albumin occurs in severe cases
- Serology – see specific viral infections below.

Management of acute viral hepatitis.

There is no specific treatment for typical viral hepatitis. Although hospitalization may be required for severe illness, most patients do not require hospital care.

A high calorie diet is desirable, and because many patients may experience nausea late in the day, the major calorie intake is best tolerated in the morning. Protein intake should be restricted.

Drugs capable of provoking adverse reactions and drugs metabolized by the liver should be avoided.

In fulminant hepatitis, the goal of therapy is to support the patient by maintenance fluid balance, support of circulation and respiration, control of bleeding, correction of hypoglycemia and treatment of other complications of the comatose state in anticipation of liver regeneration and repair

Liver transplantation is indicated in chronic hepatitis with cirrhosis and hepatocellular carcinoma.

Table 16: Summary of viral hepatitis and the route of transmission.

Disease	Etiology	Route of transmission
Hepatitis A	HAV	Fecal-oral
Hepatitis B	HBV	Vertical(intrapartal intrauterine), Blood transfusion
Hepatitis C	HCV	Blood transfusion, sexual exposure
Hepatitis D	HDV	Percutaneous
Hepatitis E	HEV	oral

Hepatitis A

Etiology

Hepatitis A is a member of the picornavirus family. It can be isolated from the stools of infected patients.

Epidemiology

The prevalence rate in developing countries reaches 100% by the age of five years. Transmission is by person-to-person contact and spreads by the fecal-oral routes. There is no mother to fetus transmission during pregnancy. Fecal excretion of the virus occurs late in the course and just before the onset of symptoms. Persons with HAV are contagious for about a week after the onset of jaundice.

Incubation may take as long as four weeks.

Pathology

There is centrilobular necrosis with increased cellularity over the portal areas. The lobular architecture is normal. Areas of infiltration by monocytes and polymononuclear leukocytes are also documented. With fulminant hepatitis, there is total destruction of the parenchyma. Three months after the onset of the acute hepatitis, the liver is

morphologically normal. There is a rise in the level of SGPT and SGOT. The height of elevation does not correlate with the extent of hepatocellular injury and has little use in predicting the prognosis. A drop in the level of SGPT and a rise of bilirubin with prolonged prothrombin time are evidences of poor prognosis.

Clinical Features

Childhood infections are usually asymptomatic in children under five or present with mild non-specific manifestations.

Usually children will have history of contact with jaundiced individuals. Most of the symptomatic children present with abrupt onset of fever, malaise, nausea, emesis, anorexia and abdominal discomfort. Jaundice is usually subtle in young children. The duration of illness is less than one month. After this, appetite, exercise tolerance and the feeling of well-being return. There is no chronicity and fulminant illness and recovery is almost complete.

Diagnosis

Diagnosis involves serology: Ig M anti-hepatitis A virus antibodies are detected at the onset of symptoms and help for diagnosis of acute infection. Thereafter, raised levels of the Ig G persist for life.

Raised levels of bilirubin, SGOT, and SGPT and prolonged prothrombin time are also diagnostic.

Differential Diagnosis

Neonates:

- Physiologic jaundice

- Hemolytic disease of the newborn

- Sepsis

Infants and older children:

- Sepsis

Malaria

Relapsing

Prevention

Precautions like hand washing and careful handling of feces and contaminated materials of infected children is an important step in the disease control.

Hepatitis B virus

Etiology

The hepatitis B virus is a DNA virus that belongs to the hepadna viridae. It has three group of antigens:

- Surface antigen (hepatitis B_s antigen, HBsAG)
- Core antigens (Hepatitis B_c antigen)
- Hepatitis B_e antigen.

The virus replicates in the liver, lymphocytes, spleen and other organs.

Epidemiology

The hepatitis B virus is highly prevalent in the Sub-Saharan Africa. The majority of the infections in children are asymptomatic. The most important mode of transmission in children is perinatal exposure to a hepatitis B_sAntigen and hepatitis B_eAntigen positive mother. Both intra uterine and intrapartal routes are possible. Breast-feeding is not a risk factor. Other risk factors like intravenous drugs, blood contacts and sexual contact are also possible ways.

In children with chronic hepatitis B infection, Hb_sAg becomes positive for six or more months and is associated with progression to chronic liver disease and hepatocellular carcinoma. The incubation period ranges from 45 to 160 days (average 100 days).

Pathology and pathogenesis

The acute response is the same as that of hepatitis A infection. Histologic changes persist in hepatitis with B, C, D types and may progress to chronic liver disease. Unlike the others, injury is by immune mediated mechanisms. After infection of the hepatocytes, viral antigens appear on the surface. (Hb_sAg, Hb_cAg). The mechanism for chronic hepatitis is not yet known.

Clinical features

Children are usually asymptomatic. Symptoms are more severe with greater rate of fulminant hepatitis than that of Hepatitis A virus

The illness is preceded by arthralgia and skin rashes, lethargy, anorexia, and malaise. Jaundice is present in 25% of cases. Other extra hepatic manifestations are polyarthritis, glomerulonephritis and aplastic anemia.

The common physical findings are jaundiced skin, mucous membrane and sclera, tender hepatomegaly and splenomegally.

In the usual course of resolving HBV infection, symptoms are present for 6 to 8 weeks. Chronic hepatitis occurs and chronic active form results in cirrhosis and hepatocellular carcinoma.

Diagnosis

Hb_sAg is the first serologic marker of infection. Its rise coincides with the onset of symptoms. Hb_eAg is present during the acute phase and it indicates a highly infectious state. Hb_sAg level falls before the end of symptoms. Ig M anti Hb_cAg rises early after infection and persists for many months before being replaced by Ig G anti Hb_cAg which persists for years. Ig M anti HB_cAg is not found in perinatal HBV infections.

Anti Hb_c Ag is the most valuable single serologic marker of acute HBV infection because it is present almost as early as Hb_sAg and continues to be present later in the course of the disease when Hb_sAg has disappeared.

Only Anti Hb_sAg is present in persons immunized with hepatitis B vaccine. AntiHb_sAg and AntiHB_cAg are detected in persons with resolved infection.

Management

Management is the same as that of the general management of viral hepatitis.

Complications

Complications include:

- Acute fulminant hepatitis
- Chronic hepatitis
- Cirrhosis

- Hepatocellular Carcinoma

Prevention

Prevention involves the immunization of neonates. More than 90% of infants who acquire the infection perinatally will become chronic carriers. There is lesser risk of acquiring chronic carrier state with age.

Infants born to Hb_sAg positive mothers should receive the vaccine at birth, in the first month and at 6 months of age.

4.7 EPIDEMIC TYPHUS

Goitom G/Yesus

Definition:

Epidemic typhus is a louse borne febrile illness caused by *rickettsia prowazekii*.

Epidemiology

Typhus is characteristically seen during times of poor hygienic practices associated with crowding, war, famine and civil conflicts. Most cases of epidemic typhus occur sporadically but epidemics have occurred in Africa, especially in Ethiopia, Nigeria, and Burundi.

Pathophysiology

Transmission is through human lice or head lice, which become infected by feeding on the blood of an infected person. The rickettsia multiply in the gut of the lice and are passed into their feces. Transmission to a susceptible human host occurs through skin abrasions or perforations and the conjunctiva. After entry through the skin, the rickettsia get access to blood vessels.

Clinical manifestations

The incubation period is usually less than 14 days. The clinical manifestations include:

- Most patients have fever, severe headache, abdominal tenderness and pink or erythematous rash over the trunk.
- Less common manifestations are non productive cough, photophobia, abdominal pain, tinnitus, constipation.

Diagnosis

Clinical diagnosis may be impossible because there are many diseases such as malaria, typhoid fever, meningococemia and relapsing fever which have similar clinical manifestations. No laboratory tests establish the diagnosis.

Treatment

Treatment includes chloramphenicol 50-100 mg/KG/24hr PO in four divided doses for 10 days, nursing care and symptomatic supportive therapy.

Exercise

1. List the factors, which lead to high prevalence of epidemic typhus?
2. What is the vector for transmission of the disease?

References

1. K.C.Watson, Enteric fever, D.B JELLIFFE, J.P. STANFIELD, Diseases of children in the subtropics and tropics, 3rd edition, 1986
2. T.G. Cleary, Enteric fever, Behrman, Kliegman, Jenson, Nelson Text Book of Pediatrics, 16th edition, 2000.

4.8 RELAPSING FEVER

Goitom G/Yesus

Definition

Relapsing fever is an arthropod borne infection characterized by recurrent episodes of fever and is caused by *borrelia* species

Epidemiology

Louse borne fever (epidemic relapsing fever)

Louse borne fever is caused by *borrelia recurrentis* and transmitted from person to person by the human body louse (pediculus humanus). Human infection occurs as a result of crushing lice during scratching, allowing infected hemolymph to enter through

abraded skin. Louse borne disease occurs usually in epidemics, often associated with typhus.

Epidemics are associated with war, poverty, famine, prison settings and poor personal hygiene. This form of relapsing fever occurs more commonly during winter. The major endemic focus of the disease is the high lands of Ethiopia.

Tick borne fever (endemic relapsing fever)

Tick borne fever is caused by *borrelia duttoni*. After ingestion of an infective meal, spirochets invade all tissue of their arthropod hosts, including salivary glands and reproductive tracts. Human infection occurs when saliva or feces is released by the tick during feeding thereby permitting spirochetes to penetrate the skin and mucous membrane.

Rodents are the principal reservoirs. Infected ticks get access to human dwellings on the rodent host. Human contacts are unnoticed because these ticks are nocturnal feeders, have painless bite and detach immediately. Tick borne relapsing fever occurs sporadically in many tropical and subtropical countries.

Pathogenesis

The cyclic nature of relapsing fever is explained by the ability of *borrelia* organisms to continually undergo antigenic variations. Multiple variants evolve simultaneously during the first relapse, with one type becoming predominant. During febrile episodes, spirochetes enter the blood stream, induce development of specific antibodies, undergo lysis and phagocytosis ending up with remission of fever. The numbers of relapses in untreated patients depend on the number of antigenic variants of the infecting strain.

Clinical manifestation

Louse borne disease has a longer incubation period, longer periods of pyrexia, fewer relapses and longer remission periods than tick borne disease. The incubation period is 2 to 14 days. Common clinical features are:

Sudden onset of fever, headache, myalgia, arthralgia, nausea, and vomiting;

Additional symptoms that may occur later include: abdominal pain, cough, bleeding manifestations (**epistaxis**, hemoptysis, hematuria, and hematemesis);

CNS manifestations include lethargy, stupor, meningismus, convulsion, peripheral neuritis, and cranial nerve palsy.

Skin manifestations may include diffuse erythematous, macular, or petechial rash;

Severe form of manifestations include: myocarditis, hepatic failure, and disseminated intravascular coagulopathy.

The initial symptomatic period ends in a crisis in 4 to 10 days, marked by abrupt diaphoresis, hypothermia hypotension, bradycardia, profound muscle weakness, and prostration. This crisis is called *Jarisch-Herxheimer* reaction and may occur within 1 to 2 hours of antibiotic therapy.

Diagnosis

Diagnosis involves the demonstration of spirochetes in thin or thick blood smear stained with Giemsa or Wright stain.

Treatment

Treatment involves the following:

- Supportive
 - Keep IV line opened
- Antibiotics
 - Procaine penicillin 400, 000 IU IM
 - Check blood film after 24 hours and, if negative, give chloramphenicol 50mg/Kg/24hr in four divided doses
 - Patients should be deloused immediately after admission by shaving the head and removing all clothing to be sprayed with insecticide or be boiled.

Prevention

Prevention involves health education on hygienic practice.

Exercise

1. What is the cause of relapsing febrile episodes in relapsing fever?
2. How does *Jarisch-Hrxheimer* manifest clinically?

References

1. K.C.Watson, Enteric fever, D.B JELLIFFE, J.P. STANFIELD, Diseases of children in the subtropics and tropics, 3rd edition, 1986.
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4.9 MALARIA

Goitom G/Yesus

Definition

Malaria is an acute and chronic protozoan illness characterized by paroxysms of fever, chills, sweats, fatigue, anemia and splenomegally.

Etiology

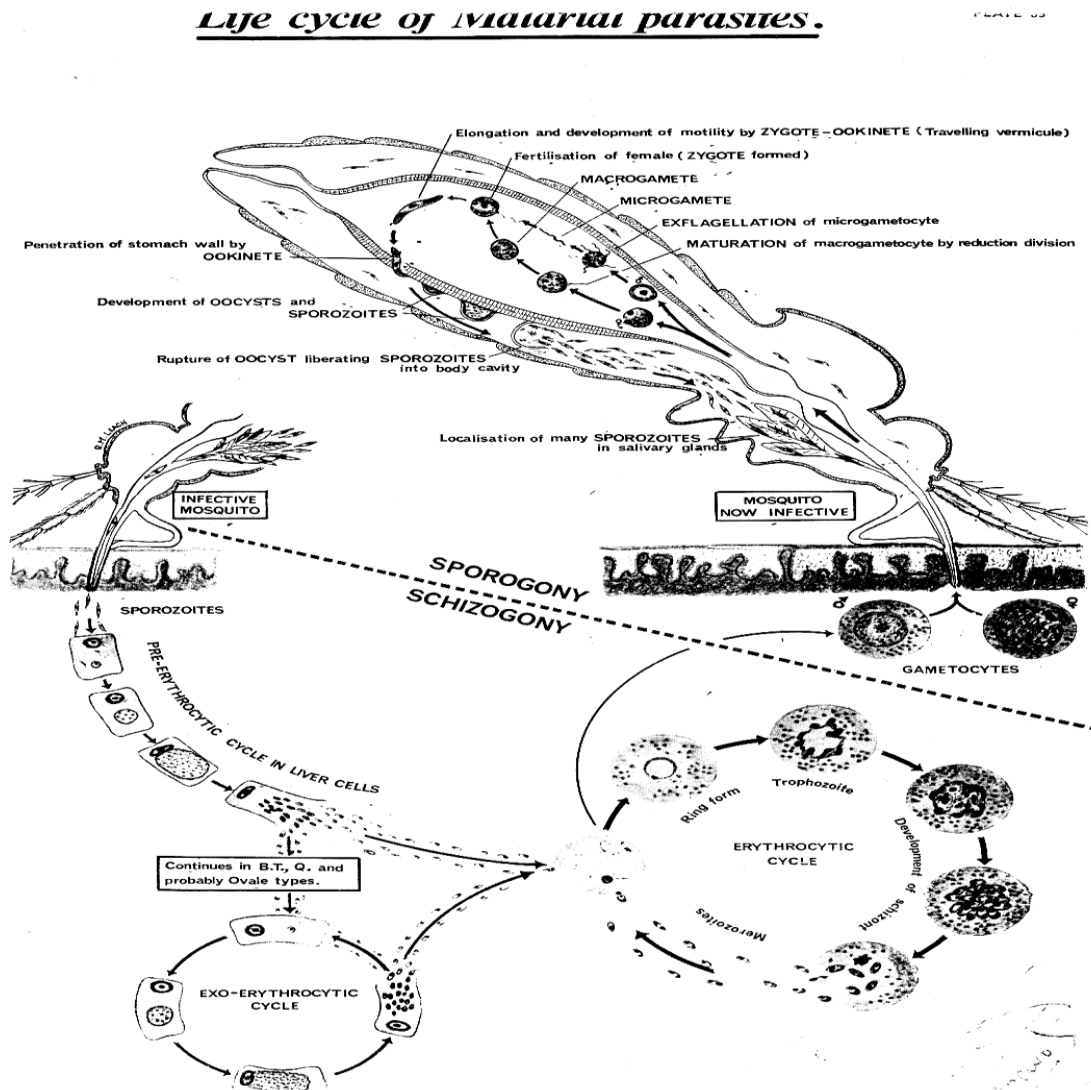
Four species of *plasmodium* are known to cause malaria: *P. falciparum*, *P. malariae*, *P. ovale*, and *P. vivax*. *Plasmodium* species exist in variety forms and have a complex life cycle that enables them to survive in different cellular environment in human host (asexual phase) and anopheles mosquito (sexual phase).

Epidemiology

Malaria is of overwhelming importance in the developing world today with an estimated 300 million cases and over 1 million deaths each year. Most malarial deaths occur in infants and young children. Malaria exists where an effective vector (anopheles mosquito) breed in nature and where human carriers of sexual forms of the parasite are available to these mosquitoes. The principal areas of transmissions are Africa, Asia, and South America.

In Ethiopia, malaria affects 75% of the population and over five million clinical cases are reported annually. In 2000-2001, malaria was the number one cause of outpatient attendance, the number one cause of hospital admission and the third highest cause of deaths in hospitals.

Malaria is transmitted to humans by anopheles mosquitoes; but it can also be transmitted through blood transfusion and from pregnant woman to her fetus. In highly endemic areas, *P.falciparum* infection of indigenous children in the first 5 years of life causes severe and potentially fatal illness. In areas of low endemicity, however, where the immunity of indigenous population is low, severe infection occur in all age groups including adults.



Pathogenesis

The symptoms and pathology are caused by the asexual erythrocytic stage of the malarial parasite. Four important pathologic processes have been identified in patients with malaria: fever, anemia, immunopathologic events and tissue anoxia resulting from cytoadherence of infected erythrocytes.

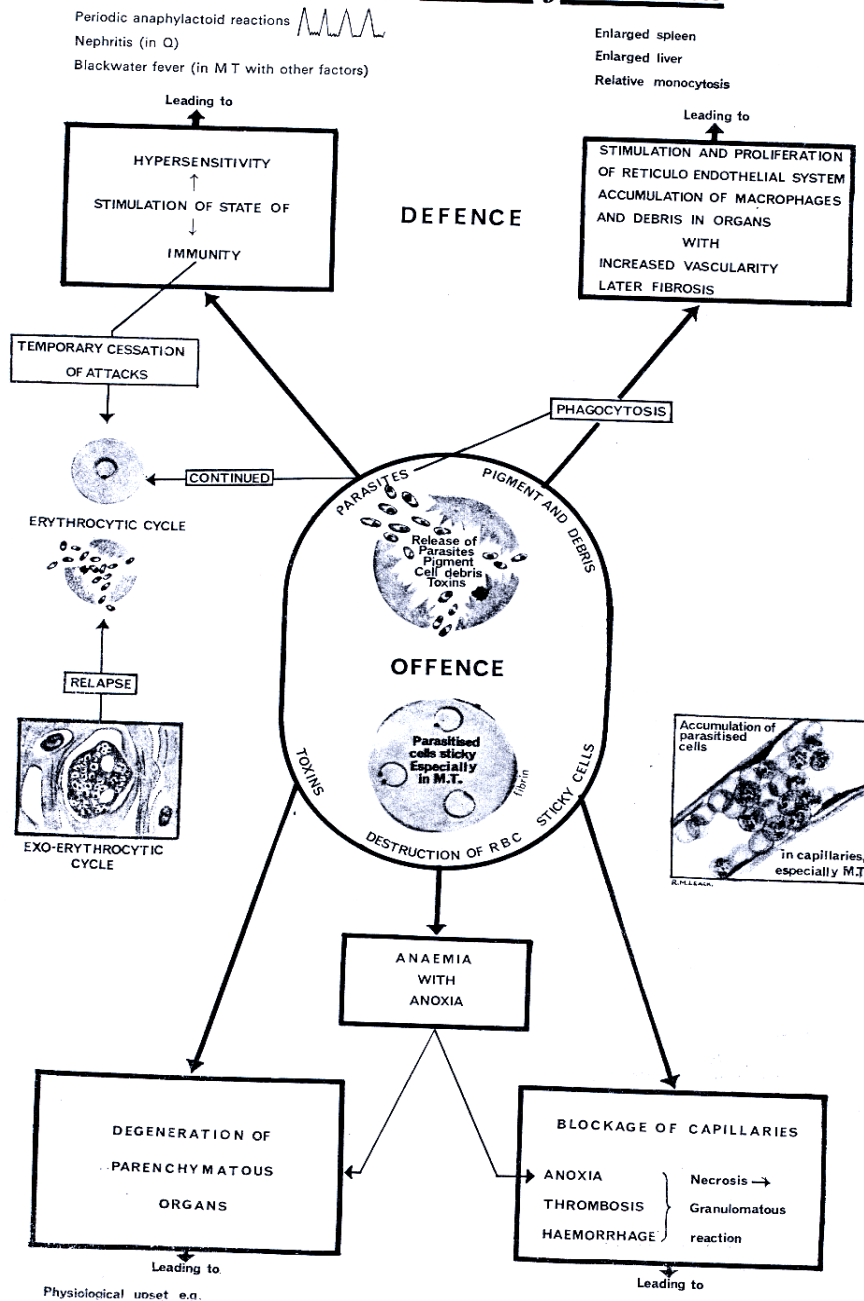
- Fever occurs when red blood cells rupture and release merozoites into circulation.
- Anemia is caused by hemolysis, sequestration of red blood cells in the spleen and other organs, and suppression of the bone marrow.

- Immunopathologic events include immune complex formation, immune suppression, and release of inflammatory mediators such as Tumor necrosis factor which may be responsible for many of the pathologic features of the disease.
- Cytoadherence to vascular endothelium which occurs in *P. falciparum* may lead to obstruction of blood flow and capillary damage with resultant vascular leakage of protein and fluid, edema and tissue anoxia in the brain, heart, lungs, intestine and kidney.

The human body responds to plasmodium infection by production of partial immunity. This averts subsequent severe form of infection, but eradication or prevention of future infection is not achieved.

The Pathogenesis of Malaria

PLATE 42



Clinical manifestation

The usual incubation periods are as follows: *P. falciparum*, 9 to 14 days, *P. vivax*, 12 to 17 days, *P. ovale*, 16 to 18 days, and *P. malariae*, 18 to 40 days. Classic manifestation includes febrile paroxysms alternating with periods of fatigue, but otherwise relative wellness. Symptoms associated with febrile paroxysms include high fever, rigors, sweats and headache, as well as myalgia, back-pain, abdominal pain, nausea, vomiting, diarrhea, pallor and jaundice.

Paroxysms correspond with the rupture of schizonts that occurs every 48 hours with *P. vivax* and *P. ovale* and results in daily fever spikes; and occurs every 72 hours with *P. malariae* and results in alternate day or every third day fever spike. Periodicity is less apparent with *P. falciparum* and mixed infections. Long-term relapse occurs in *P. vivax* and *P. ovale* due to release of merozoites from the exo-erythrocytic source in the liver.

P. falciparum malaria is the most severe form and is associated with more intense parasitemia. The case fatality rate reaches 30% in non-immune infants. The diagnoses of *P. falciparum* malaria constitutes of medical emergency.

P. falciparum infects immature and mature RBC, whereas *P. malariae* infects only mature RBC. *P. falciparum* is most commonly associated with serious complications although milder or asymptomatic infection occurs in those who are partially immune.

The severe manifestations of *P. falciparum* malaria are as follows:

- Cerebral malaria: unarousable coma not attributable to any other causes lasting 30 minutes or more after a generalized convulsion.
- Severe anemia: a hematocrit level of less than 15%, or hemoglobin less than 5gm/dl.
- Hypoglycemia: blood glucose level of less than 40mg/dl.
- Generalized convulsion: more than 2 within 24hrs
- Renal failure: urine output less than 12ml/Kg/24 hours failing to improve after rehydration.
- Pulmonary edema
- Shock

- Spontaneous bleeding (from gums, nose, intestine etc. and/or laboratory evidence of disseminated intra vascular coagulation.
- Acidosis
- Hemoglobinuria
- Impairment of consciousness less marked than unarousable coma
- Prostration
- Jaundice
- Hyperpyrexia
- Hyperparasitemia

The commonest complications in children are cerebral malaria, convulsions, severe anemia, and hypoglycemia.

Diagnosis

Diagnosis involves the identification of organisms on Wright or Giemsa stained smears of peripheral blood. The thick smear is used for the detection of parasites and the thin smear for the detection of the species of parasite.

A single negative blood smear finding does not exclude malaria, it may be necessary to repeat the smears as frequently as every 4 to 6 hours per day in order to establish a diagnosis.

In ill patients with change of sensorium, do a lumbar puncture to rule out meningitis.

Differential diagnosis

Differential diagnosis includes viral infection such as hepatitis, sepsis, pneumonia, meningitis, encephalitis, endocarditis, and gastroenteritis.

Treatment

Uncomplicated malaria

- First-line treatment

The first-line treatment of *P. falciparum* malaria is sulfadoxinr-pyrimethamine. For the treatment of malaria caused due to *P. vivax*, *P. mlariae* or *P. ovale*, the first-line drug of choice is chloroquine.

- Second-line treatment

If a *P. falciparum* positive patient returns to the laboratory with fever or history of fever between the third day and fourteenth day after treatment with sulfadoxinr-pyrimethamine, check whether the patient has vomited the drug or had diarrhea after treatment.

If the blood film is positive for asexual parasites and the above conditions are excluded, give quinine orally. For details of dosages of oral quinine according to age and body weight, see table below.

Treatment of severe P. falciparum malaria

- Prevention or early detection and treatment of convulsion, hypoglycemia, and hyperpyrexia.
- Give prophylactic anticonvulsant - phenobarbital 5-10mg/Kg
- Start antimalarial chemotherapy with intravenous quinine (see below for dosage and administration)
- Measure and monitor urine out put

Blood transfusion for severe anemia or symptomatic congestive heart failure.

Treatment Schedules

1a. Chloroquine

Tablets of 150 mg base or syrup 50 mg base per 5 ml.

Total dose of 25 mg base per kg over 3 days (10 mg base per kg on Days 1 and 2, and 5 mg base per kg on day 3).

Side effects include:

Dizziness, skeletal muscle weakness, mild gastrointestinal disturbances (nausea, vomiting, abdominal discomfort and diarrhoea) and pruritus. Pruritus may be severe but is usually over within 48-72 hours.

Contra-indications include:

- Persons with known hypersensitivity
- Persons with a history of epilepsy
- Persons suffering from psoriasis.

Table 17: Chloroquine treatment schedule

Weight (kg)	Age (years)	Day 1	Day 2	Day 3
5-6	< 4 months			
	Tablets	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
	Syrup	5ml	5ml	2.5 ml
7-10	4-11 months			
	Tablets	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
	Syrup	7.5ml	7.5 ml	5ml
11-14	1-2 years			
	Tablets	1	1	$\frac{1}{2}$
	Syrup	12.5ml	12.5ml	7.5ml
15-18	3-4 years			
	Tablets	1	1	1
	Syrup	15ml	15ml	15ml
19-24	5-7years			
	Tablets	1 1/2	1 1/2	1
	Syrup	20ml	20ml	15ml
25-35	8-10 years	2 $\frac{1}{2}$	2 $\frac{1}{2}$	1
36-50	11-13 years	3	3	2
50+	14+ years	4	4	2

1b. Sulfadoxine-pyrimethamine:

Tablet containing 500mg sulfadoxine plus 25mg pyrimethamine or injection having the same composition per 2.5 ml.

Table 18: Sulfadoxine-pyrimethamine treatment schedule

Weight (Kg)	Age (Years)	Number of tablets	Intramuscular Injection (ml)
5-6	2-3 months	$\frac{1}{4}$	0.6
7-10	4-11 months	$\frac{1}{2}$	1.2
11-14	1-2	$\frac{3}{4}$	1.8
15-18	3-4	1	2.5
19-29	5-9	1 $\frac{1}{2}$	3.8
30-39	10-11	2	5
40-49	12-13	2 $\frac{1}{2}$	6.2
50+	14+	3	7.5

Side effects include gastrointestinal disturbances, nausea and vomiting. Skin reactions can occur with repeated doses ranging from a mild form to very severe, like erythema multiforme or Stevens Johnson syndrome, which are rare but can be fatal. Patients treated with this drug should be carefully followed-up. Other sever rarer side effects include anemia, leucopenia and, in G6PD deficient persons, intravascular haemolysis.

Contra-indications include:

Malaria prophylaxis either alone or in combination.

Persons with a previous history of skin rash after using any form of sulfa drug.

Infants less than two months and

Persons with hepatic or renal dysfunction.

1c. Quinine

Treatment is with quinine 8 mg base/kg 3 times daily for 7 days

Table 19: Quinine treatment schedule

Weight (Kg)	Age (Years)	Oral (tablets) Dosage to be given daily for 7 days	
		200mg salt	300 mg salt
4-6	2-4 months	$\frac{1}{4}$	-
6-10	4-12 months	$\frac{1}{3}$	$\frac{1}{4}$
10-12	1-2 years	$\frac{1}{2}$	$\frac{1}{3}$
12-14	2-3 years	$\frac{3}{4}$	$\frac{1}{2}$
14-19	3-5 years	$\frac{3}{4}$	$\frac{1}{2}$
20-24	5-7 years	1	$\frac{3}{4}$
25-35	8-10 years	1 $\frac{1}{2}$	1
36-50	11-13 years	2	1 $\frac{1}{2}$
50+	14+	3	2

Side effects include dizziness, ringing in the ears, blurred vision and tremors known collectively as “cinchonism”. At the above dosages, these symptoms are not severe enough to stop treatment and subside spontaneously when administration of the drugs ends. Hypoglycaemia may be caused by quinine.

There are no contra-indications to the oral administration of the drug with in the above dosage.

1d. Primaquine

Treat with primaquine 0.25 mg base per kg daily for 14 days.

Table 20: Primaquine treatment schedule

Weight (Kg)	Age (Years)	Number of tablets	
		7.5 mg tablet	15 mg tablet
19-24	5-7	$\frac{3}{4}$	-
25-35	8-10	1	$\frac{1}{2}$
36-50	11-13	1 $\frac{1}{2}$	$\frac{3}{4}$
50+	14+	2	1

Side effects such as anorexia, nausea, vomiting, abdominal pain and cramps are dose related and relatively rare at daily doses up to 0.25 mg base/kg. They may also be accompanied by vague symptoms such as weakness and uneasiness in the chest.

Contraindications include the following:

- Pregnancy
- Children under four years
- Any condition that predisposes to granulocytopenia, including active rheumatoid arthritis and systemic lupus erythematosus.

Intravenous Quinine administration

Dosage- start with 20mg/Kg IV over 4hrs, followed by 10mg/Kg diluted in 10ml/Kg isotonic fluid over 4hrs repeated 8 hourly till patient can swallow, then oral tablets 10mg/Kg 8 hourly to complete 7 days or a single dose of Fancidar.

4.10 Leishmaniasis

Goitom G/Yesus

Definition

Leishmaniasis are a group of zoonotic diseases caused by intracellular protozoan parasites of the genus leishmania which are transmitted by phlebotomine sandflies.

Ethiology

In Ethiopia, leishmaniasis is caused by *leishmania donovani* (visceral leishmaniasis two lands), *L. Ethiopia* (cutaneous leishmaniasis, high lands) and occasionally *L. Major* and *L. tropica*.

Epidemiology

An estimated 10 to 50 million people are affected worldwide in endemic tropical and subtropical regions. Visual leishmaniasis occurs in Ethiopia mainly in arid and semiarid lowlands below 1300 meters altitude. Endemic areas extend across the border with neighboring countries (Sudan).

The age /sex-specific pattern indicates infants and children under the age 5, adult above 50 and women of childbearing age to be at relatively low risk which reflects an outdoor exposure to transmission.

Cutaneous leishmaniasis has a prevalence rate of 5.5 to 40 per 1,000 according to surveys done in endemic areas which are situated at 14000 to 2700 meters of altitude. Children are mainly affected with peak incidence seen at 10 – 20 years of age.

Pathogenesis

Clinical features

Classically three major forms of leishmaniasis are described: visceral leishmaniasis, cutaneous leishmaniasis and mucosal leishmaniasis

Visceral leishmaniasis (VL, kala-azar)

After exposure to the organism, the child may have completely asymptomatic infection or develop mild constitutional symptoms (malaise, diarrhea), intermittent fever and weight loss. Most children improve without therapy but one third develop active kala-azar. In Ethiopian patients splenomegaly and fever are most common (98%), wasting (84%) and ascites are not common.

Most (90%) die due to intercurrent infection (pneumonia and diarrhea), a result of the lowered cell mediated immunity. VL is recognized as an opportunistic infection associated with HIV. The typical hepatosplenomegaly is absent but GI involvement is prominent

Cutaneous leishmaniasis (CL)

Three forms of cutaneous leishmaniasis are described in Ethiopian patients

- Localized cutaneous leishmaniasis (LCL)
- Non-self healing disseminated or diffuse (DCL)
- Mucocutaneous (MCL)

Localized cutaneous leishmaniasis is mainly seen in children as a single lesion on exposed skin: face, forearms. The typical lesion is a reddish plaque with irregular border and shallow central crater. It heals spontaneously after 8 to 10 months leaving a hyperpigmented scar.

Diagnosis

Demonstration of the amastigotes in specimens (skin lesion, spleen bone – marrow – or lymph node aspirates) using Giemsa stain makes definitive diagnosis. Direct agglutination test is rapid and sensitive test. VL can have anemia, leucopenia, thrombocytopenia, elevated hepatic transaminases and hyperglobulinemia.

Treatment

Drug therapy involves pentavalent antimony compounds. Sodium stibogluconate 20ml/kg/24 hour IV or IM for 20 to 28 days is used with cure rate of 90 to 100% (LCL), 50 to 70% (ML), and 80 to 100% (VL). Alternative treatment includes pentamidine, amphotericin B, deoxycholate and ketoconazole.

Prevention

Preventive measures consist of community based residual insecticide spraying, control of reservoir hosts, chemotherapy and avoidance of exposure to the nocturnal sandfly using permethrin-impregnated mosquito nettings.

4.11 SCHISTOSOMIASIS (SCHISTOSOMA)

Tsinuel Girma

Definition

Schistosomiasis is an infectious disease caused by schistosoma, flukes or trematodes that parasitize the blood stream.

Epidemiology

Schistosoma infects more than 200 million people worldwide, primarily children and young adults. Prevalence increases with age to a maximum at 10 to 20 years of age.

Opening of irrigation fields and heavy dependence on stream water for domestic use increases spread. Most low lands (altitude between 1500 meters less than 2000 meters) in Ethiopia are endemic to *S.mansoni*. Surveys done in Ethiopia among 27,658 people showed overall prevalence of schistosoma is 25% with children predominantly affected (up to 90% in some communities)

Etiology

Five schistosme species infect humans: *S.haematobium*, *S. mansoni*, *S. japonicum*, *s. intercalatum* and *S. mekongi*. Mainly *S. mansoni* and *s. haematobium* are prevalent in Ethiopia. Snails act as an intermediate host.

Pathogenesis

Humans are infected through contact with water contaminated with cercariae (the infective stage of the parasite). The Cercaria penetrates the skin, migrates to lung and finally reaches the liver.

Sexually mature adult worms from the liver migrate to their specific sites characteristic of each species: *S. mansoni* to inferior mesenteric veins and *S. haematobium* to the venous beds of the bladder and ureters.

Chronic granulomatous injury occurs at egg deposition sites: intestine (*S. mansoni*) urinary bladder and ureter (*S. haematobium*). Cell mediated immunity plays a significant role in tissue injury.

Clinical manifestation

Symptoms occur primarily in heavily infected children.

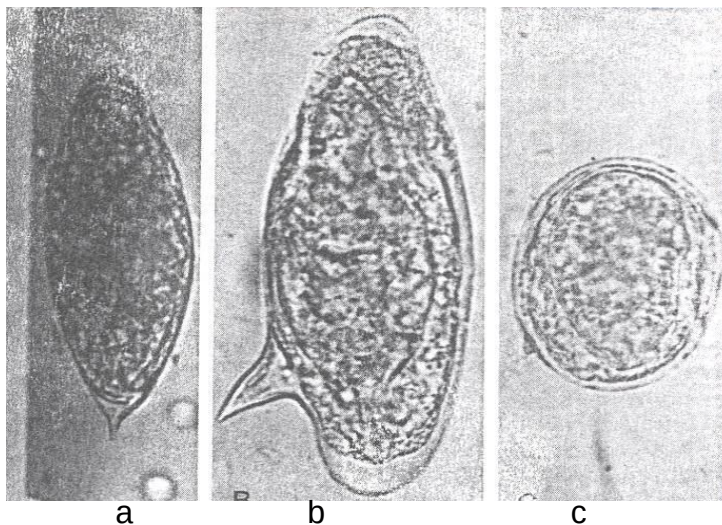
Acute schistosomiasis (katamaya fever) occur 4 to 8 weeks after exposure mimicking a serum sickness like syndrome manifested by fever, chills, sweating, lymphadenopathy, hepatosplenomegally and eosinophilia.

Children with chronic schistosoma mansoni may have bloody diarrhea and colicky abdominal pain. The intestinal phase may pass unnoticed and the syndrome of hepatosplenomegaly, portal hypertension, ascites and hematemesis may be the initial presenting symptoms.

Symptomatic children with chronic schistosomiasis haematobium usually complain of frequency, dysuria and terminal hematuria. Long standing infection results in chronic renal failure, bacterial infection and bladder cancer.

Diagnosis

S. mansoni is diagnosed by light microscopy of stool which shows characteristic eggs (see fig 5.1) *S. haematobium* is identified from urine.



a-*S. haematobium* b-*S. mansoni* c- *S. japonicum*

Figure 19: Schistosomiasis types

Treatment

The recommended treatment for schistosomiasis is praziquantel 40mg /kg in a single or divided doses.

Prevention

Prevention involves improved sanitary conditions, reducing human-water contact, molluscicides and the treatment of infected individual help in disease control

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CHAPTER FIVE

DISEASES OF THE RESPIRATORY SYSTEM

5.1 Upper respiratory tract infection

Habtamu Fekadu

Otitis media

Definition

Otitis media is an inflammation of the middle ear. It may be acute if symptoms for less than 2 weeks, or chronic if symptoms for more than 2 week.

Etiology

Otitis media is caused by *streptococcus pneumonia*, *hemophilus influenzae* and *moraxella catarrhalis*

Epidemiology

Ear pain and/or discharge is one of the most common presenting symptoms to outpatient departments in under 5 children. About one third of children will have at least one episode of acute otitis media by 3 years of age.

Pathogenesis

The two important factors are eustachian tube dysfunction and URTI. Both lead to obstruction of the tubes and serous fluid collection in the middle ear. If there is an associated entry of pathogenic organisms, suppurative otitis media may follow.

Clinical manifestations

Acute amifestations include otalgia, fever, ear discharge hearing loss. Infants may present with irritability, diarrhoea, vomiting and pulling of the ear.

Diagnosis

Clinical digostic signs include>

- Ear discharge
- Visualization of tympanic membrane: red membrane, loss of normal light reflex.

Treatment

Treatment involves the following:

1. Antibiotics for 5 days
 - Cotrimoxazole. Or
 - Amoxicillin or
 - Ampicillin
2. Wicking of the ear with dry cotton for discharging ear.
3. Analgesic - Acetaminophen

Complications

- Mastoiditis
- Hearing loss
- Intracranial (meningitis, brain abscess)

Acute tonsillopharyngitis

Definition

Acute tonsillopharyngitis is an Infection of the upper respiratory tract with principal involvement of the throat and tonsils.

Etiology

Acute tonsillopharyngitis is mostly viral as parts of URTI

Group A B-hemolytic streptococcus is the most common and the most important bacterial agent after 2 years of age.

Clinical

The clinical features of acute tonsillopharyngitis are as follows:

Viral: insidious onset

Other signs of URTI (runny nose, sneezing, coryza etc.)

Contact history with family member with URTI.

Streptococcal: is more common in children older than 2 years and peaks in children between 4 and 7 years of age. They may present with:

- Headache, abdominal pain, vomiting, high grade fever, sore throat.

- Diffuse redness and exudates, petechiae over the palates, and, tender cervical lymphadenopathy.

Most strongly suggestive: - Age 5 to 15 years.

- Clinical evidence of acute pharyngitis
- Absence of URTI

Complications

Complications include the following:

Suppurative: Otitis media

- Retropharyngeal and peritonsillar abscess

Non suppurative - Acute glomerulonephritis, Rheumatic fever.

Treatment

Treatment may include:

- Benzathine penicillin
- Oral penicillin
- Analgesics - paracetamol.

Nasopharyngitis (common cold)

Definition

Nasopharyngitis is a viral infection of the nasopharyngeal mucosa.

Etiology

The commonest cause of nasopharyngitis is *rhinovirus*. Other causes are *adenovirus*, *influenza*, *parainfluenza* viruses etc

Epidemiology

Nasopharyngitis is frequent and seasonal. Person to person transmission is usually airborne.

Clinical features

The clinical features include:

- Runny nose, often with fever, sneezing and cough.
- Sometimes accompanied by conjunctivitis.

Always check the tympanic membranes of an infant with a cold.

Complications

The complications of nasopharyngitis include:

- Acute otitis media
- Pneumonia

Treatment

Treatment may involve the following:

- Sooth the throat and relieve the cough with safe home remedies like tea and honey
- Nasopharyngeal drops using a syringe filled with normal saline (or drop clean water mixed with salt using cotton), 4 to 6 times a day.
- Treat fever
- Advise the mother to return immediately if the child develops fast breathing or difficulty of breathing or if the child becomes sicker.
- Follow up in 5 days

5.2. Acute upper airway obstruction

Tsinuel Girma

Extrathoracic (upper airway) obstruction is exacerbated during inspiration because the pressure inside the airway becomes very negative compared to the atmospheric pressure outside, causing the airways to collapse on inspiration. This physiologic mechanism (stridor, the difficulty during inspiration) explains what happens in cases of obstructions above the thoracic outlet.

Croup is a term encompassing different group of acute, mostly infectious conditions characterized by brassy cough, inspiratory stridor, hoarseness of voice and different degrees of respiratory distress secondary to laryngeal obstruction.

Classification

Upper airway obstruction can be classified based on mechanism (etiologic) of obstruction.

1. Extrinsic airway compression and narrowing

Malformations/syndromes craniofacial anomalies (e.g., Pierre-Robin syndrome)

Masses

Enlarged lymph nodes

Enlarged tonsils

Trauma

Neck and chest injury

Infection

Peritonsillar abscess

Retropharyngeal abscess

Diphtheria

2. Intrinsic airway narrowing and/or collapse

Malformation

Laryngotracheomalacia

Laryngeal stenosis

Masses

Foreign body in the airway

Laryngeal polyps or papillomatosis

Trauma

Intubation

Burns

Neck injury

Infection

Tracheitis, bacterial

Epiglottitis

Laryngotracheobronchitis, viral

Other Allergic reaction: e.g., anaphylaxis, angioneurotic edema

Vocal cord paralysis

Laryngospasm

Laryngotracheobronchitis (viral croup)

Epidemiology

Viral croup mainly affects children between the age of 3 months to 5 years of age. Males are more affected than females. The rate of viral croup increases in cold season. Fifteen percent (15%) of cases have a positive family history.

Etiology

Parainfluenza viruses account for 75% of cases. Other viruses such as adenoviruses, respiratory syncytial virus and measles virus can also be involved in few cases. Rarely, mycoplasma pneumoniae and diphtheria can be isolated.

Clinical manifestations

The degree of airway obstruction in children is severe compared to adults for the following reasons:

- Small size of the airways
- Loosely attached mucous membrane
- Abundant mucous glands of the airways
- Frequent respiratory infections

The typical patient presents with a history of upper respiratory infection several days before onset and/or other family members with common cold, low grade fever, intermittent stridor and mild brassy cough. As the disease progresses, the patient will have continuous stridor, signs of severe respiratory distress, decreased air entry, restlessness and lethargy. The degree of severity can be assessed and followed using croup score (see below).

Throat examination is best deferred, since the risk of inducing laryngeal spasm is very high.

Diagnosis

Diagnosis is mainly clinical. Lateral neck X-ray may show subglottic narrowing but because of the risk of sudden laryngeal obstruction it should be done in the presence of professionals with skills of emergency respiratory support.

Differential Diagnosis

- Epiglottitis
- Bacterial tracheitis
 - Foreign body aspiration
 - Retropharyngeal /peritonsillar abscess

Treatment

Treatment includes the following:

- Supportive - Oxygen
 - Intravenous fluids
 - Close monitoring
- Dexamethasone 0.5mg/kg i.m. repeat if no improvement after 4 to 6 hours.
- Racemic epinephrine
- Home -cold steam (soothe throat or reduce obstruction)
- Tracheostomy – if the distress gets worse, despite the above treatments or if there are signs of impending respiratory failure at any time.

Complications

- Involvement of lower air ways
- Bacterial superinfection is rare
- Respiratory failure

Prognosis

Prognosis is excellent, there is usually no residual effect

Epiglottitis

Definition

Epiglottitis is an acute inflammatory (infectious) process involving the epiglottis and surrounding structures.

Epidemiology

Children between the ages of 2 and 7 years are affected and the peak incidence occurs at about 3 and a half years of age. The male to female ratio is 3:2. There is no seasonal variation.

Etiology

H.influenzae type b causes almost all cases of epiglottitis. Rarely *streptococcus pneumoniae* and *streptococcus pyogenes* can lead to epiglottitis.

Clinical manifestations

Classically epiglottitis starts suddenly with rapid progression to complete obstruction. Patients are toxic with high grade fever, sore throat, dysphagia, tachycardia, restlessness, drooling of saliva and stridor.

Big children hyperextend their neck and sit leaning forward.

Throat examination, without adequate respiratory support preparation, should be avoided since it causes sudden reflex laryngeal spasm.

Diagnosis

Diagnosis is mainly clinical. Blood culture and lateral neck X-Ray help for diagnosis but they are time consuming and add little to the immediate management. Laryngoscope shows cherry red epiglottis if it is done with proper preparation for respiratory support or intubation.

Management

The principles of management are:

- Routine tracheostomy/nasotracheal intubation
- Oxygen
- Chloramphenicol 75 -100mg/kg /dose i.v for 7 to 20 days.

Foreign Body Aspiration

Tsinuel Girma

Most foreign bodies aspirated into the air ways are cleared by mucoliliary system or cough reflex. Review of seventy nine children treated at the Ethio-Sewedish Children's Hospital showed a male to female ratio of 2:1 4 and a mean age 3 years. Another review, in the same hospital showed an average of 6 cases per year: bean seeds accounting for nearly half of aspired objects.

Aspiration is frequent in children with neurological abnormality (poor coordination of swallowing), cleft palate, impaired consciousness and age groups of 9 months to 5 years.

The presentation varies depending on the site of blockage, degree of obstruction and time of presentation after aspiration which is usually delayed. A review of cases in Black Lion hospital showed that only 10% of cases presented within 24 hours of aspiration.

Laryngeal foreign bodies

Clinical manifestations

Cough (croupy and horse) and progressive obstruction manifested by inspiratory stridor, respiratory distress and aphonia are seen soon after aspiration. If immediate measures are not taken obstruction aggravated by local inflammatory reaction may lead to death.

Diagnosis

Direct Laryngoscopy confirms diagnosis and helps to remove the foreign body.

In a child suspected of laryngeal foreign body lateral and anteroposterior neck X-ray help in the diagnosis.

Bronchial foreign bodies

Clinical manifestations

Children seen shortly after aspiration will have history of choking, cough, wheeze and blood streaked sputum. The degree of obstruction determines the type of obstruction and presentation. If the first episode is overlooked or underestimated, latent periods of

days to months occur before patients develop recurrent pneumonia or intractable wheezing (“asthma”).

There are three types of possible obstruction to the airways.

- Stop valve (complete) - there is no entry or exit of air distal to obstruction resulting in collapse.
- Ball valve - there is only entry of air distal to obstruction resulting in emphysema (distention).
- Bypass valve - both entry and exit of air distal to obstruction is possible.

Diagnosis

On chest X-ray (76% of cases show abnormality) opaque objects are clearly seen. If non-radiopaque object is inhaled, indirect signs such as mediastinal shift, collapse or overinflation are evident.

Treatment

Any child suspected of foreign body aspiration should be immediately referred to facilities with bronchoscopy for confirmation of diagnosis and removal.

Emergency Management of Foreign Body Aspiration

If a child is actively choking and a foreign body suspected, follow the following procedure

In unconscious child younger than 1 year. a combination of five back blows and five chest thrusts are given until successful. For those above 1 year a series of five abdominal thrusts (Heimlich maneuver) with lying or standing position is administered

Prognosis

Late presentation, delayed diagnosis and intervention as well as missed diagnosis are major causes of prolonged morbidity and high mortality.

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5.3 PNEUMONIA

Goitom G/Yesus

Definition

Pneumonia is an inflammation of the parenchymal structure of the lungs, such as the alveoli and the bronchioles.

Etiology

Bacteria, viruses and non-infectious agents result in pneumonia. The most common causes, in developing countries, are bacterial agents especially *streptococcus pneumonia*, followed by *haemophilus Influenza*. *Staphylococcus aureus* and gram negatives (*E.coli*, *Klebsiella*, etc) are other causes in neonates and immunocompromized.

Epidemiology

Pneumonia is responsible for 70-80% deaths that occur due to acute respiratory infections worldwide. In Ethiopia, pneumonia is the cause of one third of infant mortality and one fifth of under 5 mortality.

Pneumonia is more severe in developing countries due to the high prevalence of risk factors such as:

- Malnutrition, including Vitamin A and D deficiencies
- Overcrowding and indoor-air pollution
- Intercurrent infections such as measles, whooping cough, malaria and diarrhea and

- Immune-deficient states

Severe forms of pneumonia are commonly encountered in children between the age of 6 months and 3 years.

Pathogenesis

The normal lung is sterile. Most of the agents are inhaled from the air breathed or aspirated into the periphery of the lung from the upper air ways. After entry to the lung, initially, a reactive oedema occur which supports proliferation of the organisms and aids in their spread into adjacent portions of the lung. The disease may have a lobar, patchy or diffuse distribution. The disease may be complicated in some by development of lung abscess, collapse of the lung, and empyema

Clinical manifestations

The clinical features of pneumonia resulting from most of the causative agents are similar. The common manifestations in bacterial pneumonia are:

- In infants, initial manifestations may be mild like upper respiratory tract infection. Cough, sudden onset of fever, and signs of respiratory distress. Respiratory distress is manifested with rapid and difficult of breathing, nasal flaring, intercostal retraction, chest indrawing and cyanosis.
- Other possible findings are crepitation, diminished breath sounds, bronchial breathing and dullness on percussion.

It should be noted that fast breathing and chest indrawing are important clinical findings in the assessment of severity of pneumonia.

- A child with cough and fast breathing (look below for cut of points of breathing) but with out chest indrawing or difficulty of breathing is classified as having *Pneumonia*.
- A child with fast breathing and chest indrawing, grunting, cyanosis or any danger signs such as lethargy, unconsciousness or convulsion is classified as having *Severe Pneumonia*.

Table 21: Cut off for fast breathing

If the child is:	The child has fast breathing if you count:
2 months up to 12 months	50 breaths per minute or more
12 months up to 5 years	40 breaths per minute or more

Note: The child who is exactly 12 months old has fast breathing if you count 40 breaths per minute or more.

Diagnosis

Diagnosis is mainly reached on the basis of clinical features.

A chest X-ray may be helpful in patients who fail to respond to treatment to check for complications such as pleural effusion, pneumatocele, atelectasis, or abscess formation.

Treatment

- Children with severe pneumonia should be admitted for in patient care
- Supportive: include administration of oxygen nasally and hydration to replace insensible water loss.
- Vitamin A.
- Antibiotics
 - Children with pneumonia are treated at out patient level with cotrimoxazole or amoxycillin orally for 5 days
 - Advise parents to come back if there is worsening.
 - Children with severe pneumonia are given:
 - Intravenous antibiotics - crystallin penicillin as a first line drug
 - Assess response after 48 to 72 hours, if there is no improvement or if there is worsening, add chloramphenicol

Exercise

1. What are the common etiologic agents of pneumonia?
2. List risk factors which predispose children to develop severe pneumonia.
3. Describe the cut off respiratory rates for fast breathing.
4. What are the criteria for severe pneumonia?
5. Describe the treatment of severe pneumonia.

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5.4 BRONCHIOLITIS

Goitom G/Yesus

Definition

Acute bronchiolitis is a common disease of the lower respiratory tract of infants, resulting from inflammatory obstruction of the small airways.

Aetiology and epidemiology

Acute bronchiolitis is a predominantly viral illness. About 50% of the disease is due to respiratory syncytial virus (RSV). Other causative organisms are parainfluenza 3 viruses, mycoplasma, some adenoviruses and occasionally other viruses.

Bronchiolitis occurs most commonly in male infants between the ages of 3 to 6 months who have not been breastfed and live in crowded conditions. The source of the viral infection is usually a family member (older child or adult) with a minor respiratory illness.

Pathophysiology

Acute bronchiolitis is characterized by bronchiolar obstruction due to:

- oedema,
- accumulation of mucus and cellular debris, and
- Invasion of the smaller bronchial radicles by virus.

The bronchiolar wall thickening affects air flow, impairing the normal exchange of gases in the lung, which will result in hypoxemia early in the course of the disease.

Clinical manifestations

Most affected infants have:

- History of exposure to older children or adults with a minor respiratory disease within the week preceding the onset of illness. The first symptoms are a mild upper respiratory tract infection, with serous nasal discharge and sneezing, lasting for several days. These maybe accompanied by decreased appetite and fever. Then there will be a gradual development of respiratory distress characterized by wheezy cough, dyspnoea and irritability. Infants may have difficulty of sucking.
- On examination there will be tachypnea, hyper-expanded chest, severe respiratory distress, a prolonged expiratory phase and wheezing, which is usually audible.
- Chest X-ray shows hyper-inflation of the lungs and an increased anteroposterior diameter on lateral view.

Diagnosis

Diagnosis is mainly clinical

Differential diagnosis

- Bronchial asthma
- Bacterial bronchopneumonia
- Congestive heart failure

Treatment

Treatment is supportive:

- The child should be placed in an atmosphere of cool and humidified oxygen to relieve hypoxemia, reduce insensible water loss from tachypnea, and also relieve the dyspnoea, cyanoses, anxiety and restlessness.

- Oral intake must often be supplemented or replaced by parenteral fluids. Antibiotics should be given if there are secondary bacterial infections like bacterial pneumonia.

Exercise

1. What is the common chest x-ray finding in bronchiolitis?

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5.6 Bronchial Asthma

Tsinuel Girma

Definition

Bronchial asthma is a diffuse obstructive lung disease with hyper reactivity of the airways and reversibility of air flow obstruction spontaneously or with treatment.

Epidemiology

Worldwide 100 to 200 million cases of asthma are seen yearly accounting for 40,000 to 50,000 deaths. Bronchial asthma was reported in 2.8% of school children in Addis Ababa while 18.2% had at least one episode of wheezing in the last 12 months. It is rare in rural setting. Increasing industrialization and urbanization is associated with high incidence of asthma.

Children are more affected than adults, but a study in Jimma demonstrated a lower rate in children. Before puberty, the ratio of males to females with asthma is 2:1 but later it occurs equally. Most children (85%) will have their first symptoms before 5 years of age. Children with severe chronic asthma and steroid dependence have high (95%) chance of becoming adult asthmatics.

Risk factors include:

- Allergy (inherited) like eczema, urticaria or hay fever
- Polygenic inheritance

If one parent has asthma, there is a 25% risk of asthma in their child. If both parents are affected there is a 50% chance.

- LBW
- Poverty, frequent respiratory infections

Etiology

There is no well defined single etiologic agent. Multiple factors interplay to disturb the mechanisms that maintains the balance between contraction and relaxation of bronchial smooth muscle.

Autonomic, immunologic, endocrine, infectious and psychosocial factors interplay.

Pathophysiology

The airway inflammation results from different chemical mediators.

Mediators are released from local mast cells through:

- i. IgE mediated specific allergens
- ii. Non specific stimulation

Mediators and their effects

Early phase (10 to 30 minutes)

- preformed mast cells products (histamine, leukotrienes, etc)
- these mediators produce bronchoconstriction, excessive mucus production and mucosal edema

Late phase (after 6 hours)

- Local mucosal infiltration with eosinophils, lymphocytes and neutrophils.
- There will be synthesis of mediators from prostaglandins
- Mediators result in worsening of the obstruction and hyperresponsiveness

The airway obstruction and hypoxia is more pronounced in children for the following reasons:

- Small size of airways
- Abundant mucous glands
- Soft and easily deformable rib cage
- Poor collateral ventilation of alveoli

Diagnosis

Before initiation of any form of treatment the following information is needed:

- a) Establishing the diagnosis
- b) Identify the triggering factors
- c) Assess degree of severity

Establishing the diagnosis

Classic symptoms of asthma are recurrent acute or insidious onset of cough and wheezing typically worse at night and triggered by upper respiratory infections. Patients may show spontaneous improvement or rapid response to therapy. Patients may have family history of asthma.

Findings on physical examination depend on severity and presence of acute attack during examination. Patients will have varying degree of respiratory distress and wheeze.

Few patients may have atypical presentation such as:

- Persistent cough with no wheezes
- Wheezing with no cough
- Isolated tachypnoea with no wheezes

Assess degree of severity

Establishing the severity of asthma by grading will help in selecting treatment plans and have objective parameters during followup. There is little experience and evidence based practice with regard to this in Ethiopia. One method of grading is shown in table 22.

Table 22: Evaluation of severity of an asthma attack

(From the international Consensus)

Sign	Mild	Moderate	Severe	Imminent arrest
Breathlessness	On walking, can lie down	On talking, prefers to sit up	On lying down	
Speaking	Phrases	Parts of phrases	Words	Cannot speak
Level of consciousness	May be agitated	Usually agitated	Always Agitated	Sleepy or Confused
Breathing rate	Increased	Increased	Increased	
Accessory muscle use	No	usually	usually	Paradoxical movement
Wheezing	Moderate End of expiration	strong	Very strong	absent
Pulse rate	Increased	Increased	Increased	Bradycardia

The presence of several parameters in the table, but not necessarily all, indicates the general classification of the exacerbation.

Identifying triggering factors

The list of triggering factors is long but some common cause is:

- Viral upper respiratory infection mainly due to RSV and para influenza viruses.
- Tobacco smoke exposure
- Change in climate
- Exercise
- Drugs (e.g. ASA)
- Chemicals (e.g. malathion insecticides)

Differential Diagnosis

It is very common to get a child with respiratory complaint and wheeze. Many conditions can have overlapping symptoms with asthma, but frequently seen are:

- Infectious – pneumonia, bronchiolitis
- Congestive heart failure
- Foreign body aspiration
- Tuberculosis

Diagnosis

Diagnosis of asthma is mainly clinical with limited information obtained from investigation.

Microscopic examination of sputum shows eosinophilia. Chest X-ray is not routinely done unless there is suspicion of associated pneumonia or complication like atelectasis.

Therapeutic trials are occurring with bronchodilators.

Management

The management is divided in two phases with different objectives:

- Management of acute exacerbations or attacks.
- Daily or long term management

Management of acute asthma (attack)

Attacks are intermittent spells of severe symptoms of asthma caused by airflow obstruction.

Objectives of treatment are to:

- prevent death from severe airway obstruction
- restore best level of activity and optimal lung function
- prevent relapse by treatment of inflammation
- Principles of treatment
- First determine severity of attack (see table below)
- Give best treatment available

- close follow –up of the response to the emergency drugs

Mild to moderate attack of asthma

Epinephrine 0.01 ml per kg sc of the 1:1000 (1mg/ml) ampoule, given at intervals of 20 minutes for 2-3 times relieves the bronchospasm rapidly in the majority of the cases. The disadvantage is its short duration of action and its effect on the heart (tachycardia etc).

If the patient shows complete response, long acting beta adrenergic is given. Salbutamole is recommended in a dose of 0.1mg/kg divided in to three doses. Prescribing low dose of prednisolone for 5-7 days helps for rapid resolution of inflammation and reduce relapse.

Admission criteria

Admit a child if there is:

- no response or incomplete response to adrenaline
- status asthmatics (progressive respiratory distress despite administration of the usually effective drugs (sympathomimetics)
- severe attack on presentation
- Risk of death from asthma is high in those with:
 - Recent emergency admission
 - Use of steroids
 - Steroid dependent asthma
 - Sudden onset of severe obstruction
 - Unreliable family for follow up

Principles of treatment consists of:

- 1) Oxygen through nasal prongs
- 2) Aminophylline 4mg/kg i.v slowly over 20 minute every 6hours
- 3) Hydrocortisone 4-6mg/kg i.v every 6 hourly for 48-72 hours
- 4) Start salbutamole when patient starts to take po properly
- 5) Treat and prevent dehydration which may result from respiratory loss

Long term management

Objectives of treatment include:

- To reduce or avoid acute attacks
- To reduce limitation of activity
- Minimal or no use of bronchodilators

Principles of treatment (depends on the degree of severity)

- avoid allergens
- use bronchodilators according to the need
- give anti-inflammatory therapy in selected cases

Essential drugs in Ethiopia are:

- Salbutamol 0.1mg /kg /day three times daily.
- Prednisolone 2mg/kg/24 hours

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CHAPTER SIX

DISEASES OF THE CARDIOVASCULAR SYSTEM

6.1 Approach to an infant or child with congenital heart disease

Habtamu Fekadu

Objectives

By the end of this topic, the reader should be able to:

- Describe the magnitude and type of congenital heart disease
- Recite the commonest types of congenital heart disease
- Describe the common manifestations
- Know how to approach systematically a child suspected of having congenital heart disease
- Describe the principle of management

General Considerations

Congenital heart disease (CHD) occurs in approximately 8 of 1,000 live births. In 10-15% of cases there are associated congenital abnormalities.

They are classified as:

- 1 Acyanotic - there is no cyanosis
e.g. ventricular septal defect, persistent ductus arteriosus and coarctation of aorta
2. Cyanotic- which present with cyanosis
e.g. tetralogy of fallot and transposition of great arteries,

Ventricular septal defect is the commonest of all CHD and tetralogy of fallot is the most common cyanotic CHD.

The causes of most congenital heart disease is unknown. For most, it is multifactorial, whereby genetic predisposition, chromosomal abnormalities and environmental or maternal conditions play a role.

The mode of inheritance is polygenic.

Genetic predisposition – the risk of recurrence is 2-6% (the general risk is 0.8%) if a parent or a sibling is affected.

Chromosomal abnormalities – 50% of children with down's syndrome have CHD.

Environmental or maternal conditions- maternal diabetes, rubella drugs (lithium, ethanol)

Clinical Presentation

The manifestations and age of onset varies depending on the type of heart defects. Most are asymptomatic at birth and the majority start to be symptomatic in the first 1 or 2 years of life. Newborns present either with shock, which usually occurs within the first two weeks of life, or respiratory distress or cyanosis or the combination of these symptoms and signs.

Infants and children present with symptoms of congestive heart failure, such as tachypnea, tachycardia, chest retractions, nasal flaring and wheezing. The heart failure is usually present during the first weeks of life (2 to 4 months) as the pulmonary vascular resistance falls after birth or it may be triggered later by an intercurrent infection.

Other symptoms are feeding difficulty (become dyspneic or sweating excessively while sucking), irritability, failure to thrive and recurrent pneumonia. Those with cyanotic CHD present with cyanosis, cyanotic spells, squatting and clubbing with long standing cyanotic CHD. Older children will have exercise intolerance.

Evaluation of an Infant or Child with Congenital Heart Disease

If an infant or child develops the above symptoms and signs, the following diagnostic approach for the diagnosis of CHD helps to narrow the differential diagnosis. Perform a full cardiovascular examination, in particular evaluate for the femoral pulse (for coarctation of the aorta), character of the heart sounds, the type of murmur, cyanosis, oxygen saturation, chest x-ray (to see whether the pulmonary blood flow is increased, normal or decreased and the cardiac size and shape) and electrocardiography.

Once CHD is suspected:

Step one – Determine whether it is cyanotic or acyanotic by looking at the tongue and if possible oxygen saturation using pulse oximetry.

Step two - Identify clinical findings such as respiratory distress, shock, pulse (normal, decreased in legs or decreased in all limbs), second heart sound (single or splitted) or murmur (harsh loud systolic or continuous and the location of the murmur).

- Do chest x-ray to assess cardiac size and shape and whether the pulmonary vascular markings are increased, normal or decreased.

Step three – Based on the above findings, try to identify the most likely diagnosis and differential diagnosis. The final diagnosis is made by echocardiography and cardiac catheterization.

Step four – Start management. These patients are best managed at hospital level so refer the patient as early as possible.).

Therapeutic measures

- Position the patient with the head and trunk elevated if the child is in distress or breathless.
- Give oxygen if the patient is hypoxic, cyanotic or in distress.
- If the child is in heart failure, start digoxin and diuretic (furosemide).
- Refer the patient as early as possible for better diagnosis and management.
- Bed rest, and salt restriction

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6.2 Rheumatic fever

Habtamu Fekadu

Objectives

By the end of this topic, the reader should be able to:

- Know the magnitude and risk factors for rheumatic fever.
- Describe the pathogenesis of rheumatic fever.
- Describe the major mode of presentation and consequences.
- Know the principle of treatment and prevention of rheumatic fever.

Definition

Rheumatic fever is a non suppurative acute inflammatory complication of group A streptococcal pharyngeal infections, characterized mainly by carditis, arthritis and chorea appearing alone or in combination, with residual chronic heart disease as possible sequel of the carditis.

Epidemiology

In the early 1960s, acute rheumatic fever and its major complication, rheumatic valvular heart disease, were frequent worldwide. Nowadays it almost disappeared in the western world, however it is still prevalent in developing countries.

Climate (high temperature), poverty, crowding, poor housing conditions and inadequate health services are the major risk factors for the frequent occurrence of acute rheumatic fever in developing countries (Shaper 1972). It is frequent in children aged between 5 and 15 years because they are susceptible to Group A streptococcal pharyngitis.

Approximately 20% of all sore throats are due to Group A streptococcal and only few (0.5 to 3%) develop acute rheumatic fever. The risk depends on the host factors (age, genetic predisposition and carriers), if untreated, and the streptococcal strain (M type).

Streptococcal skin infection does not result in acute rheumatic fever.

Pathogenesis

The exact pathogenesis remains unexplained but the most accepted is an abnormal immune response to some still unidentified component of Group A streptococci. The

antibodies produced against the streptococci antigen cross react and cause immunologic damage to the heart valves, muscle and pericardium (pancarditis), joint synovium (arthritis), nervous tissue (sydneham chorea), skin (erythema marginatum) and subcutaneous tissue (subcutaneous nodule).

The latent period, i.e. the duration between sore throat and acute rheumatic fever, is usually 1 to 3 weeks.

Clinical presentation

Symptoms develop after 1 to 5 weeks following streptococcal pharyngitis.

There is no single specific manifestation rather carditis, arthritis, sydneham chorea, erythema marginatum and subcutaneous nodule appear alone or in various combination. The onset is insidious.

1. Carditis – occurs in 40 to 80% of patients

- It presents with fever, dyspnea and cough and the main signs are tachycardia, mitral regurgitation murmur and cardiac enlargement. In severe cases there are signs of heart failure
- Aortic regurgitation may occur with the mitral regurgitation
- Other signs are arrhythmia and pericarditis

2. Arthritis – is the commonest manifestation

- It is migratory and involves several larger joints (knee, ankle, elbow and wrist)
- the affected joint is swollen, red, warm and exquisitely tender. These signs disappear within 24 hours if antiinflammatory drug is given.
- It may resolve completely without any residual sequel.

3. Sydneham chorea – is purposeless jerky, repetitive movements particularly involving the hands and face.

- Early manifestations are emotional liability (easily induced crying and inattention) and deterioration in handwriting.
- Occurs at the same time or following other features. It usually occurs several months after the sore throat.

4. Erythema marginatum – is a serpiginous rash over the chest and trunk

- It disappears in hours or days.

5. Subcutaneous nodule – is small painless nodules which occur more frequently on the extensor surface of large joints (elbow, wrist and knee) and spine.

Other minor manifestations are fever and arthralgia.

Laboratory Findings

Laboratory findings may include:

- Increased acute phase reactants – elevated ESR and C-reactive protein
- CXR – may show cardiomegaly
- Evidence of preceding streptococcal infection
- elevated antistreptolysin O antibody (ASO) titer
- throat culture
- Other tests - ECG, Echocardiography

Diagnosis

No single clinical feature or test is diagnostic of acute rheumatic fever. So the modified Jones criteria which consist of 5 major and 4 minor plus evidence of preceding streptococcal infection is used for diagnosis.

Table 23 - Modified Jones criteria

Major criteria	Minor criteria
- Carditis	- Fever
- migratory polyarthritits	- Arthralgia
- Erythema marginatum	- ↑ESR or C-reactive protein
- chorea	- Prolonged PR interval
- subcutaneous nodule	

Plus

Evidence of preceding streptococcal infection
(↑antistreptolysin O antibody(ASO) titer, throat culture)

To diagnose acute rheumatic fever the following should be present:

- - 2 major or 1 major and 1 minor criterion, plus evidence of preceding streptococcal infection
- Chorea, plus evidence of preceding streptococcal infection

Clinical course

- Except for carditis, all manifestations subside without residual effect.
- The long term outcome depends on the severity of carditis. Patients with severe carditis are usually left with residual chronic rheumatic heart disease which is worsened by rheumatic recurrence.

Treatment

Treatment involves:

- 1) Bed rest – for patients with heart failure.
- 2) Treatment with of Group A streptococcal pharyngitis.
 - 10 days course of amoxicillin or a single i.m. injection of benzathine penicillin.
- 3) Anti-inflammatory drugs – to control the clinical features
 - Arthritis – aspirin 100 mg/kg/day in 4-6 divided doses
 - Carditis without heart failure - aspirin 100 mg/kg/day in 4-6 divided doses for 4 weeks.
 - Carditis with heart failure
 - Prednisolone 2 mg/kg/day for 2 to 4 weeks depending on patient response and aspirin during tapering of prednisolone and continued for 3 to 4 weeks.
 - Digoxin and diuretic for the heart failure.
 - The best parameters to follow response are symptoms of heart failure, heart rate, ESR and ,C-reactive protein
- 4) Chorea –most recover without treatment
 - Chlorpromazine and phenobarbital are helpful.

Prevention

Prevention involves the following:

1. Primary prevention: Early (within 1 week) treatment of streptococcal pharyngitis to prevent an initial attack of acute rheumatic fever.

- 10 days course of amoxicillin or single i.m. injection of Benzathine penicillin.

2. Secondary prevention:

- Prevent recurrences so as to reduce the damage to heart valves and the risk of chronic rheumatic heart disease.
- Indicated for children who have had acute rheumatic fever or have rheumatic heart disease.
- Monthly single i.m. benzathine penicillin or daily oral penicillin

References

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6.3 INFECTIVE ENDOCARDITIS

Tsinuel Girma

Definition

Infective endocarditis is an infectious disease, mostly bacterial, involving the endocardium (including the valves) or intima.

Etiology

The common causes include :

Streptococcus viridans

Staphylococcus aureus

Enterococcus

- Culture negative (10% of cases)

Pathophysiology

Infective endocarditis most often occurs as a complication of underlying congenital or rheumatic heart disease but can occur with no cardiac malformation. Rheumatic

valvular heart diseases, primarily mitral and aortic, represent common lesions in Ethiopia.

Preceding dental, urinary or intestinal procedures are identified in one third of the patients as a source of contamination of the blood and as a result the endocardium.

Vegetation is found at the site of endocardial or intimal erosion that results from high velocity blood flow. Thus children with cardiac defects like VSD, PDA and left-sided valvular lesions are at high risk of infective endocarditis.

Clinical manifestation

The onset of symptoms can be acute and severe, with high intermittent fever and prostration or insidious with isolated prolonged fever. Commonly, it varies within a range between the two forms.

Early symptoms are nonspecific and consist of low grade fever, chills, fatigue, arthralgia, headache, splenomegally, clubbing and petechial rashes. Many of the basic dermatologic manifestations develop in untreated patients. These are Osler nodes (tender pea sized Intradermal nodules in the pads of the fingers), Janeway lesions (painless small erythematous or hemorrhagic lesions on palms and soles) and bleeding beneath the nails (splinter hemorrhages).

There should be a high index of suspicion in the evaluation of a febrile child with underlying heart disease.

Diagnosis

The gold standard for diagnosis is appropriately collected blood culture. Take 3 to 5 separate samples even during afebrile periods. In patients with no prior antibiotic treatment, culture identifies etiologic agents in 90% of cases. Additional information can be obtained from echocardiography, raised ESR, leukocytosis, mild normocytic anemia, and microscopic hematuria (may have RBC casts) or a manifestation of immune complex glomerulonephritis.

Differential diagnosis

The following diseases have to be considered:

- Recurrence of rheumatic fever
- Malaria

- Tuberculosis

Treatment

Any patient with diagnostic possibility of infective endocarditis should be treated in a hospital.

Medical management should provide both specific treatment of the infection and supportive treatment for congestive heart failure.

Initial empiric antibiotic should be effective for both Gram-positive and Gram negative organism combine cloxacillin (100 mg /kg per 24 hours), ampicillin (300 mg/kg per 24 hr, IV) and gentamicine (3 to 4 mg / kg per 24 hours IV). Adjustment of drugs can be done depending on blood culture results. Therapy should continue for 4 to 6 weeks (except gentamicin for 2 weeks).

Prevention

The prevention of infective endocarditis involves appropriate treatment of infection in high risk children, proper general dental care and oral hygiene and antimicrobial prophylaxis. Oral amoxicillin can be given 1 hour before dental procedures and surgery of the upper respiratory tract. Intramuscular or IV ampicillin plus gentamicin is recommended before surgery of the genitourinary and gastrointestinal system.

CHAPTER SEVEN

DISEASE OF THE GASTROINTESTINAL SYSTEM

Dula Benti and Yezna

Objectives

By the end of the chapter, the student is expected to:

- Describe introductory concepts about diarrheal disease and portal hypertension
- Explain the epidemiology of the above diseases
- Identify the etiologic factors of these diseases
- Describe the pathology and pathogenesis of the mentioned diseases
- Explain the diagnostic modalities and clinical features of these diseases
- Describe the treatment, complications, prevention and prognosis of the above-mentioned diseases.

Introduction

Digestive system disorder comprises a range of both acquired and congenital disorders. Digestive system disorders are manifested by vomiting, dysphagia, diarrhea, anorexia, constipation, abdominal pain, abdominal distension or mass, jaundice, GI hemorrhage, etc. Many of these symptoms are common to disorders of several parts of the gastrointestinal tract as well as extra intestinal diseases. For this reason, disorders of organs outside the gastrointestinal tract should be considered in the differential diagnosis. In this short review, only few diseases of the digestive system are discussed.

7.1 Diarrheal disease

Definition

Diarrhea is defined as passage of three or more loose or watery stools per 24 hours or increase in stool frequency or liquidity that is considered abnormal by the mother.

There are three clinical types of diarrhea

1. *Acute watery* - A diarrheal episode that begins acutely and lasts for less than fourteen days with passage of watery stools without blood.
2. *Dysentery* – This is diarrhea with blood in the stool.
3. *Persistent diarrhea* - This is a diarrheal episode that starts acutely and lasts for fourteen days or longer.

Acute Diarrheal Disease

Epidemiology

An estimated 500 million children in the developing world suffer from diarrhea three or four times a year. One out of every twenty children born into the developing world dies before reaching the age of five from dehydration resulting from diarrhea. Diarrheal disease cause about 30% of infant deaths in developing countries and contribute a lot to the high prevalence of malnutrition. Dehydration from diarrhea is one of the five leading causes of death in children under five years.

Acute watery diarrheal episodes are the commonest form, accounting for 80 to 82% of the cases. Children of 6 to 11 months of age are the most commonly affected, a time that coincides with initiation of complementary feeding. Highest mortality of diarrheal disease is encountered among infants.

Risk factors for diarrhea include the following:

Behavioral:

- Improper infant and young child feeding (suboptimal breast feeding, using feeding bottles, eating food hours after cooking it etc.)
- Drinking contaminated water
- Not washing hands
- Not disposing of feces safely

Host factors:

- Malnutrition - malnourished children have frequent episodes of diarrhea and each episode is more severe and long lasting
- Measles – recent measles predisposes to diarrhea
- Immunosuppression – like HIV/AIDS

Etiology

The commonest cause of acute watery diarrhea is *rota virus* and others are *escherichia coli*, *vibrio cholerae*, *giardia lamblia*, *cryptosporidium*, etc

The commonest cause of dysentery is *shigella* and others are *campylobacter jejuni*, *entero invasive E.coli*, *entamoeba histolytica* etc.

The transmission is usually through the fecal-oral route-by ingestion of fecally contaminated water or food or direct contact with infected feces.

Pathophysiology

Microbial agents cause diarrhea by any one of the following mechanisms:

- Villous damage and epithelial cell destruction e.g *rota virus*
- Mucosal adhesion e.g *giardia lamblia*
- Mucosal invasion e.g *shigella*, *entamoeba histolytica*
- Release of toxins e.g *vibrio cholerae*, *ETEC*

Two mechanisms of watery diarrhea have been identified.

1. Secretory diarrhea

Secretory diarrhea occurs when there is increased secretion of fluid and electrolytes (sodium, chloride) due to the microbial toxins in the small intestine beyond the absorptive capacity of the bowel. This results in a net fluid and electrolytes, e.g *vibrio cholerae*, *enterotoxigenic E.coli (ETEC)*

2. Osmotic diarrhea

If there is intake of non-absorbable solute which is isotonic, there will be diarrhea with no dehydration. If there is intake of non-absorbable solute which is hypertonic, water and salts are lost from the extra cellular fluid which results in more diarrhea with hypernatremic dehydration.

Complications

The commonest complications that are expected in patients with acute watery diarrhea are:

- Dehydration and hypovolemic shock
- Electrolyte imbalance, most importantly hypokalemia, hyponatremia

- Base deficit acidosis
- Hypoglycemia
- Malnutrition

Dehydration is the main cause of death from acute diarrhea. It is associated with loss of body fluid and electrolytes mainly sodium through the diarrheal stools.

Table 24: Assessment and classification of dehydration

Clinical signs	Degree of dehydration
Two of the following signs:	
• Lethargy or unconsciousness • Sunken eyes • Not able to drink or drinking poorly • Skin pinch goes back very slowly (takes >2 sec)	Severe Dehydration
Two of the following signs:	
• Restless, Irritable • Sunken eyes • Drinks eagerly, thirsty • Skin pinch goes back y slowly (takes <2 sec)	Some Dehydration
No enough signs to classify as some or Severe dehydration	No dehydration

Other important signs of dehydration or hypovolemia are rapid and feeble /absent pulse, low blood pressure, cool and moist extremities, depressed fontanel, decreased urine output.

<u>Degree of Dehydration</u>	<u>The fluid deficit</u>
No dehydration	0 to 4% of total body weight
Some dehydration	5-10% of total body weight
Severe dehydration	10% of total body weight

Electrolyte and acid-base abnormalities include:

Hypokalemia due to loss of K^+ in diarrheic stools.

Hyponatremia due to loss of K^+ in diarrheic stools..

Acidosis due to loss of bicarbonate in diarrheic stools.

Diarrhea results in malnutrition due to reduced intake and absorption, and increased requirement associated with diarrhea.

Investigation

Because most acute watery diarrhea is caused by viruses, there is no need to do stool examination. Do stool microscopy only for dysentery.

Management of acute diarrhea

The principles of treatment focus on:

- Giving extra fluids to replace the lost fluid and electrolytes. The amount and how to give depends on the degree of dehydration.
- Continuous feeding
- Antibiotics/ Anti-protozoal – see the indications below
- Advise the mother when to return

Composition of ORS solution

It more or less goes in conjunction with the diarrheal losses:

<u>Composition</u>	<u>mmol/l</u>	<u>Essential features</u>
Na^+	90	replaces lost Na^+
K^+	20	prevents hypokalemia
Citrate/ HCO_3	10/30	correct acidosis
Glucose	111	facilitates Na absorption
Osmolality	310	Isotonia

Treat dehydration as follows:

No dehydration	Treatment plan A
Some dehydration	Treatment plan B
Severe dehydration	Treatment plan C

See Annex at the end of these lecture notes for treatment plans A, B and C

Antibiotics in acute diarrhea

Antibiotics are given only in patients with bloody diarrhea and suspected cholera with severe dehydration. They have no practical value for others because they are ineffective for the pathogens rota virus, and most of the time the etiologic pathogen is not known. Otherwise determination of the specific etiologic agents in the laboratory is impractical.

Antibacterial or antiparasitic therapy

Anti microbial agents should not be used routinely. The indications are:

1. Dysentery - oral antibiotics for 5 days should be given.
2. The first line drugs is cotrimoxazole.
3. The second line drug is nalidixic acid
4. If cholera is suspected,
 - Cotrimoxazole or erythromycin are the first line drug in younger children. Tetracycline is the drug of choice for children older than 8 years.
5. Anti-protozoals are considered if trophozoites or cyst of *giardia*, or trophozoite of *E. histolytica* are seen in stool microscopy.

Persistent Diarrhea

Definition

Persistent diarrhea is a diarrheal episode that starts acutely and lasts for fourteen days or longer.

Etiology

About 10% of acute diarrheal episodes become persistent. There is no single microbial cause although *shigella*, *salmonella* and *enteroaggregative escherchia coli*, *giardia lamblia* and *cryptosporidium* (in immunocompromized children) play a role.

Persistent diarrhea can lead to death through negative effects on nutrition status. Several studies have shown that children with persistent diarrhea are more likely to die than children with diarrhea of shorter duration. A strong relationship exists among persistent diarrhea, malnutrition and mortality

Risk factors include:

- Malnutrition
- Recent introduction of animal milk or formula
- Young age less than 18 months of age
- Immunological impairment eg. AIDS
- Recent diarrhea

Nutritional Impact

Persistent diarrhea is a nutritional disease. It usually occurs in malnourished children and is itself an important cause of weight loss and severe malnutrition. Such effects are seen due to decreased absorption of nutrients especially fat, lactose and poor intake and also giving of dilute, low energy foods in patients with persistent diarrhea.

Management

First check the level of dehydration and treat accordingly. Usually, treatment with ORS would suffice in children with persistent diarrhea. Only few patients need intravenous rehydration. Then check for blood in the stool. If so, treat for shigellosis. Do stool microscopy and if trophozoite of giardia or entamoeba are found, treat accordingly. If there is still no response, stool culture and sensitivity should be done to look for other bacterial causes of bloody diarrhea.

Nutritional therapy

The most important aspect of treatment for persistent diarrhea is proper feeding.

Goals of treatment include:

- To reduce the amount of animal milk (or lactose) in the diet temporarily

- To provide a sufficient intake of energy, protein, vitamins and minerals to improve nutritional status
- To avoid giving foods or drinks that may aggravate the diarrhea as animal milk or lactose
- To ensure that the child's food intake is adequate to correct malnutrition

For children older than six months the goals are as follows:

- Continue breast feeding
- Give only half of the usual amount of animal milk or give it as a fermented form, yoghurt (for the lactose intolerance)
- Ensure a diet of full energy food. About six times per day, give thick cereal with vegetable oil, vegetable, chicken or fish. Avoid diluted foods. Supplement the diet with folate, vitamin B₁₂, Zinc and Iron.
- If diarrhea stops after five days, give the same amount of food for one more week. Re-introduce the usual animal milk over days to full-strength diet.
- Give an extra meal each day for at least one month

Prevention of diarrhea

The presentation of diarrhea involves:

Interrupting the spread of infection

- Give only breast milk for the first 6 months
- Do not use feeding bottles
- Prepare and store food safely
- Use clean water for drinking
- Wash hands when soiled
- Dispose of feces safely

Strengthening host defenses

- Exclusive breast feeding till 6 months and continue breast
- feed at least for two years
- Start complementary feeds at age of 6 months
- Give balanced diet in adequate amount

- Immunize against measles

7.2 Portal hypertension

Definition

An increase in portal venous pressure to greater than 10 to 12 mm Hg.

Causes of Portal Hypertension

Portal hypertension results from the obstruction of portal blood flow anywhere along the course of portal venous system. It is due to prehepatic, intrahepatic or posthepatic obstruction to the flow of portal blood. In children most important causes are extrahepatic.

Extra hepatic portal hypertension may be caused by:

- Umbilical infection in neonates
- Intraabdominal infection in older children
 - Acute appendicitis
 - Primary peritonitis
- Umbilical cathetrization during the neonatal period

Intra hepatic portal hypertension may be caused by:

- Hepatocellular disease
 - Acute and chronic viral hepatitis
 - Congenital hepatic fibrosis
 - Hepatotoxicity
- Cirrhosis
- Schistosomiasis
- Biliary tract disease
- Extra hepatic biliary atresia

Post sinusoidal obstruction may be caused by:

- Budd-chiari syndrome

Idiopathic portal hypertension is another form.

Pathophysiology

The primary hemodynamic abnormality is increased resistance to portal blood flow, which results in the development of collateral for porto systemic shunts. Complications occur due to the development of a remarkable collateral circulation. Collaterals in the esophagus, stomach and anorectal region are prone to rupture and bleed under increased pressure.

Clinical features

Bleeding from esophageal varices (hematemesis and melena), with bouts of life threatening hemorrhage is the commonest form of presentation.

Jaundice, stigmata of cirrhosis, ascitis, encephalopathy, caput medusa (dilated veins around the umbilicus), splenomegaly (with hypersplenism) may occur.

If no clinical or biochemical features of liver disease are seen and liver size is normal, portal vein obstruction is the most likely cause. If enlarged, hard liver with normal hepatic function is seen, congenital hepatic fibrosis is more likely.

Diagnosis

Diagnosis may involve:

Ultrasound with doppler

- To delineate the size and architecture of the liver
- To check for patency of the portal vein
- To show the direction of flow within the portal system.

Endoscopy

- Most reliable for detecting esophageal varices.
- Stool examination for detection of ova of parasites.

Treatment

Treatment comprises two components:

- Emergency treatment of life threatening hemorrhage
- Prophylaxis for initial or subsequent bleeding

Emergency treatment may include:

- Fluid resuscitation: crystalloids followed by packed RBC
- Vitamin K, infusion of platelets or fresh frozen plasma for correction of coagulopathy
- NG tube to document and monitor ongoing bleeding in the stomach.
- Cimetidine or ranitidine to decrease the risk of bleeding from gastric erosions.
- Pharmacological treatment to decrease portal pressure with asopressin which act by increasing splanchnic vascular tone thus decreasing portal blood flow or others
- Endoscopic sclerosis of esophageal varices.
- Sangstaken blakemore tube is inserted to stop bleeding by mechanically compressing esophageal and gastric varices.
- Surgical - portocaval shunt

Prognosis

Portal hypertension secondary to intrahepatic disease has a poor prognosis. In portal vein obstruction, episodes of bleeding become less frequent and less severe with age as a collateral circulation develops.

CHAPTER EIGHT

HEMATOLOGICAL DISEASES

Assfawossen Bekel

8.1 Anemia

Objectives

By the end of this course the students are expected to:

- To be able to define anemia
- To recite the common nutritional causes of anemia
- To know the classification of the anemias
- To describe the approach to the anemic child
- To mention the clinical manifestations of childhood anemia
- To describe the treatment of the nutritional anemias

Definition

Anemia is defined as reduction of the RBC volume or hemoglobin (Hb) concentration below the range of normal values in healthy persons (see table below)

General considerations

- Anemia is a relatively common problem. Finding the underlying cause is equally important since anemia is not a specific diagnostic entity.
- Although a reduction in the amount of Hb reduces the oxygen carrying capacity of the blood, few clinical disturbances occur until the Hb. level falls below 7-8 g/dl
- Physiologic adjustment to anemia include:-
 1. Increase in cardiac output
 2. Increase O_2 extraction by tissues
 3. Shunting of blood flow to vital organs and tissues
 4. Increase in RBC 2,3 DPG concentration

There are different classification methods but a useful classification of the anemia divides them based on the RBC morphology: microcytic, macrocytic or normocytic and based on colour normochromic or hypochromic.

Approach to the child with anemia

Even though anemia in childhood has many causes, the correct diagnosis can usually be established with relatively little laboratory cost. Frequently careful history and physical examination (Hx and P/E) suggest the cause.

History (Hx):

- Dietary habit, growth and development, symptoms of chronic illnesses, malabsorption, blood loss.
- Hx of jaundice or family Hx of anemia, gallbladder disease, splenomegally or splenectomy
- Age and ethnic background
- History of malarial attack
- History of bleeding disorder.

Physical examination (P/E):

- Anthropometry: Poor growth leading to chronic disease and/ or hypothyroidism
- Congenital anomalies leading to Fanconi's anemia, Diamond-Blackfan syndrome
- Look for pallor in the mucus membrane, nail beds, and palm.
- Petechiae and purpura
- Lymphadenopathy and hepatosplenomegally
- Evidence of chronic or recurrent infections

Laboratory studies

The initial laboratory evaluation of the anemic child generally consists of CBC with differential, platelet count, peripheral morphology, stool exam, blood film and reticulocyte count.

The following algorithm shows how this limited laboratory information together with undertaking a history and physical exam may lead to a specific diagnosis or focus additional investigations to a limited diagnostic category.

Table 25: Mean and ranges of normal hematologic values in infancy and childhood

Age	Hemoglobin (g/dl)		Hematocrit (%)		Reticulocte (%)	NCV (fl)	Leukocytes (WBC/mm ³)		Neutrophils (%)	Lymphocytes (%)	Eosinophils (%)
	Mean	Range	Mean	ange	Mean	Lowest	Mean	Range	Mean	Mean	Mean
Cord	16.8	13.7 -20.1	55	45- 65	5	110	18,000	(9-30x10 ³)	61 (40-80)	32	2
b/d	16.5	13.0 -20.0	50	42-66	1		12,000	(5-21x10 ³)	40	63	3
2week	12	9.5-14.5	36	31-41	1		12,000	6-18x10 ³)	30	48	2
3 mo's	12	10.5 -14.0	37	33-42	1	70.74	10,000	(6-15x10 ³)	45	48	2
6mo-6yr	13	11.0 -16.0	38	34-40	1	76-80	8,000	(4.5-13.5x10 ³)	55	38	2
7-12yr											
adult											
female	14	12-16	42	37-47	1.6	80			55 (35-70)	35	3
male	16	14-18	47	42-52		80	7,500	(5-10x10 ³)			

Fl = femtoliters; MCV = mean corpuscular volumes

WBC = white blood cells

Iron Deficiency Anemia

Introduction and epidemiology

- Iron deficiency anemia is the most common Hematologic disease of infancy and childhood.

The body of newborn infant contains about 0.5gm of Iron, whereas the adult content is 5gm. To make up for this discrepancy together with the normal loss of iron by shedding of cells, an average of 1mg of iron must be absorbed each day during the first 15 yrs of life. Absorption is only 10% and takes place in the duodenum. A diet containing 10 mg of iron/day is necessary for optimal nutrition.

Those who are at greatest risk include:

Children 6-24 months

Adolescents (especially females)

Special risk groups: low birth weight and premature

Etiology

Anemia results from a decreased iron store which may be caused by low birth weight and unusual perinatal hemorrhage. It results in increased requirements (children 6-24/12, adolescents).

It may be caused by blood loss (especially in older children).

Clinical manifestations Mild iron deficiency is usually asymptomatic

With moderate to severe anemia, findings include:

- Pallor, fatigue irritability, anorexia, pica
- Tachycardia, cardiac dilatation and systolic murmurs
- Adverse effects on behavior and cognition (decreases IQ)

Laboratory findings

In progressive iron deficiency, a sequence of biochemical and hematological events occur as follows:

- ↓Hemossiderin → ↓ferritin → ↓serum iron → ↑TIBC and

% Saturation

↓Hb, microcytosis, hypochromia ← ↓FEP ↓

Poikilocytosis and RDW

- The bone marrow is hypercellular with erythroid hyperplasia
- Hemossiderin can't be demonstrated in marrow specimens by Prussian blue staining.
- Reticulocyte count is usually normal but may be elevated

Differential Diagnosis

- Hypochromic, microcytic anemias
 - * Lead poisoning → coarse basophilic stippling of the RBCs
→ ↑Serum lead level
- Anemia of chronic illnesses

Treatment

- The regular response of iron deficiency anemia to adequate amount of iron is an important diagnostic and therapeutic feature. Oral administration of simple ferrous salts (sulfate, gluconate, fumarate) provides inexpensive and satisfactory therapy.
- Treatment may also include:
 - 6mg/kg/day of elemental iron in 3 divided doses for 2-3 months.
 - treatment of underlying causes
 - blood transfusion is indicated only when the anemia is severe (< 4 gm/dl) or when the patient has decompensated
 - Reticulocytosis begins within 48-72 hrs and peaks at 5-7 days.

Prevention

Prevention involves the following:

- Prevention and treatment of infestation (deworming) by intestinal parasites
- Supplemental iron for young children above 6 months and LBW babies
- Prevention of low birth weight and prematurity
- Prevention and treatment of causes of chronic blood loss

8.2 Megaloblastic Anemias

Megaloblastic anemia is a macrocytic anemia caused by deficiencies of vitamin B₁₂, folic acid (FA) or both.

- The RBCs at every stage of development are larger than normal and have an open, finely dispersed unclear chromatin and an asynchrony between maturation of the nucleus and cytoplasm.
- Both FA and vitamin B₁₂ are cofactors required in the synthesis of nucleoproteins

Etiology

Megaloblastic anemia is caused by

- Inadequate dietary intake (vitamin B₁₂, FA)
- Malabsorption (vitamin B₁₂, FA)
- Increased requirements (growth, hemolysis)
- Inborn errors of metabolism – vitamin- B₁₂

- Rarely, deficiency of intrinsic factor – Vitamin B₁₂

Clinical manifestations

Clinical manifestations include:

- the usual clinical features of anemia- pallor, fatigue, etc.
- Mild jaundice due to ineffective erythropoiesis
- Smooth and cherry red tongue
- Hemorrhage due to thrombocytopenia- in advanced cases
- Peripheral neuropathy, ataxia (esp. in B₁₂ def.)

Laboratory Findings

Laboratory findings include:

- Macrocytosis with aniso- and poikilo-cytosis on peripheral smear
- Large neutrophils with hyper segmented nuclei
- Neutropenia and thrombocytopenia- in advanced cases
- Bone marrow exam → Erythroid hyperplasia with erythroid and myeloid precursors.
- ↓In RBCs and serum levels of B₁₂ and FA level
- Schilling test (B₁₂)

Differential Diagnosis

Differential diagnosis is as follows:

- Hemolytic Anemias
- Bone marrow failure syndromes (e.g. Fanconi's anemia)
- Liver disease, Hypothyroidism, etc

Treatment

Treatment involves administering:

- Folic Acid- orally /parenterally: 1-5 mg/ 24hr, for 3-4 wks
- Vitamin B₁₂ – preferably parenterally

Questions

1. List the risk groups for iron deficiency anemia in children.
2. Discuss the approach to the anemic child
3. Discuss the treatment of iron deficiency anemia
4. Describe the laboratory finding of a child with megaloblastic anemia

References

1. Nelson text book of pediatrics, 16th edition, 2000
2. William W.Hay, et al, Current pediatric diagnosis and treatment, 15th edition, 2001
3. A.V.Hottbrand, et al, Essential hematology, 4th edition, 2001

8.2 Bleeding Disorders

Objectives

By the end of this course, the students are expected:

- To describe the haemostatic mechanisms
- To describe the approach to the bleeding child
- To recite the laboratory studies in a bleeding child
- To be able to list the important causes /mechanism of bleeding disorders
- To discuss about the primary abnormalities in hemophilia, clinical presentation and management.
- To describe the clinical presentations, and underlying problems in ITP.

General Considerations

When blood vessels are injured, the clotting process maintains vascular integrity or causes blood flow to cease through injured vessel.

If clotting is impaired hemorrhage occurs; if it is excessive thrombosis and its complication occur. Once clotting is initiated, anticoagulants must confine the clotting process to the site of injury only. The clot must then be physiologically lysed to re-establish blood vessel patency.

The classic haemostatic mechanism includes:

- The vascular response
- Platelet adhesion and aggregation
- Clot formation and stabilization
- Limitation of clotting to the site of injury by regulatory anticoagulants
- Re-establishment of vascular patency through fibrinolysis and vascular healing.

The vascular endothelium is the primary barrier against hemorrhage; when small vessels are transected it undergoes vasoconstriction to limit the local hemorrhage. Platelets are also essential for the control of hemorrhage from small blood vessels.

More extensive injury (involvement of large blood vessels) requires the participation and co-ordination of the full haemostatic process to provide a firm, stable, fibrin clot.

Generally bleeding disorders may occur as a result of:

- Qualitative or quantitative abnormalities of platelets
- Qualitative or quantitative abnormalities in plasma coagulation factors
- Vascular abnormalities

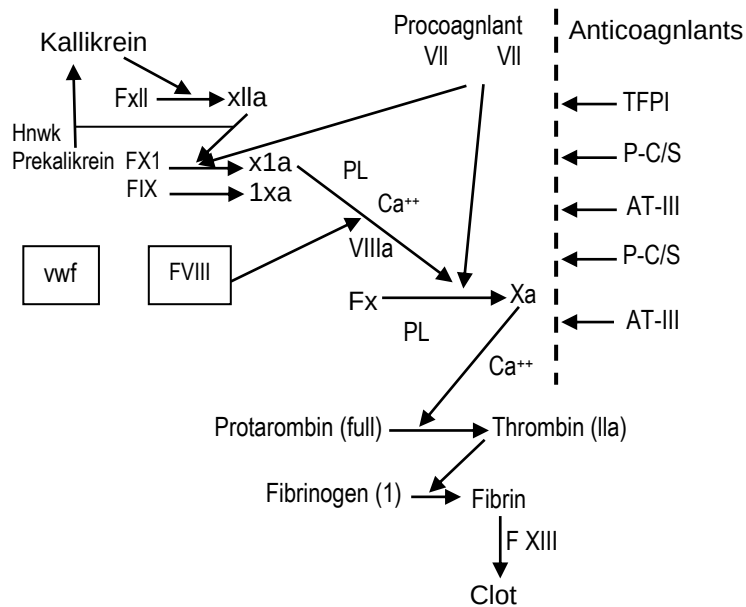


Figure 20: The clotting cascade with sequential activation and amplification of clot formation

Terms:

HMWK = high molecular weight Kininogen

PL= Phospholipid

Ca⁺⁺= calcium

TF= tissue factors

TFPI= tissue factor pathway inhibitor

P-C/S= protein C and S

AT- III= antithrombin III

Approach to the Bleeding Child

The clinical history (Hx) provides the most useful information:

History

- Site /sites of bleeding
- Severity and duration of hemorrhage
- The age of symptom onset
- Was the bleeding spontaneous or after trauma?

- Was there a previous personal or family history of similar problems? Family pedigree.
- Do symptoms correlate with the degree of injury/ trauma?
- History of drug intake.
- Are there lumps with bruising for which there is minimal trauma?
- Was there increased bleeding during surgery or dental procedures?
- History of delayed/slow wound healing
- Menstrual history in post pubertal females

Physical Examination

- Should focus on whether symptoms are primarily associated with the mucous membrane or skin (mucocutaneous bleeding) or the muscle or joints (deep bleeding)
- Look for petechiae, ecchymoses, hematomas, hemarthroses mucous membrane bleeding.
- Hepatosplenomegally
- Search for evidence of underlying disease

Laboratory tests

Laboratory tests may involve:

- Screening tests: platelet count, bleeding time (BT), PT, PTT
- If screening tests are normal, I do thrombin time,
- If screening tests are abnormal, factor assay depending on the clinical findings

In a patient with an abnormal bleeding history and a positive family history, normal screening tests should not preclude further laboratory evaluation

Bleeding time

- Assesses the function of platelets and blood vessels
- The normal range is between 4 and 8 minutes

Platelet count

- The normal platelet count is between 150 to 450 x 10⁶/L
- Thrombocytopenia < 150,000/mm³

- Patients with platelet count $> 50,000/m^3$ rarely have very significant clinical bleeding

PTT (Partial thromboplastin time)

- Measures the intrinsic pathway
- Normal ranges vary between laboratories

PT (Prothrombin time)

- Measures the extrinsic pathway
- In most laboratories, the normal of ranges between 10 and 13 sec

Thrombin time (TT)

- Measures the final step of the clotting cascade in which fibrinogen is converted to fibrin
- Normal TT varies between labs but is usually b/n 11 and 15 sec.

Idiopathic Thrombocytopenia Purpura (ITP)

Definition

Idiopathic thrombocytopenia purpura is a primary quantitative platelet disorder characterized by an autoimmune destruction and clearance from the circulation.

Acute Idiopathic thrombocytopenia purpura (ITP)

Etiology and pathogenesis

One to four weeks following exposure to common viral infections, a small number of children develop an antibody directed against the platelet surface following which antibody coated platelets are recognized by the fc receptors on the splenic macrophages, ingested and destroyed. A preceding history of viral illness is described in 5 to 65% of cases (e.g. EBV infection)

Clinical manifestations

The classic presentation of ITP is that of an apparently healthy 1 to 4 year old child who has a sudden onset of generalized petechiae and purpura. Often there is bleeding from the gums and mucous membrane. It may be the initial presentation of SLE, HIV

infection, rarely, lymphoma. A physical examination is normal except for petechiae and purpura. Seventy to eighty percent of cases have spontaneous resolution. Less than one percent develops intracranial hemorrhage. Ten to 20% develop chronic ITP.

Laboratory findings

Except for thrombocytopenia, other blood tests are normal.

A bone marrow examination, when done, reveals normal granulocytic and erythrocyte series with characteristically normal or increase no of megakaryocytes

Treatment

Most children don't require treatment. At the present time, there is no consensus regarding the management of acute ITP. Therapy doesn't appear to affect the natural history of the disease but appears to be capable of inducing a more rapid rise in platelet count.

Initial treatment options include: -

- Prednisone

Indications: - 1. Patients with significant bleeding

(I.e. hematuria, hematochezia, **epistaxis**)

2. Platelet count $<10,000/\text{mm}^3$

Dose: 1- 4 mg/kg/day for 2-3 weeks or until platelet count is $>10,000/\text{mm}^3$

- Intravenous immune globulin (IVIG)
- Intravenous Anti- D therapy

Platelet transfusion is not indicated in ITP unless life threatening hemorrhages occur.

Prognosis

70-80% will have spontaneous resolution within 6 months time. 10-20% will develop chronic ITP. Less than 1% develops intracranial hemorrhage.

Chronic ITP

Chronic ITP is characterized by:

- Persistent thrombocytopenia for more than 6 months following acute ITP. Features associated with the development of chronic ITP include: female sex, age of less than 1 year, and more insidious onset of ITP.
- Careful evaluation for associated problems should be done (e.g. SLE, HIV, etc)
- Splenectomy is one of the treatment options after careful consideration.

Questions

1. Discuss the difference in clinical presentation between ITP and hemophilia
2. Discuss the approach to the bleeding child

Hemophilia A (factor VIII deficiency) and Hemophilia B (factor IX deficiency)

General considerations

- Hemophilia A and hemophilia B are the most common and serious congenital coagulation factor deficiencies that result in severe inherited bleeding disorder.
- Hemophilia has been known since biblical times when Talmudic writings permitted the avoidance of circumcision when there was a repeated history of death from circumcision bleeding in male siblings.
- The symptoms of hemophilia A and B are virtually identical.

Genetics and classification

Hemophilia occurs in 1 out of 5000 males, with 85% having factor VIII deficiency and 10 to 15% having factor IX deficiency. The disease occurs predominantly in males and is inherited in an X-linked recessive manner.

Hemophilia is classified into three levels, based on level of clotting factors:

1. Mild - more than 5% (5u/dl) of factor VIII is present in the blood
2. Moderate - 1 to 5% (1-5u/dl)
3. Severe - less than 1 % (< 1u/dl)

Pathophysiology

- Factor VIII and IX participate in a complex required for activation of factor X.
- In hemophilia A/B, clot formation is delayed and not robust.
- The clot that is formed may be friable and re-bleeding occurs during physiological lyses of clots or with minimal trauma.

Clinical manifestations

Neither factor VIII nor factor IX crosses the placenta, thus bleeding symptoms may be present from birth or occur in the fetus but they usually go unnoticed in the newborn. It is only when the child begins to crawl and walk that mobility causes the initiation of the symptoms which include:

- Easy bruising, intramuscular hematomas and hemarthroses
- Persistence of bleeding from minor traumatic lacerations of the mouth
- The commonest presentation of hemophilia is hemarthroses, which may be induced by minor trauma but often occurs spontaneously.
- Uncontrolled bleeding during circumcision.
- Life threatening bleeding in the hemophilic patient is caused by bleeding in to vital structures (CNS and upper air way) or by exsanguinations (external, GI or iliopsoas hemorrhage)

Treatment

Treatment involves:

- Prevention of trauma
- Psychosocial support (families and child)
- Avoid Aspirin and other NSAIDs
- Replacement therapy (blood, plasma, cryoprecipitate)
- Desmopressin (DDAVP) for mild hemophilia A

Complications

Complications include:

- Chronic joint destruction
- Risk of transfusion- transmitted infections

References

1. Nelson text book of pediatrics, 16th edition, 2000
2. William W.Hay, et al, Current pediatric diagnosis and treatment, 15th edition, 2001
- 3 A.V.Hottbrand, et al, Essential hematology, 4th edition, 2001

8.3 Leukemia

Objectives

By the end of this course the student is expected:

- To describe the primary abnormality in leukemia
- To know the epidemiology of leukemia
- To discuss the common clinical manifestations and investigation
- To list the principles of management
- To enumerate the prognostic factors

Definition

The leukemias are a group of malignant diseases arising from the bone marrow in which genetic abnormality in a hematopoietic cell give rise to a clonal proliferation of cells of the lymphoid and myelogenous series resulting in disruption of normal marrow function and ultimately in marrow failure.

Epidemiology

Leukemia is the most common childhood malignancy accounting for more than one-third of pediatric malignancies.

- ALL– 77% ; AML- 11%; CML <5%
- The peak age for ALL is 4 year with 85% of the patients diagnosed between 2 and 10 years.

Etiology

The precise cause of leukemia remains unknown. Both genetic and environmental factors like drugs, ionizing radiations, viruses, etc, appear to be involved in the development of the disease.

Acute lymphoblastic leukemia (ALL)

Acute lymphoblastic leukemia (ALL) accounts for 77% of the leukemias

Classification

Classification is based on:

- Morphologic appearance of the blasts – L1, L2, L3
- Immunophenotype
 - B - Progenitor
 - T - Progenitor

Clinical manifestations

Clinical manifestations are related to decreased bone marrow production of red blood cells, white blood cells and platelets because of leukemic infiltration of the bone marrow.-Include:

- pallor, petechiae, purpura, fever, bone and joint pain
- Hepatosplenomegaly and lymphadenopathy
- Fever (mostly due to infection) is a common clinical presentation
- Rarely, signs suggestive of increased intracranial pressure secondary to leukemic infiltration of the meninges.

Laboratory findings and diagnosis

Laboratory findings and diagnosis involve the following:

- Single or pancytopenias (anemia, thrombocytopenia and neutropenia) occurs in 99% of the cases.
- Most patients with ALL present with a total WBC count of $<10,000/\text{mm}^3$.
- Immature lymphocytes (blasts) in the peripheral blood.
- The diagnosis of ALL is strongly suggested by peripheral blood findings suggestive of bone marrow failure and it is confirmed by bone marrow examination.
- Other important studies include chest x-ray (mediastinal mass), CSF- analysis, and renal function test and serum uric acid level.

Differential diagnosis

Differential diagnosis includes:

- Infectious mononucleosis

- Aplastic anemia, and myelofibrosis
- Malaria
- Disseminated tuberculosis
- HIV infection

Treatment

Treatment includes:

Supportive treatment:

- Transfusion with blood products
- Nutritional support
- Treatment of infections, when present

Drug therapy

For patients with average/ standard risk groups i.e. those with favorable prognosis (age between 1 and 10 years, initial WBC count more than 100,000/mm and L1 and L2 morphology)

- Induction remission: 4 to 6 weeks
(Vincristine, prednisone and L-asparaginase)
- CNS- prophylaxis and/or therapy: 4 to 6 weeks
(Intrathecal methotrexate)
- Maintenance treatment – for up to 2 to 3 years. (Mercaptopurine, methotrexate and intermittent dose of prednisone and vincristine)

Remission is defined by less than 5% of blasts in the bone marrow and return of neutrophils and platelet count to near normal levels after 4 to 5 weeks of treatment, is 98%.

The long term survival is more than 80% after 5 years.

Prognosis

Poor prognostic factors include:

- Age less than 1year and more than 10 years
- Initial WBC count of more than 100,000/mm³
- L3 morphology
- clinical CNS leukemia at diagnosis
- Mediastinal mass
- T- cell immunophenotype
- Certain chromosomal translocations (e.g. t(9,22), t (4;11)
- Relatively slower remission

Relapse

- 10% relapse after completion of treatment in the subsequent 4 years. Bone marrow is the commonest site of relapse; other common sites include the CNS and testes but any site can be affected with relapse.

Exercise

1. Describe the clinical presentation of a child with ALL.
2. Discuss the laboratory workup and diagnosis of ALL.

8.4 LYMPHOMA

Objectives

By the end of this course the student is expected:

- To describe the epidemiology and pathology of the lymphomas
- To be able to list the important clinical manifestations
- To recite the diagnostic laboratory work up
- To briefly mention the treatment modalities and prognosis

Definition and general consideration

The term lymphoma refers to malignant proliferation of lymphoid cells, usually in association with and arising from lymphoid tissues (i.e. lymph nodes, thymus and spleen). They are the third most common cancers in children.

There are two major categories of lymphoma: Hodgkin's and non- Hodgkin's lymphoma

Hodgkin's Disease (HD)

Epidemiology

The epidemiology of Hodgkin's disease is as follows:

- Arises in lymphoid tissues
- Accounts for 5% of cancers in children < 15 years
- Three epidemiological forms occur :
 - Childhood form (< 14 yrs old)
 - Young adult form (15-34 yrs old)
 - Older Adult form (55-74 yrs old)

There is a 4:1 male predominance in the first decade and the frequency is equal during adolescence when the majority of the cases occur.

Etiology

The exact cause of Hodgkin's disease remains unknown.

- Genetic predisposition, viral infections especially ebsteinbarrvirus, pre-existing immunodeficiency, etc are the associated risk factors in the development of the disease

Pathology

The Reed-Sternberg cell, a large (15 to 45 microm in diameter) cell with multiple or multilobulated nuclei is considered the hall mark of Hodgkin's disease.

Four histological types are present

- Lymphocyte predominant (10 to 20% of cases) it has the most favorable outcome.
- Nodular sclerosing (40 to 60% of cases)
- Mixed cellularity (20 to 40% of cases)
- Lymphocyte depleted (5to 10% of cases) it has least favorable outcome

Spread occurs via:– lymphatic system to adjacent lymph nodes

Hematogenous: - liver, spleen, bone,

Various degrees of immune impairment (especially in cell mediated immunity) are seen in Hodgkin's disease.

Clinical Manifestation

Clinical manifestations include the following:

- Painless, firm, cervical or supraclavicular adenopathy is the most common presenting sign.
- The growth rate of lymph nodes is variable and involved nodes may wax and wane for weeks or months
- Inguinal and axillary sites- uncommon but still possible
- Anterior mediastinal mass is often present (50%)
- Rarely, hepatosplenomegally
- Sometimes patients might present with airway obstruction, pleural/pericardial effusion, hepatic dysfunction or bone marrow infiltration (pancytopenia) and rarely with nephrotic syndrome
- Systemic symptoms include fever, drenching night sweat, weight loss (B-symptoms).

Diagnosis

- Diagnosis involves undertaking a history and physical exam to investigate:
 - Persistent, painless, unexplained, rubbery adenopathy unassociated with an obvious underlying inflammatory or infectious process.
 - Lymph node biopsy
 - Staging – 4 stages (I-IV) according to Ann Arbor staging system
 - Studies necessary for clinical staging of HD include CBC, ESR, LFT, chest and abdominal CT, CXR and/or BM-biopsy

Differential Diagnosis

Differential diagnoses include leukemia, tuberculosis and HIV infection

Treatment

The treatment is largely determined by disease stage, the patient's age at diagnosis, the presence or absence of B- symptoms and the presence of bulky nodal disease. It may involve chemotherapy and radiotherapy

Prognosis

Cure rates are 90% in the early stage and 60-70% in the advanced stage

Poor prognostic factors include advanced stages, B- symptoms, failure to achieve remission and early relapse.

Non- Hodgkin Lymphoma (NHL)

Epidemiology

In children, as opposed to adults, NHL is a high grade, rapidly growing and diffuse neoplasm.

It results from malignant clonal proliferation of lymphocytes of T- or B-, or indeterminate cell origin.

A special type of lymphoma, Burkitt's lymphoma, was first described in 1958 as "a sarcoma involving the jaws in African children". It is most frequent in equatorial Africa

where it accounts to 50% of childhood cancers. There is a strong epidemiologic association between Epstein Barr virus and Burkitt's lymphoma. Unlike HD, the incidence of NHL increases throughout life

Etiology, pathology and pathogenesis

EBV infection has a major role in the pathogenesis of the disease even though the actual mechanism remains unknown.

Pre- existing immunodeficiency (congenital/acquired) also predisposes to the development of the disease

Three histological subtypes exist:

- Lymphoblastic (usually of T- cell origin)
- Large cell (LCL, of T-, or B-, or indeterminate cell origin)
- Small non-cleaved cell lymphoma (SNCL, Burkitt's and non-Burkitt's subtypes, B-cell origin)

Clinical manifestations

Clinical manifestations are as follows:

- The presenting signs and symptoms vary with the disease site and extent which in turn correlate with histologic subtype
- Lymphadenopathy, mediastinal mass, pleural effusion, ascites hepatosplenomegally
- Cough, dyspnea, swelling of the face
- Abdominal pain, abdominal distension, vomiting, constipation
- Burkitt's lymphoma presents with large jaw and/or abdominal mass with /without CNS involvement

Laboratory findings

- Features of tumor lyses syndrome (increased serum uric acid, K and phosphorous)
- Increased LDH

Diagnosis

Is made after biopsy with histology, immunophenotyping and cytogenetic studies

Differential Diagnosis

- Leukemia
- Tuberculosis
- HIV infection

Treatment and prognosis

- Chemotherapy
- About 90% of patients with limited stage NHL are cured with current chemotherapeutic regimens.

Questions

1. Mention the difference in clinical presentation between childhood Hodgkin's and non-Hodgkin's lymphoma
2. Discuss the risk factors and clinical presentation of Burkitt's lymphoma.

References

1. Nelson text book of pediatrics, 16th edition, 2000
2. William W. Hay, et al, Current pediatric diagnosis and treatment, 15th edition, 2001
3. Philip Lanzkowsky, pediatric oncology, 1983
4. A.V. Hottbrand, et al, Essential hematology, 4th edition, 2001

CHAPTER NINE

DISEASES OF THE RENAL SYSTEM

Mekete and wogene

Objectives

At the end of this chapter students should:

- Understand the diagnosis and management of urinary tract infection and differentiate upper UTI from cystitis.
- Identify common causes of acute glomerulonephritis and understand its diagnosis and management.
- Understand the diagnosis and management of nephrotic syndrome
- Understand pathophysiology and common causes of ARF (Acute Renal Failure).
- Understand the diagnosis ARF and its complications.
- Understand principles of management of ARF and its complications.

9.1 Introduction

Anatomy

The kidneys lie in the retroperitoneal space slightly above the level of the umbilicus. The kidney has an outer layer, the cortex, which contains the glomeruli, proximal and distal tubules, and collecting ducts, and an inner layer, the medulla, which contains the straight portions of the tubules, the loops of Henle, the vasa recta and the terminal collecting ducts.

The blood supply to each kidney usually consists of a main renal artery that arises from the aorta; multiple renal arteries may occur. Each kidney consists of approximately 1 million nephrons (glomeruli and associated tubules). The glomerular network of specialized capillaries serves as the filtering mechanism of the kidney.

Glomerular filtration

As the blood passes through the glomerular capillaries, the plasma is filtered through the glomerular capillary walls. The ultra filtrate is cell free and it contains all the

substances in the plasma (electrolytes, glucose, urea, creatinin, phosphate and low molecular weight proteins) except proteins having a molecular weight of 68,000 or more such as albumin and globulins.

The glomerular filtration rate (GFR) may be estimated by measurement of serum creatinin level. Creatinin is a product of muscle metabolism. Its production is relatively constant, and its excretion is primarily through glomerular filtration, although tubular secretion may become important in renal insufficiency. In contrast to the concentration of blood urea nitrogen, which is affected by state of hydration and nitrogen balance, the serum creatinin level is primarily influenced by the level of glomerular function. However, the serum creatinin level does not rise above normal until the GFR falls by 30 to 40%.

9.2. Urinary Tract Infection (UTI) in Children

Definition

UTI refers to a group of conditions in which there is growth of bacteria within the urinary tract.

Etiology and epidemiology

Approximately 3.5% of girls and 1% of boys develop UTI. During the first year of life, the male: female ratio is 2.8-5.4: 1. This ratio becomes 1: 10 beyond the age of 1 to 2 years of age. In girls the first UTI usually occurs by the age of 5yr, peaks during infancy and toilet training. In boys most UTIs occur during the first year of life.

The etiologic agents in UTI are mainly colonic bacteria. E.coli, klebsiella, and proteus are among the commonest causes. E.coli causes 75-90% of UTIs in females.

Pathogenesis

Nearly all UTIs (except in neonates) are ascending infections, the origin being the colon. The bacteria arise from the fecal flora, colonize the perineum and enter the bladder via the urethra. In uncircumcised boys, the bacteria arise from the flora beneath

the prepuce. In some cases, the bacteria ascend to the kidney to cause pyelonephritis. In rare cases, renal infection may occur by hematogenous spread. UTI in neonates, however, is mostly hematogenous in origin.

The pathogenesis of UTI is based in part on the presence of bacterial pili and fimbriae on the bacterial surface. Host risk factors for developing UTI include female sex, uncircumcised male, toilet training, constipation, urethral instrumentation, anatomic abnormalities, neuropathic bladder, pinworm infestation and others.

Classification of UTI There are three forms of UTI: pyelonephritis, cystitis and asymptomatic bacteriuria.

Asymptomatic bacteriuria: This is a benign condition characterized by presence of positive urine culture without any manifestations of infection and occurs almost exclusively in females.

Cystitis: Cystitis indicates involvement of the urinary bladder. Cystitis does not result in renal injury.

Pyelonephritis: Refers to involvement of renal parenchyma and it may result in renal injury called pyelonephritic renal scarring. When there is no parenchymal involvement, the condition is termed pyelitis.

Clinical Manifestations

Newborns with UTI may present with fever, hypothermia, poor feeding, failure to thrive or sepsis. Infants may have malodorous urine, irritability and poor feeding.

Symptoms of cystitis in older children include dysuria, urgency, frequency, suprapubic pain, incontinence and malodorous urine. Isolated cystitis does not cause fever.

Pyelonephritis presents with any or all of the following: abdominal or flank pain, malaise, fever, nausea, vomiting, and occasionally diarrhea. Patients may appear toxic.

Diagnosis

Although culture of properly collected urine is mandatory for confirmation of UTI, it may be suspected based on patients' presentation and urinalysis.

Table 26: Urine Culture Interpretation in UTI

Method of collection	Quantitative culture:UTI present
Urine by Suprapubic Puncture	Any number of bacteria
Catheterization	Growth of urinary pathogens in any number(the exception is up to $2-3 \times 10^3$ CFU/ml of coagulase negative staphylococcus)
Midstream clean-void	Febrile infants or children usually have $\geq 50 \times 10^3$ CFU/ml of a single urinary pathogen, but infection may be present with counts from 10×10^3 CFU/ml.
Midstream clean-void	Symptomatic patients usually have $\geq 10^5$ CFU/ml of a single urinary tract pathogen.
Midstream clean-void	Asymptomatic patients: at least two specimens on different days with $\geq 10^5$ CFU/ml of the same organism.

*CFU = Colony Forming Units

Urinalysis

Urinalysis should be done from the same specimen and pyuria (>5 WBC/HPF) suggests infection, although infection can occur in the absence of pyuria. Pyuria can be present without UTI. Microscopic hematuria is common in acute cystitis. WBC casts in the urine suggest renal involvement.

Other Investigations

With acute renal infection, leukocytosis, neutrophilia, and elevated ESR and C-reactive protein are common.

Repeated UTI needs further investigation to exclude possible structural abnormalities.

Methods of urine collection include:

- For toilet trained children
 - Midstream clean catch sample
 - Catheterization
 - *Suprapubic puncture
- In infants
 - Catheterization
 - Application of adhesive, sealed, sterile collection bag
 - *Suprapubic puncture

*Bacteriologic results are more reliable, but the procedure should only be considered in a specialized pediatric setting.

Treatment

Acute cystitis should be treated with a 3 to 5 day course of therapy with one of the following antibiotics:

- Trimethoprim - Sulfamethoxazole (6-12mg TMP and 30-60mg SMX /kg/24hr)
- Nitrofurantoin- 5-7 mg/kg/24hr in 3 to 4 divided doses
- Amoxicillin - 50 mg/kg/24 hrs

Pyelonephritis is treated by a 14 days course of either

- Ceftriaxone (50-75mg/kg/24hr), or
- Ampicillin (100mg/kg/24hr) and gentamycin
(3 - 5 mg/kg/24hr)

Prognosis

Prognosis is excellent with lower UTI but is guarded with pyelonephritis.

9.3 Acute Glomerulonephritis (AGN)

Glomerulonephritis may have an acute or chronic presentation. Primary renal diseases presenting as acute GN include PSGN, IgA nephropathy, membranoproliferative GN and idiopathic rapidly progressive GN. AGN can also occur in the course of various systemic illnesses.

There are five major clinical presentations of GN: acute nephritic syndrome, nephrotic syndrome, rapidly progressive GN, asymptomatic abnormalities of the urinary sediment (hematuria, proteinuria) and chronic GN. Acute nephritic syndrome is characterized by the abrupt onset of hematuria, with RBC casts or dysmorphic RBCs and proteinuria (usually non-nephrotic range), temporally associated with impairment of renal function.

Acute Poststreptococcal Glomerulonephritis (APSGN)

Definition.

APSGN is a classic example of nephritic syndrome, which is characterized by sudden onset of gross hematuria, edema, hypertension, proteinuria and renal insufficiency.

Etiology and Epidemiology

PSGN is the commonest cause of postinfectious GN. The condition follows infection of the throat or the skin with nephritogenic strains of group A Beta-hemolytic streptococci. During cold weather, PSGN commonly follows streptococcal pharyngitis whereas during warm weather the GN generally follows pyoderma. Although epidemics have been described, the condition is most commonly sporadic. It is most common in children aged between 5 and 12 years of age.

Pathogenesis of PSGN

The pathogenetic mechanisms leading to renal damage are generally not well understood. Although the exact mechanisms of immune complex formation are not clear, evidences such as depression in serum C3 level and morphologic studies suggest immune complex mediated glomerular injury.

Clinical manifestations

Typical patients with acute PSGN develop an acute nephritic syndrome 1 to 2 weeks after an antecedent streptococcal pharyngitis or 3-6 weeks after streptococcal pyoderma.

The severity of renal involvement varies from asymptomatic microscopic hematuria to acute renal failure. Patients commonly have nonspecific symptoms such as malaise, lethargy, abdominal or flank pain and fever.

Edema mainly develops from salt and water retention. Hypertension is seen in 60% of cases. In severe cases, patients may develop encephalopathy and/or congestive heart failure from severe hypertension and acute renal failure.

Nephrotic syndrome may develop in 10 to 20% of cases.

Laboratory findings

Urinalysis demonstrates hematuria, usually with RBC casts, proteinuria of ++ and less, and polymorphonuclear cells.

The level of complement factor C3 is usually reduced.

Renal function test (blood urea nitrogen and serum creatinin level) may be normal or elevated.

Confirmation of the diagnosis requires clear evidence of invasive streptococcal infection. A positive throat culture may support the diagnosis or may simply represent a carrier state. On the other hand, a rising antibody titre to streptococcal antigen (s) confirms a recent streptococcal infection. The ASO (anti-streptolysin O) titre is commonly elevated after a pharyngeal infection but rarely rises after pyoderma. The best antibody titre to document cutaneous infections is anti-DNAase B titre.

Renal biopsy should be considered only in selected cases.

Complications

Complications of acute PSGN are those of acute renal failure and include hypertension, heart failure, encephalopathy, hyperkalemia and metabolic acidosis.

Treatment

There is no specific therapy for APSGN. The management is directed towards treating complications of ARF.

Although there is no evidence that antibiotics change the course of the illness, a 10 days treatment with systemic penicillin is generally recommended.

Prevention

Early systemic antibiotic therapy for streptococcal throat or skin infection does not eliminate the risk of GN. Family members of patients with AGN should be cultured for group A Beta- hemolytic streptococci and treated if culture positive.

Course and Prognosis

The acute phase generally resolves within 6-8 weeks.

Urinary protein excretion and hypertension usually normalize by 4-6 weeks after onset; however, microscopic hematuria may persist for 1-2 year after initial presentation.

Complete recovery occurs in *more than* 95 % of children with APSGN *and* recurrences are extremely rare

9.4 Nephrotic Syndrome

Definition

This is a syndrome characterized by heavy proteinuria ($>40\text{mg/m}^2/\text{hr}$), hypoalbuminemia ($< 2.5\text{g/dl}$), edema and hyperlipidemia.

Etiology

Nephrotic syndrome may occur as a result of any form of glomerular disease and may be associated with a variety of extra renal conditions.

Approximately 90% of children with this condition have some form of the idiopathic nephrotic syndrome. In the remaining 10%, the syndrome is secondary to some form of glomerulonephritis. Other possible causes include tumors (lymphoma, leukemia), infections (hepatitis B, malaria, etc), drugs (isoniazid), systemic disease (systemic lupus erythematosus, juvenile rheumatoid arthritis, diabetes) and congenital nephrotic syndrome.

Idiopathic Nephrotic Syndrome (minimal change disease)

Idiopathic nephrotic syndrome accounts for approximately 90% of nephrosis in childhood. It occurs in three morphologic patterns: minimal change disease (85%), focal segmental glomerulosclerosis (10%) and mesangial proliferative (5%).

Etiology and Epidemiology

The cause is unknown

It is primarily a pediatric problem, being 15 times more common in children than in adults.

Incidence of nephrotic syndrome is 2-3/100,000 children/year

The condition most commonly occurs between the ages of 2 and 8 years of age but it is reported as early as 6 months and throughout adulthood. Male to female ratio is approximately 2:1

Pathophysiology

Nephrotic syndrome results from alterations in the selective permeability of the glomerular basement membrane. The protein loss is generally more than 2g in 24 hours (normal loss would be less than 150mg in 24 hours). The protein lost is primarily composed of albumin resulting in hypoalbuminemia. Edema appears when serum albumin falls below 2.5 g/dl, however, the mechanism of edema formation in nephrotic syndrome is incompletely understood.

The serum level of almost all lipids is increased in nephrotic syndrome as a result of increases in lipoprotein synthesis and low levels of lipoprotein lipase.

Clinical Manifestations

Attacks may follow URTI

It usually presents with pitting edema, initially noted in periorbital area and in the lower extremities. The edema becomes generalized with time.

Some children present with hypotension secondary to significant shift of fluid from intravascular to third space and they may rarely develop renal failure

With massive pleural effusion and ascites, patients may develop dyspnoea.

Unlike nephritic syndrome, gross hematuria and hypertension are uncommon.

Diagnosis

Urinalysis reveals proteinuria (+3 or +4 on dipstick).

Serum albumin level is generally < 2.5g /24 hr.

The serum cholesterol and triglyceride levels are generally high

C3 level is often normal

- Renal biopsy is generally not indicated for children between 1 and 8 years of age, however, it is indicated in children with features that make minimal change disease less likely. These are:

- hematuria
- hypertension
- renal insufficiency
- hypocomplementemia
- age <1yr or >8yr

Complications

Infection is the major complication of nephrosis. Contributing factors include hypogammaglobulinemia, decreased bactericidal activity of leukocytes and edema fluid acting as a culture medium.

The commonest infectious complication in nephrotic syndrome for unknown reasons is spontaneous bacterial peritonitis. *S.pneumoniae* is the commonest cause of spontaneous bacterial peritonitis in these patients. Other infections include pneumonia, cellulitis, sepsis, and UTI.

Patients with nephrosis also exhibit increased tendency for arterial and venous thrombosis (2-5%).

Treatment

General measures

- Children with mild to moderate edema can be managed as outpatients.

Salt should be restricted (until the edema resolves)

Diuretics can be used cautiously to reduce the edema. In mild cases, chlorothiazide or spironolactone can be used but in severe cases with respiratory distress secondary to massive pleural effusion and ascites or severe scrotal edema furosemide (P.O, or IV with poor response to P.O treatment) can be administered with careful monitoring of renal function and serum electrolytes.

Steroids

More than 95% of children with minimal change disease, ~50% with mesangial proliferation and ~20% with FSGS respond to steroids.

In children with presumed minimal change disease prednisolone should be administered in a dose of $60\text{mg}/\text{m}^2/24\text{hr}$ or, $1\text{-}2\text{mg}/\text{kg}/24\text{hr}$ (maximum 80 mg/d) in 3 to 4 divided dose for 4 to 6 weeks. Eighty to ninety percent of patients respond to steroid therapy with the median time to remission of 10 days. Remission is defined by proteinuria of trace or negative for 3 consecutive days.

After the initial 4 to 6 week course, the prednisolone dose should be tapered to $40\text{ mg}/\text{m}^2/\text{d}$ and given every other day as a single morning dose. The alternate-day dose is continued for the next 2 to 3 months and discontinued without tapering the dose.

Patients with proteinuria of 2+ or more after 8 weeks of steroid treatment are said to have steroid resistant nephrosis.

Relapses (3-4+ proteinuria plus edema), should be treated with prednisolone daily in divided dose as above until they enter remission. The prednisolone dose is then changed to alternate-day dosing and tapered over 1 to 2 months.

Patients who relapse while on alternate-day treatment or within 28 days of stopping prednisolone are termed steroid dependent. Patients who responded well to steroid therapy but relapse 4 or more times in 12 months period are termed frequent relapsers.

Cytotoxic drugs

Cytotoxic agents such as cyclophosphamide can be considered in frequently relapsing nephrotic syndrome or if patients suffer severe corticosteroid toxicity. They may also help in some steroid resistant cases.

Prognosis

Most children with steroid responsive nephrotic syndrome have repeated relapses until the end of second decade of life when the disease resolves spontaneously.

9.5 Acute Renal Failure (ARF)

Definition

Acute renal failure (ARF) is a clinical syndrome in which a sudden deterioration in renal function results in inability of the kidneys to maintain fluid and electrolyte homeostasis.

Etiology

Causes of ARF are conventionally classified as prerenal, intrinsic renal and postrenal

a) Prerenal causes

- Hemorrhage
- Dehydration
- Sepsis
- Cardiac failure and others

b) Intrinsic renal causes

- Glomerulonephritis, such as PSGN, and rapidly progressive GN.
- Acute tubular necrosis (ATN)
- Hemolytic uremic syndrome
- Acute interstitial nephritis

c) Post renal causes

- Posterior urethral valve
- Urolithiasis

Pathogenesis

Prerenal causes of ARF produce decreased renal perfusion via decrease in the total or effective circulating blood volume. The result of decreased perfusion is decrease in GFR and cortical blood flow. There is no evidence of renal damage. If the underlying cause of the renal hypoperfusion is reversed promptly, renal function returns to normal. If hypoperfusion persists, intrinsic renal parenchymal damage may develop.

Intrinsic renal causes lead to ARF through different mechanisms, some of which are not well understood. There is evidence of renal parenchymal damage. Acute tubular necrosis occurs most often in critically ill infants and children who have been exposed to nephrotoxic and/or ischemic insults. The typical pathologic process of ATN is tubular cell necrosis.

For postrenal causes to result in renal failure, the obstruction should be bilateral if both kidneys are functioning or it should be complete obstruction of a single functioning kidney. In general, relief of the obstruction results in recovery of renal function except in patients with associated renal dysplasia or prolonged urinary tract obstruction.

Clinical Manifestations

In ARF, signs and symptoms may predominantly be that of the precipitating disease, and hence findings related to common cause should be looked for.

Clinical findings related to the renal failure include pallor, decreased urine output, edema, hypertension, vomiting and lethargy.

Renal failure may be oliguric (urine output <0.5 ml/Kg/hour <400 ml /m²) or non-oliguric (as in aminoglycoside nephrotoxicity).

Diagnosis

The diagnosis of ARF is established by detecting elevated bun (blood urea nitrogen) and creatinin levels. Considerable information however can be obtained from urinalysis in addition to clinical findings:

- Presence of hematuria, proteinuria and RBCs or granular casts on urinalysis suggests intrinsic ARF, in particular glomerular disease.
- The presence of WBCs and WBC casts with low-grade hematuria and proteinuria, suggests tubulointerstitial disease. Urinary eosinophils may be present in children with drug induced tubulointerstitial nephritis.

Urinary indices (specific gravity, osmolality, urine Na) may be useful in differentiating prerenal ARF from intrinsic ARF.

Other Laboratory investigations can be done to detect the underlying cause.

Complications

Metabolic

- Hyponatremia
- Hyperkalemia
- Hypocalcemia, hyperphosphatemia
- Hypermagnesemia
- Hyperuricemia

Metabolic acidosis

Cardiovascular

- Pulmonary edema
- CHF
- Hypertension
- Arrhythmias
- Pericarditis

Neurologic

- Coma
- Seizures

Hematologic

- Anemia
- Coagulopathies & bleeding diathesis

Treatment

Initial measures

Patients with ARF should be admitted to a hospital for careful monitoring.

Determination of the intravascular volume (hydration) status is of critical importance when initially evaluating a patient with ARF. If there is no evidence of volume overload or cardiac failure, intravascular volume should be expanded by IV administration of isotonic saline, 20 ml/kg over 30 minutes. Severe hypovolemia may require additional fluid boluses. After volume resuscitation, hypovolemic patients generally void within 2 hours. Failure to pass urine suggests the presence of intrinsic or postrenal ARF.

Diuretic therapy should be considered only after the adequacy of circulating blood volume has been established. Furosemide can be administered as a single IV dose of 2 to 4 mg/kg. If there is no urine even after catheterization, diuretics should be discontinued and fluid restriction becomes essential.

Specific complications of ARF

Fluid overload (CHF, hypertension, pulmonary edema)

treatment with salt and water restriction

- Diuretics - Furosemide
- Beta blockers and vasodilators can also be considered for treatment of hypertension in addition to the above measures

Hyperkalemia (serum K $>6\text{mEq/L}$)

- hyperkalemia may lead to cardiac arrhythmia, cardiac arrest, and death.
- treat with- K- restriction or sodium polystyrene sulphonate
- When it is $>7\text{mEq/L}$, it requires emergency measures:
- Regular insulin with glucose (50%)
- Calcium gluconate
- NaHCO_3

Acidosis

Acidosis is rarely severe

Treat with bicarbonate in severe acidosis

($\text{pH} < 7.15$ or serum bicarbonate $<8\text{mEq/L}$).

Hypocalcaemia

- Treat with lower serum phosphorus by giving phosphate-binding antacids, this increases the serum calcium level.
- If tetany develops, administration of IV Ca carbonate is needed.

Hyponatremia

Treat with fluid restriction.

- hypertonic saline (3%) for symptomatic cases or when serum Na is $<120\text{mEq/L}$

Seizures

Treatment involves controlling attacks using diazepam

- Treat underlying cause: electrolyte abnormalities

Fluid and caloric management

Patients with relatively normal intravascular volume should be restricted to an amount of fluid equal to insensible losses ($400\text{ml/m}^2/24\text{hr}$) plus urine output only.

- In general, glucose-containing fluids (10-30%) without electrolytes are used as maintenance fluids.
- Moderate protein restriction (to avoid accumulation of nitrogenous wastes) and increasing caloric intake are generally recommended.

Treat underlying causes

- Peritoneal dialysis
- Dialysis may be considered for severe complications not responding to conservative measures

Exercise

1. A 7 year old male child presented with history of urinary frequency, dysuria and suprapubic pain of 3 days duration. There is no history of fever. Urinalysis revealed 11 WBCs and 6RBCs per HPF.
 - a) What is the most likely diagnosis?
 - b) How do you confirm the diagnosis?
2. Discuss clinical and laboratory features that differentiate nephrotic syndrome from nephritic syndrome.
3. Outline the management of a patient with idiopathic nephrotic syndrome.
4. List 5 complications of ARF.

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CHAPTER TEN

DISEASES OF THE NERVOUS SYSTEM

Solomon Amsalu

10.1 Bacterial Meningitis

Objectives

By the end of this section, readers should be able to:

- Describe the causes of bacterial meningitis in children
- Discuss the clinical manifestations of meningitis
- Discuss the diagnostic methods in meningitis
- Discuss the management of meningitis

Introduction

Bacterial meningitis indicates inflammation of the meninges as a result of bacterial infection.

Bacterial meningitis is one of the most potentially serious infections in infants and older children.

The possibility of bacterial meningitis should be considered in all febrile infants with altered mental status or evidence of other neurologic dysfunction.

Etiology

- The first 2 months of life – group B *streptococcus*, gram negative enteric bacilli and *listeria monocytogenes*. Occasionally – *Hemophilus influenzae* (both type b and non-typable strains) and other pathogens found in older patients.
- 2 months to 12 years of age – *Hemophilus influenzae* type b, *Streptococcus pneumoniae* and *Neisseria meningitidis*
- In children more than 12 years old – *neisseria meningitidis* and *streptococcus pneumoniae*

Alterations in host defenses due to anatomic defects or immune deficits increase the risk of meningitis from less common pathogens such as *pseudomonas aeruginosa*,

staphylococcus aureus, *Staphylococcus epidermidis*, *salmonella* and *listeria monocytogenes*.

Epidemiology

The risk is greatest among infants 1 - 12 months of age. 95% of cases occur between 1 months and 5 years of age.

The major risk factor is lack of immunity to specific pathogens associated with young age.

The mode of transmission is probably person-to-person contact through respiratory tract secretions or droplets.

H. influenzae type b

- Most commonly occurs between 2 months and 2 years of age.
- Peak incidence occurs in infants 6 to 9 months of age.
- Otitis media, HIV infection and CSF leaks increase the risk of *H. influenzae* meningitis.

Streptococcus pneumoniae

- The risk is greater among blacks than whites, especially among blacks with sickle cell anemia.
- Other risk factors include otitis media, sinusitis, pneumonia, CSF otorrhea, or rhinorrhea and splenectomy

Neisseria meningitides

- Epidemics are usually caused by group A and C.
- In the absence of epidemic most infections are due to group B.

Pathology

The brain is hyperemic, edematous and contains exudate in the subarachnoid space.

Thrombosis may occur in meningeal veins or venous sinuses. Destructive changes can take place in the cortex. Inflammation of spinal nerves and roots produces meningeal signs, and inflammation of cranial nerves produce cranial neuropathy.

Increased intracranial pressure is due to

- Cell death (cytotoxic brain edema)

- Cytokine induced increased capillary vascular permeability (vasogenic cerebral edema) and,
- Increased hydrostatic pressure (interstitial cerebral edema) following obstructed reabsorption of CSF in the arachnoid villus or obstruction of the flow of fluid with in or exiting from the ventricles.

Hydrocephalus is uncommon after the neonatal period. It often takes the form of communicating hydrocephalus. Less often obstructive hydrocephalus develops.

Pathogenesis

Bacterial meningitis is usually the result of bacteremia from an upper respiratory infection.

Invasion by direct extension is less common but may occur from purulent infections of the paranasal sinuses, mastoids, or rarely from abscesses contiguous to the meninges.

Bacteria may also gain entry to the CNS through penetrating wounds, lumbar puncture, neurosurgical procedures or midline dermal sinus tracts.

Clinical Manifestations

The mode of onset has two predominant presentations:

- Sudden onset with rapidly progressive manifestations of shock, purpura, DIC and reduced level of consciousness is a dramatic and often fatal presentation of meningococcal sepsis with meningitis; it may evolve to death within 24 hours.
- Subacute presentation - more often, meningitis due to *H. influenzae* type b and *pneumococcus* (and some cases of meningococcal meningitis) is preceded by several days of upper respiratory tract or gastrointestinal symptoms.

The signs and symptoms of meningitis are related to:

- Non specific findings associated with a systemic infection or
- Manifestations of meningeal irritation like nuchal rigidity, back pain, Kerning sign and Brudzinski sign. In young infants these signs may not be evident.

Increased ICP is suggested by headache, emesis, bulging fontanel or diathesis of the sutures, oculomotor or abducens nerve paralysis, stupor, coma or signs of herniation.

Focal neurologic sign may be due to vascular occlusion or abscess formation. It is most commonly seen in patients with pneumococcal meningitis.

Seizures that occur on presentation or within the first 4 days of onset are usually of no prognostic significance. Seizures that persist after the fourth day of illness and those that are difficult to treat are associated with poor prognosis.

Diagnosis

The diagnosis of acute pyogenic meningitis is confirmed by CSF analysis, which reveals microorganisms on Gram stain and culture, a neutrophilic pleocytosis, elevated protein and reduced glucose concentrations.

Lumbar puncture (LP) should be performed when bacterial meningitis is suspected.

Contraindications for an immediate LP include

- Evidence of increased intracranial pressure
- Severe cardiopulmonary compromise
- Infection of the skin overlying the site of LP

If an LP is delayed, an immediate empirical therapy should be initiated. The CSF may appear turbid. The cell count is usually greatly elevated ($>1000/\text{mm}^3$) and reveals a neutrophilic predominance. CSF protein is elevated usually $>100\text{mg/dl}$ and glucose is decreased ($<40\text{ mg/dl}$ or two third of the blood glucose).

Gram's stain reveals an organism in 70% of cases.

Blood culture may reveal the responsible bacteria in 80 -90% of cases.

A latex particle agglutination test is used to detect bacterial antigen in the CSF.

Treatment

Antibiotic therapy

Prompt treatment of bacterial meningitis with an appropriate antibiotic is essential. The initial selection always should be made before definitive cultures are available.

Ampicillin and chloramphenicol are effective for initial treatment of bacterial meningitis in infants and children.

Once an organism is isolated and antimicrobial susceptibility is determined, antimicrobial therapy can be modified.

Suggested antibiotics and duration of therapy

- *H. influenzae* type b: chloramphenicol 100 mg/kg/d for 7 - 10 days.
- *S. pneumoniae*: aqueous penicillin G 300,000 IU/kg/d for 10 - 14 days.
- *N. meningitidis*: aqueous penicillin G 300,000 IU/kg/d for 5 - 7 days.

If no etiologic agent is identified the initial antibiotic therapy should be continued for 10 to 14 days.

Gram-negative bacillary meningitis (*E. coli* and *P. aeruginosa*) should be treated for three weeks. Initial therapy might include combination of ampicillin and gentamycin or ceftiaxone.

Corticosteroids

Rapid killing of bacteria in the CSF effectively sterilizes the meningeal infection but releases toxic cell products after cell lysis that precipitates the cytokine mediated inflammatory response.

The use of intravenous dexamethasone 0.15 mg/kg/dose every 6 hours for 2 days is indicated for children greater than 6 weeks of age. Use of corticosteroids is associated with less fever, lower CSF protein and lactate levels and a reduction on permanent auditory nerve damage. Most experience with dexamethasone treatment has been with *H. influenzae* type b infection.

Corticosteroids appear to have maximum benefit if given just before antibiotics and should be administered within 1 to 2 hr of antibiotics.

Supportive care

- Supportive care involves: Vital sign monitoring and neurologic assessment should be made frequently during the 1st 72 hrs.
- Intravenous fluid administration should be restricted to one half to two thirds of maintenance, until it can be established that increased ICP and SIADH is not present.
- Shock must be treated aggressively to prevent brain and other organ system dysfunction.

Increased ICP should be treated with:

- Endotracheal intubation and hyperventilation (in order to maintain P_{CO_2} approximately 25 mmHg).
- Intravenous furosemide and mannitol osmotherapy may reduce ICP.
- Immediate therapy for seizures includes intravenous diazepam.

Serum glucose, calcium and sodium level should be monitored.

After immediate management of seizure, patients should receive phenytoin (15 - 20 mg/kg loading dose, 5 mg/kg/d maintenance).

Complications

Neurologic complications include seizures, increased ICP, cranial nerve palsies, brain abscess, hydrocephalus, stroke, herniation and subdural effusion.

Syndrome of inappropriate secretion of antidiuretic hormone may exacerbate cerebral edema or produce hyponatremic seizures.

Prognosis

With appropriate therapy mortality is 1 to 8%. The highest mortality rates are observed with pneumococcal meningitis. Severe neurodevelopmental sequelae may occur in 10 to 20 % of patients.

Poor prognostic factors include:

- Age under 6 months
- More than 10^6 colony forming units of bacteria /ml of CSF
- Seizure occurring more than 4 days into therapy
- Coma or focal neurologic signs on presentation

Sensorineural hearing loss is the most common sequelae of bacterial meningitis. Other common sequelae include mental retardation, seizures, and delay in the acquisition of language and visual impairment.

Prevention

N. meningitides

Chemoprophylaxis is recommended for all close contacts with rifampin 10 mg/kg/dose every 12 hr (maximum dose of 600 mg) for 2 days.

Meningococcal quadrivalent vaccine against groups A, C, Y and W135 is recommended for high-risk children older than two years and during epidemics.

Hemophilus influenzae type b

Chemoprophylaxis should be given for all household contacts with rifampin 20 mg/kg/24 hours (maximum dose of 600 mg) given once each day for 4 days.

Vaccination with H. influenzae type b vaccine is recommended

Streptococcus pneumoniae

No chemoprophylaxis or vaccination is required for normal hosts, as secondary cases rarely occur.

Polyvalent pneumococcal vaccine is recommended for cases with asplenia, splenectomy and nephrotic syndrome.

10.2 Aseptic Meningitis

Aseptic meningitis is an inflammatory process of the meninges. It is a relatively common occurrence and is caused by a large number of different etiologic factors.

The CSF is characterized by mild mononuclear predominance, normal or slightly elevated protein and the absence of microorganism on Gram stain and on routine culture.

CSF glucose is usually normal but may be decreased in certain viral infections like mumps.

In the majority of instances, the illness is self-limited; however, with some etiologies the resulting disease may be severe, protracted, recurrent and progressive and lead to disability and death.

Viruses are the cause of most cases of aseptic meningitis. Enteroviruses account for approximately 85% of all cases of aseptic meningitis.

Other etiologies include mycobacterium tuberculosis, fungi, protozoa, parasites and malignancy.

Hospitalization is usually necessary because of the possibility of treatable bacterial disease and the frequent need of fluid therapy for dehydration.

Treatment is symptomatic.

EXERCISE

1. What are the leading etiologies of meningitis in children?
2. What findings on CSF are suggestive of bacterial meningitis?
3. How do you treat acute bacterial meningitis in the absence of culture?

10.3 Coma

Solomon Amsalu

Objectives

By the end of this section the reader should be able to:

- Discuss main causes of coma
- Discuss the approach to a comatose patient

Levels of impaired consciousness

Levels of impaired consciousness include:

Lethargy: A state of decreased responsiveness to stimuli.

Obtundation: A state of much more decreased responsiveness to stimuli.

Stupor: Abnormal response to a given stimuli.

Coma: A state of unconsciousness from which the child can not be aroused by stimuli

The conscious state is maintained by the central reticular formation, which extends from the medulla oblongata through the pons, midbrain and dorsal hypothalamus to the reticular nuclei of the thalamus. These interconnected nuclei act as an alerting system that keeps the cerebral cortex in a state of wakefulness when incoming stimuli are received.

Disturbances of arousal and awareness may be the result of structural lesions, diffuse encephalopathies and status epilepticus (Table 27).

Table 27 Causes of coma

Structural	Intracranial mass lesions
Metabolic	Anoxic- ischemic coma Hypoglycemia Hyperglycemia Hyponatremia Hepatic dysfunction Renal failure
Toxins	Lead Sedative and hypnotic drugs
Trauma	Head injuries
Infection	Meningitis, encephalitis, severe generalized infection
Non convulsive status epilepticus	

Assessment of impaired consciousness

A modification of the Glasgow coma scale (Table 27) is a useful tool for the grading of the degree of coma and the severity of the insult in infants and in children.

A coma score of less than five is associated with a grave prognosis, while a score five to eight may indicate a better prognosis in the child than in the adult.

Table 28: Glasgow coma scale

Eye opening		
Spontaneous	4	
To Voice	3	
To pain	1	
None		
Verbal response		
Older Children		Infants & young children
Oriented	5	Appropriate words; smiles fixes, and follows 5
Confused	4	Consolable crying 4
Inappropriate	3	Persistently irritable 3
Incomprehensible	2	Restless, agitated 2
None	1	None 1
Motor response		
Obeys	6	
Localizes pain	5	
Withdraws	4	
Flexion	3	
Extension	2	
None	1	

Proper evaluation of the comatose patient involves obtaining a history, performing a rapid directed physical and neurological examination and obtaining certain laboratory studies, while the patient's airway and vital signs are protected. An attempt is made to delineate the cause of coma in the hope of finding a treatable process.

ABCs

Before obtaining a history and physical examination make sure that the patients airway, breathing and circulation (ABCs) are stable.

History

Ask for history of:

- Head trauma
- Ingestion of certain house hold product and drugs in toxic amounts
- Medical illnesses like seizure disorder, diabetes, chronic renal disease, chronic liver disease, cardiac disease or recent febrile illness.

Physical examination

- Do meticulous physical examination
- Observe for signs of trauma and posture of the patient
- Presence of seizures, signs of liver disease and stiff neck.

Localization

1- Motor response to pain

- Decorticate posturing (arm flexion with leg extension). Implies hemisphere or diencephalon dysfunction. May be caused by destructive lesions or metabolic derangement.
- Decerebrate posturing (extension of legs and arms) implies dysfunction of midbrain or upper pons on a structural or metabolic basis.

2- Respiratory pattern

- Cheyne - Stokes respiration (a crescendo-decrescendo breathing pattern with apneic pauses in between) implies bilateral hemisphere dysfunction or diencephalic lesion with an intact brain stem.
- Central neurologic hyperventilation (rapid deep breathing) usually indicates damage between midbrain and pons.
- Apneustic breathing consists of a prolonged inspiration followed by an expiratory pause and is usually a sign of pontine lesion.
- Ataxic (irregular or agonal) breathing is usually a pre-terminal event signifying disruption of medullary centers.
- Coma with hyperventilation frequently signifies a metabolic derangement

3- Examination of pupils

- Reactive pupils in a comatose patient without other responses suggest a metabolic cause.
- Damage to the midbrain usually produces mid dilated pupils that do not react to light but may fluctuate in size.
- Pontine damage produces pin point pupils that react to bright light when viewed with a magnifying glass.
- A unilaterally fixed, dilated pupil is seen with damage to the third nerve (table 3).

Table 29: Rostral caudal progression of coma

Area involved	Level of consciousness	Pupils	Muscle tone	Respiration
1. Diencephalon (thalamus/hypo thalamus)	Agitation lethargy Obtundation	Responds to light briskly	Some purposeful Movement in response to pain; combative Movement, Decorticate	Yawning and sighing→ Cheyne Stokes
2- Midbrain	Stupor → coma	Mid-position fixed	Decerebrate	Neurogenic hyperventilation.
3- Pons	Coma	Pin point	Decerebrate	Apneustic
4- Medulla	Coma	Mid position fixed	Flaccid	Ataxic

Laboratory Studies

Laboratory Investigations should be individualized but must address structural, infectious, and metabolic causes of coma. Investigations include:

- Blood: Complete blood count, blood film, serum electrolytes, calcium, phosphorus, glucose, creatinine, ammonium level, liver function studies and a toxic screen.
- CSF: Must be analyzed if meningitis or encephalitis is suspected.
- Urine: Glucose, ketones and further studies as indicated
- EEG: May be indicated if seizures are suspected etiology of coma
- Skull X-ray
- CT or MRI - If mass lesions are suspected

Treatment

- Establish a clear airway. Begin bag and mask resuscitation if patient is not breathing spontaneously
- Check carotid pulse, if it is absent, carry out external cardiac massage.

- Cannulate vein for administration of parenteral fluids, blood and drugs as indicated.
- Control seizures, Phenytoin is least likely to cloud consciousness.
- Remember that lack of oxygen and glucose for brain metabolism can lead to permanent damage; Use these whenever it is necessary.
- Identify the specific cause of coma and correct the problem in a safe and controlled fashion.

Exercise

1. What are the main causes of coma in the pediatric patient?
2. What are the initial steps to be taken in a comatose patient?
3. How are the motor response, respiratory pattern and pupillary examination helpful in neuro-localization?

10.4 Seizure Disorders

Objectives

By the end of this section the reader should be able to:

- Describe how to approach a child with seizure
- Describe the different types of seizure disorders
- Discuss treatment of seizure disorders

Definitions

A seizure is a sudden involuntary disturbance of brain function manifested by impairment or loss of consciousness, abnormal motor activity, behavioral abnormalities, sensory disturbances or autonomic dysfunction.

Epilepsy is recurrent seizure unrelated to fever or an acute cerebral insult.

Prevalence

Seizures are common in pediatric age group (3-5%).

- Aura – automayism ,epigastric pain, etc. The aura of seizure is very short.
- During the seizure
- the type of movement, duration, associated loss of consciousness, associated symtoms and whether it can be stimulated or stopped with strain
- After the seizure stops
- Postictal headache, confussion, sleep
- Focal neurologic deficit (Todd's pralysis)

Pseudoseizures (or hysterical seizure) tend to occur in public where the patient can gain attention, injuries are rare, and the description of the seizures are often bizarre.

Once you have decided this a siezure, determine the type of the seizure based on the description of the seizure from the history and if possible using EEG. See the classification above for the type of the seizures.

3. Identify the cause of the seizure using:

- History
 - Hx of birth injury,
 - Perinatal Hx—for asphyxia, jaundice, meningitis
 - Hx of encephalitis, meningitis
 - Hx of head injury
 - Hx of drug intake
 - Hx of metabolic or systemic diseases
 - Family Hx of seizure or neurologig diseases
- Physical examination
 - congenital malfomation
 - focal neurologic deficit (hemiplegia, monoplegia
 - Signs of systemic diseases
 - Investigations – based on Hx and physical findings, selectand do the following tests.
 - complete blood count
 - VDRL

- Biochemical tests (blood glucose, serum calcium, electrolytes, etc)
 - EEG
 - Neuroradiological – skull x-ray, CT

After the above assessment, one decides on the type and cause of the seizure, and plan treatment.

Partial Seizures

Simple partial seizures (SPS)

- Motor activity is the most common symptom of SPS
- Characterized by asynchronous clonic or tonic movements
- Tends to involve the face, neck and extremities
- Automatism do not occur
- Some patients have aura
- Duration - 10 to 20 seconds
- No postictal phenomenon

Complex partial seizures (CPS)

- An aura is present in approximately one third of the children
- Automatism occur in 50 to 75% of cases
- Automatism consists of confused, repetitive, purposeless behavior during which the patient has limited awareness of his actions and no memory of the event following the seizures.
- The average duration of CPS is 1 to 2 minutes.

Generalized Seizures

Absence seizures

Simple (typical) absence (petit mal) seizures Characterized by sudden cessation of motor activity with a blank facial expression

- More common in females
- Rare before the age of 5 years
- Never associated with aura
- Rarely persist longer than 30 seconds
- Not associated with postictal state
- Frequently associated with automatism

- Hyperventilation for 3 to 4 minutes routinely produces an absence seizure.
- Electroencephalogram /EEG/shows typical 3/sec. spike and generalized wave discharge

Complex (atypical) absence seizure

- Has associated motor component
- Frequency of seizures per day is usually less than simple absence

Generalized tonic clonic seizures

- Common
- Patients suddenly lose consciousness, eyes roll back, entire body musculature undergoes tonic contractions, and they become cyanotic in association with apnea. The clonic phase of the seizure is heralded by rhythmic clonic contractions alternating with relaxation of all muscle groups.
- Loss of sphincter control is common
- Postictally, it is followed by 30 minute to 2 hour deep sleep

Myoclonic Epilepsy

- Characterized by repetitive seizures consisting of brief, often symmetric muscular contraction with loss of body tone. It may cause a forward fall resulting in facial injury.

Febrile seizures

- They are the most common seizure disorder in childhood
- Incidence approaches 3 to 4 % in young children
- They occur between 9 months and 5 years of age
- The peak age of onset 14 to 18 months
- They usually develop when the core temperature reaches 39°C or more
- Seizure type is generalized tonic clonic of few seconds to 10 minutes duration

Atypical febrile seizure

- Seizure persisting for more than 15 minutes
- Repeated convulsions for several hours or days
- Focal seizure

Risk factors for development of epilepsy

- Positive family history of epilepsy
- Onset before 9 mo of age
- Atypical febrile seizure
- Delayed developmental milestones
- Abnormal neurologic findings

Risk of epilepsy

- With several risk factors - 9%
- No risk factor - 1%

Treatment

Treatment involves:

- Control fever
- Oral diazepam reduces risk of recurrence and it should be give at the onset of each febrile illness (usually for 2-3 days).

Investigation

- Complete blood count
- Fasting blood sugar, serum calcium, magnesium and serum electrolytes (phosphorus, sodium and potassium)
- Skull X ray
- EEG

Treatment of seizure /epilepsy

First afebrile convulsion with negative family history and Normal examination & EEG:

- 70 % of them will not have another convulsion
- Anticonvulsant is not recommended

If it is recurrent seizure, begin anticonvulsant therapy.

Start with a single drug, increase the dose until seizure control is achieved or undesirable side effects develop.

A single anticonvulsant is adequate in most children.

Table 30: Drug of choices for afebrile seizure

Type of seizure	Drug of choice
Partial	Carbamazepine
Absence	Ethosuximide, clonazepam
Grandmal	Phenytoin, phenobarbital
Myoclonic	Clonazepam, sodium valproate

When to stop anticonvulsant

Two seizure free years for a patient with no risk factors

Risk factors for recurrence

- Age more than 12 years at onset
- Neurologic dysfunction
- History of prior neonatal seizures
- Numerous seizures before control is achieved

Chance of recurrence with no risk factor is approximately 20 to 25%.

Drug withdrawal should be gradual over a period of 3 to 6 months.

Neonatal seizures

Generalized tonic clonic convulsions do not occur in the first month of life because arborization of axons and dendritic processes as well as myelination is incomplete in the neonatal brain.

Types

- Focal seizures

Consists of rhythmic twitching of muscle groups particularly extremities and face. It is often associated with localized structural lesion, infections and subarachnoid hemorrhage.

- Multifocal clonic seizures

Many muscle groups are involved

- Tonic seizures

Characterized by rigid posturing of the extremities and trunk

- Myoclonic seizures

Brief focal or generalized jerks of extremities

- Subtle seizures

Consist of chewing motion, excessive salivation, apnea and pedaling movements.

EEG Classification

Clinical seizures with a consistent EEG event

- Focal clonic
- Focal tonic
- Some myoclonic seizures

Likely to respond to anticonvulsant

Clinical seizures with inconsistent EEG event

- Generalized tonic seizure
- Subtle seizure
- Some myoclonic seizures

May not respond to anticonvulsant

Electrical seizures with absent clinical seizure

- May be seen in comatose infants.

Etiology

The most common cause is hypoxic ischemic encephalopathy.

Other causes include metabolic, infectious, traumatic, structural and hemorrhagic disorders.

Investigation

Investigation is geared to the etiology and may involve blood glucose and serum electrolytes or a lumbar puncture

Treatment

Treat the cause

Administer glucose and oxygen.

Administer anticonvulsant

- Phenobarbitone – loading dose of 20mg/kg iv and maintainance 4 to 6 mg/kg. If no response, repeat 10mg/kg.
- Phenytoin- loading dose of 20mg/kg and maintainance dose of 4 to 6 mg/kg.
- Caciium gluconate – if it is difficult to control with above treatment especially for infants who are asphyxiated, preterms, and infants of diabetic mothers

Prognosis

The prognosis depends on the primary cause of the disorder

Status Epilepticus

Status epilepticus is a continuous convulsion lasting longer than 30 minutes, or the occurrence of serial convulsions between which there is no return of consciousness. Generalized tonic clonic seizures predominate.

Etiology

There are 3 major subtypes

1. Prolonged febrile seizures is the most common cause
2. Idiopathic status epileptics

No underlying CNS lesion

Cause - sudden withdrawal of anticonvulsants

-non-compliant patients

3. Symptomatic status epilepticus

- Underlying neurologic disorder or a metabolic abnormality
- Higher mortality rate

Treatment

General measures

- Check airway, breathing and circulation
- Give oxygen (facemask)
- Insert naso-gastric tube
- Open iv line and draw blood for CBC, electrolytes, glucose and creatinine
- If there is hypoglycemia give intravenous glucose (5 ml/kg of 10 % dextrose)

Anticonvulsants

- Diazepam 0.1 - 0.3 mg/kg iv (2 mg/min) followed by
 - phenytoin 15-30 mg/kg iv (1 mg/kg/min)
- (Alternate Phenobarbital 15 - 20 mg/kg iv during 10-30 minutes)

If seizures are controlled continue with maintenance dose of phenytoin or phenobarbital.

If the seizure doesn't stop within 30 minutes treat with paraldehyde iv infusion or general anesthesia

Exercise

1. What is the difference between convulsion and epilepsy?
2. How do you clinically distinguish between absence and generalized tonic clonic seizure?
3. How do you manage a child in status epilepticus?

CHAPTER ELEVEN

POISONING

Assfawossen

Objectives

By the end of this course the student is expected to:

- Describe the epidemiology of poisoning in children
- Recite the principles of management
- Discuss the clinical presentation and management of common pediatric poisonings.

Definition

A poison is any substance that can harm or cause death when taken in to the body or absorbed.

Epidemiology and general considerations

- Poisoning in children below 5 years of age accounts for 50% of all cases. Almost all these exposures are unintentional. More than 90% occur at home and most involve only a single substance.
- Ingestion is the most common route (76%), with dermal, ophthalmic, and inhalation each occurring in about 6% of cases.
- About 60% involve non drug products, most commonly cosmetics, personal care products, cleansing substances, plants, foreign bodies and hydrocarbons.
- More than 75% of pediatrics poisoning exposures can be managed without direct medical intervention, because either the product involved is not inherently very toxic or the quantity of the material involved is not sufficient to produce significant toxic effects.
- Poisoning exposures in children 6-12 years of age are much less common (4%). Toxic exposures in adolescents are primarily intentional (suicide/abuse) or occupational.

Approach to the management of the poisoned child

I. History should include: -

Description of toxins

- Product names, ingredients and concentrations

Magnitude of exposure

- Amount ingested and/or length of contact time

Time of exposure

Progression of symptoms

Medical history

- Underlying diseases and/or concurrent drug intake can increase the susceptibility for toxic effects.

Criteria for hospital admission

- Exposure to caustic/corrosive substances
- Exposure to an amount sufficient to anticipate manifestation
- Exposure to a substance that causes delayed symptomatology or prolonged ones dose e.g slow release preparations, amanita phalloides mushrooms, methanol, ethylene glycol.
- When symptoms are already present
- Suspicion of chemical mal-treatment of children or intentional overdose

NB: A history of significant toxic exposure usually requires observation for at least 6 hours.

Initial medical care

- Initial attention should focus on life support, rapid assessment and resuscitation (e.g. treatment of shock, seizures and dysrhythmias)

Physical examination

A brief physical exam including v/s, pupillary size and reaction to light, mental status etc. should also be done pending complete exam after resuscitation and stabilization of the patient.

Definitive care

1. Preventing absorption

Dermal and ocular

- Flush the affected area with tepid water
- Soap and water can be used for dermal exposures

Inhaled toxin

- Move the patient away to fresh air and/or give O₂

GIT decontamination

NB: most liquid toxins are almost completely absorbed within 30 minutes of ingestion and most solid forms within 1 to 2 hours. GI-decontamination beyond this time is unlikely to be of value.

I. Emesis

The only emetic routinely used is syrup of ipecac. The onset of vomiting is usually 20 to 30 minutes after dosing with vomiting occurring in 90 to 95% of patients. It removes only one-third of stomach contents

Contra indications include: -

- Loss of protective air way reflexes of coma/convulsion
- Ingestion of caustics/corrosive
- Ingestion of petroleum distillates and hydrocarbons
- Ingestion of substances likely to produce abrupt depression of consciousness or early onset of seizure
- Prior significant vomiting/hematemesis
- Children less than 6 months old.

ii. Gastric lavage

Although used for many years, objective data do not document its efficacy particularly in children. It is time consuming and even under the best of circumstances removes only a fraction of gastric contents. It should only be used in older children and only in selected situations. The contraindications are almost similar to emesis

iii. Activated charcoal (AC)

Its use has increased dramatically over the past two decades as data demonstrating its efficacy has accumulated. AC is specially prepared to have a large adsorptive surface area. Many but not all toxins are adsorbed on its surface some of which include: heavy metals, iron, cyanide, hydrocarbons, alcohols, caustics and corrosives.

The usual dose is 10 to 30g for a child and 30 to 100 g for adolescent and adults.

iv. Cathartics

Cathartics are commonly used in conjunction with activated charcoal to hasten the clearance of AC toxin complex although no evidence shows that this is of value.

Commonly used cathartics include: sorbitol, MgSO_4 , etc

v. Whole bowel irrigation (WBI)

- Whole bowel irrigation involves instilling large volumes of a polyethylene glycol electrolyte solution in the stomach to cleanse the entire GIT. It should be used with caution in young children because of the risk of fluid and electrolyte imbalance.

Enhancing elimination

- Enhancing elimination is useful for only few toxins

a) Diuresis and/or iron trapping

- Alkalinization of urine
Increases excretion of weak acids e.g. salicylates, phenobarbitone
- Acidifying urine
Increases excretion of weak bases e.g. amphetamine

b) Dialysis

Hemo-/peritoneal-dialysis for selected agents

II. Laboratory Evaluation

- Includes at least toxicological screening and determination of blood levels

11.1 Organophosphates (OPs)

OPs are insecticides most commonly used at home or in the environment for eradicating insects. Most pediatric poisonings are accidental e.g. include – malathion, parathion, fenthion

Mechanism of action and pathophysiology

OP insecticides bind to and phosphorylate carboxylic esterase enzymes, including RBC cholinesterase (acetyl cholinesterase or true cholinesterase) plasma cholinesterase (pseudocholinesterase) and neurotoxic esterase (CNS).

The phosphorylation is time dependent and relatively irreversible; the process renders the enzymes incapable of degrading the neurotransmitter. Excessive Acetylcholine (ACh) accumulates in the neuro-effector junctions in the skeletal muscles system, in the autonomic and CNS produces first stimulatory and then inhibitory effects on neurotransmission.

Clinical manifestation

The sign and symptoms of a particular OP poisoning depend on the balance between stimulation of muscarinic and nicotinic receptors. Mixed clinical syndrome is common.

1. Stimulation of the muscarinic receptors produces a constellation of sign and symptoms that are best remembered with the mnemonic: **DUMBELS**

Defecation

Urination

Miosis

Bradycardia, bronchorrhea, bronchospasm

Emesis

Lacrimation

Salivation

2. Stimulation of the nicotinic receptors causes release of epinephrine and nor epinephrine producing a clinical picture best remembered by the mnemonic:

MATCH

Muscle weakness and fasciculation

Adrenal medulla activity increase

Tachycardia

Cramping of skeletal muscle

Hypertension

Typically the patient presents anxious and diaphoretic with tachycardia, miosis, excessive salivation, copious vomiting and profusely watery diarrhea by systems, the signs include the following:-

- HEENT: salivation, lacrimation, miosis, blurred vision
- Respiratory system: dyspnea, tachypnea, bronchorrhea, bronchospasm, and pulmonary edema
- CVS: tachycardia, bradycardia, hypotension, arrhythmias, conduction block
- GIT: nausea, vomiting, diarrhea, abdominal cramps
- Neurologic: anxiety, tremulousness, ataxia, muscle fasciculation, slurred speech, seizures, choreoathetosis, diminished level of consciousness with confusion and rarely coma in severe cases.

Diagnosis

Diagnosis can be made using four criteria:

- i) Appropriate history of exposure to an insecticide
- ii) Sign and symptom of excessive muscarinic and nicotinic stimulation
- iii) Depressed plasma and RBC cholinesterase
- iv) Response to atropine and pralidoxime therapy

Management

1. General measures

General supportive care should be provided including fluid and electrolyte replacement and intubation and artificial ventilation when necessary.

2. GI – decontamination

- Activated charcoal
- Gastric lavage

3. Dermal decontamination

Removal of clothes, washing with soap and water

4. Antidote administration

- a. Atropine is a competitive antagonist of Ach. Give 0.01 mg/kg IV every 15' until signs of atropinization occur(mydriasis,decreased secretion,tachycardia)
- b. Pralidoxime chloride (2 PAM) chemically breaks the bond between the organophosphate and the enzyme. It reverses both muscarinic and nicotinic effects of OP-poisoning. It should be administered in all cases of known/suspected OP-poisonings. 25 – 50 mg/kg IV should be given over 30-60 minutes time. it can be repeated in 1hr and then every 8-12 hrs until improvement is noted

11.2 Carbamate insecticides

Carbamate insecticides produce similar but mild form of toxicity to that of OP-poisonings.

Two important differences distinguish carbamates from OP-poisonings:

- i) Carbamate toxicity is typically short lived. Spontaneous hydrolysis of the carbamylated acetyl cholinesterase complex regenerates enzymatic activity, usually within 24 hrs.
- ii) Little or no CNS toxicity occurs because of inability of carbamates insecticides to cross the blood brain barrier.

Treatment

Some as OP-poisonings except that pralidoxime is not recommended in carbamates poisonings.

11.3 Hydrocarbons

Hydrocarbons include a wide array of chemical substances containing hydrogen and carbon atoms only and found in thousands of commercial products. Lighter fluid, paint thinners and removers, some furniture polishes, cleaning agents, solvents, various automotive products and ordinary fuels are common examples.

Pathophysiology

Hydrocarbons can be absorbed after ingestion, inhalation or dermal contact. Most cases involve accidental imbibing by young children. Abuse by inhalation also occurs in adolescents.

The most important toxic effects of hydrocarbons is aspiration pneumonitis. Aspiration occurs at the time of ingestion when choking and gassing occurs but can also be secondary to vomiting that commonly occurs after ingestion.

Small amounts (less than 1ml) of compounds with low viscosity such as mineral spirits, naphtha, kerosene, gasoline and lamp oil spread rapidly across surfaces and cover large areas of the lungs when aspirated and cause significant injury.

Pneumonitis doesn't result from dermal absorption of ingestion without aspiration. Other less common toxic effects of hydrocarbons include hepatotoxicity, renal toxicity, dysrhythmias and sudden death. Chronic abuse of volatile hydrocarbons can also lead to cerebral atrophy, neuropsychological changes and peripheral neuropathy.

Clinical and laboratory manifestation

Manifestations include

- Transient, mild CNS depression
- Coughing, mild to severe respiratory distress
- Fever and leukocytosis. The fever can stay for as long as 10 days
- Abnormality on CXR

Treatment

- Treatment is supportive and involves oxygen, intubations and mechanical ventilation when indicated. Antibiotics are not indicated routinely

11.4 Acetaminophen

Acetaminophen is the most widely used analgesic and antipyretic commonly available at home. Acetaminophen toxicity results from the formation of a highly reactive intermediate metabolite, N-acetyl p-benzoquinonemine (NAPQI) which in large amount causes hepatocellular necrosis and damage. The acute toxic dose in children less than 12 years of age is about 200mg/kg clinical and laboratory manifestation.

If untreated, patients who have acutely overdosed pass through four stages of toxicities.

Table 31: Stages of N-acetyl p-benzoquinonemine toxicity

Stage	Time after ingestion	Characteristics
I.	½ - 24 hr	Anorexia, nausea, vomiting, malaise, pallor, diaphoresis.
II.	24 - 48 hr	Resolution of above; RUQ abdominal pain and tenderness; elevated bilirubin, PT, hepatic enzymes; oliguria.
III.	72 - 96 hr	Peak liver function abnormalities; anorexia, nausea, vomiting, malaise may reappear.
IV.	4 day – 2 wk	Resolution of hepatic dysfunction or complete liver failure.

- Plasma level should be measured after 4 hours of ingestion and level plotted on monogram to determine whether antidote treatment is indicated.
- LFT including enzymes, bilirubin and PT should be followed daily

RUQ - right upper quadrant

PT - prothrombin time

Treatment

Treat with:

- Activated charcoal
- Antidote – N – acetylcysteine (NAC)

11.5 Salicylates

The incidence of salicylate poisoning has declined, particularly in young children, as the use of alternative antipyretics has increased.

Pathophysiology

Salicylates directly or indirectly affect most organ systems by:

1. Uncoupling oxidative phosphorylation
2. Inhibiting Krebs cycle enzymes
3. Inhibiting amino acid synthesis
4. Decreasing platelet adhesiveness
5. Increasing pulmonary capillary permeability

Stimulation of respiratory center → Hyperventilation → Respiratory alkalosis → compensatory alkaluria → ↑ urinary excretion of both K- & Na- HCO_3

↓ Serum K → exchange of H^+ for K^+ in the renal tubules → “paradoxical aciduria”

Clinical and laboratory manifestations

Chronic toxicity leads to signs and symptoms that mimic flu or other febrile illness.

Acute toxicity is manifested by::

- Nausea and vomiting 2^o to gastric irritation
- Dehydration and metabolic acidosis eventually develop
- CNS changes such as agitation, restlessness and confusion are important signs of serious toxicity
- Coma may develop 2^o to cerebral edema
- Pulmonary edema or hemorrhage may develop in more severe cases
- Hyperglycemia or hypoglycemia, particularly in infants
- Hepatotoxicity occurs both from chronic exposure and acute over dosage/toxicity.

Laboratory investigations

- Serial serum salicylate levels and urine pH and volume should be monitored
- Other plasma pH, glucose, K^+ , other electrolytes, clotting studies and liver function test

Treatment

- Rehydration and correction of electrolyte abnormalities
- Activated charcoal
- Alkalinization of urine using iv HCO_3
- Hemodialysis in severe cases

11.6 Co-Poisoning (co)

Co:- is a tasteless and colorless gas produced by incomplete combustion of carbon compounds. It binds to Hb, displacing O₂, with an affinity for Hb about 250 x that of O₂. It impairs the ability of Hb to release O₂ to tissues. It binds to cytochrome oxidase in tissues, impeding O₂ utilization

Clinical presentation

The clinical presentation of co-poisoning is only approximately correlated to the measured carboxy-Hb level. Early symptoms are usually misdiagnosed as flu-like illnesses. Signs and symptoms include:

- Levels of 10-20%: headache, dyspnea on exertion and weakness predominate
- Levels of 20 -30%: severe headache and nausea may appear
- Levels of 30-40%: severe headache, nausea, vomiting and impaired judgment
- Levels of 50-60%: confusion, syncope, seizures and coma predominate
- Levels of more than 60-70%: coma, seizure, cardio respiratory depression, death

Laboratory analysis

- Determination of carboxy-Hb. level in blood and expired air
- chest x-ray (30% of patients have an abnormal finding);
- EKG

Treatment

- General supportive care
- 100% O₂ by mask
- Hyper baric O₂ therapy

Prognosis

- 10% develop long term neuropsychiatric complications

Exercise

1. Discuss the management principles of poisoning.
2. Mention the clinical manifestation of a child with CO poisoning
3. Describe the management principles of a child with op-poisoning

References

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2. William W.Hay, et al, Current pediatric diagnosis and treatment, 15th edition, 2001
3. Peter Vicelli, Hand book of medical toxicology, 1993

ANNEX

Annex 1

Treatment plan A: Treat diarrhea at home

Counsel the mother on the 3 rules of home treatment:

Give extra fluid, continue feeding, when to return

1. GIVE EXTRA FLUID (as much as the child will take)



Tell the mother to breastfeed frequently and for longer at each feed.

- If the child is exclusively breastfed, give ORS or clean water in addition to breast milk.
- If the child is not exclusively breastfed, give one or more of the following: ORS solution, food-based fluids (such as soup, rice water, and yoghurt drinks), or clean water.

It is especially important to give ORS at home when:

- The child has been treated with plan B or plan C during this visit.
- The child cannot return to a clinic if the diarrhea gets worse.



TEACH THE MOTHER HOW TO MIX AND GIVE ORS. GIVE THE MOTHER TWO PACKETS OF ORS TO USE AT HOME.



SHOW THE MOTHER HOW MUCH FLUID TO GIVE IN ADDITION TO THE USUAL FLUID INTAKE.

Up to 2 years 50 to 100 ml after each loose stool

2 years or more 100 to 200 ml after each loose stool

Tell the mother to:

- Give frequent small sips from cup.
- If the child vomits, wait 10 minutes. Then continue, but more slowly.
- Continue giving extra fluid until the diarrhea stops.

**2. CONTINUE FEEDING
3. WHEN TO RETURN } SEE MOTHER'S CARD**

2. Treatment plan B: Treat some dehydration with ORS

Give in clinic recommended amount of ORS over 4-hour period

⇒ **DETERMINE AMOUNT OF ORS TO GIVE DURING FIRST 4 HOURS.**

Age*	Up to 4 months	4 months up to 12 months	12month up to 2 years	2 years up to 5 years
Weight	< 6 kg	6 -< 10 kg	10 -<12kg	12 - 19kg
Amount in ml	200-400	400-700	700-900	900-1400

*Use the child's age only when you do not know the weight. The approximate amount of ORS required (in ml) can also be calculated by multiplying the child's weight (in kg) by 75.

- If the child wants more ORS than shown, give more.
- For infants under 6 months who are not breastfed, also give 100-200 ml clean water during this period.

⇒ **SHOW THE MOTHER HOW TO GIVE ORS SOLUTION.**

- Give frequent small sips from a cup.
- If the child vomits, wait 10 minutes. Then continue, but more slowly.
- Continue breastfeeding whenever the child wants.

⇒ **AFTER 4 HOURS:**

- Reassess the child and classify the child for dehydration.
- Select the appropriate plan to continue treatment.
- Begin feeding the child in clinic.

⇒ **IF THE MOTHER MUST LEAVE BEFORE COMPLETING TREATMENT:**

- Show her how to prepare ORS solution at home.
- Show her how much ORS to give to finish 4-hour treatment at home.
- Give her enough ORS packets to complete rehydration. Also give her 2 packets as recommended in plan A.

1. GIVE EXTRA FLUID
2. CONTINUE FEEDING
3. WHEN TO RETURN

Treatment plan C: treat severe dehydration quickly

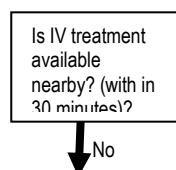
⇒ FOLLOW THE ARROWS. IF THE ANSWER IS "YES", GO ACROSS. IF "NO", GO DOWN.

- Start IV fluid immediately. If the child can drink, give ORS by mouth while the drip is set up. Give 100 ml ringer's lactate solution (or, if not available normal saline), divided as follows.

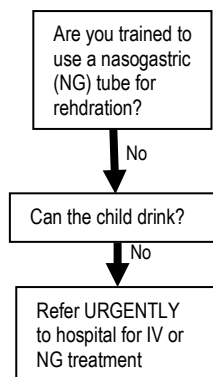
Can you give intravenous (IV) fluid immediately?	Yes	Age	First give 30 ml/kg in:	Then give 70 ml/kg in:
		Infants (under 12 months)	1 hour*	5 hours
		Children (12 months up to 5 years)	30 minutes*	2 1/2 hours

*Repeat once if radial pulse is still very weak or not detectable.

- Reassess the child every 1-2 hours. If hydration status is not improving, give the IV drip more rapidly.
- Also give ORS (about 5 ml/kg/hour) as soon as the child can drink: usually after 3-4 hours (infants) or 1-2 hour (children).
- Reassess an infant after 6 hours and a child after 3 hours. Classify dehydration. Then choose the appropriate plan (A, B or C) to continue treatment.



- Refer **urgently** to hospital for IV treatment.
- If the child can drink, provide the mother with ORS solution and show her how to give frequent sips during the trip.



- Start rehydration by tube (or mouth) with ORS solution: give 20 ml/kg/hour for 6 hours (total 120 ml/kg).
- Reassess the child every 1-2 hours:
 - If there is repeated vomiting or increasing abdominal distension, give the fluids more slowly.
 - If hydration is not improving after 3 hours, send the child for IV therapy.
- After 6 hours, reassess the child. Classify dehydration. Then choose the appropriate plan (A, B or C) to continue treatment.

NOTE: If possible, observe the child for atleast 6 hours after rehydration to be sure the mother can maintain hydration giving the child ORS solution by mouth.

Drug	Supplied	Dose	Comments
Section 1			
Anesthetics			
Atropine	Vial: 0.1 mg/mL, 0.4 mg/mL, 0.6 mg/mL	Preoperative: 0.01 mg/kg i.v., i.m., p.o. Maximum dose 0.4 mg. CPR bradycardia: 0.05 mg/kg every 2-5 minutes.	Well absorbed. May be given any route including via endotracheal tube. Overdose- agitation, urinary retention, photophobia, tachycardia.
Diazepam	Tablet: 2 mg, 5 mg, 10 mg Ampoule: 10 mg/2 mL Liquid: 1 mg/mL	Sedation: 0.1-0.3 mg/kg i.v. 0.5-0.8 mg/kg p.o. Status epilepticus: 0.3 mg/kg i.v. 0.5 mg/kg p.r.	Irritating to veins. Do not mix with other intravenous drugs. Respiratory depression in high doses. In status epilepticus, follow with a long-acting drug such as phenobarbitone.
Lidocaine	Vial: 1% (10 mg/mL),	Arrhythmias: 1 mg/kg load, 20-50 µg/kg per minute infusion. Therapeutic range 4.5-21 µmol/L. Local infiltration: Maximum doses: 7 mg/kg (with adrenaline), 4.5 mg/kg (without adrenaline).	In high dose, associated with seizures, hypotension, heart block. Beware exceeding maximum dose when infiltrating a large area.
Oxygen		1-5 L/min by nasal prongs or mask.	Neonats: prologed high doses associated with retineal changes. Keep O2 saturateion <95% in permature infants. Chance of suppression of hypoxic drive very rare in children.

Section 2**Antirheumatic
and analgesic
drugs**

Acetysalicylic acid	Chewable: 80 mg Tablet: 325 mg, 650 mg suppository: 150 mg, 650 mg	Antipyretic: 10-15 mg/kg per dose P.O. every 4-6 hours. Maximum dose 60-80 mg/kg per 24 hour. Antirheumatic: 60-100 mg/kg per 24 hour : 6 hourly.	Bleeding risk, particularly with platelet and coagulation disorders. Severe poisoning in overdose. Monitor levels when using high dose. Therapeutic range 1.5-2.0 mmol/L.
Ibuprofen	Tablet: 200 mg, 300 mg, 400 mg, 600 mg	Antipyretic: 20 mg/kg per 24 hour - 8 hourly p.o. Antirheumatic: 20-35 mg/kg per 24 hour - 8 hourly p.o.	Gastritis, platelet dysfunction, decreased renal function in small children.
Morphine	Tablet: 100 mg, 500 mg	10-20 mg/kg per dose p.o. or p.r. every 4-6 hours.	Severe liver toxicity in overdose, particularly if mixed with alcohol. Do not exceed maximum daily dose.
Paracetamol	Suppository: 100 mg Syrup: 125 mg/mL	Maximum 90 mg/kg per 24 h.	

Section 3**Antiallergics**

Chlorpheniramine	Tablet: 4 mg Liquid: 0.5 mg/mL Ampoule: 10 mg/mL	0.35 mg/kg per 24 hour - 4 doses, p.o., i.v. or i.m.	Sedating, hypotension if i.v. given quickly, Movement disorder with cumulative, high doses. Many alternative drugs.
Adrenalin	Tablet; 10 mg, 20 mg	Physiological replacement: 14-20 mg/m ² per 24 hour - 3	4 mg hydrocortisone is equivalent to 1 mg prednisone.
Hydrocortisone			

	Ampoule: 100 mg, 250 mg, 500 mg Cream; 0.55, 1.0% Ophthalmic: 2.5% Suppository: 10 mg, 25 mg	doese p.o. Topical: Apply 2-3 times daily sparingly. Anti-inflammatory: 0.8-4.0 mg/kg per 24 hour - 4 doses i.v. or p.o.	Hypertension, immunosuppression, electrolyte abnormalities in prolonged use. significant absorption may occur with topical use on a raw skin.
Prednisone	Tablet: 1 mg, 5 mg, 50 mg Liquid: 2 mg/mL, 5 mg/mL	Physiologic replacement: 4-5 mg/m ² per 24 hour - 2 doses p.o. Anti-inflammatory: 0.5-2.0 mg/kg per 24 hour - 2 doses p.o. Wean to alternate day doses if possible for long-term treatment.	Prednisolone better for hepatic disease since not metabolized in liver. usual steroid side effects in chronic use.
Section 4 poison antidoes			
Activated charcoal	Powder	1-2 g/kg in water or sorbitol given p.o.	Do not give with emetic or will absorb the drug. If giving by nasogastric tube in comatose child, ensure nasogastric tube is definitely in the stomach and not lung.
Naloxone	Ampoule: 0.4 mg/mL, 1.0 mg/mL	0.05-0.1 mg/kg per dose. Maximum 2 mg/dose. Repeat 3-5 minutes until effect achieved. If no response after 3 doses, sedation probably not	Half life only 1-2 hours. Repeat doses will be required before original narcotic overdose has worn off.

due to narcotic overdose.

Section 5

Anticonvulsants

Clonazepam	Tablet: 0.5 mg, 2 mg	Initially, 0.01-0.05 mg/kg per 24 hour - 3 doses p.o. Increase at weekly intervals to maximum 3 mg/kg per 24 hour.	Increase gradually. Do not stop suddenly, Drowsiness common, also slurred speech, ataxia and salivation.
Diazepam	Capsul: 250 mg	20-30 mg/kg per 24 hour - 1-2 doses p.o. Increase dose slowly at 4-7 day intervals. Maximum dose 1500 mg/24 h.	Nausea, gastritis. Rare bone marrow depression and hepatitis. Therapeutic range 280-700 µmol/L.
Ethosuximide	Liquid: 50 mg/mL		
Phenobarbital	Tablet: 15 mg, 30 mg, 60 mg, 100 mg Liquid: 5 mg/mL Ampoule: 30 mg/mL, 120 mg/mL	Status epilepticus: After control with diazepam, load with 18 mg/kg by slow i.v. infusion. Maximum dose, if seizures continue, 30 mg/kg. Chronic dose: 5 mg/kg per day - 2 doses p.o. but needs adjusting with age.	Respiratory depression, hypotension. Therapeutic range 65-170 µmol/L.
Phenytoin	Tablet: 50 mg ; capsule: 100 mg; Suspension: 6 mg/mL, 25 mg/mL Ampoule: 50 mg/mL	Status epilepticus: After control, load with 18 mg/kg by slow i.v. infusion. Maintenance: 5-10 mg/kg per 24 hour - 2 doses p.o. Absorption very variable in children.	Therapeutic range 40-80 µmol/L. Risk of hypotension is appreciable if given by fast infusion. Chronic effects include hirsutism, gum hyperplasia, rash, ataxia. Suspension very variably absorbed.

Valproic acid	Capsule: 125 mg, 250 mg, 500 mg Liquid: 50 mg/mL	initially, 10 mg/ kg per 24 hour - 3 doses p.o. Increase 5-10 mg/kg weekly to maximum range 30-60 mg/kg per 24 hour.	Therapeutic range 350-700 μ mol/L., Valproic acid may affect levels of other anticonvulsants if added. Gastritis, tremor, rare hepatitis.
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Section 6.1.1. Intestinal anthelminthics

Albendazole	Tablet: 400 mg, 200 mg	Hydatidosis or cysticercosis: 15 mg/kg per day - 2 doses p.o. for 28 days. Maximum 400 mg b.i.d. Hookworm or ascariasis: 400 mg once p.o.	Occasionally diarrhea, abdominal pain, migration of Ascaris through mouth.
Levamisole	Tablet: 150 mg, 50 mg	Hookworm or Ascaris: 2.5 mg/kg once p.o.	Low toxicity, low cost.
Mebendazole	Tablet: 100 mg Suspension : 20 mg/mL	Hookworm, Ascaris, Trichuris: 100 mg b.i.d x 3 days p.o. for all ages over 2 years.	Occasional diarrhea and abdominal pain.
Niclosamide	Tablet: 500 mg	Tapeworms: <2 yr: 500 mg once p.o. 2-6 yr: 1 g > 6 yr: 2 g	Minor side effects. Drug not absorbed.
Praziquantel	Tablet: 150 mg, 600 mg	S.mansoni: 40 mg/kg per day - 2 doses. s.japonicum: 60 mg/kg per day - 3 doses. Cysticercosis: 50 mg/kg per	Frequently headache, dizziness. Occasionally abdominal pain, fever, sweating, rash, Severe headache associated with

		day - 3 doses for 2-4 weeks.	treatmetn of
		Tapeworms: 20 mg/kg p.o. once.	neurocysticercosis.
		Flukes: 75 mg/kg per day - 3 doses p.o. for 1 day.	
Pyrantel pamoate	Tablet: 250 mg Suspension: 50 mg/mL	Hookworm or ascariasis: 11 mg/kg once p.o. maximum 1 g.	Occasional abdominal pain, headache, rash, fever.
Section 6.1.2			
Antifilarials			
Diethylcarbamazine	Tablet: 100 mg, 50 mg	Filariasis, pulmonary eosinophilia: 6 mg/kg per day - 3 doses for 21 days. Start with 0.5 mg/kg on day 1 and increase to full dose over 4-5 days as tolerated.	Drug effects include headache, nausea, vomiting, fever, rare encephalopathy. Rapid killing of large filarial loads associated with fever and allergiv reactions that may be severe.
Ivermectin	Tablet: 6 mg, 3 mg	O.Volvulus: 150 µg/kg p.o. once. Repeat every 6-12 months.	Occasional Mazzotti reactions due to reaction to death of microfilariae with fever, facial swelling, joint pain.
Section 6.2			
Antibacterials			
Amoxicillin	Capsule: 500 mg, 250 mg Tablet: 250 mg, 125 mg Suspension: 50 mg/ mL	40 mg/kg per 24 hour - 3 doses p.o.	Absorbed as ampicillin. Achieves better serum levels than oral ampicillin, Side effects as for ampicillin.
Ampicillin	Powder: 125	Severe infections:	Amoxicillin preferred for oral

	mg, 250 mg, 500 mg, 1 g, 2g Capsule: 500 mg, 250 mg Suspension: 50 mg/mL	200-400 mg/kg per 24 hour - 4 doses i.v. Maximum 2 g. Repeated 6 hourly.	therapy Except when treating diarrhea caused by shigella or Salmonella. Rashes common, particularly with EBV infection.
Amoxicillin and clavulanic acid	Tablet: 500 mg, 250 mg suspension: 125 mg/mL. 250 mg/mL	40 mg/kg per 24 hour - 3 doses p.o.	Clavulanic acid extends coverage to include beta-lactamase producing H. influenzae and E. Coli. Main side effect is gastrointestinal upset and diarrhea.
Benzyle penicillin	Powder: 1mIU, 5 mIU, 10 mIU	Sever infections: 200000 u/kg per 24 hour - 3 doses i.v.	For injection only. Well tolerated. Rare sensitivity reactions and serum sickness.
Cefotaxime	Powder: 500 mg, 1 g, 2g	100-200 mg/kg per 24 hour - 3 doses i.v.	Rare sensitivity reactions and serum sickness.
Chloramphenicol	Capsule: 250 mg Suspension: 50 mg/mL Powder: 1 g Eye drops: 1%	50 - 75 mg/kg per 24 hour - 4 doses i.v. or p.o.	Oral route well absorbed. Dose related and idiosyncratic bone marrow depression.
Cloxacillin	Capsule: 500 mg, 250 mg Suspension: 25 mg/mL Powder: 500 mg, 1 g, 2 g	100-200 mg/kg per 24 hour - 4 doses i.v.	Oral form bitter. Long-term, high dose therapy associated with urinary cast formation.

Doxycycline Erythromycin	Tablet: 250 mg Capsule: 250 mg Suspension: 50 mg/mL Powder: 500 mg, 1g Eye ointment: 0.5%	Oral: 30-40 mg/kg 24 hour - 3 doses p.o. Intravenous: 20-50 mg/kg per 24 hour - 4 doses i.v.	Well absorbed by oral route. Intravenous form is very irritating and needs diluting and slow infusion. Gastrointestinal upset with oral form. Do not give i.m.
Gentamicin	Vial: 20 mg, 80 mg Eye ointment: 0.3%	7.5 mg/kg per 24 hour - 3 doses i.v. or i.m.	Ototoxic and nephrotoxic in high doses.
Metronidazole	Tablet: 250 mg Suspension: 50 mg/mL Vial: 500 mg in 100 ML	C. difficile: 20 mg/kg per 24 hour 4 doses i.v. or p.o. Amebiasis: 35-50 mg/kg per 24 hour - 3 doses p.o. x 10 days. Giardiasis: 15 mg/kg per 24 hour - 3 doses for 5 days. Intravenous: 30 mg/kg per 24 hour 3-4 doses	Frequently nausea, headache and metallic taste. Occasional "antabuse" effect with alcohol.
Sulfamethoxazole /trimethoprim	Tablet: 400 mg/80 mg, 200 mg/40 mg Suspension: 40 mg/8 mg per mL	PCP. Toxoplasmosis: 80 mg/16 mg SMX/TMP per kg/24 hour - 3 or 4 doses p.o. or i.v. 2-3 weeks> Prophylaxis: 40 mg/8mg SMX/TMP per kg - 2 doses p.o. 3 d/wk.	Only use i.v. form if p.o. not tolerated

	Vial: 80 mg/16 mg per mL		
Rifampin	Capsule: 300 mg, 150 mg	Leprosy: >35 kg: 600 mg supervised monthly 20-35 kg: 450 mg supervised monthly 12-20Kg: 300 mg supervised monthly < 12 kg: 150 mg supervised monthly Tuberculosis: 10 mg/kg daily unsupervised 15 mg/kg: 3 d/wk supervised Meningitis prophylaxis: 10 mg/kg per 24 hour - 2 doses p.o. for 4 days. Maximum 600 mg/d.	Discolors body fluids. May cause liver impairment in long-term use. Reduce dose if liver disease. See "leprosy."

Section 6.2.4

Antituberculosis

Ethambutol	Tablet: 100 mg, 400 mg	Tuberculosis: 15 mg/kg daily unsupervised p.o. 30 mg/kg 3 d/wk supervised.	Optic neuritis, red/green color blindness, peripheral neuritis. Reserved for use in children old enough for eye testing. See "Tuberculosis."
Isoniazid	Tablet: 300 mg, 100 mg	Tuberculosis: 5-10 mg/kg 9maximum 300 mg) P.O. daily unsupervised. 15 mg/kg p.o. 3 d/wk supervised 9maximum 600 mg).	Hepatitis rare in children. Peripheral neuropathy, prevented with pyridoxine daily, also rare in children. Occasional hypersensitivity reactions.

			See "Tuberculosis."
Pyrazinamide	Tablet: 500 mg	Tuberculosis: 35 mg/kg p.o. daily unsupervised. 50 mg/kg p.o. 3 d/wk supervised.	Hepatitis, Jaundice, hyperuricemia, diabetes.
Rifampin			See section 6.2.3. See "tuberculosis."
Streptomycin	Powder: 500 mg, 1 g	Tuberculosis: 15-20 mg/kg per 24 hour - 2 doses i.m. Maximum 1 g/d.	Ototoxic and nephrotoxic. See "Tuberculosis." Also drug of choice for brucellosis in conjunction with doxycycline.
Griseofulvin	Tablet: 250 mg, 125 mg Suspension: 25 mg/mL	15 mg/kg per 24 hour once p.o. Treatment course 4-6 weeks.	No effect given topically. Well tolerated. Take with food-absorption improved.
Ketoconazole	Tablet: 200 mg Suspension: 20 mg/mL	5-10 mg/kg per 24 hour p.o. - 2 doses.	Risk of hepatitis in courses longer than 14 days. Interferes with hepatic metabolism of several other drugs.
Nystatin	Suspension: 100 000 u/mL Cream: 100 000 u/g	Apply cream liberally to affected area. Suspension: 200 000 u/dose p.o. 4 times daily.	Not absorbed when given p.o.
Sodium Stibogluconate	Bottle: contains equivalent of 100 mg Sb/mL	20 mg/kg Sb base i.m. or i.v. for 3-4 weeks	Numerous side effects. If giving i.v., use slow infusion or risk of thrombosis. ECG changes and conduction

Chloroquine	Tablet: 100 mg, 150 mg (Chloroquine base) Syrup: 10 mg/mL (chloroquine base)	Treatment: 10 mg/kg base one then 5 mg/kg base at 6, 24 and 48 hours. Prophylaxis: 5 mg/kg base (maximum 300 mg base) once weekly.	Doses calculated as chloroquine base. Not for prophylaxis if epilepsy. P. falciparum resistance widespread. Corneal opacities and retinal damage in chronic high dose, dermatoses, see "Malaria." do not give with halofantrine.
Mefloquine	Tablet: 250 mg	Malaria prophylaxis treatment: 25 mg/kg (maximum 1.5 g) - 1 or 2 doses. 15-19 kg: 1/4 tab weekly 20-30 kg: 1/2 tab weekly 31-45 kg: 3/4 tab weekly >45 kg: 1 tab weekly	Do not give with, or after, halofantrine; risk of severe arrhythmias. GI effects and a range of neuropsychiatric problems from insomnia to rare convulsions and psychosis. Not for use in patients with epilepsy.
Primaquine	Tablet: 15 mg, 7.5 mg	0.3 mg/kg base per day for 14 days.	Hemolysis in patients with G-6PD. Calculated as active base. Only for eradication of P.vivax and P.ovale. Occasional methemoglobinemia.
Quinine salts	Tablet: 300 mg, 200 mg (sulfate) Ampoule: 300 mg/mL (dihydrochloride)	Oral: 25 mg/kg per 24 hour - 3 doses p.o. for up to 7 days in partially resistant areas. If quinine resistance suspected, follow by: doxycycline, clindamycin or	Not for prophylaxis, only treatment. Side effects common at usual dose cinchonism (tinnitus, headache, nausea, visual disturbance). Monitor glucose: hypoglycemia common.

fransidar.

Dihydrochloride:

20 mg/kg loading dose in 5% dextrose over 4 hours, then

10 mg/kg over 2-4 hours.

Repeat 8 hourly until oral therapy can be started.

Arrhythmias, especially if given

by fast i.v. push. See 'Malaria

"Do not give with halofantrine.

Section 10**Antianemia**

Folic Acid

Tablet: 1 mg,
5 mg

Vial: 5 mg/mL

Deficiency: 0.5 - 1.0 mg/kg per
24 hour p.o.

Maintenance:

<1 yr: 30-45 µg/24 h

1-3 yr: 100 µg/24 h

4-6 yr: 200 µg/24 h

2-10 yr: 300 µg/24 h

> 10 yr: 400 µg/24 h

Will not prevent neurologic
abnormalities due to vitamin
B₁₂ deficiency.

Iron

Gluconate,

Tablet: 300

mg (35 mg

Fe)

Syrup: 60

mg/mL (7 mg

Fe), Sulfate

Tablet: 300

mg (60 mg

Fe)

Syrup: 30

mg/mL (6 mg

Fe)

Drops: 125

Deficiency: 3-6 mg(Fe)/kg
per 24 hour - 2 or 3 doses
p.o.Prophylaxis: 1-2 mg (Fe)/kg
per 24 hour - 1-2 doses
p.o. (maximum 15 mg/24 h
fe).Calculated dose depends on
iron base. Liquid may stain
teeth. Absorption decreased
by tetracycline and antacides.
Give with meals to decrease
gastrointestinal upset. Very
toxic in overdose.

	mg/mL (25 mg Fe)		
Vitamin K ₁	Ampoule: 1 mg/0.5 mL, 10 mg/mL	Neonatal prophylaxis: 0.5-1.0 mg.i.m., S.C. or i.v. Sever liver disease: 5 mg/24 h i.v. or i.m. Deficiency: 1-2 mg/dose i.v. 2-5 mg/dose p.o.	Intravenous dose rarely associated with allergic reactions. The risk is less with vitamin K ₁ compared to other vitamin K preparations.

Section 11

Plasma and substitutes

Dextran 70	500 mL bags of 6% solution	Resuscitate with a maximum of 20 mL/kg.	Several starch and gel based plasma substitutes exist. Dextran 70 is as effective as fresh plasma for resuscitation of severe shock in burns and sepsis. rare allergic reactions.
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Section 12

Cardiovascular

Captopril	Tablet: 6.25 mg, 12 mg, 25 mg, 50 mg, 100 mg	Children: Start 0.6 mg/kg per 24 hour - 3 doses p.o. Increase to effect. Maximum 150 mg/d.	Start with small test dose and observe for hypotension.
Digoxin	Tablet: 62.5 µg, 125 µg, 250 µg Syrup: 50 µg/mL Ampoule: 50 µg/mL, 250 µg/mL	Digitalizing dose: Give as 1/2, 1/4, 1/4 separated by 8 hours. < 2 yr: 40 µg/kg load - 3 doses p.o. > 2 yr: 30 µg/kg load - 3 doses p.o. Maintenance dose:	Well absorbed orally. Intravenous dose is 75% of p.o. dose. Do not give i.m. Reduce dose in renal failure. Toxicity is enhanced by hypokalemia. High doses - visual disturbances, arrhythmias.

		Start 12 hours after loading dose completed. <2 yr: 10-15 µg/kg per 24 hour - 2 doses p.o. >2 yr: 5-10 µg/kg per 24 hour - 2 doses p.o.	
Adrenalin	Ampoule: 1/1000 (1 mg/mL) 1/10000 (0.1 mg/mL)	Anaphylaxis: Use 1/1000 strength given s.c. 0.01 mL/kg per dose (maximum 0.3 mL). repeat every 10 minutes. CPR: Use 1/10000 strength. give i.v. 0.1 mL/kg per dose. Repeat every 3-5 minutes. Inotrope: Mix 0.3 mg adrenalin/kg in 50 mL D5W. 1mL/h = 0.1 µg/kg per minute. Dose range 0.1-0.5 µg/kg per minute.	Preferably give infusion via central line. Absorbed intraosseously or via endotracheal bolus. Hypertension, arrhythmias, nervousness, headache.
Hydralazine	Tablet: 10 mg, 25 mg, 50 mg Powder: 20 mg/mL	Acute hypertension: 0.1-0.5 mg/kg per dose i.m. or i.v. 4-6 hourly. Chronic: 0.75-3.5 mg/kg per 24 h-3-4 doses. Start low and increase as needed.	Oral dose may take 3-4 days for full effect. Intravenous dose should be given slowly. High dose associated with lupus-like syndrome.
Methyldopa	Tablet: 125 mg, 250 mg Vial: 50 mg/mL	Acute hypertension: 5-10 mg/kg per dose 4-6 hourly. Maximum 65 mg/kg per 24 h. Chronic: 10 mg/kg per 24 h-3-4 doses. Increase every 2	Positive Coombs test common. fever, liver dysfunction and hemolysis.

		days as needed to maximum 65 mg/kg per 24 hour.	
Nystatine	Rinse: 1%	Head lice:	Avoid contact with eyes and
Permethrin	Skin cream: 5% Tube: 30 g	Apply rinse after shampooing. Cover scalp then washoff after 10 minutes. Scabies: < 2 yr: 1/4-1/8 tube 2-5 yr: 1/4 tube 5-12yr: 1/2 tube >12 yr: 1 tube	raw skin. Care required for treating children <2 yrs because of risk of absorption, Stinging and rash, May require second treatment in 1 week. Prolonged itching may occur despite successful treatment.
		Massage cream on skin from head to foot. Wash off after 12 hours.	
Silver sulfadiazine	Cream: 1%, Jar 500 g Tube; 30 g	Once daily burn treatment- less frequent for other wounds.	If large areas covered, three may be significant sulfonamide and silver absorption.

Section 14**Diagnostic
agents****Section 15****Disinfectants**

Chlorhexidine	Solution: 5% for dilution	Apply liberally, topically.	Not for irrigating body cavities. Avoid eyes. Occasional sensitivity.
Povidone/Iodine	Solution: 10%	Apply undiluted topically as a surgical scrub.	Absorption through large wounds may be associated with acidosis, electrolyte abnormalities and high iodine levels. Contraindicated in

premature infants.

Section 16**Diuretics**

Furosemide	Tablet: 20 mg, 40 mg Liquid: 10 mg/mL Ampoule: 10 mg/mL	Oral: 3-6 mg/kg per 24 hour 3-4 doses p.o. Intravenous: 0.5-1.0 mg/kg per dose i.v. Infusion: 0.1-1.0 mg/kg per hour. Maximum dose 6 mg/kg.	Hypokalemia, alkalosis, dehydration, hyperuricemia. Ototoxicity in high dose.
Hydrochlorothiazide	Tablet: 25 mg, 50 mg	1-2 mg/kg per 24 hour - 2 p.o. Maximum 100 mg/d.	Monitor electrolytes in prolonged dose.
Spironolactone	Tablet: 25 mg, 100 mg	1-4 mg/kg per 24 hour - 2-4 doses p.o.	Possible K ⁺ retention. Beware if using potassium supplementaiton. contraindicated in renal failure.

Section 17**Gastrointestinal**

Aluminum hydroxide	Tablet: 500 mg Liquid: 4%	Adult: 1-2 tabs, 4 times daily. Child: Up to 5 mL 3 times daily.	Aluminium accumulation does not occur if renal function normal. Constipation.
Cimetidine	Tablet: 200 mg, 300 mg Liquid: 60 mg/mL Ampoule: 150 mg/mL	Oral: 20-40 mg/kg per 24 hour - 4 doses before meals. Intravenous: 20-40 mg/kg per 24 hour - 4 doses.	Diarrhea, rash, gynecomastia, dizziness. May increase serum levels of phenytoin and theophylline.
Magnesium hydroxide	Liquid: 5.5%	Child: 5 mL 4 times daily.	Laxative.
Metoclopramide	Tablet: 5 mg,	GE reflux: 0.1 mg/kg per	Cisapride has fewer side

	10 mg	dose before meals.	effects as a motility agent.
	Liquid: 1 mg/mL	Maximum 0.5 mg/kg per 24 hour.	Chronic tardive dyskinesia seen with long-term use.
	Ampoule: 5 mg/mL	Antiemetic: 1-2 mg/kg per dose 4-6 hourly. Maximum 10 mg/kg per 24 hour. Give with diphenhydramine, 1 mg/kg per dose to prevent dystonia.	
Section 18			
Hormones			
Intermediate insulin	Vial: 40 u/mL, 100 u/mL	Given in mixture with soluble insulin. May be mixed in one syringe. Ratio is usually: 2/3 dose intermediae and 1/3 soluble.	Several preparations but all longer acting (onset 1-2 hours, peak >4 hours). Local reactions, fat hypertrophy and hypoglycemia in overdose. Not for use in infusion.
Soluble insulin	Vial: 40 u/mL	Varies but starts 0.25 u/kg per 24 hour. usually final dose 0.75-1.0 u/kg per 24 hour s.c. Infusion: 0.1 u/kg per hour for treatment of DKA.	Several preparations but all short acting (onset 30-60 minutes, peak 2-4 hours). Local reactions, fat hypertrophy and hypoglycemia in overdose.
Chloramphenicol eye drops	Drops: 0.5% Ointment: 1.0%	Apply 4 times a day.	Stinging sensation. Very rare aplastic response.
Gentamicin eye drops	Drops: 0.3% Ointment: 0.3%	Apply 4 times a day.	well tolerated.
Section 25			
Antiasthma			
Aminophylline	Vial: 25 mg/mL	Load: 6 g/kg i.v. over 20 minutes. Reduce dose by 50% if child taking oral theophylline. Infusion: 0.5-1.0 mg/kg per	theophylline in stabilized form for intravenous use. Only give i.v. since irritating. Numerous side effects.

		hour.	Check level if available. Keep 55-110 $\mu\text{mol/L}$. Required infusion dose depends on many variables including age. Arrhythmias, seizures, gastrointestinal upset. device-may be made easily with a plastic cup or bottle. with practice, may be used by children of 1-2 years of age.
Adrenalin	Ampoule: 1/1000 (1 mg/mL) 1/10 000 (0.1 mg/mL)	For acute asthmatics with very poor air entry. Subcutaneous: 1/1000 strength. Give 0.01 mL/kg per dose (maximum 0.3 mL). Repeat every 10 minutes. Intravenous: Only in dire emergency. Use 1/10 000 strength. Give 0.1 mL/kg per dose (maximum 1 mL). Dilute and give slowly. Nebulizer: Dilute 1 mL of 1/1000 to 4 mL with saline. May give continuously initially until controlled. Monitor closely. Also for bronchiolitis.	Nebulized adrenalin is a strong bronchodilator. Its alpha effects may improve air entry in children <1 year of age who do not respond to salbutamol. In emergency, nebulized sdrenalin may be given safely if monitored carefully. Aerosol treatmetn is often of some value in small children with bronchiolitis.
Salbutamol	Tablet: 2 mg, 4 mg Syrup: 2 mg/5 mL Inhaler: 100	Acue: Nebulized salbutamol may be given continuously in emergency. Cut back to 2-4 hourly when controlled. Exercise asthma: 1	Oral form of little use in wheezy children. Systemic effects-tremor, hyperactivity common. Chronic abuse and overdose associated with sever side

	µg Nebulizer: 2.5 mg 5.0 mg	inhalation 20 minutes before sputum. Give with spacing device. Chronic: 1 puff twice a day to be given before anti- inflammatory medication.	effects -arrhythmias, increased risk of fatal attacks. Should be used with an anti-inflammatory in most cases.
Section 27 Vitamins and minerals Ascorbic acid (Vitamin C) Glucose	Tablet: 100 mg, 250 mg, 500 mg 5%, 10%: 500 mL, 1000 mL 25%, 50%: 25 mL	Scurvy: 100-200 mg/24 h p.o. at least 2 weeks. Maintenance: 35-45 mg/24 hour p.o. Hypoglycemia: (25%, 50%) 0.5 -1.0 mL/kg i.v. Maintenance: (5%, 10%) For maintenance fluid requirements, use 10% in neonates. Add Na ⁺ (20 mEq/L) and K (15 mEq/L) per liter to avoid electrolyte abnormalities.	Diarrhea with very large doses. Follow acute treatment for hypoglycemia with maintenance glucose. Do not use glucose solutions alone for fluid management or risk of serious electrolyte abnormalities. Numerous premixed glucose and electrolyte solutions exist.
Iodine	oil: 480 mg/mL Capsule: 200 mg	Single treatment of 1mL, capsule p.o. provides sufficient iodine for many months.	Not for long-term treatment or risk of serious side effects. Iodine deficiency better treated with mass dietary supplementation programs.
Nicotinamide	Tablet: 50 mg	Pellagra: 50 mg b.i.d.p.o for 1 week	Nutritional rehabilitation necessary after initial treatment.
Pyridoxine (Vitamin B6)	Tablet: 25 mg, 100 mg Ampoule: 100	Deficiency: 50 mg/24h - 2 doses. Isoniazid neuropathy	Chronic treatment with high doses can cause a neuropathy (>2 g/24 h).

	mg/mL	prophylaxis: 1-2 mg/kg per 24 h p.o seizures: 50 mg i.v. single dose under EEG monitoring, then 200 mg/24 h for 2 weeks.	Pyridoxine is widespread. Deficiency indicates severe malnutrition. Isoniazid neuropathy is rare in children.
Retinol (Vitamin A)	Tablet: 10 000 U Capsule: 200 000U Oil: 100 000 u/mL Ampoule: 50 000 u/mL	Malnutrition: 200 000 units p.o. if evidence of keratomalacia. Repeat in 2 days. Maintenance: 200 000 unit capsules 6 monthly to children in high risk areas. Diarrhea, measles: 200 000 units on admission.	Dangerous in overdose. Not for high dose chronic treatment. Birth defects associated with high maternal doses. Do not exceed 10 000 units daily in women of reproductive age. Single dose treatment improves outcome in diarrhea and measles.
Riboflavin (vitamin B ₂)	Tablet: 5 mg	maintenance: 1-2 mg/24 h.	Isolated deficiency rare-usually part of global malnutrition. usually forms part of a multivitamin preparation.
Sodium chloride	0.9%: 500 mL, 1000 mL 3%: 10mL vials	0.9%: 10-30 mL/kg may be given safely for fluid resuscitation. Larger volumes may be necessary but require close monitoring. 3%: Each mL contains 0.5 mEq Na ⁺ . Give in boluses of 2-5 mL/kg for symptomatic hyponatremia. Replace slowly.	Hypernatremia, fluid overload in overdose. Do not attempt to replace Na ⁺ deficit rapidly.
Thiamin	Tablet: 50mg,	Beriberi: 1 mg/kg per 24 hour	Rare allergic reactions to

(Vitamin B1)	25 mg	p.o. or i.v.	intravenous form.
	Ampoule: 100 mg/mL	Maintenance: 1-2 mg/24 h.	